MUHS final-year BPTh (IV BPTh), 2012 syllabus

Community Physiotherapy

Complete exam notes in syllabus order: health promotion, women's health, geriatrics, rehabilitation, disaster management, industrial health, the project synopsis and exam skills.



Built from	20 MUHS question papers, Summer 2016 to Summer 2026 (code 51414)
Arrangement	The order of the MUHS syllabus, so the notes follow the order your college teaches
Contents	27 pages with past questions, worked long cases, answer skeletons and self-tests
Content checked	21 October 2026
Online version	https://hrushikesh-shinde7.github.io/bpt-study-material/community/

Educational notes for exam preparation. Not medical advice. Written in original words from current textbooks and dated guidelines; each page lists its sources.

How to use these notes

1. **Follow your classes.** The parts are in the order of the MUHS syllabus. Each part opens with its pages and their page numbers.
2. **Read a topic once**, slowly, including the "Why?" boxes.
3. **Close it and recall.** Write the keywords and draw the diagrams on paper from memory, then compare.
4. **Test yourself** with the questions at the end of each topic. Cover the answer before you look.
5. **Come back** on day 1, day 7 and day 21. Testing yourself again beats re-reading.

> **What the real paper looks like (20 papers, 2016–2026)**
>
> 80 marks in 3 hours. **No MCQs**, even though the syllabus printed a scheme with them.
>
> - Short answers: any 5 of 6 at **3 marks**, and any 5 of 6 at **7 marks**.
> - Long answers: two questions, each any 1 of 2 at **15 marks**. Most long cases are a person in a community setting (a worker in pain, a pregnant or postnatal woman, an older person, a person with disability) or a planning question; answers need ICF, goals, treatment with doses, and ergonomic, home or community measures.
>
> Practical: long case 50, project synopsis 25, journal 5. See the exam skills pages at the end.

How each topic is laid out

Each topic opens with its syllabus heading, priority, the sessions it was asked in (LAQ = 15-mark long answer, 7m and 3m = short answers; S = Summer, W = Winter, for example S24 = Summer 2024), the usual mark split, and the guidelines it was checked against. Page references such as (p. 123) point to related pages in this book. Abbreviations are listed at the end.

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Part 1

Health promotion

10 hours theory, 15 field

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1(a, b) Health, disease and the health care delivery system in India: the three tiers, the PHC and community physiotherapy

> These are short-answer topics (9 of 20 papers), but they are the frame for the whole subject: where in India's health system a person meets care, and where a physiotherapist fits. Learn the three tiers with their population numbers and the functions of a PHC exactly.

Syllabus	Ch 1(a) WHO definition of health and disease; 1(b) health delivery system, three tier
Priority	**Tier 2.** In 9 of 20 papers (2016–2026): 2 seven-mark and 7 three-mark questions.
Asked as	7m S22 · W25 · 3m S17 · S19 · S20 · S21 · S24 · W24 · S25
Usual split	7 (functions of the PHC; principles and scope of community physiotherapy); 3 (WHO definition; PHC setting; delivery system; levels; socioeconomic factors; role of PT)
Checked against	WHO Constitution (1948); Declaration of Alma-Ata (1978); Indian Public Health Standards population norms (2012); Ministry of Health and Family Welfare updates on Ayushman Arogya Mandirs. Full list under Sources (p. 10). Content checked 20 October 2026.

1 Health and disease

WHO definition of health (S19, 3 marks).

> **Definition (WHO Constitution, 1948)**
>
> "Health is a state of complete physical, mental and social well-being and not merely the absence of disease or infirmity."

- **Dimensions of health:** physical, mental, social, and often added spiritual, emotional and vocational.
- **Disease:** a disturbance of the structure or function of the body or mind, with signs and symptoms, that departs from the state of health.
- **Health is a continuum** from positive health through ill-health to death; the physiotherapist works along all of it, not only with disease.
- **Determinants of health:** biology and genes, behaviour and lifestyle, environment, socioeconomic conditions, health services.

Levels of prevention

Level	Aim	Physiotherapy example
Primordial	Prevent risk factors from arising in the population	Active schools and playgrounds, healthy lifestyle in childhood
Primary	Prevent disease in healthy people (health promotion and specific protection)	Fitness programmes, ergonomics, antenatal exercise, fall-prevention exercise
Secondary	Early diagnosis and prompt treatment	Screening posture in schools, developmental screening, early back care
Tertiary	Limit disability and rehabilitate	Stroke, amputee and cardiac rehabilitation, CBR

Disability prevention uses the same idea: see disability (p. 127).

2 Primary health care

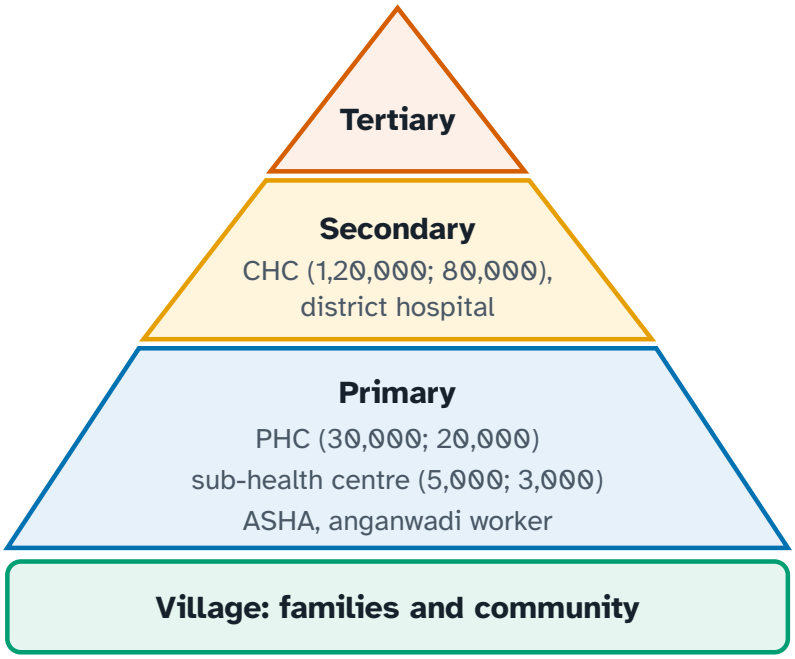
Describe the primary health care setting (S17, 3 marks).

- **Definition (Alma-Ata, 1978):** essential health care based on practical, scientifically sound and socially acceptable methods and technology, made universally accessible to individuals and families in the community through their full participation, at a cost the community and country can afford; the first level of contact with the health system, as close as possible to where people live and work.
- **Principles:** equitable distribution, community participation, intersectoral coordination, appropriate technology, focus on prevention and promotion.
- **Eight elements:** health education; food supply and nutrition; safe water and sanitation; maternal and child health including family planning; immunisation; control of locally endemic diseases; treatment of common diseases and injuries; essential drugs.

3 The three-tier health care delivery system in India

Health care delivery system in India (S21) and levels of health care delivery (S25), 3 marks each.

Three tiers of public health care in rural India



Population served by one centre in plains; second number for hilly, tribal and desert areas (Indian Public Health Standards). Medical colleges and national institutes form the tertiary level.

Unit	Population and staff	Services
Village level	ASHA (about one per 1,000), anganwadi worker	Link workers: identification, home visits, health education, referral
Sub-health centre	5,000 (3,000 in hilly and tribal areas); ANM and male health worker; now with a community health officer as an Ayushman Arogya Mandir	Maternal and child care, immunisation, family planning, first aid, screening for common NCDs
Primary health centre	30,000 (20,000); medical officer, nurses, pharmacist, laboratory technician; 4–6 beds; referral unit for about six sub-centres	Section 4
Community health centre	1,20,000 (80,000); 30 beds; specialists (surgeon, physician, obstetrician, paediatrician); first referral unit for four PHCs	Specialist care, emergency obstetric care, X-ray, laboratory; physiotherapist in many states

Unit	Population and staff	Services
District hospital	Whole district	Specialist and inpatient care, physiotherapy department, DEIC for children, DDRC links
Tertiary	Medical colleges, AIIMS, national institutes	Super-specialist care, teaching, research

- **Ayushman Arogya Mandirs** (formerly Health and Wellness Centres, Ayushman Bharat): sub-centres and PHCs upgraded to give comprehensive primary health care in 12 service packages, including non-communicable disease screening, elderly and palliative care, mental health, eye, ENT, oral and emergency care.
- **Urban:** urban PHCs and urban Ayushman Arogya Mandirs, municipal and district hospitals; large private sector.

4 Functions of the primary health centre

Functioning of primary health centres (S22, 7 marks).

1. Medical care: outpatient and basic inpatient treatment of common illnesses and injuries; referral.
2. Maternal and child health: antenatal, delivery and postnatal care; newborn care; family planning.
3. Immunisation.
4. Prevention and control of communicable diseases (tuberculosis, malaria, leprosy) and screening for non-communicable diseases (hypertension, diabetes, common cancers).
5. National health programmes carried out locally.
6. Health education and promotion; nutrition.
7. Safe water and environmental sanitation.
8. Collection and reporting of vital statistics and disease data.
9. Basic laboratory services and essential drugs.
10. Training and supervision of ANMs, health workers and ASHAs.
11. School health; elderly, palliative and mental health care under the Ayushman Arogya Mandir packages.

5 Physiotherapy in the health care system

Role of PT in health care delivery systems (S24), socioeconomic factors affecting PT services in India (S20, W24), 3 marks each.

Level	Role of the physiotherapist
Primary	Health promotion and fitness, ergonomic and back-care education, antenatal and postnatal classes, screening (posture, development, falls risk), training ASHAs and anganwadi workers, home programmes, CBR
Secondary	Outpatient and inpatient physiotherapy at CHCs and district hospitals: fractures, stroke, chest care, post-operative care, early intervention
Tertiary	Specialised ICU, cardiac, neuro, paediatric, sports and rehabilitation services; teaching and research

Socioeconomic factors affecting physiotherapy services in India

- **Cost:** much of health care is paid out of pocket; repeated sessions and lost wages are hard for daily earners.
- **Distribution:** physiotherapists are concentrated in cities and private clinics; few government posts at CHCs and PHCs.
- **Distance and transport** to services, especially for disabled and elderly people in villages.
- **Awareness and literacy:** many people do not know what physiotherapy offers or think rest is the cure; preference for traditional healers.
- **Gender and family roles:** women may not get permission or time to attend; preference for female therapists.
- **Housing and environment:** crowded homes, floor-level living and Indian toilets affect home programmes.
- **Insurance and schemes:** limited coverage of outpatient rehabilitation (PM-JAY mainly covers hospital care).

- **Solutions:** CBR, camps, tele-rehabilitation, physiotherapists in Ayushman Arogya Mandirs and CHCs, simple home programmes, community awareness.

6 Principles and scope of community physiotherapy

Principles and scope of community physiotherapy (W25, 7 marks).

- **Definition:** the application of physiotherapy to the whole community, with a preventive and promotive emphasis, to optimise health and function where people live and work.

Principles

- Prevention and health promotion first; then treatment and rehabilitation.
- Equity and access: services for rural, poor, elderly, women and disabled people.
- Community participation and ownership.
- Holistic, biopsychosocial approach using the ICF.
- Teamwork and intersectoral coordination (health, education, social welfare, industry).
- Appropriate, low-cost technology and local resources.
- Education and empowerment for self-management.
- Evidence-based, measured outcomes; sustainability.

Scope

- Health promotion and fitness across ages (schools, colleges, workplaces) (see fitness (p. 11) and obesity (p. 26)).
- Women's health: antenatal, postnatal, menopause, pelvic floor (see pregnancy (p. 43)).
- Geriatrics: fitness, falls, home care (see ageing (p. 101)).
- Industrial health: ergonomics, injury prevention, return to work (see ability management (p. 179)).
- Disability and rehabilitation, CBR (see CBR (p. 150)).
- Non-communicable disease prevention and management (diabetes, hypertension, COPD, heart disease).
- Disaster management (see disaster (p. 167)); sports and school health; palliative and home care.
- Research, policy and advocacy.

7 Exam corner

Past questions (MUHS, 2016–2026)

7 marks

Functioning of primary health centres (S22); principles and scope of community physiotherapy (W25).

3 marks

Primary health care setting (S17); WHO definition of health (S19); socioeconomic factors and PT services in India (S20, W24); health care delivery system in India (S21); role of PT in health care delivery systems (S24); levels of health care delivery (S25).

Keywords to underline

complete physical, mental and social well-being Alma-Ata 1978 eight elements primary, secondary, tertiary sub-centre 5,000 PHC 30,000 CHC 1,20,000 Ayushman Arogya Mandir prevention first equity

Answer skeleton: health care delivery system in India (3 marks)

1. **[Tiers, 1]** Primary, secondary, tertiary.
2. **[Units with numbers, 1.5]** Sub-centre, PHC, CHC, district hospital, medical college.

3 **[Recent change, 0.5]** Ayushman Arogya Mandirs.

Draw these

- The pyramid of three tiers with population norms (section 3)

Viva questions

1. Give the WHO definition of health.
2. What population does a PHC serve in plains and in tribal areas?
3. How many beds does a CHC have?
4. Name the eight elements of primary health care.
5. What is an Ayushman Arogya Mandir?
6. Give two socioeconomic barriers to physiotherapy in villages.

Linked practical spots

- During field visits, note the staff and services at a PHC and the referral path to the district hospital for your journal.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Levels of health care delivery. (S25)**

Answer

Primary: first contact at village, sub-centre (5,000; 3,000) and PHC (30,000; 20,000), now Ayushman Arogya Mandirs. Secondary: CHC (1,20,000; 80,000; 30 beds, four specialists) and district hospitals. Tertiary: medical colleges and specialist institutes. Referral moves up; follow-up moves down.

3 m **How do socioeconomic factors affect physiotherapy services in India? (S20, W24)**

Answer

Out-of-pocket cost and lost wages; urban and private concentration of physiotherapists; distance and transport; low awareness and literacy; cultural beliefs and traditional healers; gender norms; home environment; limited insurance for outpatient rehabilitation. Solutions: CBR, camps, tele-rehabilitation, posts at CHCs and AAMs.

7 m **Functioning of primary health care centres. (S22)**

Answer

Define PHC and its population and staff; list functions: medical care and referral, maternal and child health, family planning, immunisation, communicable and NCD control, national programmes, health education and nutrition, water and sanitation, vital statistics, laboratory and drugs, training and supervision of workers, school health, AAM packages (elderly, palliative, mental health). Add the place of physiotherapy (education, screening, home programmes).

MCQ **A PHC in a plain area serves a population of about:**

A. 5,000

B. 30,000

C. 80,000

D. 1,20,000

Answer
B.

MCQ **The Alma-Ata declaration on primary health care was made in:**

A. 1948

B. 1978

C. 1995

D. 2005

Answer
B.

MCQ **The first referral unit for four PHCs is the:**

A. Sub-centre

B. Anganwadi

C. Community health centre

D. Medical college

Answer
C.

Last recall check. Write the WHO definition, the four levels of prevention, the Alma-Ata definition and eight elements, the pyramid with numbers, and ten functions of a PHC.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Community Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. World Health Organization. Constitution of the World Health Organization, 1948.
3. Declaration of Alma-Ata. International Conference on Primary Health Care, Alma-Ata, USSR, 6–12 September 1978.
4. Ministry of Health and Family Welfare. Indian Public Health Standards (2012): population norms for sub-centres, PHCs and CHCs.
5. Ministry of Health and Family Welfare. Updates on Ayushman Arogya Mandirs (comprehensive primary health care, 12 service packages).

1(c) Physical fitness: components and tests, energy systems, aerobic and anaerobic exercise, exercise prescription, inactivity and bed rest

> Fitness was asked in 12 of 20 papers, always as short answers: FITT, aerobic versus anaerobic, energy systems, fitness tests, circuit training, bed rest. These are the building blocks of every exercise prescription you write in a long case, so learn the numbers exactly.

Syllabus	Ch 1(c) Physical fitness: components, assessment, principles of training, exercise prescription, effects of inactivity
Priority	**Tier 1.** In 12 of 20 papers (2016–2026): 9 seven-mark and 7 three-mark questions.
Asked as	7m S16 ×3 · W16 · W18 · W22 · S25 · W25 ×2 · 3m W16 · S18 · S19 · W20 · S21 · W24 · S26
Usual split	7 (energy systems; field tests; principles and phases; circuit training; bed rest; sedentary lifestyle; exercise in hypertension); 3 (FITT; aerobic vs anaerobic; diabetic precautions; fitness tests)
Checked against	ACSM position stand on quantity and quality of exercise (Garber et al., 2011); ACSM position stand on exercise and hypertension (Pescatello et al., 2004) and ACSM FITT recommendations for hypertension (2019 pronouncement); network meta-analysis of exercise and blood pressure (Edwards et al., 2023); American Diabetes Association position statement on physical activity (Colberg et al., 2016); WHO guidelines on physical activity and sedentary behaviour (2020) and fact sheet (2024); SBRN terminology consensus (Tremblay et al., 2017); Adult Compendium of Physical Activities (2024); NSCA guidance on test order and work-to-rest ratios. Full list under Sources (p. 24). Content checked 21 October 2026.

1 Physical fitness and its components

- **Physical activity:** any bodily movement by skeletal muscles that uses energy. **Exercise:** planned, structured, repeated activity done to improve or keep fitness. **Physical fitness:** a set of attributes that people have or achieve that relates to the ability to perform physical activity (Caspersen et al., 1985).

Type	Components
Health-related	Cardiorespiratory endurance, muscular strength, muscular endurance, flexibility, body composition
Skill-related	Agility, balance, coordination, speed, power, reaction time

2 Fitness tests

Adult health-related and skill-related fitness tests (S26, 3 marks); field tests used in athlete training (S16, 7 marks).

Component	Tests
Cardiorespiratory endurance	YMCA 3-minute step test, Rockport 1-mile walk, Cooper 12-minute run, six-minute walk test, submaximal cycle test (YMCA or Åstrand)
Muscular strength	One-repetition maximum (1RM), handgrip dynamometer
Muscular endurance	Push-up test, curl-up (partial sit-up) test
Flexibility	Sit-and-reach, shoulder flexibility (back scratch)
Body composition	BMI, waist, skinfolds, bioimpedance (see body composition (p. 28))
Agility	Illinois agility run, T-test, 4 × 10 m shuttle run
Balance	Stork (one-leg) stand, Y-balance or star excursion test
Coordination	Alternate-hand wall toss
Speed	30–50 m sprint

Component	Tests
Power	Vertical jump, standing broad jump, medicine-ball throw
Reaction time	Ruler-drop test

Field tests in athlete training (7 marks in S16)

A field test is a test done away from the laboratory that needs neither extensive training nor expensive equipment (National Strength and Conditioning Association, NSCA). A good test is valid: it measures what it claims to measure. It is also reliable: it gives the same result when repeated. Physiotherapists use field tests for pre-season screening, to set training loads, to track progress and to decide return to play after injury.

Common field tests by fitness component

Test (component)	How it is done	What it shows; limits
20 m multistage shuttle run, or beep test (aerobic capacity)	Run between two lines 20 m apart, reaching each line by the beep. Speed starts at **8.5 km/h** and rises by **0.5 km/h** each minute (Léger et al., 1988). Stop when the runner can no longer keep pace, usually after missing the line twice.	The last stage reached predicts VO_2max, the maximal oxygen uptake (**r = 0.90** in adults). A whole squad can run at once at little cost. Practice, motivation and the surface change the score.
Yo-Yo intermittent recovery test (repeated intense running)	Run 2 × 20 m shuttles at rising speeds, with **10 s** of jogging (2 × 5 m) after each. Level 1 starts at **10 km/h** and level 2 at **13 km/h** . Stop after failing twice to reach the line in time; the score is the total distance.	The ability to repeat intense running with short recovery, as in football and other team sports (Bangsbo et al., 2008). It relates only moderately to VO_2max (**r = 0.70** for level 1), so it tests more than aerobic capacity.
Cooper 12-minute run (aerobic capacity)	Run or walk as far as possible in 12 minutes on a measured track. Estimate VO_2max in ml/kg/min as **22.351 × km − 11.288** .	Cooper (1968) found a correlation of **0.897** with treadmill VO_2max in 115 Air Force men. It needs good pacing, and motivation and weather affect it.
Sprints over 10–40 m (speed)	Sprint from a standing start, timed with electronic gates or a stopwatch. Record split times, for example at 10 m and 20 m in a 40 m sprint.	The first 10 m shows acceleration; the later splits show maximal speed. Hand timing is less accurate than timing gates.
Running-based anaerobic sprint test, RAST (anaerobic power)	Six maximal 35 m sprints with **10 s** rest between them. For each sprint, power in watts = body mass × distance^2 ÷ time^3.	Peak power, mean power and a fatigue index. It is a field substitute for the laboratory Wingate test, with moderate agreement (**r = 0.46–0.63**) (Zagatto et al., 2009).
Vertical (Sargent) jump and standing broad jump (power)	Vertical: mark the standing reach, jump from both feet and touch the highest point; the score is the difference, best of 3. Broad: two-foot take-off and landing, measured to the back of the heels, best of 3.	Leg power. Peak power in watts = **60.7 × jump (cm)** + **45.3 × mass (kg)** − 2055 (Sayers et al., 1999). Technique and arm swing change the score.
Illinois agility run, T-test, 505 test (change of direction)	Illinois: start lying face down, then run a 10 m × 5 m course that weaves through four centre cones. T-test: sprint 9.14 m (10 yards) forward, shuffle 4.57 m left, 9.14 m right and 4.57 m back to the middle, then run backwards to the start; no crossing of the feet; best of 3. 505: a 10 m run-in, then time 5 m to a line, the turn and 5 m back, for each leg.	Speed of change of direction. The T-test is highly reliable (intraclass correlation **0.98** ; Pauole et al., 2000). Illinois, aged 16–19: excellent is under **15.2 s** for boys and **17.0 s** for girls. These tests have no reaction to a signal, so they do not test true agility (Sheppard and Young, 2006).
1RM and handgrip (strength)	1RM: warm up, then do single lifts with rising loads and full rest, finding the maximum within 3–5 attempts. Grip: squeeze a hand dynamometer.	Maximal strength. The 1RM needs good technique and a trained spotter.

Test (component)	How it is done	What it shows; limits
Push-up and curl-up (muscular endurance)	Push-ups: as many as possible in a row without rest (men on the toes, women on the knees in the American College of Sports Medicine, ACSM, protocol). Curl-ups: done to a metronome beat.	Local muscular endurance. Protocols and norms differ between manuals, so use one manual throughout.
Sit-and-reach (flexibility)	Sit with the knees straight and the feet against a box; reach forward slowly and hold; record the best of the trials.	Moderate validity for hamstring length (**r = 0.67**) but poor for the low back (**r = 0.26**) (Mayorga-Vega et al., 2014). Arm and leg length affect it.

Order and conditions of testing

Test the least tiring items first, so fatigue from one test does not spoil the next. The NSCA order is:

1. Non-fatiguing tests: height, weight, flexibility, skinfolds, vertical jump.
2. Agility tests (T-test, Illinois).
3. Maximal power and strength tests (1RM).
4. Sprint tests.
5. Local muscular endurance tests (push-ups).
6. Fatiguing anaerobic tests (RAST).
7. Aerobic capacity tests (beep test, Yo-Yo, Cooper run), last or on another day.

- **Standardise:** the same warm-up, surface, footwear, time of day and order each time; record the weather.
- **Tester:** trained, gives clear instructions and a demonstration, allows full rest between maximal attempts.
- **Repeat** at set points (pre-season, mid-season, after injury) and compare with the athlete's own earlier scores.

Advantages of field tests	Limits of field tests
Cheap, with simple equipment	Less precise and less controlled than laboratory tests (surface, wind, heat)
Test many athletes at once, quickly	Scores depend on motivation, pacing and practice
Done in the sport's own setting, so they are sport-specific and realistic	Most give predicted values (for example VO_2max), not measured ones
Easy to repeat through the season	Hand timing and judging add error

3 Energy systems, aerobic and anaerobic exercise, and the MET

Energy systems (W16, 7 marks); aerobic and anaerobic exercise with examples and the MET (W22, 7 marks); aerobic vs anaerobic (W24) and what is aerobic exercise (S21), 3 marks.

Which energy system leads, by length of effort



All three systems work all the time; the length and intensity of the effort decide which one leads. Orange and yellow are anaerobic, blue is aerobic.

Energy systems in detail (7 marks in W16)

ATP (adenosine triphosphate) is the only fuel a muscle can use directly. Stored ATP lasts only a few seconds of maximal contraction, so three systems rebuild it all the time.

System	How it rebuilds ATP	Rate, capacity and limits
Phosphagen (ATP–PC)	Creatine phosphate passes its phosphate to ADP, helped by the enzyme creatine kinase. One step, no oxygen.	Fastest rate but the smallest store, so it leads in all-out efforts of a few seconds. Creatine phosphate is rebuilt aerobically within minutes of rest.
Fast (anaerobic) glycolysis	Glucose or glycogen is split to pyruvate in the cytoplasm, and pyruvate becomes lactate. Net **2 ATP** per glucose, or **3** from muscle glycogen.	Fast, with a medium store. Acid builds up in hard efforts and adds to fatigue.
Oxidative (aerobic)	Pyruvate and fatty acids are burnt in the mitochondria (Krebs cycle and electron transport chain) to carbon dioxide and water. About **30–32 ATP** per glucose (older books say 36–38); fat gives much more per molecule.	Slowest rate but the largest store, so it leads at rest and in efforts longer than a few minutes.

Main energy system by length and intensity of effort (NSCA)

Length of effort	Intensity	Main system
0–6 s	Extremely high	Phosphagen
6–30 s	Very high	Phosphagen and fast glycolysis
30 s to 2 min	High	Fast glycolysis
2–3 min	Moderate	Fast glycolysis and oxidative
Over 3 min	Low	Oxidative

- **Rate against capacity:** the faster a system makes ATP, the less it can make in total. The phosphagen system is fastest and smallest; the oxidative system is slowest and largest.
- **Fuel mix:** the respiratory exchange ratio (carbon dioxide out ÷ oxygen in) is about **0.7** when burning fat and **1.0** when burning carbohydrate. Carbohydrate use rises as intensity rises.
- **Lactate threshold:** the intensity at which blood lactate starts to rise sharply. It begins at about **50–60% of VO_2max** in untrained people and **70–80%** in trained athletes (NSCA).
- **Oxygen deficit:** at the start of exercise oxygen uptake lags behind demand, so the anaerobic systems cover the gap.

- **EPOC (excess post-exercise oxygen consumption):** the extra oxygen used after exercise to restore the body to its resting state: rebuild creatine phosphate, clear lactate, and bring temperature and heart rate down.

Interval training for each system (NSCA work-to-rest ratios)

System trained	Work bout	Work : rest
Phosphagen	90–100% of maximal power for 5–10 s	**1:12 to 1:20**
Fast glycolysis	75–90% for 15–30 s	**1:3 to 1:5**
Fast glycolysis and oxidative	30–75% for 1–3 min	**1:3 to 1:4**
Oxidative	20–30% for more than 3 min	**1:1 to 1:3**

Aerobic and anaerobic exercise (7 marks in W22)

Feature	Aerobic exercise	Anaerobic exercise
Definition	Rhythmic, continuous exercise of large muscle groups in which energy comes mainly from oxygen-using pathways	Short, intense exercise in which energy comes mainly from ATP–PC and glycolysis without enough oxygen
Intensity and length	Low to moderate, minutes to hours	High, seconds to about 2 minutes
Fuel and by-product	Fat and glucose; carbon dioxide and water	Creatine phosphate and glucose; lactate
Fibres	Type I (slow-twitch)	Type II (fast-twitch)
Lactate threshold	Mostly below it, so it can be kept up for long	Above it; lactate and acid build up, so it can last only briefly
Adaptations	More capillaries, mitochondria and enzymes; bigger stroke volume; higher VO_2max; lower resting heart rate and BP	Strength, power, speed, muscle size, tolerance to lactate
Examples	Brisk walking, jogging, cycling, swimming, dancing	Sprinting, weightlifting, jumping, high-intensity intervals

The MET and how to use it (7 marks in W22)

- **MET (metabolic equivalent):** the energy used sitting quietly at rest, **3.5 ml O_2/kg/min** (about 1 kcal per kg per hour). Activities are graded as multiples of this.
- **Bands (Compendium of Physical Activities):** sedentary **1.0–1.5 METs**, light **1.6–2.9**, moderate **3.0–5.9**, vigorous **6.0 or more**.
- **Energy cost:** kcal per minute = METs × 3.5 × body mass (kg) ÷ 200. A 60 kg person walking briskly (4.8 METs) uses 4.8 × 3.5 × 60 ÷ 200, about **5 kcal a minute**, or about 150 kcal in 30 minutes.
- **Weekly volume:** MET-minutes = METs × minutes. Brisk walking for 30 minutes on 5 days gives 4.8 × 150 = **720 MET-min**. ACSM advises **500–1000 MET-min a week** (Garber et al., 2011).
- **Uses:** grade and prescribe intensity; express weekly volume; compare activities; estimate energy use for weight control; choose activities by their MET cost in cardiac rehabilitation.
- **Limit:** 3.5 ml/kg/min is an average. In 769 adults measured resting uptake averaged 2.6 ml/kg/min, so the standard MET overestimated it by about 35% (Byrne et al., 2005). Treat MET values as estimates and check effort with heart rate or RPE (rating of perceived exertion).

MET values of common activities (Adult Compendium of Physical Activities, 2024)

Activity	METs
Hatha yoga	2.3
Walking slowly, 4.0 km/h, level ground	3.0

Activity	METs
Sweeping floors	3.3
Climbing stairs slowly	4.5
Walking briskly, 5.6–6.3 km/h	4.8
Circuit training, moderate effort	5.0
Cycling for leisure, 16–19 km/h	6.8
Running, about 10 km/h	9.3

More detail, including the intensity table, is on exercise physiology in CVR.

4 Principles of training and exercise prescription

General principles of exercise and its phases (S16, 7 marks); principles of aerobic training (S19) and the FITT principle (W20), 3 marks; principles of exercise prescription (W25, 7 marks).

Principles of training

- **Overload:** the body adapts only when stressed beyond its usual level.
- **Progression:** increase the load gradually (usually duration first, then frequency, then intensity; about 10% a week).
- **Specificity:** the body adapts to the exact type of training (aerobic training builds endurance, not strength).
- **Reversibility:** gains are lost when training stops ("use it or lose it").
- **Individuality:** people respond differently; prescribe for the person's age, fitness, disease and goals.
- **Recovery:** adaptation happens during rest; too little rest leads to overtraining.

FITT-VP for healthy adults (ACSM)

FITT	Aerobic	Resistance
Frequency	5 or more days a week moderate, or 3 or more days vigorous	2–3 days a week for each major muscle group
Intensity	Moderate 40–59% heart rate reserve (RPE 12–13) or vigorous 60–89% (RPE 14–17)	60–70% of 1RM for beginners (8–12 repetitions); 40–50% for older or very unfit people
Time	30–60 minutes a day moderate (150 a week) or 20–60 minutes vigorous (75 a week)	2–4 sets of 8–12 repetitions
Type	Rhythmic activity of large muscle groups: walking, cycling, swimming	Multi-joint exercises for all major muscle groups, opposing groups balanced
Volume and progression	About 500–1000 MET-minutes a week; progress gradually	Increase load when the top of the repetition range is easy

- **Flexibility:** 2–3 days a week, each stretch held 10–30 seconds (30–60 in older adults), repeated to total about 60 seconds.
- **Neuromotor (balance, agility):** 2–3 days a week, especially for older adults.
- **Heart rate target (Karvonen):** resting HR + (maximum HR − resting HR) × intensity.

Steps of exercise prescription (W25)

1. Screen: history, risk factors, medicines, red flags; medical clearance when needed.
2. Test fitness (section 2) and set goals with the person.
3. Write the FITT-VP for aerobic, resistance, flexibility and balance work.
4. Teach warm-up, cool-down, self-monitoring (talk test, RPE, pulse) and warning signs.
5. Progress, review and retest every 4–6 weeks; adjust for adherence and enjoyment.

Phases

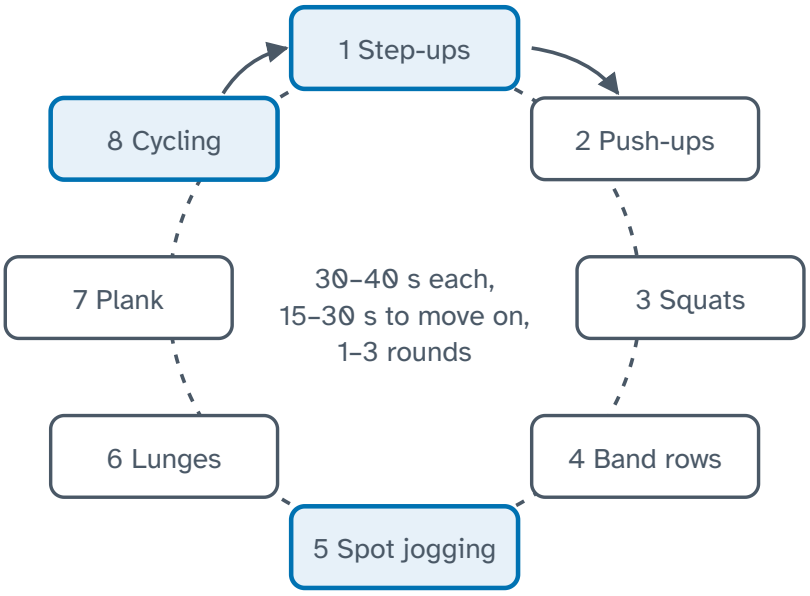
Phase	What it involves
Warm-up (5–10 minutes)	Light aerobic activity and dynamic movements; raises heart rate, muscle temperature and joint mobility; reduces injury and heart strain
Conditioning (20–60 minutes)	The main aerobic, resistance or sport work at the prescribed intensity
Cool-down (5–10 minutes)	Light activity so heart rate and BP fall gradually; prevents blood pooling and dizziness
Stretching (at least 10 minutes)	Static stretches when muscles are warm

A training programme also has stages: an **initial stage** (first few weeks, low intensity, building the habit), an **improvement stage** (the next few months, steady progression) and a **maintenance stage** (keeping gains long term).

5 Circuit training

Define aerobic exercise and explain circuit training (S25, 7 marks).

A circuit of eight stations



Blue stations are aerobic; the others alternate upper body, lower body and trunk so each muscle group rests while another works.

Definition and method (7 marks in S25)

- **Definition:** a series of exercises (stations) done one after another with little rest, working different muscle groups and adding aerobic stations; the whole set is repeated.
- **Origin:** developed by R. E. Morgan and G. T. Adamson at the University of Leeds in 1953, and described in their book *Circuit Training* (1957).
- **Stations:** about **10–15 exercises**, set out so that upper body, lower body, trunk and aerobic work alternate.
- **Work:** **30–40 s** or **12–15 reps** at each station, at **40–60% of 1RM** for a weight circuit.
- **Rest:** **15–30 s** while moving to the next station.
- **Rounds:** **1–3 circuits**, about 30 minutes in all, with a warm-up before and a cool-down after.
- **Progress** by adding a round, lengthening work time, shortening rest or adding load, one at a time.

Types of circuit

Type	How it is set up	Main effect
Circuit weight training	Light weights, many repetitions, short rest. A classic study used 10 stations at 40–55% of 1RM, as many repetitions as possible in 30 s, 15 s to change, 3 circuits, 3 days a week (Wilmore et al., 1978)	Muscular endurance and strength, with a modest aerobic gain
Aerobic circuit	Aerobic bouts (step-ups, skipping, cycling) between resistance stations, or a very light load with the heart rate kept at 70–80% of maximum	Cardiorespiratory fitness as well as muscular endurance
High-intensity circuit	12 body-weight exercises, 30 s each with 10 s to change, repeated 2–3 times (Klika and Jordan, 2013)	Fitness in a short time for healthy, fit adults
Circuit class therapy	Task-practice stations (sit-to-stand, stepping, walking, reaching) for a group of patients, supervised by therapists	Mobility and walking, for example after stroke
Sport-specific circuit	Skills and movements of the sport at each station	Fitness for the demands of that sport

What it achieves

- **Aerobic fitness:** early circuit weight training studies raised VO_2max by an average of 5.1% in men and 8.0% in women (Gettman and Pollock, 1981). When the heart rate was kept at 70–80% of maximum (20% of 1RM, 10 stations, 40 s work and 20 s rest, 3 rounds), VO_2max rose 11%, close to the 12% after endurance training (Kaikkonen et al., 2000).
- **Energy cost:** about **5.0 METs** at moderate effort, **6.0** with body-weight exercises and **7.5** when vigorous with minimal rest (Compendium, 2024).
- **Patients:** in a Cochrane review after stroke (17 trials, 1297 people), circuit class therapy improved the six-minute walk distance by about **61 m** and gait speed by **0.15 m/s** (English et al., 2017).

Advantages	Disadvantages
Strength, muscular endurance and some aerobic fitness in one short session	Smaller strength gains than heavy, low-repetition training
Many people at once with little equipment: schools, workplaces, cardiac rehabilitation phase III, older adults with chair stations	Needs space, set-up time and supervision
Varied and enjoyable, so people keep coming	Technique slips as fatigue builds

Precautions: teach correct technique first; avoid breath-holding; adapt stations for joint problems; monitor heart rate or RPE in patients.

6 Inactivity: sedentary lifestyle and bed rest

Hazards of a sedentary lifestyle (W18, 7 marks); deconditioning effects of bed rest (S16, 7 marks; W16, 3 marks).

Hazards of a sedentary lifestyle (7 marks in W18)

Two different things make up a sedentary lifestyle, and the exam answer should define both.

- Sedentary behaviour: any waking behaviour with an energy use of **1.5 METs or less** while sitting, reclining or lying, such as desk work, driving and watching television (Tremblay et al., 2017).
- Physical inactivity: not doing enough physical activity to meet current recommendations (Tremblay et al., 2017).
- They are separate risks. A person can meet the activity guidelines and still sit for most of the day, and long sitting still harms metabolic health (Panahi and Tremblay, 2018).

Size of the problem

- **World:** **31%** of adults (1.8 billion) were insufficiently active in 2022, up from 23% in 2000. The share is projected to reach about 35% by 2030 (Strain et al., 2024; WHO).
- **India:** about **49%** of adults were insufficiently active in 2022: 57% of women and 42% of men (WHO Global Health Observatory).
- **ICMR-INDIAB study:** in four Indian regions, **54.4%** of adults were inactive and only 13.7% highly active. Inactivity was higher in towns (65.0%) than in villages (50.0%), and about 92% did no recreational activity at all (Anjana et al., 2014).
- **Deaths:** people who are insufficiently active have a **20–30%** higher risk of death than those who are sufficiently active (WHO). Worldwide, inactivity causes about 6% of coronary heart disease, 7% of type 2 diabetes, 10% each of breast and colon cancer, and 9% of premature deaths, more than 5.3 million deaths in 2008 (Lee et al., 2012).

Health effects by system

System	Hazards	Evidence or mechanism
Heart and vessels	Coronary heart disease, stroke, hypertension	High sitting time: cardiovascular death hazard ratio **1.18** (Biswas et al., 2015)
Metabolism	Type 2 diabetes, obesity, high triglycerides and low HDL cholesterol, metabolic syndrome	High sitting time: diabetes hazard ratio **1.91** (Biswas et al., 2015). Idle muscle takes up less glucose and fat; in rats, inactivity removed about 90–95% of muscle lipoprotein lipase activity, the enzyme that clears blood fat (Bey and Hamilton, 2003)
Cancer	Colon, breast and endometrial cancer	High sitting time: cancer death hazard ratio **1.17** (Biswas et al., 2015)
Muscles, bones and joints	Weak, wasted muscles (sarcopenia), osteoporosis, stiffness, low back and neck pain from static posture, knee pain with weight gain	Reversibility: unused tissue loses strength and mass. Breaking up sitting reduces back discomfort and fatigue in office workers (Buckley et al., 2015)
Mind and brain	Depression, anxiety, poor sleep, lower cognitive function	Sedentary behaviour: depression relative risk **1.25** (Zhai et al., 2015)
Fitness and function	Low VO_2max, strength and flexibility; in old age, falls and loss of independence	Deconditioning, as with bed rest below but slower

- **All-cause death:** high sitting time raised the hazard ratio to **1.24**, and the effect was larger in people who did little exercise (Biswas et al., 2015).
- **Dose:** risk rises more steeply above about **6–8 h a day** of sitting and **3–4 h a day** of television (Patterson et al., 2018).
- **Can exercise cancel it?** In over 1 million adults, the least active who sat more than 8 h a day had the highest risk (hazard ratio 1.59). About **60–75 min a day** of moderate activity seemed to remove the extra risk of long sitting. Television for 3 h or more a day stayed harmful except in the most active quarter (Ekelund et al., 2016).

Physiotherapy prescription: sit less, move more

- **Meet the guideline:** **150–300 min a week** of moderate or 75–150 min of vigorous activity, plus muscle strengthening on **2 or more days** (WHO, 2020).
- **Limit sitting:** WHO (2020) advises adults to limit sedentary time and replace it with activity of any intensity, including light activity. It sets no numerical limit. People who sit a lot should aim above the recommended activity levels.
- **Break up sitting often:** in overweight adults, a **2-minute walk every 20 minutes** over 5 hours lowered the rise in blood glucose after a drink by 24–30% and insulin by 23% (Dunstan et al., 2012).
- **Desk workers:** build up to at least **2 h a day** of standing and light walking in working hours, then **4 h a day**. Alternate sitting and standing work, with a sit-stand desk if possible, and avoid long static standing too (Buckley et

al., 2015). See desk workers (p. 203).

- **Practical habits:** stand for phone calls, walk to a colleague instead of phoning, take the stairs, hold walking meetings, walk or cycle part of the commute, and do household chores or stretches during television breaks.
- **Behaviour change:** set small goals, use a step counter or phone reminder, and enlist the family or workplace; workplace and school activity programmes and community awareness.

Deconditioning effects of bed rest

System	Effects
Heart and circulation	Higher resting heart rate, smaller stroke volume, lower VO_2max, postural hypotension (plasma volume falls), deep vein thrombosis
Lungs	Lower lung volumes, basal collapse, retained secretions, pneumonia
Muscles	Wasting and weakness of antigravity muscles; a classic estimate is about 1–1.5% strength loss a day of strict bed rest
Bones and joints	Bone loss and high urine calcium; contractures and stiffness
Skin	Pressure sores over the sacrum, heels, hips
Metabolic	Insulin resistance, protein loss
Kidney, bladder and gut	Urinary stasis, infection and stones; constipation, poor appetite
Nervous system and mind	Poor balance, confusion, low mood, poor sleep, sensory deprivation

Prevention: avoid unnecessary bed rest; sit out and walk early; breathing exercises; ankle pumps; active and resisted exercise; positioning and turning; standing and weight-bearing as early as possible. Full table with mechanisms on bed rest in CVR.

7 Exercise in diabetes and hypertension

Precautions when prescribing fitness exercise for a diabetic (S18, 3 marks); effects of exercise in a hypertensive person (W25, 7 marks).

Diabetes: precautions

- **Dose:** at least 150 minutes a week of moderate to vigorous aerobic activity over at least 3 days, with no more than 2 days in a row without activity; resistance training 2–3 non-consecutive days a week (ADA, 2016).
- **Low sugar (hypoglycaemia):** check blood glucose before and after exercise if on insulin or sulphonylureas; carry fast sugar; take carbohydrate or reduce insulin as advised; do not inject insulin into the exercising limb; lows can come 6–15 hours later and the risk lasts up to 48 hours.
- **High sugar:** do not exercise with high blood glucose and ketones.
- **Feet:** inspect daily, good footwear and socks; non-weight-bearing exercise (cycling, swimming) if there is an ulcer.
- **Eyes:** with severe or unstable retinopathy avoid vigorous and heavy resistance exercise, jumping, jarring, head-down positions and breath-holding.
- **Autonomic neuropathy:** use RPE rather than heart rate; avoid quick position changes; drink enough; avoid heat.
- Exercise at a regular time, often 1–2 hours after a meal; exercise with a partner who knows the signs of hypoglycaemia.

Hypertension: effects of exercise (7 marks in W25)

Hypertension is common in India: **35.5%** of adults in the ICMR-INDIAB survey (Anjana et al., 2023). In NFHS-5 (2019–21), 21.3% of women and 24.0% of men aged 15 and over had a raised BP or were on treatment. Exercise is a first-line lifestyle treatment for all of them.

- **During exercise:** in dynamic aerobic work, systolic BP rises with the workload while diastolic BP stays about the same or falls a little, because the working muscles' vessels dilate. Heavy isometric holds and breath-holding (Valsalva) raise both sharply.
- **After one session:** BP stays lower for up to **22 hours** (post-exercise hypotension) (Pescatello et al., 2004).
- **After training:** resting BP falls most in people with hypertension, by about **5–8 mmHg**, against 2–4 mmHg with prehypertension and 1–2 mmHg with normal BP (US Physical Activity Guidelines Advisory Committee, 2018). In hypertensive groups, endurance training lowered BP by **8.3/5.2 mmHg** (Cornelissen and Smart, 2013).
- **Other gains:** fewer cardiovascular events, better fitness and mood, less need for drugs.

Fall in resting BP by type of training: network meta-analysis of 270 trials (Edwards et al., 2023)

Training	Fall in systolic/diastolic BP
Aerobic	**4.5/2.5 mmHg**
Dynamic resistance	**4.6/3.0 mmHg**
Combined aerobic and resistance	**6.0/2.5 mmHg**
High-intensity interval training	**4.1/2.5 mmHg**
Isometric (wall squat, handgrip)	**8.2/4.0 mmHg**; the isometric wall squat ranked best for systolic BP and running for diastolic

How exercise lowers BP

- **Nerves and hormones:** after endurance training, plasma noradrenaline fell by 29% and plasma renin activity by 20% (Cornelissen and Fagard, 2005).
- **Vessels:** systemic vascular resistance fell by 7.1% (Cornelissen and Fagard, 2005). Endothelial function (flow-mediated dilatation) improves with aerobic, resistance and combined training (Ashor et al., 2015). Aerobic training lowers arterial stiffness; resistance training alone does not (Ashor et al., 2014).
- **Metabolism:** weight and waist fall and insulin sensitivity improves (Cornelissen and Fagard, 2005).

Prescription (ACSM FITT for hypertension)

FITT	Aerobic	Resistance
Frequency	**5–7 days a week**	**2–3 days a week**
Intensity	Moderate: **40–59%** of heart rate reserve, RPE 12–13	**60–70% 1RM**, progressing to 80%; 40–50% for older people and beginners
Time	**30 min or more** a day, continuous or in bouts	**2–4 sets of 8–12** for 8–10 exercises
Type	Walking, cycling, swimming	Machines, free weights or bands for the major muscle groups

Add flexibility work on 2–3 days a week, each stretch held 10–30 s and repeated 2–4 times.

Classic answer vs current practice: resistance and isometric exercise in hypertension

Classic	Current	Write in the exam
Aerobic exercise only; avoid weights and isometric exercise because they raise BP.	Dynamic resistance training lowers resting BP about as much as aerobic training, and isometric training lowered it most in meta-analyses. The danger is breath-holding and maximal straining, not resistance itself.	"Aerobic exercise on most days plus resistance training 2–3 days a week at moderate load, breathing out on effort; no breath-holding or maximal lifts."

Precautions

- Do not start exercise if resting BP is very high (above **200/110 mmHg**) until it is controlled. During an exercise test, stop if BP rises above **250/115 mmHg** (American Heart Association, 2013).
- Avoid breath-holding and heavy isometric straining at maximal effort.
- Long warm-up and cool-down: vasodilators and alpha-blockers cause dizziness after exercise.
- Use RPE with beta-blockers, which blunt the heart rate; watch for dehydration and low potassium with diuretics. See hypertension in CVR.

8 Exam corner

Past questions (MUHS, 2016–2026)

7 marks

Field tests used in athlete training (S16); general principles of exercise and its phases (S16); deconditioning effects of bed rest (S16); energy systems (W16); hazards of a sedentary lifestyle (W18); aerobic and anaerobic exercise with examples and the MET (W22); aerobic exercise and circuit training (S25); effects of exercise in hypertension and principles of exercise prescription (W25).

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3 marks

Deconditioning effects of bed rest (W16); precautions for a diabetic exercising (S18); principles of aerobic training (S19); FITT (W20); what is aerobic exercise (S21); aerobic vs anaerobic (W24); adult health-related and skill-related fitness tests (S26).

Keywords to underline

health-related and skill-related ATP–PC, glycolysis, aerobic 1 MET = 3.5 ml/kg/min overload, specificity, progression, reversibility FITT-VP heart rate reserve warm-up and cool-down circuit field test: valid, reliable least tiring first, aerobic last sedentary behaviour ≤1.5 METs break up sitting post-exercise hypotension 5–8 mmHg hypoglycaemia

Answer skeleton: principles of exercise prescription (7 marks)

1. **[Definition and aim, 1]** What exercise prescription is; health and fitness goals.
2. **[Principles, 2]** Overload, progression, specificity, reversibility, individuality, recovery.
3. **[FITT-VP, 3]** Aerobic, resistance, flexibility and balance with numbers.
4. **[Safety, 1]** Screening, warm-up and cool-down, monitoring, warning signs.

Draw these

- Energy systems on a time line (section 3)
- A circuit of eight stations (section 5)
- Phases of a session as a heart-rate curve: rise, plateau, fall

Viva questions

1. Name the five health-related components of fitness.
2. Which energy system powers a 100 m sprint?
3. What is one MET?
4. Calculate a 60% heart rate reserve target for a 40-year-old with a resting HR of 70.
5. What is the principle of reversibility?
6. Why must a diabetic on insulin carry sugar while exercising?

Linked practical spots

- Perform the YMCA step test, sit-and-reach, push-up and curl-up tests; set out a beep test, T-test and Illinois course; calculate Karvonen targets and MET-minutes; set up a circuit for a group.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **FITT principle of aerobic training. (W20)**

Answer

Frequency: 5 or more days moderate or 3 or more vigorous. Intensity: 40–59% HRR (RPE 12–13) moderate, 60–89% vigorous. Time: 30–60 minutes (150 a week) moderate or 20–60 (75 a week) vigorous. Type: rhythmic large-muscle activity such as walking, cycling, swimming. Add volume and progression (FITT-VP).

3 m **Precautions while prescribing fitness exercise for a diabetic patient. (S18)**

Answer

Check glucose before and after if on insulin or sulphonylureas; carry fast sugar; no exercise with high glucose and ketones; avoid injecting the exercising limb; watch for delayed hypoglycaemia; foot inspection and footwear, non-weight-bearing with ulcers; avoid jarring, head-down and breath-holding with severe retinopathy; RPE with autonomic neuropathy; hydration.

7 m **Energy systems. (W16)**

Answer

ATP is the only direct energy for contraction and is stored for seconds. ATP–PC: creatine phosphate rebuilds ATP without oxygen; fastest, smallest store; all-out efforts of a few seconds (sprint, jump). Fast glycolysis: glucose to lactate, net 2 ATP; leads from about 30 s to 2 min (400 m). Oxidative: carbohydrate and fat burnt in mitochondria, about 30–32 ATP per glucose; slowest, largest store; at rest and beyond 2–3 min (walking, marathon). All overlap; intensity and duration decide the lead system (NSCA table). Add lactate threshold, oxygen deficit and EPOC, then training: work-to-rest 1:12 to 1:20 for phosphagen, 1:3 to 1:5 for glycolysis, continuous or 1:1 to 1:3 for aerobic. Draw the time line.

7 m **Discuss the effects of an exercise intervention on a hypertensive individual. (W25)**

Answer

During exercise systolic BP rises, diastolic stays level; post-exercise hypotension for up to 22 hours; after training resting BP falls 5–8 mmHg in hypertension; all types work, isometric most (Edwards 2023); mechanisms (noradrenaline and renin down, lower vascular resistance, better endothelium, less stiffness, weight and insulin sensitivity); prescription (aerobic 5–7 days at 40–59% heart rate reserve for 30 min or more, resistance 2–3 days at 60–70% 1RM); precautions (resting BP above 200/110, Valsalva, drugs: beta-blockers, vasodilators, diuretics; long cool-down).

MCQ **Which is a skill-related component of fitness?**

A. Flexibility

B. Body composition

C. Agility

D. Muscular endurance

Answer
C.

MCQ **Moderate-intensity activity is:**

A. Below 3 METs

B. 3–5.9 METs

C. 6–8 METs

D. Above 10 METs

Answer
B.

MCQ **A 400 m run is powered mainly by:**

A. ATP–PC

B. Anaerobic glycolysis

C. Fat oxidation

D. Protein

Answer
B.

Last recall check. Name the components of fitness with one test each, draw the energy-system time line, write FITT-VP with numbers, and list six effects of bed rest.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Community Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Caspersen CJ, Powell KE, Christenson GM. Physical activity, exercise, and physical fitness: definitions and distinctions for health-related research. *Public Health Rep* 1985;100:126–31.
3. Garber CE, et al. American College of Sports Medicine position stand: quantity and quality of exercise for developing and maintaining cardiorespiratory, musculoskeletal, and neuromotor fitness in apparently healthy adults. *Med Sci Sports Exerc* 2011;43:1334–59.
4. Pescatello LS, et al. American College of Sports Medicine position stand: exercise and hypertension. *Med Sci Sports Exerc* 2004;36:533–53.
5. Colberg SR, et al. Physical activity/exercise and diabetes: a position statement of the American Diabetes Association. *Diabetes Care* 2016;39:2065–79.
6. World Health Organization. Guidelines on physical activity and sedentary behaviour, 2020; Bull FC, et al. World Health Organization 2020 guidelines on physical activity and sedentary behaviour. *Br J Sports Med* 2020;54:1451–62. WHO Physical activity fact sheet, 26 June 2024.
7. Haff GG, Triplett NT, eds. Essentials of Strength Training and Conditioning (NSCA). 4th ed. Champaign, IL: Human Kinetics; 2016. Definition of a field test, order of testing, energy systems by duration, work-to-rest ratios and lactate threshold, checked through study reproductions of the NSCA text, not the book itself.
8. Léger LA, Mercier D, Gadoury C, Lambert J. The multistage 20 metre shuttle run test for aerobic fitness. *J Sports Sci* 1988;6:93–101.
9. Bangsbo J, Iaia FM, Krustrup P. The Yo-Yo intermittent recovery test: a useful tool for evaluation of physical performance in intermittent sports. *Sports Med* 2008;38:37–51; starting speeds and recovery jog from Schmitz B, et al. The Yo-Yo intermittent tests: a systematic review and structured compendium of test results. *Front Physiol* 2018;9:870.

10. Cooper KH. A means of assessing maximal oxygen intake: correlation between field and treadmill testing. *JAMA* 1968;203:201–4 (prediction equation checked through Topend Sports and a second reproduction).
11. Zagatto AM, Beck WR, Gobatto CA. Validity of the running anaerobic sprint test for assessing anaerobic power and predicting short-distance performances. *J Strength Cond Res* 2009;23:1820–7.
12. Sayers SP, et al. Cross-validation of three jump power equations. *Med Sci Sports Exerc* 1999;31:572–7.
13. Pauole K, et al. Reliability and validity of the T-test as a measure of agility, leg power, and leg speed in college-aged men and women. *J Strength Cond Res* 2000;14:443–50 (course layout checked through Topend Sports and the ACE T-test protocol sheet). Illinois agility norms checked through two web reproductions (Topend Sports; King Fahd University fitness test page); the original tables were not seen.
14. Sheppard JM, Young WB. Agility literature review: classifications, training and testing. *J Sports Sci* 2006;24:919–32.
15. Mayorga-Vega D, Merino-Marban R, Viciana J. Criterion-related validity of sit-and-reach tests for estimating hamstring and lumbar extensibility: a meta-analysis. *J Sports Sci Med* 2014;13:1–14.
16. Herrmann SD, et al. 2024 Adult Compendium of Physical Activities: a third update of the energy costs of human activities. *J Sport Health Sci* 2024;13:6–12, with activity values from pacompendium.com; Ainsworth BE, et al. 2011 Compendium of Physical Activities. *Med Sci Sports Exerc* 2011;43:1575–81.
17. Byrne NM, et al. Metabolic equivalent: one size does not fit all. *J Appl Physiol* 2005;99:1112–9.
18. OpenStax. Anatomy and Physiology 2e, sections 10.3 and 10.5 (ATP stores, glycolysis yield, fibre types). Modern oxidative ATP yield from Hinkle PC. P/O ratios of mitochondrial oxidative phosphorylation. *Biochim Biophys Acta* 2005;1706:1–11, and Berg's Biochemistry (checked through a summary).
19. Morgan RE, Adamson GT. Circuit Training. London: G. Bell; 1957 (the 1953 Leeds origin checked through secondary summaries).
20. Wilmore JH, et al. Physiological alterations consequent to circuit weight training. *Med Sci Sports* 1978;10:79–84. Circuit parameters also from Agudo-Ortega A, et al. Acute neuromuscular fatigue in circuit strength training: effects of varying work-to-rest durations. *Front Sports Act Living* 2026;8:1761646.
21. Kaikkonen H, et al. The effect of heart rate controlled low resistance circuit weight training and endurance training on maximal aerobic power in sedentary adults. *Scand J Med Sci Sports* 2000;10:211–5 (also the source for the averages from Gettman LR, Pollock ML. Circuit weight training: a critical review of its physiological benefits. *Phys Sportsmed* 1981;9:44–60).
22. Klika B, Jordan C. High-intensity circuit training using body weight: maximum results with minimal investment. *ACSM's Health Fit J* 2013;17(3):8–13 (checked through secondary summaries).
23. English C, Hillier SL, Lynch EA. Circuit class therapy for improving mobility after stroke. *Cochrane Database Syst Rev* 2017; (6):CD007513.
24. Tremblay MS, et al. Sedentary Behavior Research Network (SBRN): Terminology Consensus Project process and outcome. *Int J Behav Nutr Phys Act* 2017;14:75; Panahi S, Tremblay A. Sedentariness and health: is sedentary behavior more than just physical inactivity? *Front Public Health* 2018;6:258.
25. Strain T, et al. National, regional, and global trends in insufficient physical activity among adults from 2000 to 2022: a pooled analysis of 507 population-based surveys with 5.7 million participants. *Lancet Glob Health* 2024;12:e1232–43. India estimate (2022, age-standardised) from the WHO Global Health Observatory, which holds the same model's country values.
26. Anjana RM, et al. Physical activity and inactivity patterns in India: results from the ICMR-INDIAB study (Phase-1) [ICMR-INDIAB-5]. *Int J Behav Nutr Phys Act* 2014;11:26.
27. Lee IM, et al. Effect of physical inactivity on major non-communicable diseases worldwide: an analysis of burden of disease and life expectancy. *Lancet* 2012;380:219–29.
28. Biswas A, et al. Sedentary time and its association with risk for disease incidence, mortality, and hospitalization in adults: a systematic review and meta-analysis. *Ann Intern Med* 2015;162:123–32.
29. Ekelund U, et al. Does physical activity attenuate, or even eliminate, the detrimental association of sitting time with mortality? A harmonised meta-analysis of data from more than 1 million men and women. *Lancet* 2016;388:1302–10.
30. Patterson R, et al. Sedentary behaviour and risk of all-cause, cardiovascular and cancer mortality, and incident type 2 diabetes: a systematic review and dose response meta-analysis. *Eur J Epidemiol* 2018;33:811–29.
31. Zhai L, Zhang Y, Zhang D. Sedentary behaviour and the risk of depression: a meta-analysis. *Br J Sports Med* 2015;49:705–9.
32. Bey L, Hamilton MT. Suppression of skeletal muscle lipoprotein lipase activity during physical inactivity: a molecular reason to maintain daily low-intensity activity. *J Physiol* 2003;551:673–82.
33. Dunstan DW, et al. Breaking up prolonged sitting reduces postprandial glucose and insulin responses. *Diabetes Care* 2012;35:976–83.
34. Buckley JP, et al. The sedentary office: an expert statement on the growing case for change towards better health and productivity. *Br J Sports Med* 2015;49:1357–62.
35. Anjana RM, et al. Metabolic non-communicable disease health report of India: the ICMR-INDIAB national cross-sectional study (ICMR-INDIAB-17). *Lancet Diabetes Endocrinol* 2023;11:474–89; International Institute for Population Sciences. National Family Health Survey (NFHS-5) 2019–21: India fact sheet. Mumbai: IIPS; 2021.
36. Physical Activity Guidelines Advisory Committee. 2018 Physical Activity Guidelines Advisory Committee Scientific Report, Part F, Chapter 5. Washington, DC: US Department of Health and Human Services; 2018; Pescatello LS, et al. Physical activity to prevent and treat hypertension: a systematic review. *Med Sci Sports Exerc* 2019;51:1314–23.
37. Cornelissen VA, Smart NA. Exercise training for blood pressure: a systematic review and meta-analysis. *J Am Heart Assoc* 2013;2:e004473.
38. Edwards JJ, et al. Exercise training and resting blood pressure: a large-scale pairwise and network meta-analysis of randomised controlled trials. *Br J Sports Med* 2023;57:1317–26.
39. Cornelissen VA, Fagard RH. Effects of endurance training on blood pressure, blood pressure-regulating mechanisms, and cardiovascular risk factors. *Hypertension* 2005;46:667–75; Ashor AW, et al. Exercise modalities and endothelial function: a systematic review and dose-response meta-analysis of randomized controlled trials. *Sports Med* 2015;45:279–96; Ashor AW, et al. Effects of exercise modalities on arterial stiffness and wave reflection: a systematic review and meta-analysis of randomized controlled trials. *PLoS One* 2014;9:e110034.
40. American College of Sports Medicine. FITT recommendations for hypertension (adapted from ACSM's Guidelines for Exercise Testing and Prescription and the 2019 ACSM pronouncement) and patient handout "Exercising your way to lowering your blood pressure" (2021); checked through these ACSM documents, not the book.
41. Fletcher GF, et al. Exercise standards for testing and training: a scientific statement from the American Heart Association. *Circulation* 2013;128:873–934.

1(c) Obesity in adults and children: body composition, BMI, grades, causes and weight management

> Obesity was asked in 14 of 20 papers, with 10 long cases about a housewife, banker, lecturer, schoolboy or bride-to-be who needs to lose weight. Every answer has the same three parts: measure (BMI, waist, body fat, fitness), explain (why she gained weight and what it does to health), and prescribe (exercise with FITT, diet with a dietitian, and habits she can keep).

Syllabus	Ch 1(c) Physical fitness: obesity in children and adults, body composition, weight management (health promotion)
Priority	**Tier 1.** In 14 of 20 papers (2016–2026): 10 long cases, 8 seven-mark and 6 three-mark questions.
Asked as	LAQ S16 · W16 · S19 · S20 · W20 · S21 · W21 · W22 · W24 · S25 · 7m S16 · W16 · S20 · W20 · W21 · S24 · W24 · S26 · 3m W16 · W17 · W22 · S23 · S24 · S25
Usual split	15 (BMI 3, functional diagnosis 5, PT 7; or fitness status 5, programme 10; or exercise prescription 5, lifestyle 5, ergonomics 5; or goals 5 + 7 + diet tips 3); 7 (body composition methods; grades and significance; causes of childhood obesity)
Checked against	WHO BMI classification; Asian Indian consensus (Misra et al., 2009) and the revised definition of obesity in Asian Indians (Misra et al., 2025); IAP growth charts (2015); ACSM position stand on physical activity for weight loss (Donnelly et al., 2009); Khelo India school fitness assessment; CSEP push-up norms. Full list under Sources (p. 35). Content checked 20 October 2026.

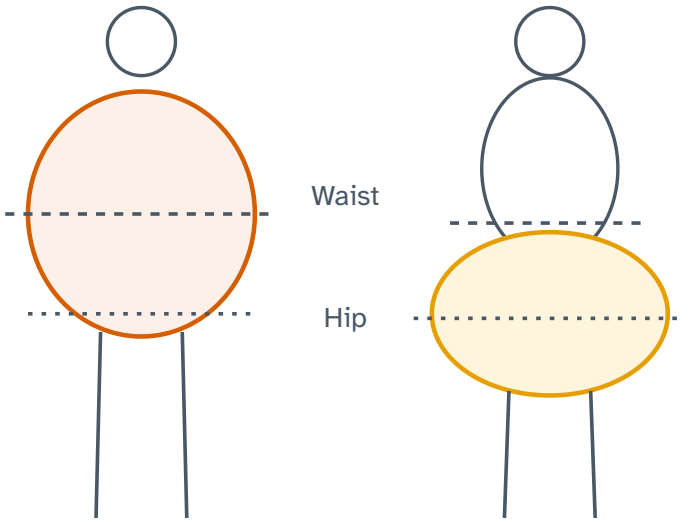
1 Definition and types

Define obesity and list its types: 3 marks in W16 and S25.

- **Obesity** is abnormal or excessive fat accumulation that may impair health (WHO). It is screened with the body mass index and waist, and confirmed, where possible, by body fat.
- **BMI** = weight (kg) ÷ height (m)^2. Example: 68 kg and 1.60 m gives 68 ÷ 2.56 = 26.6 kg/m^2.

Basis	Types
Fat distribution	**Android** (apple, central, upper-body; commoner in men; higher risk of diabetes and heart disease) and **gynoid** (pear, hips and thighs; commoner in women)
Fat cells	**Hyperplastic** (more fat cells; starts in childhood; harder to reverse) and **hypertrophic** (bigger fat cells; adult onset)
Cause	**Primary** (simple, exogenous: energy intake more than use) and **secondary** (endogenous: hypothyroidism, Cushing's syndrome, PCOS, drugs such as steroids, genetic syndromes such as Prader–Willi)
Severity	By BMI grade (section 2)

Apple and pear: where the fat sits, and where to measure



Waist (long dashes): midway between the lowest rib and the top of the iliac crest, at the end of a normal breath out. Hip (short dashes): the widest part of the buttocks. Abdominal fat (orange) carries more risk than hip and thigh fat (yellow).

2 Grades of obesity and their significance

Grades of obesity and their clinical significance: 3 marks in W17, 7 marks in S24; BMI and its uses: 3 marks in W22 and S23.

Category	WHO (kg/m²)	Asian Indian, 2009 (kg/m²)
Underweight	Below 18.5	Below 18.5
Normal	18.5–24.9	18.5–22.9
Overweight	25–29.9	23–24.9
Obese	Class I 30–34.9; class II 35–39.9; class III 40 or more	25 or more

The revised Indian definition (2025)

- Any BMI over 23 kg/m² counts as obesity in Asian Indians; there is no separate "overweight" step.
- **Stage 1:** extra fat without effects on organ function or daily activities.
- **Stage 2:** BMI over 23 plus a large waist (90 cm or more in men, 80 cm or more in women) or waist-to-height ratio over 0.5, plus limits in daily activities (for example knee pain, breathlessness) or an obesity-related disease (for example type 2 diabetes).
- BMI grades in the same paper: I 23–24.9, II 25–27.5, III 27.6–32.4, IV 32.5 or more. Body fat over 25.5% in men and over 38% in women is obesity.

Clinical significance

- Risk of diabetes, high blood pressure, heart disease and death rises with each grade; at the same BMI, Asian Indians carry more body fat and abdominal fat, so the cut-offs are lower.
- Grade guides treatment: lifestyle for all; drugs and closer medical care in higher grades or with disease; bariatric surgery for severe obesity with disease.
- Waist adds what BMI misses: a person with normal BMI and a large waist is still at risk.
- **Uses of BMI:** simple, cheap screening of individuals and populations; tracking change; deciding treatment. **Limits:** does not separate fat from muscle (a muscular athlete scores high), says nothing about fat distribution, needs age charts in children.

3 Body composition and anthropometry

Methods of measuring body composition (S16, W16, 7 marks); anthropometric assessment in obesity (S26, 7 marks); fat mass and fat-free mass (S24, 3 marks).

- **Fat mass:** all the body's fat, both essential fat (in nerves, bone marrow, organs; a little needed for health, more in women) and storage fat under the skin and around the organs.
- **Fat-free mass:** everything else: muscle, bone, organs and water (about three-quarters water). It sets the resting metabolic rate, so weight loss should remove fat and keep fat-free mass (resistance training, enough protein, no crash diets).
- **Body fat percentage** = fat mass ÷ body weight × 100.

Method	How it works	Comment
BMI, waist, hip, waist-to-hip and waist-to-height ratios	Tape and scale	Field methods; waist-to-hip increased risk at 0.90 or more in men and 0.85 or more in women (WHO); waist-to-height over 0.5
Skinfold thickness	Calipers at set sites (for example biceps, triceps, subscapular, suprailiac), then equations give body density and % fat	Cheap and portable; needs a trained tester; less accurate in severe obesity
Bioelectrical impedance (BIA)	A small current passes faster through water-rich lean tissue than through fat	Quick; affected by hydration, food and exercise before the test
Underwater (hydrostatic) weighing	Archimedes' principle: fat floats, lean tissue sinks, giving body density	The old reference method; needs a tank and full breath-out
Air displacement (Bod Pod)	Body volume from displaced air	Accurate, comfortable, costly
DEXA	Two low-dose X-ray energies separate fat, lean tissue and bone	Accurate, regional fat and bone density; costly
CT and MRI	Images of fat areas	Measure visceral fat; research use

Anthropometric assessment in obesity (S26)

1. Weight (light clothes, no shoes, calibrated scale) and height (stadiometer); BMI.
2. Waist circumference (site in the figure); hip circumference; waist-to-hip and waist-to-height ratios.
3. Skinfolds at standard sites; mid-upper-arm and neck circumference (neck for sleep apnoea risk).
4. In children: BMI-for-age on IAP charts (section 5).
5. Record the same time of day, the same side and the mean of two readings; repeat monthly.

4 Causes, set-point theory and effects on health

Set-point theory (S20, 7 marks); obesity and its effects on adult health (W20, 7 marks).

- **Causes:** energy intake more than energy use: high-calorie food, sugary drinks, eating out; sedentary work and screen time; poor sleep and stress; genetics and family habits; pregnancy and caesarean recovery without return to activity; menopause; drugs (steroids, some antidepressants); hormonal disease (hypothyroidism, PCOS); stopping sport or smoking; ageing.
- **Set-point theory:** the brain (hypothalamus, with signals such as leptin from fat) defends a "set" body weight like a thermostat. When a person diets, hunger rises and metabolism falls, so weight returns towards the old level; this explains regain after crash diets. Slow weight loss with regular exercise and lasting habits is thought to help the body settle at a lower weight.

System	Effects of obesity in adults
Metabolic	Insulin resistance, type 2 diabetes, high cholesterol, metabolic syndrome, fatty liver, gallstones, gout
Heart and vessels	Hypertension, coronary heart disease, heart failure, stroke, varicose veins, deep vein thrombosis
Lungs	Breathlessness on effort, obstructive sleep apnoea, obesity hypoventilation
Joints	Knee and hip osteoarthritis, low back pain, flat feet, plantar heel pain
Women	PCOS, infertility, pregnancy problems (gestational diabetes, pre-eclampsia), stress incontinence
Cancer	Breast (after menopause), colon, endometrial and others
Mind and life	Low self-esteem, depression, stigma, poor fitness and quality of life, shorter life

See also obesity in CVR for metabolic syndrome and medical treatment.

5 Childhood obesity

Causes of childhood obesity (S20, W21, W24, 7 marks); causes, ill effects, school fitness screening and health promotion (W20, long answer); 14-year-old boy who eats fast food (W21, long case).

- **Measure:** BMI-for-age on the IAP 2015 charts for 5–18 years: overweight from the line equal to an adult BMI of 23, obesity from the line equal to 27. Under 5 years, use WHO standards.
- **Causes:** fast food, sugary drinks and snacks; large portions; too little play and sport; screen time; short sleep; parents with obesity and family eating habits; urban living with no safe play space; academic pressure; emotional eating; rarely endocrine or genetic disease; low or high birth weight, mother's diabetes in pregnancy, formula feeding.
- **Ill effects:** early type 2 diabetes, high blood pressure and cholesterol, fatty liver, sleep apnoea, asthma; knee pain, flat feet, knock knees, slipped capital femoral epiphysis; early puberty and PCOS in girls; teasing, low self-esteem, depression; obesity continues into adult life.

School fitness screening (W20)

Component	Test (Khelo India, 9–18 years)
Body composition	BMI (plus waist)
Cardiorespiratory endurance	600 m run or walk
Speed	50 m dash
Abdominal strength	Partial curl-ups
Upper-body strength	Push-ups (boys), modified push-ups (girls)
Flexibility	Sit and reach

Plan: consent from parents, screening at the start of the year by trained teachers and physiotherapists, a report card for each child, referral of children with obesity, high blood pressure or very low fitness, and repeat testing at the end of the year.

Health promotion for overweight children

- **Activity:** at least 60 minutes a day of moderate to vigorous activity, mostly aerobic, with vigorous and muscle- and bone-strengthening activity on 3 days a week (WHO); daily physical education and sport periods; active travel to school.
- **Food:** no junk food or sugary drinks in canteens; fruit and home food in tiffin; water bottles; family meals.
- **Habits:** limit recreational screen time; regular sleep.
- **Whole family** changes habits together; no blame or teasing; praise effort, not weight.
- Teacher and parent education; posters and competitions; fun, non-competitive games for children who dislike sport.

6 Weight management: exercise and lifestyle

Exercise prescription and lifestyle change appear in almost every obesity long case (S19, S21, S20, W24, W22, S25).

Exercise prescription (FITT) for weight loss

FITT	Aerobic	Resistance
Frequency	At least 5 days a week	2–3 days a week, not on consecutive days
Intensity	Moderate (40–59% of heart rate reserve; can talk, not sing), progress to vigorous	Moderate load, 8–12 repetitions, 2–3 sets
Time	30 minutes a day building to 60 (150 minutes a week, then more than 250 for real weight loss and to keep it off)	8–10 exercises for the major muscle groups
Type	Brisk walking, cycling, swimming, water exercise, dancing; low impact if joints hurt	Body weight, bands, weights; keeps fat-free mass

- **Heart rate target (Karvonen):** target = resting HR + (maximum HR − resting HR) × intensity. For a 27-year-old with resting HR 80: maximum 220 − 27 = 193; reserve 113; 40–59% gives about 125–147 beats a minute.
- **Evidence:** 150–250 minutes a week prevents weight gain and gives modest loss; more than 250 minutes a week gives clinically important loss, especially with a diet (ACSM, 2009).
- **Diet** (with a dietitian): a deficit of about 500 kcal a day loses about 0.5 kg a week; fewer refined carbohydrates, sugar and fried food; more vegetables, pulses and protein; regular meals.
- **Target:** 5–10% of body weight over 3–6 months; then maintenance.
- **Lifestyle:** more daily steps (stairs, walking during calls), less sitting, 7–8 hours of sleep, stress management, a food and activity diary, family support.
- **Precautions:** screen for heart disease, diabetes and high blood pressure; low-impact work for painful knees (cycling, water exercise, quadriceps strengthening; see section 8); avoid exercising in heat; good footwear; monitor blood sugar in diabetes.

Why crash dieting fails

- Rapid loss is mostly water and muscle, not fat; metabolism slows (set point); hunger rises.
- Fatigue, dizziness, poor concentration, hair loss, constipation, irregular periods, gallstones, nutrient deficiency; risk of disordered eating.
- Weight returns, often with more fat ("yo-yo" dieting).

7 Worked long case: weight loss before marriage

From real papers (W22, 3 + 5 + 7; S25, 5 + 10)

> "A 27-year-old girl has been referred to your physiotherapy clinic for weight reduction. She is 5 ft 3 inches and weighs 68 kgs and has a waist circumference of 99 cms. Her body fat percentage is 50%, and YMCA sit and reach test values are -3 cms. She can do 12 modified push-ups and 34 squats in a minute. She is getting married in 3 months and has started crash dieting for her weight loss. Comment on her fitness status and write about her fitness management program." (S25)

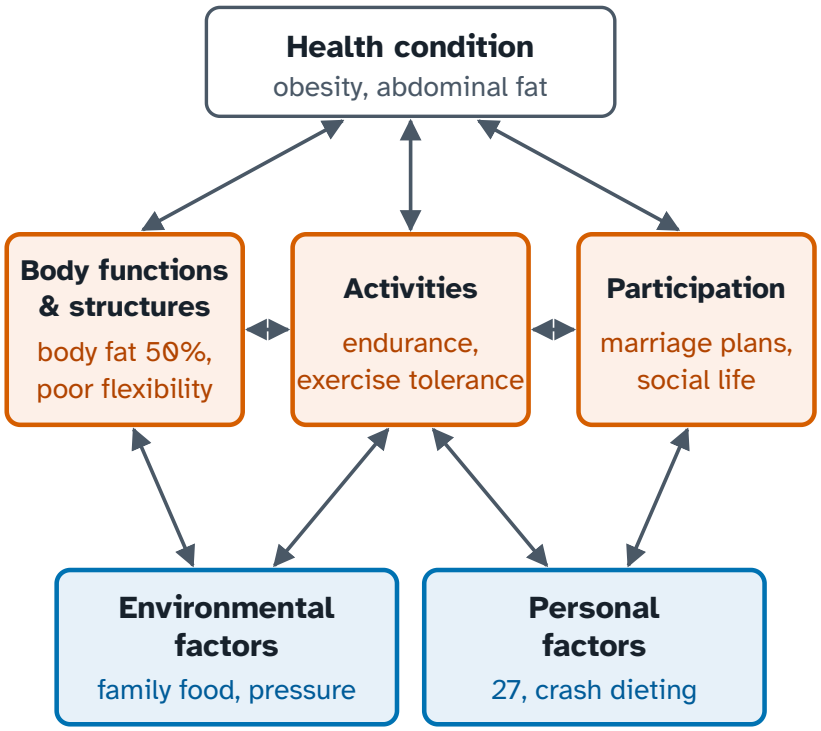
Spot the clues before you write

5 ft 3 in, 68 kg: BMI 26.6. waist 99 cm, body fat 50%: abdominal obesity with very high fat. sit and reach −3 cm: poor flexibility. 12 modified push-ups: fair. marriage in 3 months: a deadline, realistic goals. crash dieting: stop it and explain why.

Fitness status

Measure	Her value	Meaning
BMI	68 ÷ (1.60)² = 26.6 kg/m²	Overweight by WHO; obese by the Asian Indian 2009 cut-off (25 or more); obesity grade II by the 2025 Indian grades
Waist	99 cm; waist-to-height 99 ÷ 160 = 0.62	Abdominal obesity (women 80 cm or more; ratio over 0.5)
Body fat	50%	Very high (obesity in women is over 38% by the 2025 Indian definition)
Flexibility	Sit and reach −3 cm	Poor: cannot reach her toes
Muscular endurance	12 modified push-ups; 34 squats a minute	Push-ups fair for women aged 20–29 (good is 15–20, CSEP); squats recorded as a baseline to retest
Cardiorespiratory fitness	Not given	Test it: six-minute walk, YMCA step test or Rockport walk; check resting HR and BP

The ICF model filled for this case



Draw the six boxes with two-way arrows, then write two or three points from the case in each box.

ICF component	From this case	Clinical reasoning
Health condition	Obesity with abdominal fat (BMI 26.6, waist 99 cm, body fat 50%); rule out thyroid disease and PCOS	Excess fat for her height with central distribution
Body functions and structures	High body fat, poor hamstring and back flexibility, fair muscular endurance, unknown aerobic fitness; fatigue and weakness from crash dieting	Sedentary habits; crash diet removes muscle and water
Activity limitations	Low exercise tolerance; tires in long busy days (wedding preparation)	Low fitness, poor food intake
Participation restrictions	Anxiety about appearance before marriage; social events	Body image and social pressure
Environmental factors	**Barriers:** family and wedding food, social pressure, little time. **Facilitators:** wellness clinic, family support	Habits need family involvement

ICF component	From this case	Clinical reasoning
Personal factors	27 years, motivated, short deadline, believes in crash dieting	Motivation is high; expectations must be realistic

One-line functional diagnosis

A 27-year-old woman with obesity (BMI 26.6, waist 99 cm, body fat 50%), poor flexibility and fair muscular endurance, who is crash dieting before her marriage in 3 months.

Goals

Area	Short-term (4 weeks)	Long-term (3 months and after)
Diet	Stops crash dieting; balanced plan with a dietitian (about 500 kcal deficit)	Keeps healthy eating after marriage
Weight and waist	0.5 kg a week	About 5 kg lost (7%); waist down by 5 cm; lower body fat
Fitness	150 minutes a week of activity; sit and reach to 0 cm	More than 250 minutes a week; 15–20 modified push-ups; better aerobic test

Fitness management programme

- **Warm-up** 5–10 minutes of easy walking and mobility.
- **Aerobic** 5 days a week: brisk walking, cycling, dance or swimming; 30 minutes rising to 50–60; 40–59% of heart rate reserve (about 125–147 beats a minute), progressing to intervals of faster walking.
- **Resistance** 2–3 days a week: squats, lunges, bridges, modified push-ups, rows with a band, planks; 8–12 repetitions, 2–3 sets.
- **Flexibility** daily: hamstring, calf, hip flexor and back stretches, 30 seconds × 2–3.
- **Cool-down** and stretching.
- **Lifestyle:** a daily step target that rises each week, stairs, sleep, water, food diary; explain why crash diets fail; a plan to continue after the wedding.
- **Review** every 2–4 weeks: weight, waist, body fat, fitness tests.

Practice cases from the papers

- **S21:** 35-year-old banker, 155 cm, 96 kg (BMI 40, class III), weight gain after a caesarean, long working hours: exercise prescription, lifestyle change and ergonomics (desk set-up and breaks, see work stress (p. 203)).
- **S19:** 35-year-old housewife, 150 cm, 80 kg (BMI 35.6) who cooks for five: home-based exercise and lifestyle change that fit around housework.
- **S20 and W24:** moderately obese lecturer with type 2 diabetes and hypertension: goals, precautions (blood sugar, BP, feet) and diet tips.
- **W21:** 14-year-old boy, 64 kg, 5 ft 3 in (BMI 25.0, above the adult-equivalent 23 line): assessment with IAP charts, family-based short- and long-term plan.
- **S16 and W16:** obesity with knee pain (section 8).

8 Obesity with knee pain

S16 (50-year-old man, class I obesity, both knees) and W16 (54-year-old homemaker, 150 cm, 75 kg, knee pain worse on stairs): ICF and rehabilitation.

- **Why the knees hurt:** extra load on the joint with every step and stair, weak quadriceps, low activity, and inflammation from fat tissue; early knee osteoarthritis.
- **Exercise:** quadriceps (static quads, straight-leg raise, terminal knee extension, mini-squats within pain), hip abductors and extensors, hamstring and calf stretches; non-weight-bearing aerobic exercise (static cycling, water

walking) progressing to walking.

- **Weight loss** with diet and exercise; losing more than 5% of body weight within about 20 weeks improved disability in knee osteoarthritis (Christensen et al., 2007).
- **Joint protection:** avoid squatting and sitting cross-legged on the floor during flares, use a higher chair and western toilet, handrail on stairs (up with the good leg, down with the painful one), a stick in the opposite hand, cushioned footwear.
- **Pain relief:** heat or ice, TENS, as a bridge to exercise.

9 Exam corner

Past questions (MUHS, 2016–2026)

Long cases

50-year-old with class I obesity and knee pain (S16); 54-year-old homemaker with knee pain (W16); housewife who cooks for five (S19); obese lecturer with diabetes and hypertension (S20, W24); childhood obesity, school screening and health promotion (W20); banker after caesarean (S21); 14-year-old boy (W21); bride-to-be crash dieting (W22, S25).

7 marks

Methods of measuring body composition (S16, W16); set-point theory and causes of childhood obesity (S20); obesity and its effects on adult health (W20); causes of childhood obesity (W21, W24); grades of obesity and clinical significance (S24); anthropometric assessment (S26).

3 marks

Define obesity and its types (W16, S25); grades of obesity (W17); BMI and its uses (W22, S23); fat mass and fat-free mass (S24).

Keywords to underline

BMI = kg/m² Asian Indian cut-offs waist 90/80 cm waist-to-height 0.5 android and gynoid fat-free mass skinfold, BIA, DEXA set point FITT heart rate reserve more than 250 minutes a week 5–10% weight loss no crash diets

Answer skeleton: an obesity long case (15 marks)

1. **[Assessment, 5]** BMI with working, waist, body fat, fitness tests, risk screening; interpretation; ICF and one-line diagnosis.
2. **[Goals, 3]** Short- and long-term, measurable (weight, waist, minutes of activity, test scores).
3. **[Programme, 7]** FITT for aerobic, resistance and flexibility with heart rate target; diet with a dietitian; lifestyle; precautions; review.

Draw these

- Apple and pear shapes with waist and hip sites (section 1)
- The ICF model for the case

Viva questions

1. Calculate the BMI of a person 1.50 m tall weighing 75 kg.
2. What are the Asian Indian BMI cut-offs?
3. Where is the waist measured?
4. Name four methods of measuring body composition.
5. What is set-point theory?

6. How many minutes of activity a week are needed for meaningful weight loss?

Linked practical spots

- Measure height, weight, waist and hip on a classmate and calculate BMI and ratios; use skinfold calipers; perform the sit-and-reach and push-up tests.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Fat mass and fat-free mass. (S24)**

Answer
Body weight = fat mass + fat-free mass. Fat mass: essential and storage fat. Fat-free mass: muscle, bone, organs and water (about three-quarters water); it determines resting metabolic rate. Measured by skinfolds, BIA, DEXA, underwater weighing. Weight loss should reduce fat mass and preserve fat-free mass (resistance training, protein, no crash diets).

3 m **Body mass index and its uses. (W22, S23)**

Answer
BMI = weight (kg) ÷ height $(m)^2$. WHO: normal 18.5–24.9, overweight 25–29.9, obese 30 or more; Asian Indians: normal up to 22.9, overweight 23–24.9, obese 25 or more. Uses: screening, grading, tracking, treatment decisions, population surveys. Limits: does not separate fat from muscle or show fat distribution.

7 m **Discuss five causes of childhood obesity. (W21)**

Answer
Define (BMI-for-age, IAP adult-equivalent 23 and 27 lines). Causes: fast food and sugary drinks; inactivity and screen time; short sleep; family obesity and habits; urban environment without play space; academic stress and emotional eating; rarely endocrine or genetic disease; early-life factors (maternal diabetes, birth weight). Add one line on prevention for each.

LAQ **A 35-year-old banker, 155 cm, 96 kg, gained weight after a caesarean six years ago; long working hours. Exercise prescription, lifestyle modification and ergonomic advice. (5 + 5 + 5, S21)**

Answer
BMI 96 ÷ 1.55^2 = 40.0 (class III). Screen BP, sugar, thyroid, joints. **Exercise:** low-impact aerobic 5 days a week, 20–30 minutes building to 60, 40–59% HRR; resistance 2–3 days, 8–12 reps; stretches; progress to more than 250 minutes a week; check abdominal wall (diastasis) and back. **Lifestyle:** dietitian-led 500 kcal deficit, food diary, sleep, stress, walking breaks and stairs, home-cooked lunch. **Ergonomics:** workstation set-up, chair with lumbar support, screen at eye level, feet supported, micro-breaks every 30–60 minutes with desk exercises.

MCQ **The waist circumference cut-off for abdominal obesity in Indian women is:**

A. 70 cm

B. 80 cm

C. 88 cm

D. 102 cm

Answer
B. 90 cm in men.

MCQ **The body composition method based on Archimedes' principle is:**

A. BIA

B. Skinfold

C. Hydrostatic weighing

D. DEXA

Answer
C.

MCQ **Weekly physical activity linked with clinically important weight loss is:**

A. 60 minutes

B. 100 minutes

C. 150 minutes

D. More than 250 minutes

Answer
D.

Last recall check. Calculate a BMI, give both sets of cut-offs, list four body composition methods, explain set-point theory, and write a FITT prescription with a heart rate target.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Community Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. World Health Organization. Obesity: preventing and managing the global epidemic (TRS 894), 2000; Waist circumference and waist–hip ratio: report of a WHO expert consultation, 2008.
3. Misra A, et al. Consensus statement for diagnosis of obesity, abdominal obesity and the metabolic syndrome for Asian Indians. *J Assoc Physicians India* 2009;57:163–70.
4. Misra A, Vikram NK, Ghosh A, Ranjan P, Gulati S, et al. Revised definition of obesity in Asian Indians living in India. *Diabetes Metab Syndr* 2025;19:102989.
5. Indian Academy of Pediatrics Growth Charts Committee. Revised IAP growth charts for height, weight and body mass index for 5- to 18-year-old Indian children. *Indian Pediatr* 2015;52:47–55.
6. Donnelly JE, et al. American College of Sports Medicine position stand: appropriate physical activity intervention strategies for weight loss and prevention of weight regain for adults. *Med Sci Sports Exerc* 2009;41:459–71.
7. World Health Organization. Guidelines on physical activity and sedentary behaviour, 2020.
8. Christensen R, Bartels EM, Astrup A, Bliddal H. Effect of weight reduction in obese patients diagnosed with knee osteoarthritis: a systematic review and meta-analysis. *Ann Rheum Dis* 2007;66:433–9.
9. Sports Authority of India. Khelo India fitness assessment for schools: administration manual (age group 9–18+).
10. Canadian Society for Exercise Physiology. Physical Activity Training for Health (CSEP-PATH) resource manual: push-up norms.

Part 2

Women's health

20 hours theory, 20 field

2(a–c)	Women in India: social issues, legal rights and health promotion	37
2(d, e)	Pregnancy: physiological changes, discomforts and antenatal care	43
2(e)	Labour and postnatal care	52
2(d)	Menopause and osteoporosis	63
2(f)	Urogenital dysfunction: incontinence, prolapse and the pelvic floor	70

2(a–c) Women in India: social issues affecting physical health, legal rights and benefits, and health promotion for rural women

> This part was asked in 5 of 20 papers: social issues affecting women's physical function three times, plus two long answers on legal rights and on reaching rural pregnant women. The physiotherapist's angle is always the same: name the social issue, show how it reaches the body (anaemia, early pregnancy, heavy work, no toilet, no time to exercise), and say what the physiotherapist and the law can do about it.

Syllabus	Ch 2(a) Women in India; 2(b) social issues having impact on physical function; 2(c) legal rights and benefits related to health
Priority	**Tier 2.** In 5 of 20 papers (2016–2026): 2 long answers and 3 seven-mark questions.
Asked as	LAQ W19 · S22 · 7m S21 · W21 · W24
Usual split	15 (social issues 7, legal rights and benefits 8; or approach to the community 5, community management 10); 7 (social issues; health promotion for rural women)
Checked against	National Family Health Survey 5 (2019–21); Sample Registration System maternal mortality bulletin 2019–21; Census of India 2011; texts of the Acts listed (Maternity Benefit Amendment 2017, MTP Amendment 2021, PWDVA 2005, POSH 2013, PCPNDT 1994); Ministry of Women and Child Development scheme pages. Full list under Sources (p. 42). Content checked 20 October 2026.

1 Women's health in India: key numbers

- **Anaemia:** 57% of women aged 15–49 were anaemic in NFHS-5 (2019–21), up from 53% in NFHS-4.
- **Maternal mortality:** 93 per 100,000 live births in 2019–21, down from 130 in 2014–16 (Sample Registration System).
- **Early marriage:** about 1 in 4 women aged 20–24 were married before 18 (NFHS-5).
- **Violence:** 29% of ever-married women reported spousal violence (NFHS-5).
- **Literacy:** female literacy 65% against 82% for men (Census 2011).

2 Social issues affecting women's physical function

7 marks in W21 and W24; 7 marks of the W19 long answer.

Social issue	Effect on the body and function
Gender discrimination and son preference	Girls fed less and later, less medical care: stunting, anaemia, weak bones
Early marriage and early, repeated pregnancies	Obstructed labour, fistula, prolapse, incontinence, anaemia, low bone mass, maternal deaths
Poverty and malnutrition	Anaemia, fatigue, low muscle strength, osteoporosis, poor wound healing
Heavy domestic and farm work, double burden with paid work	Back and neck pain from squatting, stooping and head-loading water and fuel; knee osteoarthritis; no time to rest or exercise
Smoke from cooking on chulhas	Chronic bronchitis, COPD, eye problems
Poor sanitation (no toilet)	Holding urine and stool, urinary infections, pelvic floor problems, unsafe trips at night
Low literacy and awareness	Late care for illness, little knowledge of exercise, posture and pelvic floor health
Restricted mobility and decision-making	Need permission or escort to visit clinics; cannot attend repeated physiotherapy sessions

Social issue	Effect on the body and function
Domestic violence and dowry	Injuries, chronic pain, depression, anxiety
Cultural limits on exercise and sunlight	Inactivity, obesity, vitamin D deficiency
Stigma around menstruation and widowhood	Poor menstrual hygiene and infections; isolation of widows, poverty in old age

3 Legal rights and benefits for women

Legal rights and benefits provided by the Government of India (8 marks of the W19 long answer).

Constitution

- Equality before law (Article 14); no discrimination by sex, with special provisions for women allowed (Article 15); equal pay for equal work as a directive principle (Article 39); just and humane conditions of work and maternity relief (Article 42); one-third of panchayat seats reserved for women (73rd Amendment, 1992); one-third of Lok Sabha and assembly seats (Women's Reservation Act, 2023, to take effect after delimitation).

Law	What it does
Maternity Benefit Act, 1961 (amended 2017)	26 weeks of paid maternity leave in establishments covered by the Act; nursing breaks; crèche in larger establishments
Medical Termination of Pregnancy Act, 1971 (amended 2021)	Safe, legal abortion up to 20 weeks, and up to 24 weeks for special categories, by registered doctors
PCPNDT Act, 1994	Bans sex determination of the fetus to prevent female foeticide
Dowry Prohibition Act, 1961	Giving or taking dowry is an offence
Protection of Women from Domestic Violence Act, 2005	Protection orders, right to live in the shared home, maintenance, protection officers
Prohibition of Child Marriage Act, 2006	Marriage of a girl under 18 is prohibited and punishable
Sexual Harassment of Women at Workplace Act, 2013	Internal complaints committees at workplaces
Hindu Succession (Amendment) Act, 2005	Daughters get equal rights in ancestral property
Equal pay (Code on Wages, 2019)	No discrimination in wages by gender

Health schemes and benefits

Scheme	Benefit
Janani Suraksha Yojana (2005)	Cash for delivery in a health facility
Janani Shishu Suraksha Karyakram (2011)	Free delivery, caesarean, drugs, diet, blood and transport in public facilities
Pradhan Mantri Matru Vandana Yojana (2017)	₹5,000 maternity benefit for the first child; an additional benefit for a second child if a girl (from 2022)
Pradhan Mantri Surakshit Matritva Abhiyan (2016)	Free antenatal check-up by a doctor on the 9th of every month
Poshan Abhiyaan and Anaemia Mukt Bharat (2018)	Nutrition, iron and folic acid, deworming for girls and women
Pradhan Mantri Ujjwala Yojana (2016)	LPG connections for poor households, reducing chulha smoke
Beti Bachao Beti Padhao (2015)	Prevents sex selection, promotes girls' education
Mission Shakti	One Stop Centres for women facing violence; Women Helpline 181
Swachh Bharat Mission	Household toilets

Women with disability also have the rights in disability policies (p. 144); senior women in senior citizens (p. 85).

4 Health promotion for rural women

Strategies of health promotion for rural women (S21, 7 marks).

1. **Know the community:** meet the sarpanch, ASHA, ANM, anganwadi worker and self-help group leaders; learn the main problems (anaemia, back pain, pelvic floor problems, pregnancy, smoke, diabetes) and the women's daily timetable.
2. **Use existing platforms:** anganwadis, Village Health, Sanitation and Nutrition Days, self-help group meetings, mahila mandals, PHC and Ayushman Arogya Mandir clinics, temples and schools.
3. **Choose the message:** nutrition and iron; antenatal and postnatal exercise; pelvic floor exercise; back care and safe ways to lift water pots, squat and carry; osteoporosis prevention; physical activity (walking groups, yoga); clean cooking fuel; toilet use; menstrual hygiene; screening for BP, diabetes and cancer.
4. **Use the right methods:** local language, pictures, demonstrations and practice, songs and role plays, short sessions at convenient times, women educators, home visits for those who cannot come.
5. **Involve men, mothers-in-law and elders** so women get permission and time.
6. **Link to schemes and services:** JSY, PMMVY, PMSMA, Ujjwala, pensions, PHC referral.
7. **Train local women** (ASHAs, self-help group leaders) to continue the programme.
8. **Evaluate:** attendance, knowledge before and after, behaviour change (exercise, iron tablets, institutional delivery), and feedback.

5 Worked long answer: reaching first-time pregnant women in a village

From a real paper (S22, 5 + 10)

> "Write your strategy to approach the rural community for health education on role of Physiotherapy in Primigravidae. Explain your community management for the same."

Strategy to approach the community (5 marks)

- Permission and partnership: PHC medical officer, gram panchayat, ASHA, ANM and anganwadi worker.
- Identify primigravidae from ASHA and anganwadi registers and antenatal records.
- Hold sessions where women already gather: anganwadi, Village Health, Sanitation and Nutrition Day, the PMSMA clinic on the 9th of the month.
- Invite husbands and mothers-in-law to one session to win family support.
- Use the local language, pictures, flip charts, videos on phones and live demonstrations; keep sessions short and practical.

Community management (10 marks)

Session	Content
1. Body changes and posture	Why the back aches; standing, sitting and lying positions; safe lifting of water pots and children; avoiding long squatting; side-lying rest (see pregnancy (p. 43))
2. Exercise	Walking 30 minutes most days; pelvic tilts, cat–camel, calf and ankle exercises; when to stop (warning signs)
3. Pelvic floor	How to find and train the pelvic floor; prevents leakage and prolapse later (see pelvic floor (p. 70))
4. Labour preparation	Breathing and relaxation, labour positions, what happens in labour, going to hospital in time (JSY, JSSK, ambulance 108) (see labour (p. 52))
5. After birth	Early pelvic floor and abdominal exercise, feeding posture, back care, spacing of children

Session	Content
Throughout	Nutrition and iron tablets, antenatal visits, danger signs (bleeding, swelling, headache, reduced baby movements)

- **Follow-up:** ASHAs trained to repeat the exercises during home visits; pictorial handouts; a WhatsApp group where phones are common.
- **Referral:** women with pain, leakage, high BP or bleeding to the PHC.
- **Evaluation:** attendance, knowledge test before and after, exercise practice, institutional delivery rate, postnatal back pain and leakage at follow-up.

6 Exam corner

Past questions (MUHS, 2016–2026)

Long answers

Impact of social issues on women's physical health and legal rights and benefits (W19, 7 + 8); approaching a rural community for health education on physiotherapy for primigravidae and community management (S22, 5 + 10).

7 marks

Health promotion strategies for rural women (S21); social issues affecting physical function of women in India (W21, W24).

Keywords to underline

anaemia 57% early marriage double burden head-loading chulha smoke no toilet Maternity Benefit Act 26 weeks PWDVA 2005 PCPNDT JSY, JSSK, PMMVY, PMSMA ASHA and anganwadi local language and demonstration

Answer skeleton: social issues affecting women (7 marks)

1. **[Introduction, 1]** Two or three numbers on women's health.
2. **[Issues and effects, 5]** Eight issues, each with its effect on the body.
3. **[Role of PT, 1]** Education, exercise, ergonomics, pelvic floor, links to schemes.

Draw these

- A flow from a social issue to a physical problem to a PT action (for example no toilet → holding urine → UTI and pelvic floor problems → pelvic floor training and education)

Viva questions

1. What percentage of Indian women are anaemic?
2. How long is paid maternity leave under the 2017 amendment?
3. What does the PCPNDT Act prohibit?
4. Name two schemes for safe delivery.
5. How does chulha smoke affect women's health?
6. Where in a village would you hold a health education session?

Linked practical spots

- Prepare a pictorial handout on back care for women who carry water; conduct a mock group session on pelvic floor exercise in the local language.

Test yourself

Cover each answer and write yours on paper first, then check.

7 m **What are the strategies of health promotion for rural women? (S21)**

Answer

Know the community and its leaders and workers; use existing platforms (anganwadi, VHSND, SHGs, PHC); choose messages (nutrition, antenatal and postnatal exercise, pelvic floor, back care, osteoporosis, activity, clean fuel, toilets, screening); use local language, pictures and demonstrations at convenient times; involve men and elders; link to schemes; train local women; evaluate.

7 m **Discuss various social issues having impact on physical function of women in India. (W21, W24)**

Answer

Discrimination and nutrition; early marriage and repeated pregnancies; poverty and anaemia; heavy domestic and farm work with double burden; chulha smoke; poor sanitation; low literacy; restricted mobility and decision-making; violence; cultural limits on exercise and sunlight; menstrual and widowhood stigma; each with its physical effect and a PT response.

LAQ **Describe the impact of social issues on physical health of women in India. Explain the legal rights and benefits provided by the Government of India. (7 + 8, W19)**

Answer

Social issues and effects (section 2). Rights: constitutional articles; Acts (Maternity Benefit, MTP, PCPNDT, Dowry Prohibition, Domestic Violence, Child Marriage, POSH, Succession, equal pay); schemes (JSY, JSSK, PMMVY, PMSMA, Poshan and Anaemia Mukt Bharat, Ujjwala, Beti Bachao Beti Padhao, Mission Shakti, Swachh Bharat); panchayat reservation.

MCQ **Paid maternity leave under the Maternity Benefit (Amendment) Act 2017 is:**

A. 12 weeks

B. 18 weeks

C. 26 weeks

D. 52 weeks

Answer

C.

MCQ **The scheme providing free antenatal check-ups on the 9th of every month is:**

A. JSY

B. PMSMA

C. Ujjwala

D. ICDS

Answer

B.

MCQ **The law that prohibits sex determination of the fetus is the:**

A. MTP Act

B. PCPNDT Act

C. Dowry Prohibition Act

D. Maternity Benefit Act

Answer

B.

Last recall check. List ten social issues with their physical effects, eight laws and eight schemes for women, and eight steps of health promotion for rural women.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Community Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. International Institute for Population Sciences. National Family Health Survey (NFHS-5), 2019–21: India fact sheet.
3. Office of the Registrar General of India. Special bulletin on maternal mortality in India 2019–21 (Sample Registration System).
4. Census of India 2011: literacy by sex.
5. The Maternity Benefit (Amendment) Act, 2017; the Medical Termination of Pregnancy (Amendment) Act, 2021; the Pre-Conception and Pre-Natal Diagnostic Techniques Act, 1994; the Protection of Women from Domestic Violence Act, 2005; the Prohibition of Child Marriage Act, 2006; the Sexual Harassment of Women at Workplace Act, 2013.
6. Ministry of Health and Family Welfare and Ministry of Women and Child Development: JSY, JSSK, PMMVY, PMSMA, Mission Shakti scheme pages.

2(d, e) Pregnancy: physiological changes, discomforts, diastasis recti, exercise guidelines and antenatal classes

> Pregnancy was asked in 13 of 20 papers, mostly as 7-mark questions: physiological changes (four times), diastasis recti (four times), exercise guidelines and contraindications, aims of obstetric physiotherapy, antenatal classes. Learn the changes system by system, because every discomfort, precaution and exercise rule follows from them.

Syllabus	Ch 2(d) Anatomical and physiological variations associated with pregnancy; 2(e) antenatal care
Priority	**Tier 1.** In 13 of 20 papers (2016–2026): 2 long answers, 9 seven-mark and 8 three-mark questions.
Asked as	LAQ W17 · S22 · 7m W16 · W18 · S20 · W20 · W21 · S22 · S24 · W24 · S25 · 3m S17 ×2 · W17 · W18 · W19 · S21 · S22 · W24
Usual split	15 (reasoning 5, functional diagnosis 5, PT 5; or changes 4, five discomforts 5, their PT 6); 7 (changes; diastasis recti; exercise guidelines; aims and principles; antenatal class)
Checked against	ACOG Committee Opinion 804 on exercise in pregnancy and the postpartum period (2020); Canadian guideline for physical activity throughout pregnancy (Mottola et al., 2018); Institute of Medicine weight-gain guidelines (2009); prevalence of diastasis recti (Sperstad et al., 2016). Full list under Sources (p. 51). Content checked 20 October 2026.

1 Physiological changes in pregnancy

7 marks in S20 and W20; six changes for 3 marks in W17; 4 marks of the S22 long answer.

System	Change	What it means for PT
Hormones	Progesterone, oestrogen and relaxin rise	Ligament laxity, smooth muscle relaxation (veins, bowel, ureters)
Heart and blood	Blood volume up by about 40–50%, cardiac output up, resting heart rate up by 10–20 beats, blood pressure dips in mid-pregnancy; dilutional anaemia; blood clots more easily	Heart rate is a poor guide to effort (use the talk test or Borg); clot risk
Supine position	From about 20 weeks the uterus presses on the inferior vena cava when she lies flat	Dizziness, low BP (supine hypotension): use side-lying or a wedge under the right hip
Breathing	Diaphragm rises, rib cage widens; tidal volume and minute ventilation rise (progesterone); FRC falls; mild breathlessness	Breathlessness on effort is normal; breathing exercises; upright positions
Musculoskeletal	Weight gain; centre of gravity moves forward; posture changes; lax ligaments of the pelvis and spine; abdominal muscles stretch and may separate; pelvic floor carries more load; feet flatten and widen	Back and pelvic girdle pain, diastasis recti, leakage, balance changes, foot pain
Kidneys and bladder	Filtration rises; uterus presses on the bladder	Frequency, urgency, stress leakage
Gut	Slower stomach and bowel; relaxed lower oesophageal sphincter	Nausea, heartburn, constipation, haemorrhoids
Metabolism	Basal metabolic rate rises; insulin resistance later in pregnancy; heat production up	Avoid overheating; gestational diabetes risk
Fluid and skin	Fluid retention; stretch marks, darker skin	Ankle swelling, carpal tunnel syndrome, varicose veins

Recommended total weight gain by pre-pregnancy BMI (Institute of Medicine, 2009)

BMI before pregnancy	Total gain
Underweight (below 18.5)	12.5–18 kg
Normal (18.5–24.9)	11.5–16 kg
Overweight (25–29.9)	7–11.5 kg
Obese (30 or more)	5–9 kg

2 Musculoskeletal dysfunctions and common discomforts

Musculoskeletal dysfunctions in pregnancy (S21, 3 marks); five discomforts and their PT (S22, 11 marks of the long answer).

Problem	Cause	Physiotherapy
Low back pain	Posture change, weight, lax ligaments, weak abdominals	Posture and body mechanics, pelvic tilts, cat–camel in four-point kneeling, back and hip strengthening, heat (not on the abdomen), side-lying with a pillow between the knees, supportive belt
Pelvic girdle pain (sacroiliac or symphysis pubis)	Laxity and uneven load on the pelvic joints	Pelvic belt, keep knees together when turning in bed and getting out of a car, avoid standing on one leg (sit to dress), short steps, stabilising exercise, crutches if severe
Diastasis recti	Stretch of the linea alba	Section 3
Carpal tunnel syndrome	Fluid retention	Night wrist splint in neutral, elevation, tendon and nerve gliding, hand pumping
Leg cramps	Uncertain; fatigue, circulation	Calf stretching before bed, ankle pumps, gentle massage, stretch the calf (pull toes up) during a cramp
Varicose veins and ankle swelling	Progesterone, pressure on pelvic veins, standing	Leg elevation, ankle pumps, compression stockings, avoid long standing, walking and swimming
Heel, foot and knee pain	Weight gain, flattened arches, lax ligaments, standing	Supportive footwear and insoles, calf and plantar fascia stretches, foot and leg strengthening, ice, rest breaks
Breathlessness and rib pain	Uterus pushes up the diaphragm; rib cage widens	Upright posture, side-flexion stretches, lateral costal breathing, rest in supported sitting
Urinary frequency and leakage	Pressure on the bladder, pelvic floor load	Daily pelvic floor training, the knack (see incontinence (p. 70))
Constipation and heartburn	Slow gut, relaxed sphincters	Walking, fibre and fluids, footstool posture; small meals, head-up sleeping
Others	De Quervain's tenosynovitis, round ligament pain, coccyx pain, meralgia paraesthetica	Splint and posture; supported positions; cushion

Hyperemesis gravidarum (W19, 3 marks)

- Severe nausea and vomiting of pregnancy with dehydration, weight loss and electrolyte imbalance, often needing hospital care; medical treatment is fluids and anti-emetics.
- **PT role:** relaxation and slow breathing to ease nausea and anxiety; acupressure wristbands at the P6 point (some women find them helpful); gentle bed and chair exercises and ankle pumps to prevent deconditioning and clots while in bed; early walking as she improves; energy conservation; posture and positioning; emotional support.

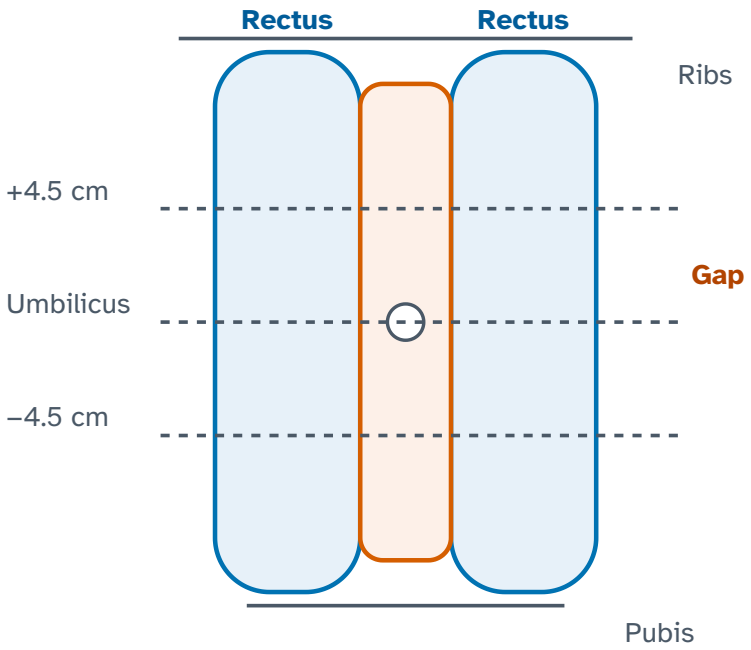
3 Diastasis recti

7 marks in W16, S22 and S24; 3 marks in S17; assessment for 3 marks in W24.

- **Definition:** separation of the two rectus abdominis muscles along the linea alba, from stretching in pregnancy.

- **How common:** in first-time mothers, 33% at 21 weeks of pregnancy, 60% at 6 weeks after birth, 45% at 6 months and 33% at 12 months (Sperstad et al., 2016).
- **Risk factors usually listed:** multiple or large babies, many pregnancies, weak abdominals, obesity, heavy lifting; the study above found no clear risk factor.
- **Effects:** abdominal doming or bulging, poor trunk support, possibly back and pelvic pain, cosmetic worry; rarely hernia.

Testing for diastasis recti: three levels



Two finger-widths or more at any level counts as diastasis (the definition Sperstad et al. used). Recti in blue, the widened linea alba in orange.

Assessment

1. Crook lying; she lifts her head and shoulders slowly until the shoulder blades just clear the bed.
2. Place the fingers across the midline at the umbilicus, 4.5 cm above and 4.5 cm below; count finger-widths between the muscle edges (or measure with calipers or ultrasound).
3. Note depth and tension of the linea alba (does it feel firm or do the fingers sink?) and any doming.
4. Record pain, back and pelvic symptoms, pelvic floor function and function (lifting, getting up).

Management

- **In pregnancy:** roll to the side to get up from lying; avoid straining and breath-holding; transversus abdominis and pelvic floor activation; good posture; avoid exercises that make the abdomen dome.
- **After birth:** transversus drawing-in with pelvic floor, pelvic tilts, side-lying and four-point exercises; progress to head lifts and curl-ups only when she can keep the abdomen flat (pre-activate transversus first); later, loaded functional training.
- Splinting the abdomen with crossed hands or a binder for comfort when coughing and getting up in the early days.
- Correct lifting, constipation care, weight management; refer for surgical repair only if a large gap with hernia or major symptoms persists after a long rehabilitation period.

4 Exercise in pregnancy

Guidelines for women exercising in pregnancy (W18, 7 marks); exercises contraindicated in pregnancy (S17, S22, 3 marks).

- **Benefits:** less gestational diabetes, pre-eclampsia and excessive weight gain, better mood and sleep, less back pain, better fitness for labour and recovery; physical activity is not associated with miscarriage or preterm birth

(Canadian guideline, 2018).

- **Dose:** at least 150 minutes a week of moderate activity over at least 3 days, ideally daily, combining aerobic and resistance exercise (Canadian guideline; ACOG). Women active before pregnancy can continue their usual activity.
- **Intensity:** moderate, by the talk test (can talk but not sing) or Borg 12–14, because heart rate is altered in pregnancy.
- **Good choices:** walking, stationary cycling, swimming and water exercise, low-impact aerobics, prenatal yoga and Pilates (not hot), light resistance training, daily pelvic floor training.
- **Precautions:** warm-up and cool-down; drink water; avoid heat and humidity; avoid lying flat for long after the first trimester (side-lying or inclined instead); avoid breath-holding and heavy straining; avoid end-range and bouncing stretches (lax ligaments); good footwear; eat enough.

List	Contents (ACOG, 2020)
Activities to avoid	Contact sports (boxing, hockey, kabaddi); activities with a high risk of falling (horse riding, skiing, gymnastics, off-road cycling); scuba diving; sky diving; hot yoga or hot Pilates; unaccustomed exercise at high altitude
Absolute contraindications to aerobic exercise	Significant heart disease; restrictive lung disease; incompetent cervix or cerclage; multiple pregnancy at risk of preterm labour; persistent bleeding in the second or third trimester; placenta praevia after 26 weeks; preterm labour; ruptured membranes; pre-eclampsia or pregnancy-induced hypertension; severe anaemia
Stop exercising if	Vaginal bleeding; regular painful contractions; leaking fluid; breathlessness before starting; dizziness; headache; chest pain; muscle weakness affecting balance; calf pain or swelling

Exercises usually listed as contraindicated in Indian textbooks: double leg raises and full sit-ups (strain the abdominal wall), prolonged supine exercise after the first trimester, deep squats and lunges at the end of range with pelvic girdle pain, ballistic stretching, heavy lifting with breath-holding, jumping and high-impact work in late pregnancy, and any of the activities in the table.

5 Obstetric physiotherapy and antenatal classes

Aims and principles of obstetric physiotherapy (W21, W24, 7 marks); early bird classes (W18, 3 marks); antenatal programme goals and class structure (S25, 7 marks).

Aims of obstetric physiotherapy

- Help the woman adapt to the physical changes of pregnancy and prevent or treat discomforts.
- Keep her fit and active safely; control weight gain.
- Protect the back, pelvis, abdomen and pelvic floor.
- Prepare her for labour: knowledge, positions, relaxation, breathing, less fear.
- Help recovery after birth and the return to fitness (see labour and postnatal care (p. 52)).
- Prevent long-term problems: incontinence, prolapse, back pain, diastasis.

Principles

- Assess first and screen for contraindications and red flags; work with the obstetrician.
- Moderate, regular exercise with precautions (section 4); progress with the trimester.
- Education and self-management: posture, body mechanics, rest positions.
- Include the partner and family; respect beliefs; group classes for support.
- Individualise: high-risk pregnancies need tailored or reduced activity.

Early bird and antenatal classes

- **Early bird class** (first trimester): changes of pregnancy, nutrition, posture and back care, safe exercise, pelvic floor, warning signs, discomforts of early pregnancy; builds healthy habits early.

- **Antenatal classes** (usually from about 28 weeks, weekly for 6–8 sessions, groups of 8–12, partner invited to some):
  - **Goals:** knowledge of pregnancy, labour and the baby; fitness and posture; relief of discomforts; pelvic floor training; relaxation, breathing and positions for labour; postnatal recovery and baby care; confidence and less fear.
  - **Structure of a session (about 90 minutes):** welcome and questions (10 minutes); a talk on one topic (20 minutes); exercise: warm-up, low-impact aerobic, strengthening, pelvic floor, stretching (30 minutes); relaxation and breathing practice (20 minutes); summary and handout (10 minutes).
  - **Topics across the course:** body changes and posture; exercise and discomforts; pelvic floor; signs of labour and when to go to hospital; stages of labour and pain relief; labour positions and breathing (partner session); caesarean section; postnatal care, feeding and baby handling.

6 Worked long case: a nurse in her third trimester

From a real paper (W17, 5 + 5 + 5)

> "A 25 years, nurse, in third trimester complains of severe heel and knee pain on standing causing difficulty in walking. She has gained nearly 20 kg of extra weight. She wishes to continue with her job till her due date which is one month away. i) Discuss the reasoning for such impairments in this community. (5) ii) Explain the functional diagnosis. (5) iii) Discuss the physiotherapy management for her."

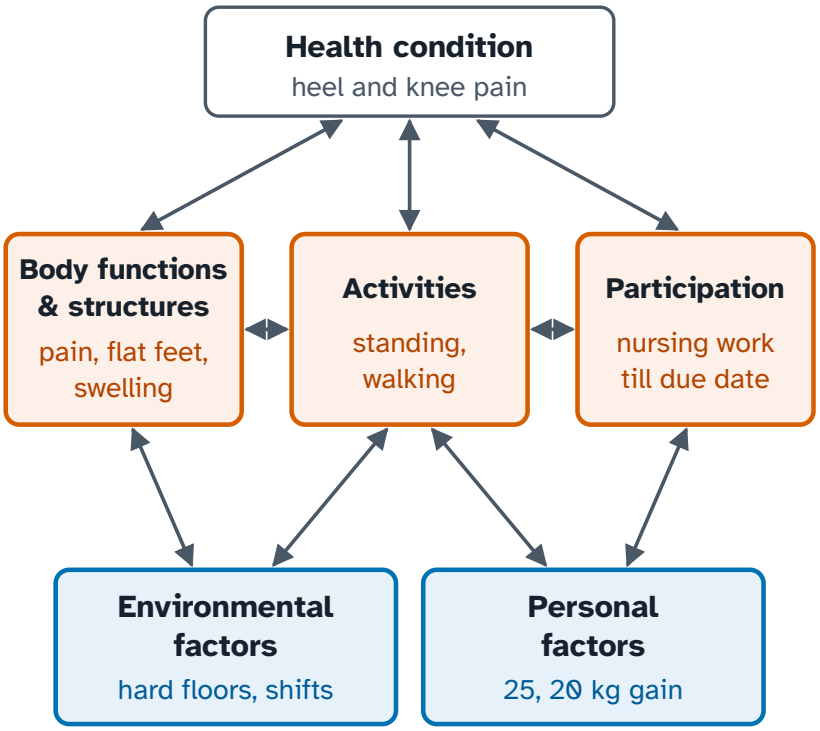
Spot the clues before you write

third trimester: most weight, most laxity, centre of gravity forward. 20 kg gain: above the 11.5–16 kg advised for a normal BMI. nurse: long standing and walking on hard floors, lifting patients. heel and knee pain on standing: overloaded feet and knees. work till due date: plan must keep her working safely.

Reasoning for the impairments (5 marks)

- Extra weight multiplies the load on the heels and knees with every step.
- Relaxin loosens the foot ligaments: the arches drop and the feet roll in, straining the plantar fascia (heel pain) and turning the knees in (knee pain).
- The centre of gravity moves forward; she leans back and locks the knees, overloading the front of the knee.
- Fluid retention and ankle swelling add pain and stiffness.
- Nursing work: hours of standing and walking on hard floors, lifting and turning patients, little rest.

The ICF model filled for this case



Draw the six boxes with two-way arrows, then write two or three points from the case in each box.

ICF component	From this case	Clinical reasoning
Health condition	Third-trimester pregnancy with excess weight gain; plantar heel pain and anterior knee pain; rule out pre-eclampsia if swelling is sudden	Load, laxity and posture
Body functions and structures	Heel pain with first steps and standing, knee pain, flattened arches and pronated feet, tight calves, ankle swelling, weak hip and thigh muscles, forward centre of gravity	Section above
Activity limitations	Standing and walking for long, stairs, lifting and turning patients, getting up from low seats	Weight-bearing tasks provoke pain
Participation restrictions	Full nursing shifts until her due date; household work	Job demands exceed what her feet tolerate
Environmental factors	**Barriers:** hard hospital floors, long shifts, patient handling. **Facilitators:** health-care workplace that can offer lighter duties, hoists, seats	Work changes reduce load
Personal factors	25 years, motivated to keep working, health knowledge as a nurse	Good adherence expected

One-line functional diagnosis

A 25-year-old nurse in her third trimester with excess weight gain, flattened arches and lax ligaments, causing heel and knee pain that limits standing and walking at work.

Management

Problem	Technique and dose	Why
Heel pain, flat feet	Supportive cushioned shoes with arch support or insoles; calf and plantar fascia stretches, 30 seconds × 3, 3 times a day; rolling the sole over a cold bottle; low-dye taping	Supports the arch and reduces fascia strain
Knee pain	Static quadriceps, straight-leg raise in side-lying or sitting, hip abductor strengthening in side-lying, 10 × 2–3 sets; avoid deep squats	Controls knee alignment without heavy load

Problem	Technique and dose	Why
Swelling	Ankle pumps 20 every hour, elevation in side-lying, compression stockings at work	Improves venous return
Work load	Ask for sitting tasks or desk duties, a stool and anti-fatigue mat, breaks every hour, no manual lifting of patients (use help and hoists), shorter shifts	She can keep working with less load
Fitness and weight	Water walking or swimming and stationary cycling at moderate intensity; healthy eating with the dietitian (no dieting in pregnancy)	Keeps fitness without impact
Preparation for labour	Pelvic floor training, relaxation and breathing, labour positions	Due in a month

Warning signs she must report: sudden swelling of the face or hands, headache, blurred vision (pre-eclampsia), bleeding, leaking fluid, reduced baby movements, calf pain.

7 Exam corner

Past questions (MUHS, 2016–2026)

Long answers

Nurse in the third trimester with heel and knee pain (W17); physiological changes in pregnancy and five discomforts with their PT (S22).

7 marks

Diastasis recti: evaluation and management (W16, S22, S24); guidelines for exercising in pregnancy (W18); physiological changes in pregnancy (S20, W20); aims and principles of obstetric physiotherapy (W21, W24); antenatal programme goals and class structure (S25).

3 marks

Diastasis recti (S17); exercises contraindicated in pregnancy (S17, S22); six physiological changes (W17); early bird classes (W18); hyperemesis gravidarum (W19); musculoskeletal dysfunctions in pregnancy (S21); assessment of diastasis recti (W24).

Keywords to underline

relaxin, blood volume up 40–50%, supine hypotension, centre of gravity forward, pelvic girdle pain, diastasis recti, two finger-widths, 150 minutes a week, talk test, absolute contraindications, warning signs, antenatal class

Answer skeleton: physiological changes in pregnancy (7 marks)

1. **[Hormones, 1]** Progesterone, oestrogen, relaxin and what they do.
2. **[Systems, 5]** Cardiovascular, respiratory, musculoskeletal, urinary, digestive, metabolic and skin changes in a table.
3. **[PT meaning, 1]** Supine hypotension, talk test, laxity and posture.

Draw these

- Diastasis recti test at three levels (section 3)
- Posture in pregnancy: centre of gravity forward, changed lumbar curve, knees locked
- Side-lying rest position with pillows under the abdomen and between the knees

Viva questions

1. Why should a woman avoid lying flat after about 20 weeks?
2. How do you measure intensity of exercise in pregnancy?

3. Name four absolute contraindications to exercise in pregnancy.
4. How do you test for diastasis recti?
5. What advice helps pelvic girdle pain when turning in bed?
6. What is an early bird class?

Linked practical spots

- Demonstrate the diastasis recti test, side-lying rest positions, pelvic tilts in standing and four-point kneeling, and pelvic floor teaching.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Enumerate exercises contraindicated during pregnancy. (S17, S22)**

Answer
Contact sports; activities with a high fall risk (riding, skiing, gymnastics); scuba and sky diving; hot yoga; prolonged supine exercise after the first trimester; double leg raises and full sit-ups; heavy lifting with breath-holding; ballistic and end-range stretching; high-impact jumping late in pregnancy. Mention that exercise is avoided altogether with absolute contraindications such as placenta praevia after 26 weeks or pre-eclampsia.

3 m **Early bird classes. (W18)**

Answer
Antenatal education in the first trimester: changes of pregnancy, nutrition, posture and back care, safe exercise, pelvic floor training, early discomforts (nausea, fatigue), warning signs. Purpose: build healthy habits early, prevent problems, reduce anxiety; later antenatal classes prepare for labour.

7 m **Diastasis recti and its management. (S24)**

Answer
Definition; prevalence (33% at 21 weeks, 60% at 6 weeks postpartum, Sperstad 2016); effects; assessment at three levels during a head lift, finger-widths, depth and doming; management in pregnancy (roll to side, avoid straining, TrA and PFM) and after birth (TrA with PFM, pelvic tilts, progress curl-ups when flat, functional training), splinting, lifting and constipation advice; surgery rarely.

LAQ **Write physiological changes during pregnancy. Explain any five discomforts during pregnancy and their physiotherapy management. (4 + 5 + 6, S22)**

Answer
Changes by system (section 1). Five discomforts with cause and PT: low back pain, pelvic girdle pain, leg cramps, varicose veins and swelling, carpal tunnel syndrome (or breathlessness, heartburn, constipation, frequency), each with specific advice and exercise (section 2). Close with safe exercise guidelines.

MCQ **Supine hypotensive syndrome in pregnancy is due to compression of the:**

A. Femoral vein

B. Inferior vena cava

C. Renal artery

D. Portal vein

Answer
B.

MCQ **Which is an absolute contraindication to aerobic exercise in pregnancy?**

A. Mild anaemia

B. Placenta praevia after 26 weeks

C. Back pain

D. Being a first-time mother

Answer
B.

MCQ **Recommended total weight gain in pregnancy for a woman of normal BMI is:**

A. 5–9 kg

B. 7–11.5 kg

C. 11.5–16 kg

D. 20–25 kg

Answer
C.

Last recall check. List the changes of pregnancy by system, five discomforts with their PT, the diastasis test, the exercise dose and four absolute contraindications.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Community Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. American College of Obstetricians and Gynecologists. Physical activity and exercise during pregnancy and the postpartum period. Committee Opinion No. 804. *Obstet Gynecol* 2020;135:e178–88.
3. Mottola MF, et al. 2019 Canadian guideline for physical activity throughout pregnancy. *Br J Sports Med* 2018;52:1339–46.
4. Institute of Medicine and National Research Council. Weight gain during pregnancy: reexamining the guidelines. Washington, DC: National Academies Press; 2009.
5. Sperstad JB, et al. Diastasis recti abdominis during pregnancy and 12 months after childbirth: prevalence, risk factors and report of lumbopelvic pain. *Br J Sports Med* 2016;50:1092–6.

2(e) Labour and postnatal care: physiotherapy in labour, the puerperium, caesarean section and postnatal complications

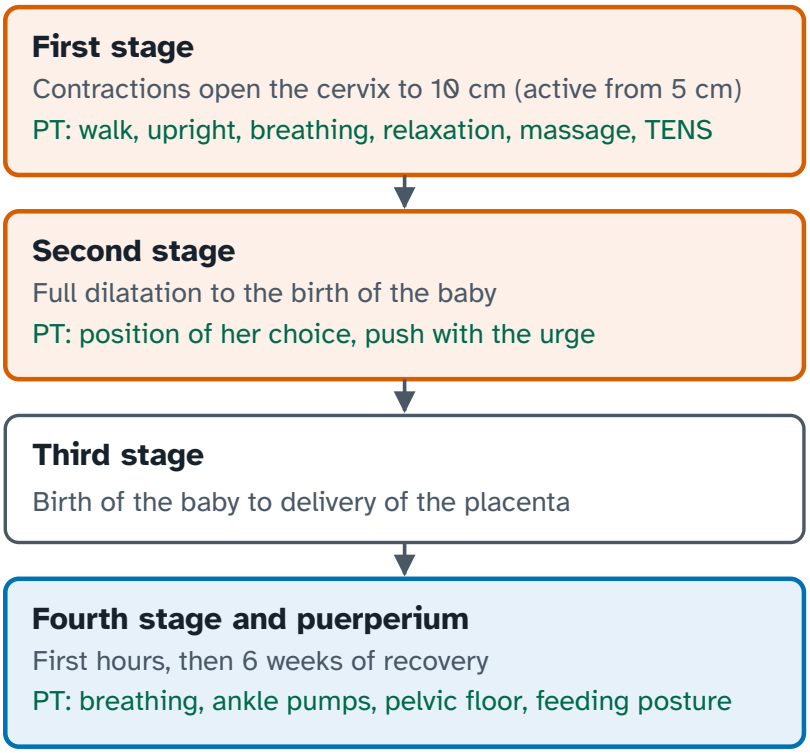
> Labour and postnatal care were asked in 14 of 20 papers, with 9 long answers. The physiotherapist's job is the same throughout: help the woman cope with labour without fear, then rebuild the pelvic floor, abdomen and back after birth while she learns to feed, lift and carry her baby without strain.

Syllabus	Ch 2(e) Antenatal and postnatal care, advice on labour positions, pain relief
Priority	**Tier 1.** In 14 of 20 papers (2016–2026): 9 long answers, 3 seven-mark and 6 three-mark questions.
Asked as	LAQ S16 · S17 · S18 · W18 · W20 · W21 · S23 · S24 · S26 · 7m S17 · S19 · S23 · 3m S17 · W19 · S21 · S22 · S24 · W25
Usual split	15 (functional diagnosis 5, PT 10; or changes 10, PT 5; or evaluation 4, management 6, health promotion 5; or complications 7, management 8); 7 (pain relief, breathing or PT in labour)
Checked against	Cochrane reviews of positions in the first stage of labour (Lawrence et al., 2013) and TENS in labour (Dowswell et al., 2009); WHO recommendations on intrapartum care (2018); Academy of Breastfeeding Medicine protocol 36 (2022); return-to-running postnatal guideline (Goom et al., 2019); StatPearls on postpartum blues. Full list under Sources (p. 62). Content checked 20 October 2026.

1 Labour: stages and changes

Physical and physiological changes of labour and the puerperium: 10 marks in the S18 long answer.

Stages of labour and what the physiotherapist does



The active first stage usually lasts up to 12 hours in a first labour and up to 10 hours in later labours (WHO, 2018).

? Text version of the flowchart

1. First stage: contractions open the cervix to 10 cm (active phase from 5 cm). PT: walking, upright positions, breathing, relaxation, massage, TENS.
2. Second stage: full dilatation to birth. PT: position of her choice, pushing with the urge.
3. Third stage: birth of the baby to delivery of the placenta.
4. Fourth stage and puerperium: the first hours, then 6 weeks of recovery. PT: breathing, ankle pumps, pelvic floor contractions, feeding posture.

Changes in labour

- **Uterus and cervix:** contractions become stronger, longer and more frequent; the cervix thins (effacement) and opens (dilatation); the baby descends and rotates through the pelvis.
- **Heart and circulation:** each contraction pushes blood back into the circulation, so cardiac output and blood pressure rise; pain and anxiety add to this.
- **Breathing:** oxygen use rises; pain causes fast breathing and over-breathing (tingling, dizziness).
- **Other:** stomach empties slowly; sweating, fluid loss and fatigue; the bladder is pressed and may not empty; the pelvic floor and perineum stretch greatly in the second stage.

2 Physiotherapy in labour

Pain relief in labour (S17), breathing in labour (S19), PT during labour (S23), each 7 marks; positions in labour (W19, 3 marks); the second stage: positions, breathing and electrotherapy (S24, long answer).

- **Aims:** reduce fear and pain, help her stay in control and mobile, help labour progress, save energy, protect the perineum, and keep the baby well oxygenated.
- **Antenatal preparation** is where most of this is taught: what happens in labour, positions, breathing and relaxation practice, partner's role (see antenatal classes (p. 43)).

Positions

- **First stage:** walking, standing and leaning forward on the partner or bed, sitting on a birthing ball, kneeling forward over a pillow or ball, all-fours (eases backache when the baby faces forward), side-lying for rest or with an epidural. Upright and mobile positions shortened the first stage by about 1 hour 22 minutes and reduced caesarean and epidural use (Lawrence et al., 2013).
- **Second stage:** the position she chooses, including upright ones such as kneeling, squatting with support, all-fours and side-lying; avoid lying flat on the back, which presses on the great vessels and narrows the pelvic outlet (WHO, 2018).

Breathing

- **Early labour:** slow, relaxed breathing through the contraction: a sigh out at the start, slow breaths in through the nose and out through the mouth, a sigh at the end.
- **Strong contractions:** lighter, slightly faster breathing at the peak, returning to slow breathing as it fades.
- **Transition** (urge to push before full dilatation): "pant-pant-blow" to avoid pushing too early.
- **Second stage:** push with the natural urge, breathing out or with short breath-holds, rather than long forced breath-holding; pant as the head crowns so the perineum stretches slowly.
- **Avoid over-breathing:** if she feels tingling or dizziness, slow down and breathe into cupped hands.
- **Why breathing helps:** focuses attention, aids relaxation, keeps oxygen supply to mother and baby, and prevents exhausting breath-holding.

Other pain relief

Method	How	Note
Relaxation	Mitchell's physiological relaxation or progressive relaxation, practised before labour; music	Recommended by WHO for women who want it
Massage and warmth	Firm sacral pressure or massage during contractions; warm packs on the back or lower abdomen	Recommended by WHO for women who want it
TENS	Two pairs of electrodes on the back: one pair either side of the spine at T10–L1, one at S2–S4; low background current, with a boost button during contractions	Little evidence that it lowers pain scores, but most women would use it again and no harms were reported (Dowswell et al., 2009); start early; not in water
Water	Warm shower or bath in the first stage	Comfort and relaxation
Support	Continuous presence of a partner or companion, encouragement	Reduces fear and the need for drugs
Medical	Epidural, nitrous oxide (Entonox), opioids	Given by doctors; PT adjusts positions (side-lying, supported sitting)

3 The puerperium and the goals of postnatal care

Changes of the puerperium (S18); three goals of postnatal care (W25, 3 marks).

- **Puerperium:** the 6 weeks after delivery while the body returns towards its non-pregnant state.
- **Uterus:** involution: the fundus is at the umbilicus after delivery, shrinks by about a finger's breadth a day, is no longer felt above the pubis by about 2 weeks and is near normal size by 6 weeks; after-pains, stronger while feeding.
- **Lochia:** red (first few days), then pink-brown, then yellow-white, lasting a few weeks.
- **Pelvic floor and perineum:** stretched, bruised or torn; episiotomy heals in 1–2 weeks; leakage is common at first.
- **Abdomen:** lax, weak muscles; diastasis recti in many women (see diastasis recti (p. 43)).
- **Joints:** ligaments stay lax for months; back and pelvic girdle pain are common.
- **Circulation:** blood volume and cardiac output fall back over weeks; blood clots more easily, so deep vein thrombosis risk stays raised for 6 weeks.

- **Breasts:** colostrum, then milk "comes in" on about day 2–4; engorgement may follow.
- **Bladder and bowel:** passing more urine in the first days; constipation and haemorrhoids are common.
- **Mind:** joy, tiredness, sleep loss; baby blues in many.

Goals of postnatal physiotherapy

1. Prevent complications: chest problems after anaesthesia, deep vein thrombosis, urinary retention, constipation.
2. Relieve pain: perineum, back, pelvis, wound, breasts.
3. Restore the pelvic floor (continence, support) and the abdominal wall.
4. Teach posture and body mechanics for feeding, lifting and carrying the baby.
5. Support emotional well-being, rest and family help.
6. Return gradually to fitness and to work, and keep a lifelong pelvic floor habit.

4 Postnatal exercise programme

Postnatal care after a normal delivery with episiotomy (S16) and for an executive who wants her fitness back after her second baby (W20).

When	Exercises and advice
Day 1 to the end of week 1	Deep breathing; ankle pumps and leg exercises (clots); gentle pelvic floor contractions from the first day, which also ease perineal swelling; transversus abdominis drawing-in and pelvic tilts; getting out of bed by rolling to the side; walking; check for diastasis recti; feeding and lifting posture
Weeks 2–6	Progress pelvic floor holds and fast contractions; bridging, side-lying leg lifts, four-point kneeling with arm and leg lifts; head lifts only if the abdominal gap does not dome; daily walks
6–12 weeks	After the 6-week check: low-impact aerobic exercise (brisk walking, cycling, swimming once bleeding has stopped), resistance training for legs, back and arms; progress abdominal work
After 12 weeks	Running and high-impact exercise only when she passes load and impact tests without leaking, heaviness or pain (Goom et al., 2019); 150 minutes a week of activity

Care of the perineum and episiotomy

- Ice pack wrapped in cloth for 10–15 minutes in the first 24–48 hours for swelling and pain.
- Gentle pelvic floor contractions to pump away swelling.
- Lie on the side to rest and feed in the first days; sit on a soft cushion and squeeze the pelvic floor before sitting down.
- Hold a clean pad on the stitches when opening the bowels; avoid straining; fibre and fluids.
- Keep the area clean and dry; report increasing pain, smell or discharge (infection).

5 Caesarean section

Complications of caesarean section (S17, 3 marks); elective caesarean for cephalopelvic disproportion: management and home programme (S17, long answer).

- **Cephalopelvic disproportion:** the baby's head is too big for the mother's pelvis, so a planned (elective) caesarean is done.
- **Complications:** bleeding; infection of the wound or uterus; deep vein thrombosis and pulmonary embolism; chest infection and atelectasis after anaesthesia; urinary retention or bladder injury; paralytic ileus and wind pain; wound breakdown, later incisional hernia, adhesions and chronic pain; spinal headache; delayed mobility and breastfeeding; in later pregnancies, scar rupture and abnormal placenta. Baby: fast breathing for a short time after birth.

Physiotherapy after caesarean section

- **Before surgery (elective):** teach breathing exercises, supported huffing and coughing, bed mobility and leg exercises.
- **Day 0–1:** deep breathing; huff or cough with a pillow pressed over the wound; ankle pumps and static quadriceps; roll to the side and push up to sit; walk early with good upright posture; gentle transversus and pelvic floor contractions (pregnancy itself loaded the pelvic floor).
- **Days 2–7:** progress walking; posture; feeding positions that keep the baby off the wound (side-lying, football hold, pillow on the lap); relief of wind pain by walking, pelvic tilts and abdominal massage.
- **Weeks 2–6:** progress abdominal and pelvic floor exercises; scar care once healed (gentle massage around the scar); no lifting heavier than the baby or straining for about 6 weeks; drive when she can do an emergency stop without pain.
- **Home programme:** breathing and leg exercises daily in the first week, pelvic floor and transversus 3 times a day, walking a little more each day, posture and lifting advice, warning signs (calf pain, fever, wound discharge, breathlessness).

6 Postnatal complications and their management

Complications of the postnatal period (S23, 7 + 8); low back pain 2 months after delivery (W18, long answer); breast engorgement (S24) and postpartum blues (S22), 3 marks each; forceps delivery with dragging back pain (W21).

Complication	Management
Perineal pain, episiotomy or tear	Ice, pelvic floor contractions, positioning, hygiene, support for defecation (section 4)
Urinary retention or incontinence	Early walking, privacy, relaxed toileting; pelvic floor training for leakage (see incontinence (p. 70))
Constipation, haemorrhoids, anal incontinence	Fibre and fluids, footstool posture, no straining; anal sphincter training after a tear
Diastasis recti	Transversus and pelvic floor activation, avoid sit-ups while the gap domes, posture, progressive loading
Low back and pelvic girdle pain, coccyx pain	Assessment below; posture, core and hip strengthening, pelvic belt for pelvic girdle pain, cushion for coccyx pain
Deep vein thrombosis	Prevention: early walking, leg exercises, fluids, stockings as advised; calf pain or swelling needs urgent medical review
Chest problems after anaesthesia	Breathing exercises, supported huffing, early mobility
Breast engorgement and mastitis	See below; fever and a red hot area suggest mastitis: refer
Wrist and thumb pain (de Quervain's), carpal tunnel	Neutral wrist in lifting and feeding, splint, ice, ergonomics
Baby blues, postnatal depression	See below; screen and refer

Low back pain after delivery (W18)

- **Differential diagnosis:** mechanical (postural) low back pain from feeding, lifting and weak core; pelvic girdle pain (sacroiliac joint or symphysis pubis); lumbar disc prolapse with nerve root pain; coccydynia (pain on sitting, after a difficult birth); dragging back pain from pelvic floor weakness or prolapse; rarely infection or haematoma at the epidural site, or osteoporosis of pregnancy with vertebral fracture.
- **Red flags:** fever, night pain, leg weakness or numbness, bladder or bowel change, saddle numbness: refer urgently.
- **Assessment:** pain site and behaviour, delivery details (instruments, epidural, baby weight), feeding and lifting habits, posture, lumbar and SIJ tests (active straight leg raise, posterior pelvic pain provocation), neurological screen, diastasis check, pelvic floor assessment, function and mood.

- **Other impairments in this population:** pelvic floor weakness (leakage, prolapse), diastasis recti, neck and upper back pain from feeding, wrist and thumb pain, fatigue and deconditioning, weight retention, low mood.

Breast engorgement (S24)

- **Definition:** painful, swollen, hard breasts, usually 2–5 days after birth as milk comes in, from milk stasis plus swelling of the breast tissue.
- **Management:** frequent, effective feeding (on demand, 8–12 times a day) with a correct latch; hand-express a little to soften the areola before latching; cold packs between feeds for swelling; gentle lymphatic drainage strokes towards the armpit, avoiding deep or forceful massage; a supportive, non-tight bra; pain relief as prescribed; refer if fever or a red, hot area appears (ABM protocol 36, 2022).

Postpartum blues (S22)

- Short-lived tearfulness, mood swings, irritability, anxiety and poor sleep in about half or more of new mothers, starting 2–3 days after birth and settling within 2 weeks; due to the sudden fall in hormones, sleep loss and adjustment.
- **Management:** reassurance that it is common, rest and sleep, family help with the baby, talking about feelings, gentle walks and relaxation.
- If symptoms last beyond 2 weeks or are severe, think of postnatal depression: screen (Edinburgh Postnatal Depression Scale) and refer; confusion or hallucinations mean postpartum psychosis, an emergency.

7 Breastfeeding posture: ergonomics for a lactating mother

Ergonomic advice for a lactating mother (S21, 3 marks); poor feeding technique in the S26 long case.

- **Bring the baby to the breast, not the breast to the baby:** a pillow on the lap raises the baby to breast height.
- Sit in a chair with back support (a small cushion behind the low back), feet flat or on a low stool, shoulders relaxed, arm supported on a pillow or armrest.
- Avoid long neck bending: look up and around between sucks; stretch the neck and upper back after feeds.
- Change positions: cradle, cross-cradle, football (rugby) hold (good after caesarean), and side-lying (good at night and with a sore perineum).
- Keep the wrists straight when holding the baby's head; support with the forearm.
- Change nappies at waist height; lift the baby close with knees bent; carry in a carrier that holds the baby high and close; do not carry on one hip for long.
- Sleep and rest when the baby sleeps; drink water during feeds.

8 Worked long case: back pain and breast heaviness after a first delivery

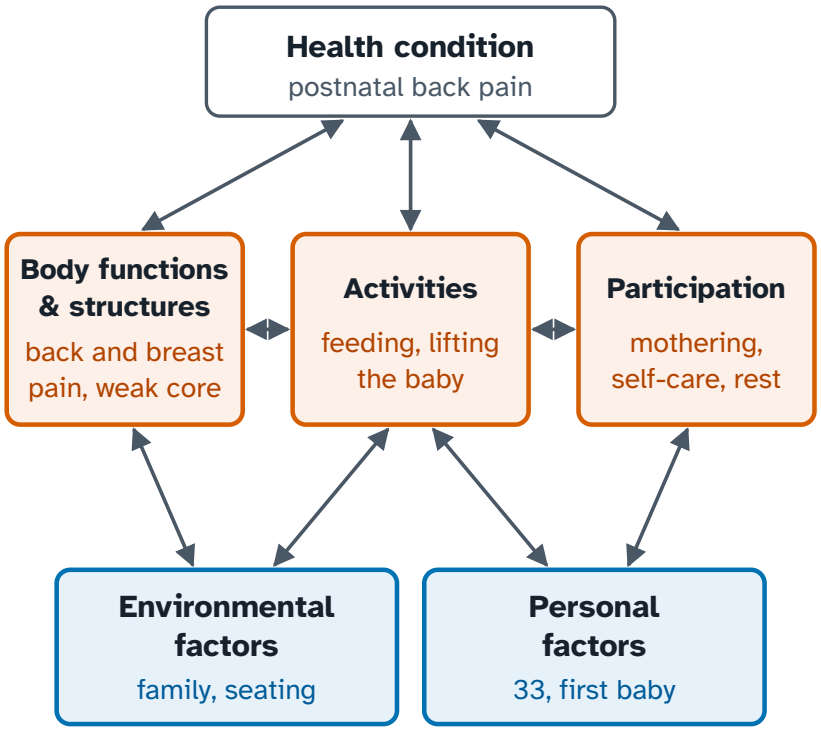
From a real paper (S26)

> "Thirty three years old mother who underwent full term normal delivery 15 days back. At present she has dull aching lower back pain, heaviness in both the breast tissues. She has become mother for the first time and found to be practicing inappropriate breast feeding techniques. Give your functional diagnosis in ICF format, plan short term and long term physiotherapy management."

Spot the clues before you write

15 days after a normal delivery: puerperium, lax ligaments, weak abdomen and pelvic floor. dull aching low back pain: postural, mechanical. heaviness in both breasts: engorgement. first-time mother, poor feeding technique: slouched posture, poor latch, needs teaching.

The ICF model filled for this case



Draw the six boxes with two-way arrows, then write two or three points from the case in each box.

ICF component	From this case	Clinical reasoning
Health condition	Postnatal mechanical low back pain with breast engorgement, 15 days after a first normal delivery	Lax ligaments, weak abdomen and poor feeding posture
Body functions and structures	Dull low back pain (for example 5/10), tight, heavy, tender breasts, weak transversus and pelvic floor, possible diastasis recti, forward head and rounded shoulders, perineal soreness, fatigue and sleep loss	Slouched feeding for hours a day; milk stasis from poor latch
Activity limitations	Feeding for long, lifting and carrying the baby, bending to bathe and change, getting up from the floor or a low bed	Repeated bending and static postures load the back
Participation restrictions	Caring for the baby comfortably, self-care, rest, enjoying motherhood	Pain and tiredness
Environmental factors	**Facilitators:** family help, lactation support at the hospital or from the ASHA. **Barriers:** floor sitting, low beds, no feeding pillow	Home set-up shapes posture
Personal factors	33 years, first baby, anxious, little knowledge of feeding technique	Teaching and reassurance change behaviour quickly

One-line functional diagnosis

A 33-year-old first-time mother, 15 days after a normal delivery, with postural low back pain and breast engorgement from poor feeding posture and technique, limiting comfortable feeding and baby care.

Goals

Problem	Short-term (1–2 weeks)	Long-term (6–12 weeks)
Breasts	Engorgement settled; correct latch and positions	Comfortable feeding in several positions
Back	Pain 5 to 2 out of 10; good feeding and lifting posture	Pain-free baby care; strong core

Problem	Short-term (1–2 weeks)	Long-term (6–12 weeks)
Pelvic floor and abdomen	Correct pelvic floor and transversus contraction	Pelvic floor good strength; abdominal gap functional
Fitness and mood	Daily short walks; rest plan	150 minutes a week; good mood and sleep routine

Management

Impairment	Technique and dose	Why
Breast engorgement	Feed on demand with correct latch; hand-express to soften the areola; cold packs between feeds; gentle lymphatic strokes towards the armpit; supportive bra	Removes milk and reduces swelling
Poor feeding posture	Chair with back support, pillow on lap, footstool, alternate cradle, football and side-lying holds	Ends the slouch that loads the back and neck
Low back pain	Heat for 10–15 minutes; pelvic tilts, transversus drawing-in 10 × 10-second holds, bridging 10 reps, cat–camel, 2–3 times a day	Relieves pain and supports the lax spine
Weak pelvic floor	Slow holds and quick contractions, at least 8 contractions 3 times a day, progressing	Recovery after vaginal birth; prevents leakage
Body mechanics	Lift the baby close with bent knees; change at waist height; avoid twisting	Reduces repeated strain
Fatigue, mood	Rest when the baby sleeps, family help, short walks, relaxation; watch for low mood beyond 2 weeks	Recovery and screening for depression

Practice cases from the papers

- **S16:** 25-year-old, 15 days after a normal delivery with episiotomy: functional diagnosis and postnatal PT (sections 4 and 6).
- **W20:** 29-year-old executive and fitness enthusiast after her second normal delivery: postnatal assessment, programme and the fitness plan in the months after (section 4, return to running after 12 weeks).
- **W21:** 26-year-old with three children, 3 weeks after a forceps delivery with an epidural, vague dragging back pain: evaluate the pelvic floor and for prolapse (PERFECT, symptoms of heaviness) and emphasise pelvic floor training (see incontinence and prolapse (p. 70)), plus health promotion.
- **W18:** 26-year-old clerk with low back pain 2 months after delivering a 3.5 kg baby: differential diagnosis, assessment and management (section 6).
- **S17:** 25-year-old after elective caesarean for cephalopelvic disproportion (section 5).

9 Exam corner

Past questions (MUHS, 2016–2026)

Labour

PT for pain in labour (S17, 7 marks); breathing exercises in labour (S19, 7 marks); positions in labour (W19, 3 marks); PT during labour (S23, 7 marks); second stage: positions, breathing and electrotherapy (S24, long answer); changes of labour and the puerperium (S18, long answer).

Postnatal

Normal delivery with episiotomy (S16), executive after second delivery (W20), forceps with epidural (W21), back pain after delivery (W18), back pain and breast heaviness (S26), long answers; complications of the postnatal period (S23); three goals of postnatal care (W25).

Caesarean and others

Complications of caesarean section (S17, 3 marks); elective caesarean (S17, long answer); ergonomic advice for a lactating mother (S21); postpartum blues (S22); breast engorgement (S24).

Keywords to underline

upright and mobile | position of her choice | slow breathing | pant-pant-blow | avoid over-breathing | TENS T10–L1 and S2–S4 | involution | pelvic floor from day 1 | diastasis recti | supported huffing | baby to breast | red flags

Answer skeleton: a postnatal long case (15 marks)

1. **[ICF diagnosis, 5]** Delivery details, impairments (pelvic floor, abdomen, back, breasts, perineum), activities, participation, factors; one-line diagnosis.
2. **[Goals, 3]** Short- and long-term, measurable.
3. **[Management, 7]** Early care (breathing, circulation, perineum), pelvic floor and abdominal programme with doses, posture and feeding ergonomics, complication care, home programme, return to fitness.

Draw these

- The stages of labour with PT in each (section 1)
- TENS electrode sites on the back (two pairs: T10–L1 and S2–S4)
- A mother feeding with a pillow on her lap and feet on a stool

Viva questions

1. What are the stages of labour?
2. Which positions help in the first stage, and why?
3. Where do you place TENS electrodes in labour?
4. When can pelvic floor exercises start after a normal delivery?
5. How do you tell baby blues from postnatal depression?
6. Why should a mother avoid sit-ups if her abdomen domes?

Linked practical spots

- Demonstrate labour breathing levels; place TENS electrodes for labour; teach supported huffing after caesarean; check for diastasis recti; set up a feeding position.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Breast engorgement and its management. (S24)**

Answer

Painful swelling of both breasts 2–5 days after birth from milk stasis and tissue swelling. Management: frequent effective feeds with correct latch, hand expression to soften the areola, cold packs between feeds, gentle lymphatic strokes towards the axilla (no deep massage), supportive bra, analgesics; refer if fever or a red hot area (mastitis).

3 m **Positions in labour. (W19)**

Answer
First stage: walking, standing leaning forward, sitting on a ball, kneeling forward, all-fours, side-lying; upright positions shorten the first stage (about 1 hour 22 minutes) and reduce caesarean and epidural use. Second stage: her choice, including kneeling, supported squatting, all-fours, side-lying; avoid lying flat on the back.

7 m **Physiotherapy management of pain during labour. (S17)**

Answer
Antenatal education to reduce fear; positions and mobility; breathing levels (slow, lighter at the peak, pant-pant-blow in transition, avoid over-breathing); relaxation (Mitchell, progressive); sacral massage and warm packs; TENS with electrodes at T10–L1 and S2–S4 and a boost button; warm water; continuous support; coordination with medical analgesia (positions with epidural).

LAQ **Discuss the various complications that can be seen in the postnatal period and their management. (7 + 8, S23)**

Answer
List with management: perineal pain (ice, pelvic floor, positioning), urinary retention and incontinence (mobility, PFMT), constipation and haemorrhoids (diet, posture), diastasis recti (TrA, avoid sit-ups while doming), back and pelvic girdle pain (posture, core, belt), DVT (prevention, urgent referral), chest complications after anaesthesia (breathing, huffing), wound problems after caesarean, breast engorgement and mastitis, de Quervain's, baby blues and depression (support, EPDS, referral). End with the home programme and 6-week review.

MCQ **The active first stage of labour, as defined by WHO (2018), starts at a cervical dilatation of:**

A. 2 cm
B. 3 cm
C. 5 cm
D. 10 cm

Answer
C.

MCQ **"Pant-pant-blow" breathing is used:**

A. In early labour
B. To resist the urge to push before full dilatation
C. In the third stage
D. After caesarean

Answer
B.

MCQ **The best feeding hold after a caesarean section is usually:**

A. Cradle hold with the baby across the wound

B. Football (rugby) hold

C. Standing

D. Prone

Answer

B. Side-lying also keeps the baby off the wound.

Last recall check. Draw the stages of labour, write the breathing levels, place TENS, list six goals of postnatal care and six postnatal complications with one treatment each.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Community Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. World Health Organization. WHO recommendations: intrapartum care for a positive childbirth experience. Geneva: WHO, 2018.
3. Lawrence A, Lewis L, Hofmeyr GJ, Styles C. Maternal positions and mobility during first stage labour. *Cochrane Database Syst Rev* 2013; (10):CD003934.
4. Dowswell T, et al. Transcutaneous electrical nerve stimulation (TENS) for pain management in labour. *Cochrane Database Syst Rev* 2009; (2):CD007214.
5. Mitchell KB, et al. Academy of Breastfeeding Medicine Clinical Protocol #36: the mastitis spectrum. *Breastfeed Med* 2022;17:360–76.
6. Goom T, Donnelly G, Brockwell E. Returning to running postnatal: guideline for medical, health and fitness professionals managing this population, 2019.
7. Balaram K, Marwaha R. Postpartum blues. In: StatPearls. Treasure Island (FL): StatPearls Publishing.

2(d) Menopause and osteoporosis: hormonal and physiological changes, types and risk factors of osteoporosis, prevention and physiotherapy

Menopause and osteoporosis were asked in 8 of 20 papers, with 3 long answers. Every question follows from one fact: oestrogen falls. That speeds bone loss, weakens the pelvic floor and vaginal tissues, adds abdominal fat and raises heart risk, and the physiotherapist answers each with exercise, education and fall prevention.

Syllabus	Ch 2(d) Anatomical and physiological variations associated with menopause; Ch 1(c iii) physical fitness in women: menopause, osteoporosis
Priority	**Tier 1.** In 8 of 20 papers (2016–2026): 3 long answers, 3 seven-mark and 4 three-mark questions.
Asked as	LAQ W18 · W24 · S25 · 7m S17 · S22 · S23 · 3m W18 · W22 · S23 · S26
Usual split	15 (hormonal changes and sequelae 8, PT with reasoning 7; or evaluation 5, exercise programme with rationale 10; or functional diagnosis, assessment, management); 7 (importance of PT in postmenopausal women; physiological variations); 3 (types, risk factors, prevention of osteoporosis)
Checked against	WHO definition and T-score criteria for osteoporosis (1994); Riggs and Melton classification of involutional osteoporosis (1986); Too Fit to Fracture recommendations (Giangregorio et al., 2014); Indian studies of age at menopause; pelvic floor and continence evidence from the incontinence page. Full list under Sources (p. 69). Content checked 20 October 2026.

1 Menopause and its hormonal changes

Hormonal changes and sequelae of menopause (W24, 8 marks); physiological variations with menopause (S23, 7 marks).

- **Menopause:** the permanent end of menstruation from loss of ovarian follicle activity, diagnosed after 12 months without a period. Indian studies report an average age of about 46 years, earlier than the about 51 years usual in Western women.
- **Perimenopause:** the years of irregular cycles and symptoms before and the first year after the last period. **Postmenopause:** the years after. **Premature menopause:** before 40; **surgical menopause:** after removal of both ovaries (sudden symptoms).
- **Hormonal changes:** oestrogen (oestradiol) and progesterone fall; inhibin falls, so FSH rises markedly and LH rises; androgens become relatively higher.

System	Effect of low oestrogen
Vasomotor	Hot flushes, night sweats, palpitations, poor sleep
Mind	Mood swings, irritability, anxiety, low mood, poor concentration
Bone	Rapid bone loss in the first years after menopause, osteoporosis and fractures (wrist, spine, hip)
Muscles and joints	Loss of muscle mass and strength, joint aches and stiffness
Genitourinary	Vaginal dryness and painful intercourse, urgency and frequency, urinary infections, stress incontinence, prolapse (weaker pelvic floor and tissues)
Heart and metabolism	Worse cholesterol profile, more abdominal fat, insulin resistance, higher blood pressure and heart disease risk
Skin and others	Thin dry skin, hair changes, weight gain

Medical treatment: hormone therapy for troublesome symptoms in suitable women (doctor's decision), vaginal oestrogen for genitourinary symptoms, and bone-protecting drugs when fracture risk is high.

2 Importance of physiotherapy for postmenopausal women

Importance of physiotherapy in postmenopausal women (S17, S22, 7 marks).

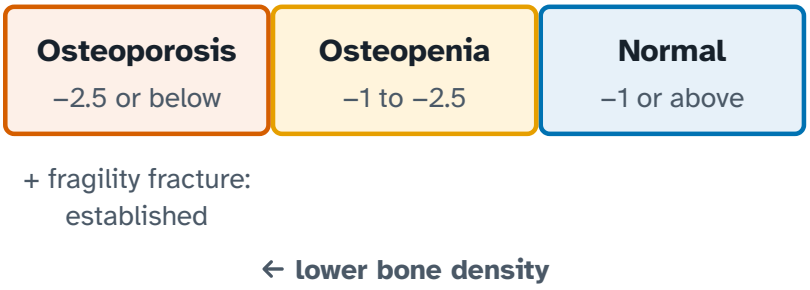
Problem	Physiotherapy
Bone loss	Progressive resistance and impact (weight-bearing) exercise; posture and back extensor training; safe movement (section 5)
Falls and fractures	Balance and strength training, home safety, footwear (see falls (p. 119))
Pelvic floor weakness	Pelvic floor muscle training, bladder training, continence advice (see incontinence and prolapse (p. 70))
Weight gain, heart and diabetes risk	Aerobic exercise 150 minutes a week, resistance training, weight management (see fitness (p. 11))
Joint and back pain	Exercise, posture, pain relief, ergonomic advice
Hot flushes, sleep, mood	Regular exercise, relaxation and paced breathing, yoga, sleep routine; layered clothing and cool rooms
Knowledge	Education on menopause, bone health, calcium and vitamin D, sunlight, avoiding smoking and alcohol, screening

3 Osteoporosis: definition and types

Types of osteoporosis (S23, 3 marks).

- **Definition (WHO, 1994):** a systemic skeletal disease characterised by low bone mass and microarchitectural deterioration of bone tissue, with a consequent increase in bone fragility and susceptibility to fracture.
- **Diagnosis:** bone mineral density by DEXA at the hip and spine, reported as a T-score (standard deviations from the young adult mean).

T-score bands (WHO)



The cut-offs were set for postmenopausal women; fracture risk also depends on age, falls and other risk factors (FRAX combines them).

Type	Features
Primary type I (postmenopausal)	Women about 50–70; oestrogen loss; mainly trabecular bone; vertebral crush and distal radius (Colles') fractures
Primary type II (senile, age-related)	Men and women over about 70; trabecular and cortical bone; hip and vertebral fractures
Idiopathic (juvenile)	Children and young adults without a known cause; rare
Secondary	Steroids, thyroid and parathyroid excess, rheumatoid arthritis, malabsorption, chronic kidney or liver disease, alcohol, long immobilisation, some drugs (anticonvulsants, heparin)
Regional	Disuse after paralysis or a plaster; complex regional pain syndrome

4 Risk factors and prevention

Predisposing factors (W22) and risk factors (S26) for osteoporosis; prevention of osteoporosis with ageing (W18). All 3 marks.

Risk factors	Examples
Cannot change	Female sex, older age, early menopause, family history of hip fracture, small thin build, previous fragility fracture, Asian or white ethnicity
Can change	Physical inactivity, low body weight, low calcium and vitamin D (little sunlight), smoking, alcohol, excess caffeine and salt
Diseases and drugs	Long-term steroids, rheumatoid arthritis, thyroid or parathyroid excess, diabetes, malabsorption, kidney disease, immobility

Prevention

- **Build peak bone mass** in childhood and youth: sport, jumping and running, enough calcium and protein.
- **Exercise throughout life:** weight-bearing and impact activity, progressive resistance training, balance training.
- **Nutrition:** calcium from milk, curd, ragi, sesame, green leafy vegetables; vitamin D from sunlight and supplements if deficient; enough protein.
- **Lifestyle:** stop smoking, limit alcohol and caffeine, keep a healthy weight.
- **Prevent falls:** balance training, home safety, vision and drug review.
- **Screen and treat:** DEXA for women at risk; medicines (bisphosphonates and others) when fracture risk is high.

5 Exercise and physiotherapy for osteoporosis

Exercise programme with rationale for a 75-year-old with osteoporosis and knee pain (S25, 10 marks).

- A multicomponent programme with **progressive resistance training and balance training** is recommended; aerobic exercise alone is not enough (Too Fit to Fracture, 2014).
- After a vertebral fracture, exercise should be planned with a physiotherapist.

Component	Exercise	Rationale
Resistance	Squats or sit-to-stand, step-ups, hip and knee strengthening, rows, wall push-ups, back extensors in prone or sitting; 2–3 days a week, 8–12 repetitions, progressing load	Muscle pull on bone stimulates bone formation; strength for function and falls
Impact (if no fracture and joints allow)	Heel drops, stamping, marching, stair climbing, brisk walking; low impact first	Loading through the skeleton maintains bone
Balance	Tandem and single-leg stance, stepping, tai chi; most days	Prevents the falls that cause fractures
Posture	Chin tucks, scapular retraction, thoracic extension, back extensor strengthening	Reduces kyphosis and vertebral load
Aerobic	Walking, cycling (alongside, not instead of, the above)	Fitness and heart health

Spine-sparing advice

- Avoid repeated or loaded forward bending and twisting of the spine (toe touches, sit-ups, lifting with a rounded back).
- Bend at the hips and knees (hip hinge) to pick things up; carry loads close; roll to the side to get out of bed.
- Avoid high-risk activities for falls; use rails and good lighting.

6 Worked long case: menopausal symptoms with rib and back pain

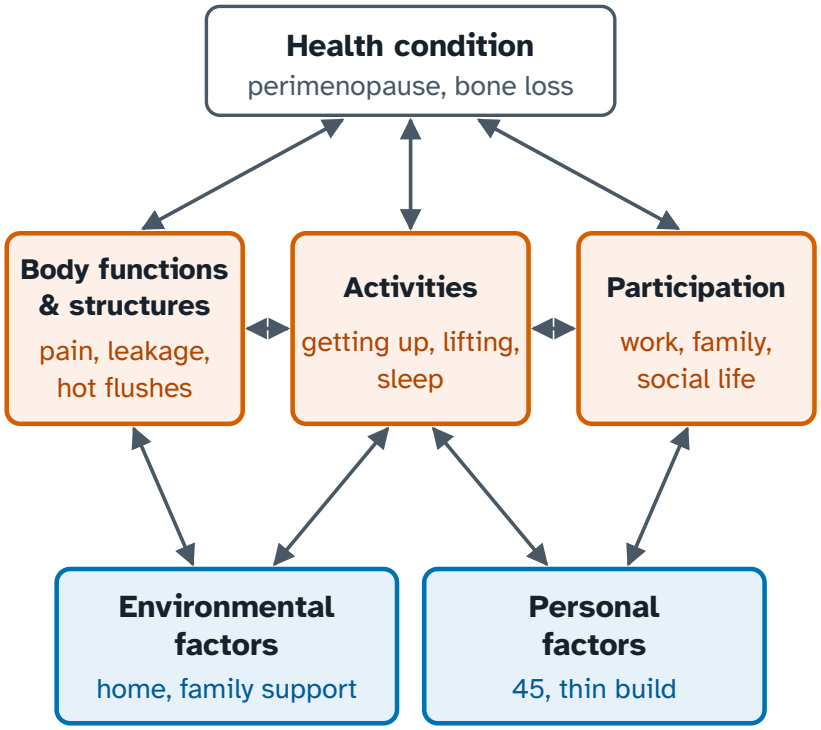
From a real paper (W18)

> "A 45 years old, female, 56 kg, 5'6 ht, complains of hot flushes, night sweats, urinary incontinence and mood swings since 3 months. She also complains of rib pain and low back pain while getting up from the bed. Explain the functional diagnosis, physiotherapy assessment and management for the same."

Spot the clues before you write

hot flushes, night sweats, mood swings at 45: perimenopause. urinary incontinence: weak pelvic floor and tissues. rib and low back pain on getting up from bed: possible osteoporotic vertebral or rib fracture; needs DEXA and an X-ray. 56 kg, 5'6": BMI about 20, a thin build, a risk factor for osteoporosis.

The ICF model filled for this case



Draw the six boxes with two-way arrows, then write two or three points from the case in each box.

One-line functional diagnosis

A 45-year-old perimenopausal woman with vasomotor symptoms, stress or mixed incontinence and mechanical back and rib pain with suspected osteoporosis, limiting getting out of bed, lifting and sleep.

Assessment

- History: menstrual history, symptoms and their timing, falls, fractures, steroid use, diet and calcium, sunlight, smoking, activity, family history; type of leakage.
- Red flags: sudden severe back pain, height loss, night pain, weight loss (refer for X-ray, DEXA, blood tests).
- Examination: posture (kyphosis, height), spinal tenderness and range, rib springing gently (avoid forceful tests), strength of back extensors and hips, balance (single-leg stance, TUG), pelvic floor (PERFECT), BMI.
- Questionnaires: menopause symptom scale, ICIQ for leakage, mood screen.

Management

Problem	Technique and dose	Why
Rib and back pain	Heat, TENS, gentle mobility; log-roll to get out of bed; avoid bending and twisting under load	Relieves pain and protects a possibly fragile spine
Bone loss	Progressive resistance (legs, hips, back extensors) 2–3 times a week; weight-bearing walking daily; impact only after fracture is excluded	Loading maintains bone
Posture	Thoracic extension, chin tucks, scapular retraction, 10 × 2 daily	Less kyphosis and vertebral load
Incontinence	Pelvic floor training, at least 8 contractions 3 times a day; the knack; bladder training if urgency	First-line treatment
Hot flushes, sleep, mood	Aerobic exercise 150 minutes a week, relaxation and slow breathing, yoga; cool bedroom, light clothing	Better sleep, mood and well-being
Risk factors	Calcium-rich diet, vitamin D and sunlight, no smoking, little alcohol; balance training and home safety	Prevents fractures

Practice case (S25)

A 75-year-old woman with osteoporosis on calcium, bilateral knee pain and difficulty on stairs (evaluation 5, exercise programme with rationale 10): evaluate knee OA and falls risk (TUG, chair stand, balance), posture and fracture history; programme from section 5 with knee-friendly loading (sit-to-stand from a higher chair, step-ups on a low step, quadriceps work), balance training, back extensors, walking, stair training with a rail, and spine-sparing advice.

7 Exam corner

Past questions (MUHS, 2016–2026)

Long answers

45-year-old with menopausal symptoms, incontinence and rib and back pain (W18); hormonal changes and sequelae of menopause with PT (W24, 8 + 7); 75-year-old with osteoporosis and knee pain: evaluation and exercise programme (S25).

7 marks

Importance of physiotherapy in postmenopausal women (S17, S22); physiological variations with menopause (S23).

3 marks

Prevention of osteoporosis with ageing (W18); predisposing factors for osteoporosis (W22); types of osteoporosis (S23); risk factors for osteoporosis (S26).

Keywords to underline

12 months without periods · oestrogen falls, FSH rises · vasomotor symptoms · genitourinary syndrome · rapid bone loss · T-score –2.5 · type I and type II · resistance and balance training · impact loading · spine-sparing · fall prevention

Answer skeleton: importance of PT in postmenopausal women (7 marks)

1. **[Changes, 2]** Hormones and their effects by system.
2. **[PT for each, 4]** Bone, falls, pelvic floor, weight and heart, pain, sleep and mood, education.
3. **[Precautions, 1]** Screen for osteoporosis and fractures; spine-sparing movement.

Draw these

- The T-score scale (section 3)

- Hip hinge versus rounded-back lifting

Viva questions

1. Define menopause.
2. Which hormone rises markedly after menopause?
3. What T-score defines osteoporosis?
4. Contrast type I and type II osteoporosis.
5. Which exercises should a woman with osteoporosis avoid?
6. Why is aerobic exercise alone not enough for bone?

Linked practical spots

- Teach the hip hinge and log-roll; measure height and occiput-to-wall distance for kyphosis; demonstrate back extensor and balance exercises.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Types of osteoporosis. (S23)**

Answer
Primary type I (postmenopausal, about 50–70, trabecular bone, vertebral and wrist fractures); primary type II (senile, over 70, trabecular and cortical, hip and vertebral fractures); idiopathic juvenile; secondary (steroids, endocrine, rheumatoid arthritis, malabsorption, alcohol, immobilisation, drugs); regional (disuse, CRPS).

3 m **Risk factors for osteoporosis. (S26)**

Answer
Non-modifiable: female, age, early menopause, family history of hip fracture, small build, previous fragility fracture. Modifiable: inactivity, low weight, low calcium and vitamin D, smoking, alcohol, caffeine. Disease and drugs: steroids, RA, thyroid or parathyroid excess, malabsorption, immobility.

7 m **Physiological variations associated with menopause. (S23)**

Answer
Definition and stages; hormonal changes (oestrogen and progesterone down, FSH and LH up); effects: vasomotor, mood and sleep, bone loss, muscle and joints, genitourinary and pelvic floor, heart and metabolism, skin and weight. Add one line on PT for each.

LAQ **Describe the hormonal changes and primary sequelae of menopause, and physiotherapy management with clinical reasoning. (8 + 7, W24)**

Answer
Hormones and sequelae by system (section 1). PT with reasoning: resistance and impact exercise for bone; balance for falls; pelvic floor training for leakage and prolapse; aerobic exercise for weight, heart and mood; posture and back extensors; relaxation for flushes and sleep; education on diet, vitamin D, lifestyle and screening.

MCQ **A T-score of −2.0 indicates:**

A. Normal bone

B. Osteopenia

C. Osteoporosis

D. Established osteoporosis

Answer
B.

MCQ **Postmenopausal (type I) osteoporosis typically causes fractures of the:**

A. Skull and ribs

B. Vertebrae and distal radius

C. Femoral shaft

D. Clavicle

Answer
B.

MCQ **For a woman with osteoporosis, which exercise should be avoided?**

A. Sit-to-stand

B. Repeated loaded sit-ups

C. Back extensor strengthening

D. Balance training

Answer
B.

Last recall check. Write the hormonal changes and their effects by system, the T-score bands, the types and risk factors of osteoporosis, and an exercise programme with spine-sparing advice.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Community Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. World Health Organization. Assessment of fracture risk and its application to screening for postmenopausal osteoporosis. Technical Report Series 843. Geneva: WHO; 1994.
3. Riggs BL, Melton LJ 3rd. Involutional osteoporosis. *N Engl J Med* 1986;314:1676–86.
4. Giangregorio LM, et al. Too Fit To Fracture: exercise recommendations for individuals with osteoporosis or osteoporotic vertebral fracture. *Osteoporos Int* 2014;25:821–35.
5. Pallikadavath S, et al. Natural menopause among women below 50 years in India: a population-based study. *Indian J Med Res* 2016;144:366–77; and Indian studies reporting a mean age at natural menopause of about 46 years.

2(f) Urogenital dysfunction: the pelvic floor, incontinence, prolapse and gynaecological conditions

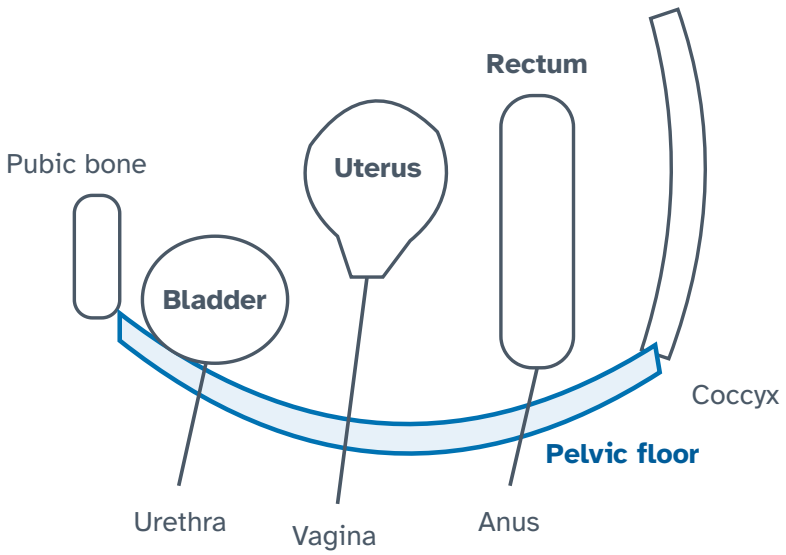
> Incontinence and prolapse were asked in 16 of 20 papers. One idea runs through all of it: the pelvic floor is a muscle sling that holds up the bladder, uterus and rectum and closes their openings, so teaching a woman to find, strengthen and time that muscle is the first treatment for leaking and for a mild prolapse.

Syllabus	Ch 2(f) Urogenital dysfunction, prolapse, incontinence, malignancy and their therapeutic interventions
Priority	**Tier 1.** In 16 of 20 papers (2016–2026): 6 long cases, 5 seven-mark and 9 three-mark questions.
Asked as	LAQ W16 · S18 · W19 · W22 · S25 · S26 · 7m S19 · S20 · S21 · W22 · W25 · 3m S16 · S18 · W18 · W19 · W20 · W21 · S23 · S24 · S25
Usual split	15 (ICF functional diagnosis 5, management with goals 10; or assessment 10, management 5; or micturition 5, assessment 5, management 5); 7 (types or grades of prolapse or stress incontinence with their PT)
Checked against	NICE guideline NG123 and quality standard QS77 on urinary incontinence and prolapse in women; Cochrane review of pelvic floor muscle training for urinary incontinence (Dumoulin et al., 2018); Cochrane review of electrical stimulation for stress incontinence (Stewart et al., 2017); POPPY trial (Hagen et al., 2014), Brækken et al. (2010) and the International Urogynecology Consultation review (Bø et al., 2022) on training for prolapse; PERFECT scheme (Laycock and Jerwood, 2001); ICS definitions; POP-Q staging; RCOG guideline on obstetric anal sphincter injury; international PCOS guideline (2023). Full list under Sources (p. 82). Content checked 21 October 2026.

1 The pelvic floor

Muscles of the pelvic floor were a 3-mark question in S16, and every answer on this page starts here.

The pelvic floor as a sling, side view



The muscles (blue) form a sling from pubis to coccyx. Three tubes pass through it, so one muscle controls the bladder, the vagina and the bowel.

Layer	Muscles
Pelvic diaphragm (deep)	**Levator ani**: pubococcygeus (with puborectalis, which loops behind the rectum), iliococcygeus; and coccygeus (ischiococcygeus)

Layer	Muscles
Deep perineal pouch	Deep transverse perineal, external urethral sphincter (with compressor urethrae and urethrovaginal sphincter)
Superficial perineal pouch	Bulbospongiosus, ischiocavernosus, superficial transverse perineal
Others	External anal sphincter; the perineal body is the central tendon where they meet

- **Nerve supply:** pudendal nerve (S2–S4) and direct branches of S3–S4 to levator ani.
- **Fibres:** mostly slow-twitch for constant support, with fast-twitch fibres that close the openings quickly when pressure rises (cough, sneeze, lift).
- **Functions:** support the pelvic organs; keep urine, wind and stool in (continence); relax to let them out; sexual function; work with the diaphragm, transversus abdominis and multifidus to control abdominal pressure and steady the spine and pelvis; guide the baby's head during birth.

2 How the bladder fills and empties

Physiology of micturition was the first 5 marks of the W19 long answer.

Nerve	Spinal level	Action
Sympathetic (hypogastric nerve)	T10–L2	Relaxes the detrusor and closes the bladder neck: **storage**
Parasympathetic (pelvic nerve)	S2–S4	Contracts the detrusor: **emptying**
Somatic (pudendal nerve)	S2–S4	Contracts the external sphincter and pelvic floor: voluntary holding

1. **Filling:** the bladder stretches at low pressure; sympathetic tone and the sphincters keep it closed; stretch signals rise to the brain; the first desire to void is felt at about half capacity (normal capacity about 400–600 ml).
2. **Holding:** the frontal cortex inhibits the pontine micturition centre until it is convenient; the pelvic floor can be contracted to hold on.
3. **Voiding:** the pontine centre switches over: the sphincters and pelvic floor relax, the detrusor contracts, and the bladder empties completely.

The same circuit explains the spinal cord injury bladder (see bladder and bowel in Neuro).

3 Urinary incontinence: types

Types of stress incontinence and their PT (S19, 7 marks).

Type	Definition (ICS)	Typical cause
Stress urinary incontinence (SUI)	Involuntary leakage on effort or exertion, or on sneezing or coughing	Weak pelvic floor and urethral support after childbirth, menopause, chronic cough, obesity, heavy lifting
Urgency urinary incontinence (UUI)	Leakage accompanied by or immediately preceded by urgency	Overactive detrusor (overactive bladder), irritants (caffeine), infection, neurological disease
Mixed	Leakage with urgency and also with exertion, sneezing or coughing	Both; treat the more troublesome first
Overflow and others	Dribbling from an over-full bladder that does not empty; continuous leakage; functional incontinence	Outlet obstruction or weak detrusor; fistula; poor mobility or cognition (cannot reach the toilet in time)

Types of stress incontinence

- **By severity (Stamey grades):** grade 1, leakage with coughing, sneezing or lifting; grade 2, with smaller rises in pressure such as walking, standing up or climbing stairs; grade 3, leakage regardless of activity or position.

- **By mechanism:** urethral hypermobility (the bladder neck and urethra drop on effort because the support is weak) or intrinsic sphincter deficiency (the urethral closure itself is weak, as after surgery or with severe atrophy).
- **PT:** pelvic floor muscle training is first-line for all; add "the knack" for grade 1, biofeedback or electrical stimulation for a very weak or unaware muscle, and lifestyle change. Grade 3 or sphincter deficiency needs the urogynaecologist (surgery, bulking agents), with PT before and after.

4 Assessment of incontinence

Principles of incontinence assessment (W19), tests for incontinence (S25, 3 marks), the PERFECT scheme (W20, 3 marks).

History

- When it leaks (cough, lift, on the way to the toilet), how much, how often, pads used; frequency by day and night; urgency; difficulty emptying; bowel habit and constipation.
- Obstetric history (number of births, big baby, long labour, forceps, tears), menopause, surgery, chronic cough, weight, drugs (diuretics), diabetes or neurological disease, fluid and caffeine intake, effect on life and sexual function.
- Red flags for referral: blood in urine, recurrent infection, pain, a palpable bladder, neurological signs, a large prolapse.

Tests

- **Bladder diary** (frequency–volume chart) for at least 3 days: fluid in, voids and volumes, leaks and their triggers.
- **Questionnaires:** ICIQ-UI short form for severity and bother.
- **Pad test:** weigh a pad worn during set activities; in the 1-hour test a gain of 1 g or more is positive.
- **Cough stress test:** leakage seen at the moment of a cough with a comfortably full bladder.
- **Q-tip test:** a cotton swab in the urethra swings more than 30° on straining in urethral hypermobility.
- **Digital vaginal examination** of the pelvic floor (with consent, by a trained physiotherapist): PERFECT and the Modified Oxford scale; perineometer or EMG biofeedback for a number.
- **Medical tests:** urine test for infection, post-void residual by ultrasound, urodynamics before surgery.

The PERFECT scheme (Laycock and Jerwood, 2001)

Letter	What is measured
P: Power	Strength of a maximal contraction on the Modified Oxford scale: 0 none, 1 flicker, 2 weak, 3 moderate with lift, 4 good against resistance, 5 strong against strong resistance
E: Endurance	Seconds a maximal contraction is held before it fades (up to 10)
R: Repetitions	Number of those holds repeated with a short rest between (up to 10)
F: Fast	Number of quick 1-second contractions in a row (up to 10)
ECT: Every Contraction Timed	The results are timed and recorded so they become her own exercise programme

Remember

P-E-R-F-ECT: Power, Endurance, Repetitions, Fast, Every Contraction Timed

A woman graded P3 E5 R4 F6 starts with 5-second holds repeated 4 times, then 6 quick contractions, and builds up from there.

5 Physiotherapy for incontinence

Kegel's exercises (S24), deferment techniques (W19), interferential therapy (W18) and continence-promoting advice (S18), each 3 marks.

Pelvic floor muscle training (Kegel's exercises)

- Described by the gynaecologist Arnold Kegel in 1948. Supervised training for at least 3 months is the first-line treatment for stress and mixed incontinence (NICE). Women with stress incontinence who trained were about eight times more likely to report cure than untreated women (Dumoulin et al., 2018).
- **Teach:** "squeeze and lift as if stopping wind and the flow of urine", without holding the breath, squeezing the buttocks or pulling the tummy in hard. Check the contraction by vaginal palpation where possible. Stopping the urine stream is only for finding the muscle once, not an exercise.
- **Dose:** slow holds of up to 10 seconds, repeated up to 10 times, and up to 10 quick 1-second contractions; at least 8 contractions 3 times a day (NICE), set from her PERFECT result.
- **Progress** from lying to sitting to standing, then during functional tasks (lifting, climbing stairs, jumping).
- **The knack:** a strong contraction just before and during a cough, sneeze or lift.

Bladder training and urge deferment (urgency and mixed)

- **Bladder training** for at least 6 weeks (NICE): void by the clock, then lengthen the gap by about 15 minutes at a time towards 3–4 hours.
- **Deferment when the urge comes:** stop and stand still or sit down (pressure on the perineum helps); do several quick strong pelvic floor contractions; breathe slowly and relax; distract the mind (count backwards); curl the toes or cross the legs; when the urge fades, walk calmly to the toilet.
- Cut down caffeine, fizzy drinks and alcohol; no "just in case" visits.

Adjuncts

Method	Use	Point to write
Biofeedback	Vaginal pressure probe (perineometer) or EMG shows the contraction on a screen	Helps women who cannot feel the muscle; not needed routinely (NICE)
Electrical stimulation	Vaginal electrode produces a pelvic floor contraction; higher frequencies (about 50 Hz) for weak muscles in SUI, low (about 10 Hz) to calm an overactive bladder	Use when she cannot contract the muscle at all, always with active training
Interferential therapy	Medium-frequency currents meet in the pelvis and produce a low-frequency effect deep in the pelvic floor with little skin discomfort; no vaginal probe is needed (details below)	Weak evidence; only alongside training; avoid in pregnancy, infection, bleeding, cancer in the area or with a pacemaker
Vaginal cones	Weighted cones held in the vagina while walking	The muscle must contract to keep the cone in

Interferential therapy in stress incontinence (3 marks in W18)

Principle: two medium-frequency currents are passed through the pelvis by crossed pairs of electrodes. Where they cross, they interfere and produce a low-frequency beat frequency deep in the tissues. Medium frequencies meet less skin resistance, so the current reaches the pelvic floor with little skin discomfort. The beat makes the pelvic floor contract without a vaginal probe.

- **Aim:** a passive contraction for a woman who cannot contract the muscle at all, so that she feels it and can then train it actively.
- **Proposed effect:** stimulation is meant to restore reflex activity of the pelvic floor, so urethral closure pressure rises and she leaks less on effort.
- **Why choose it:** it is applied over the skin, so it suits women who cannot or will not use a vaginal probe, such as some older women.

Electrode placement

Method	Where the electrodes go	Point to write

Method	Where the electrodes go	Point to write
Bipolar (Laycock and Green, 1988)	One electrode under both ischial tuberosities; one over the anterior perineum just below the pubic symphysis	The classic two-electrode set-up
Bipolar (Dumoulin, 1995)	One electrode over the anal region; one just above the pubic symphysis	In 10 continent volunteers it gave the same contraction with less current; 7 preferred it
Quadripolar (four electrodes)	Two suction electrodes above the pubis; two near the inner side of the ischial tuberosities, placed crosswise	The two currents cross in the pelvic floor (Demirtürk et al., 2008)

Parameters used in published studies

Parameter	Setting
Carrier frequency	Medium frequency, usually about **4000 Hz** on most machines; **2000 Hz** in a 2026 trial
Beat frequency	Swept **0–100 Hz** (Demirtürk et al., 2008); or one run at **10–15 Hz** and one at **50 Hz** (Sekandari et al., 2026)
Intensity	Enough for a tolerable contraction, without an excessive sustained (tetanic) contraction
Duration	**15 min** (Demirtürk); two runs of **15 min** (Dumoulin; Sekandari); **20 min** (Trivedi and Dhingra, New Delhi)
Sessions	**3 times a week**: 15 sessions over 5 weeks (Demirtürk), 12 over 4 weeks (Sekandari), 9 over 3 weeks (Dumoulin), or for 3 months (Trivedi and Dhingra)
Always with	Supervised pelvic floor exercises, the knack and a home programme

Evidence for interferential therapy

- **Small trials only.** In 40 women, interferential therapy and biofeedback-guided exercises gave similar gains in pad test, muscle strength and quality of life (Demirtürk et al., 2008).
- In 30 women, adding interferential therapy to pelvic floor exercises reduced weekly leaks more than sham current. Muscle pressure, severity and quality of life improved equally in both groups (Sekandari et al., 2026).
- An uncontrolled study of 30 women at Lady Hardinge Medical College, New Delhi, reported better urodynamic values and fewer leaks after 3 months (Trivedi and Dhingra, 2011).
- **Electrical stimulation in general** is probably better than no treatment. It has not been shown to beat pelvic floor training or to add to it; the evidence is of low quality (Cochrane review, Stewart et al., 2017).
- **NICE:** do not routinely add electrical stimulation to pelvic floor training. Consider it, or biofeedback, for a woman who cannot actively contract her pelvic floor, to help motivation and adherence.

> **Write in the exam**
>
> "Interferential therapy is an adjunct for a woman who cannot contract her pelvic floor. It is used with supervised pelvic floor training, never instead of it."

Continence-promoting advice

- Drink normally (about 1.5–2 litres a day unless advised otherwise); do not cut fluids, which concentrates urine and irritates the bladder.
- Reduce caffeine, fizzy drinks and alcohol.
- Avoid constipation and straining: fibre, fluids, a footstool on the toilet.
- Lose weight if overweight (NICE suggests weight loss when BMI is over 30); stop smoking; treat chronic cough.
- Sit fully on the toilet, relax and empty completely; do not hover.
- Do the knack before lifting, coughing or sneezing; lift with help and with the load close.
- Keep up pelvic floor exercises for life, especially in pregnancy, after birth and around menopause.

6 Uterine and vaginal prolapse

Asked as 7 marks in S20, S21, W22 and W25, 3 marks in W21, and in the W16 long case (58-year-old with grade I prolapse).

- **Definition:** descent of one or more of the front wall of the vagina, the back wall, the uterus (cervix) or the vault after hysterectomy.

Compartment	Type
Anterior	Cystocele (bladder), urethrocele (urethra), cystourethrocele
Middle (apical)	Uterine prolapse (uterovaginal); vault prolapse after hysterectomy
Posterior	Rectocele (rectum), enterocele (small bowel in the pouch of Douglas)

Degrees of uterine prolapse

Clinical degree	What is seen	POP-Q stage
First degree	The cervix comes down but stays inside the vagina	Stage I: lowest point more than 1 cm above the hymen
Second degree	The cervix comes out through the vaginal opening on straining	Stage II: within 1 cm above or below the hymen
Third degree (procidentia)	The whole uterus lies outside the vaginal opening	Stage III: more than 1 cm below the hymen; stage IV: complete eversion

- **Causes:** childbirth (many births, long labour, big baby, instruments, pushing before full dilatation), loss of oestrogen after menopause, chronic rise in abdominal pressure (cough, constipation, heavy lifting, obesity), ageing and inherited tissue weakness.
- **Symptoms:** something coming down, dragging or heaviness worse by evening, low backache, urinary leakage or incomplete emptying, difficulty emptying the bowel, discomfort in intercourse.

Physiotherapy for prolapse

- **Pelvic floor muscle training**, individually taught and supervised, for at least **16 weeks** is the first option for symptomatic POP-Q stage 1 or 2 prolapse; continue it afterwards if it helps (NICE). It reduced prolapse symptoms in the POPPY trial (Hagen et al., 2014).
- **Effects of exercise** (W25): training improved the prolapse by one POP-Q stage in 19% of women against 8% of controls, raised the bladder and rectum and reduced symptoms (Brækken et al., 2010). It also helps the leaking that often comes with prolapse, and support of the spine and pelvis. Activities that strain the pelvic floor (heavy lifting, straining at stool, repeated breath-held sit-ups, high-impact jumping) can worsen the dragging feeling, so they are modified, not every activity stopped. Mechanisms and trials are below.
- **Lifestyle:** weight loss, treat constipation and cough, safe lifting with the knack.
- **Pessary** (fitted by a doctor) for women who are unfit for or do not want surgery; PT continues with it.
- **Surgery** (vaginal hysterectomy with repair, colporrhaphy) for large prolapse, with PT before and after (section 8). Adding pelvic floor training around prolapse surgery did not improve the prolapse result in 9 of 10 trials (Bø et al., 2022). Post-operative PT is still needed for breathing, early walking and lifting advice.

How pelvic floor training helps prolapse (7 marks in W25)

The pelvic floor closes the gap (the levator hiatus) through which the bladder, uterus and rectum can bulge. Training works in two ways, and both can act at once (Bø, 2006; Bø et al., 2022).

1. **Timing (the knack):** she learns to contract the pelvic floor before and during a cough, lift or strain. Because the contraction narrows the hiatus and lifts the floor, the organs are not pushed down when abdominal pressure rises. With practice this becomes a habit.
2. **Strength training ("building structural support"):** months of near-maximal contractions make the muscle thicker and stiffer. Because a firmer floor sits higher at rest, it holds the organs up all day, not only when she squeezes.

Proof on ultrasound: after 6 months of supervised strength training, women with prolapse had thicker and shorter pelvic floor muscles, a smaller levator hiatus and a higher resting position of the bladder and rectum than controls. The hiatus also stretched less on straining, a sign of a stiffer floor (Brækken et al., 2010, 109 women).

What the main trials show

Trial	Women and training	Result
Brækken et al., 2010 (Norway)	109 women, stage I–III; 6 months of physiotherapist-led strength training and daily home exercise vs advice and the knack only	One POP-Q stage better in **19% vs 8%**; bladder and rectum raised; fewer and less bothersome symptoms
POPPY, Hagen et al., 2014 (UK, New Zealand, Australia)	447 women with symptomatic stage I–III; 5 one-to-one physiotherapy appointments over 16 weeks vs an advice leaflet	Greater fall in the prolapse symptom score at 12 months
Kashyap et al., 2013 (Chandigarh, India)	140 women, stage I–III; one-to-one training plus a self-instruction manual vs the manual alone, for 24 weeks	One-to-one training gave greater improvement in symptoms and quality of life
Wiegersma et al., 2014 (Netherlands, primary care)	287 women aged 55 or over with mild prolapse; training vs watchful waiting for 3 months	**57% vs 13%** felt better, but the score difference was below the level judged clinically important; no difference in prolapse stage

What the reviews conclude

- **Cochrane review** (Hagen and Stark, 2011): training improves prolapse symptoms and severity compared with no training.
- **Meta-analysis** of 13 trials with 2,340 women (Li et al., 2016): better symptom scores and POP-Q stage, more women reporting improvement, and stronger muscles with more endurance.
- **International Urogynecology Consultation** (Bø et al., 2022): good evidence from 11 trials that training reduces symptoms or improves the stage by one stage in stage I–III. It is recommended as first-line treatment. No serious side effects have been reported. It works only with thorough teaching and supervision.
- **Limits:** few long-term studies, with mixed results (Bø et al., 2022). A 2026 meta-analysis found benefit in the short term (up to 6 months), but no significant effect beyond 6 months or in women aged 55 or over (Cong et al., 2026).

Dose for prolapse (Bø et al., 2022): weekly individual or group sessions with a physiotherapist for at least **6 months**, plus home training. Strength work follows general training rules: **3 sets of 8–12** near-maximal contractions at least **3–4 times a week** for at least **4 months**. Then keep up **8–12 contractions 1–2 times a week**.

7 Bowel and anorectal problems in pregnancy and after birth

S18 long answer: bowel and anorectal dysfunction in the antenatal and postnatal periods, with reasons, assessment and management.

Problem	Reason
Constipation (antenatal)	Progesterone slows the bowel; iron tablets; the growing uterus presses on the bowel; less activity
Haemorrhoids	Raised pressure in pelvic veins plus straining at stool; worsened by pushing in labour
Constipation and painful defecation (postnatal)	Fear of opening the bowels with episiotomy stitches or a tear, pain relief drugs, less food and fluid in labour
Anal (faecal or flatus) incontinence and urgency	Tear of the anal sphincter (third or fourth degree), stretch injury of the pudendal nerve in a long second stage or forceps delivery, weak pelvic floor

- **Assessment:** bowel habit and diary, stool type (Bristol Stool Form Scale), straining, urgency, leakage of wind or stool, perineal pain, obstetric details (tear, episiotomy, instruments); inspection of the perineum; pelvic floor assessment; refer for anal sphincter assessment after a tear.

- **Defecation position:** feet on a footstool so the knees are above the hips, lean forward with straight back, relax the belly out, breathe out and do not strain; support the stitches with a clean pad held on the perineum.
- **Diet:** fibre, fluids, regular routine; stool softeners as prescribed after a sphincter tear.
- **Pelvic floor and anal sphincter exercises** from the first days after birth (gently), progressed over weeks; physiotherapy after repair of a sphincter tear can help (RCOG); biofeedback and electrical stimulation for persistent weakness.
- **Pain relief** for the perineum: ice packs wrapped in cloth, positioning (side-lying, cushion), early walking.
- Advice on lifting the baby, avoiding straining, and when to seek help. See labour and postnatal care (p. 52) and pregnancy (p. 43).

8 Gynaecological surgery, cancer and PCOD

Indications for hysterectomy (S23, 3 marks); a 23-year-old with PCOD, BMI 34 and back pain (S26 long case); malignancy is named in the syllabus.

Hysterectomy

- **Indications:** fibroids with heavy bleeding or pressure (the commonest), abnormal uterine bleeding not controlled by drugs, adenomyosis and endometriosis, uterovaginal prolapse, cancer of the uterus, cervix or ovary, chronic pelvic pain from pelvic inflammatory disease, and life-threatening bleeding after childbirth.
- **Types:** total, subtotal (cervix left), radical (Wertheim's, with lymph nodes) for cancer; abdominal, vaginal or laparoscopic.
- **PT after surgery:** breathing exercises and supported coughing, leg exercises against thrombosis, early walking, gentle pelvic floor and deep abdominal exercises, getting out of bed by rolling to the side, no heavy lifting for about 6 weeks, graded return to activity.

Gynaecological cancer

- Breast cancer is now the commonest cancer in Indian women and cervical cancer is second; see mastectomy and oncological surgery for the breast.
- PT problems: wound and chest care after surgery; leg lymphoedema after pelvic node removal (compression, exercise, skin care, manual lymph drainage); cancer-related fatigue (graded aerobic and strength exercise); bladder, bowel and sexual problems and vaginal narrowing after radiotherapy (pelvic floor work, dilators as advised); bone loss if the ovaries are removed (see menopause and osteoporosis (p. 63)).

Polycystic ovary disease (PCOD or PCOS)

- Diagnosed by two of three: irregular or absent ovulation, signs or tests of raised androgens (acne, hair growth), polycystic ovaries on ultrasound (international guideline, 2023).
- Linked with weight gain, insulin resistance, type 2 diabetes risk, infertility, anxiety and depression.
- **PT role:** lifestyle change is the first treatment; aerobic activity of 150–300 minutes a week at moderate intensity or 75–150 minutes vigorous, plus muscle-strengthening on 2 non-consecutive days, and less sitting; even modest weight loss improves periods and ovulation; core and back exercises and posture for back pain; stress management; diet with a dietitian. See obesity (p. 26).

S26 ICF in brief: health condition, PCOD with obesity (BMI 34) and mechanical low back pain; body functions, back pain, excess weight, low fitness, weak core, stress; activities, prolonged sitting and standing, exercise; participation, studies or work, social life, plans for pregnancy; environment, food habits at home and outside; personal, young, motivated by fertility. Goals: short term, pain relief and a daily walking habit; long term, 5–10% weight loss over 6 months, regular cycles, pain-free activity.

9 Worked long case: a woman who leaks on coughing

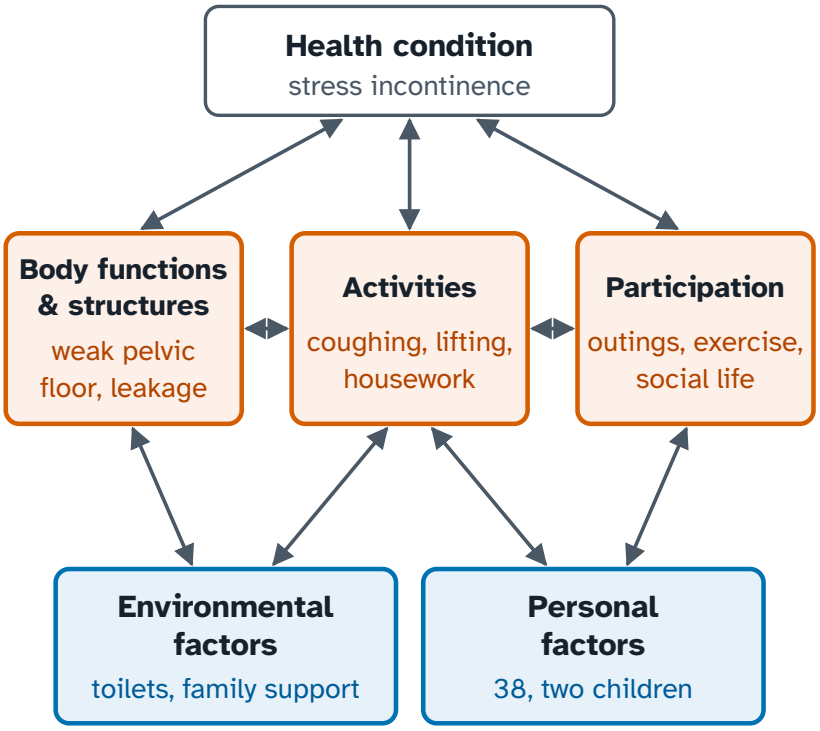
From a real paper (W22, 5 + 10 marks; repeated in S25)

> "A 38 year old woman comes to you with complaints of inability to control urination, especially during coughing & during doing house work & lifting weights. She has 2 children & is from an affluent family. Presently she has been referred to you for physiotherapy management. Give her functional diagnosis with clinical reasoning as per ICF (5 Marks) Also write her PT management with the treatment goals. (10 Marks)"

Spot the clues before you write

coughing, housework and lifting: leakage when abdominal pressure rises, so stress urinary incontinence. 2 children: vaginal births stretched the pelvic floor and its nerve. 38: before menopause, so muscle training should work well. affluent family: can attend supervised sessions; may have help at home but may also be inactive or overweight.

The ICF model filled for this case



Draw the six boxes with two-way arrows, then write two or three points from the case in each box.

ICF component	From this case	Clinical reasoning
Health condition	Stress urinary incontinence (rule out urgency and infection on the diary and urine test)	Leaks only when abdominal pressure rises
Body functions and structures	Weak pelvic floor (for example Oxford 2, endurance 4 s, 5 fast), no pre-contraction before a cough, possible bladder neck descent; leakage on the pad and cough tests	Two births stretched the muscle and its nerve; urethral support fails under load
Activity limitations	Coughing, sneezing, laughing, lifting, housework, exercise	Each raises abdominal pressure above urethral closure pressure
Participation restrictions	Avoids outings, exercise classes and social events; uses pads; embarrassment, possible effect on sexual life	Fear of leaking in public
Environmental factors	**Facilitators:** can afford supervised physiotherapy, family support. **Barriers:** unclean or distant toilets when out, pads cost	Regular supervised sessions improve results

ICF component	From this case	Clinical reasoning
Personal factors	38 years, two children, embarrassment, possibly overweight or inactive	Motivation and privacy matter for adherence

One-line functional diagnosis

A 38-year-old mother of two with stress urinary incontinence due to a weak pelvic floor, leaking on coughing, housework and lifting, which limits her outings and exercise.

Goals

Problem	Short-term (4–6 weeks)	Long-term (3–6 months)
Muscle	Correct contraction without breath-holding; Oxford 2 to 3; 6-second holds	Oxford 4; 10-second holds × 10; 10 fast
Leakage	Uses the knack; leaks halved on the diary	Dry on coughing, lifting and housework; no pads
Life	Understands the problem and the plan	Back to outings and exercise; lifelong home programme

Management

Impairment	Technique and dose	Why
Weak pelvic floor	Supervised training from her PERFECT result: slow holds and fast contractions, at least 8 contractions 3 times a day, for at least 3 months	First-line treatment for stress incontinence (NICE)
Cannot feel the muscle	Biofeedback or vaginal electrical stimulation in the first weeks	Teaches awareness, then active training takes over
Poor timing	The knack before every cough, sneeze and lift	Closes the urethra before pressure rises
Weak function under load	Progress from lying to standing, then contractions during lifting, stairs and housework	She leaks in upright, loaded tasks
Risk factors	Continence advice, weight loss if BMI over 30, treat constipation and cough, safe lifting	Reduces load on the pelvic floor
Low fitness	Low-impact aerobic exercise (walking, cycling, swimming) 150 minutes a week, core training with pelvic floor pre-contraction	Health and weight without heavy strain

Review with the bladder diary, pad test and PERFECT every 4–6 weeks; refer to the urogynaecologist if there is no improvement after 3 months of good training.

Practice case (W16 long answer)

A 58-year-old with grade I prolapse and dribbling of urine on coughing and straining for 5 months (assessment 10, management 5). Assessment: the history and tests in section 4 plus prolapse symptoms, bowel function, menopause, cough and lifting; inspection on straining; PERFECT. Management: pelvic floor training for prolapse and stress incontinence (sections 5 and 6), the knack, constipation and weight advice, and referral for a pessary or local oestrogen if needed.

10 Exam corner

Past questions (MUHS, 2016–2026)

Long cases

58-year-old with grade I prolapse and dribbling (W16, 10 + 5); bowel and anorectal dysfunction in pregnancy and after birth (S18); physiology of micturition, assessment of incontinence and objective management (W19, 5 + 5 + 5); 38-

year-old who leaks on coughing, housework and lifting (W22, S25); 23-year-old with PCOD, BMI 34 and back pain (S26).

7 marks

Types of stress incontinence and their PT (S19); PT in uterine prolapse (S20); grades of uterine prolapse and PT (S21); types and degrees of urogenital prolapse and management (W22); types of uterine prolapse and the effects of exercise (W25).

3 marks

Muscles of the pelvic floor (S16); continence-promoting advice (S18); interferential therapy in stress incontinence (W18); deferment techniques for urge incontinence (W19); PERFECT (W20); define uterine prolapse and its types (W21); indications for hysterectomy (S23); Kegel's exercises (S24); tests for incontinence in women (S25).

Keywords to underline

levator ani pudendal nerve S2–S4 pontine micturition centre stress, urge, mixed bladder diary pad test PERFECT Modified Oxford scale supervised training for 3 months the knack bladder training urge deferment beat frequency POP-Q levator hiatus first-line

Answer skeleton: a long case of incontinence (15 marks)

1. **[ICF diagnosis, 5]** Type of incontinence with reasoning, the six ICF boxes, one-line diagnosis.
2. **[Goals, 3]** Short- and long-term, measurable (PERFECT, diary, pad test).
3. **[Management, 7]** PFMT with dose, the knack, functional progression, adjuncts, bladder training if urgency, lifestyle advice, review and referral.

Draw these

- The pelvic floor sling with the three openings (section 1)
- Degrees of uterine prolapse: three small side views
- The ICF model for the case

Viva questions

1. Name the parts of levator ani and its nerve supply.
2. Which nerve contracts the detrusor, and from which segments?
3. What does each letter of PERFECT stand for?
4. What is the knack?
5. How would you tell stress from urge incontinence in the history?
6. What is the difference between second- and third-degree uterine prolapse?

Linked practical spots

- Explain how to teach a pelvic floor contraction in words; show a perineometer or biofeedback unit; read a bladder diary; name the PERFECT grades.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Deferment techniques for urge incontinence. (W19)**

Answer
When the urge comes: stop and stand still or sit down; do several quick strong pelvic floor contractions; breathe slowly and relax; distract the mind (count backwards); press on the perineum or curl the toes; once the urge passes, walk calmly to the toilet. Used within bladder training, which lengthens the gap between voids by about 15 minutes at a time.

3 m **PERFECT method of pelvic floor muscle grading. (W20)**

Answer
Laycock and Jerwood (2001), by digital vaginal examination: Power on the Modified Oxford scale 0–5; Endurance, seconds a maximal contraction is held (up to 10); Repetitions of that hold (up to 10); Fast, number of 1-second contractions (up to 10); Every Contraction Timed, so the result becomes her exercise programme.

3 m **Interferential therapy in stress urinary incontinence. (W18)**

Answer
Two medium-frequency currents cross in the pelvis and give a low-frequency beat that contracts the pelvic floor without a vaginal probe. Electrodes: bipolar, under the ischial tuberosities and over the anterior perineum (Laycock and Green), or four electrodes, two suprapubic and two by the ischial tuberosities, crosswise. Trial settings: carrier about 4000 Hz, beat 0–100 Hz, 15 minutes, 3 times a week for about 15 sessions. Small trials only; NICE: consider only if she cannot contract the muscle, always with supervised training.

7 m **Types and degrees of urogenital prolapse; explain briefly its management. (W22)**

Answer
Types by compartment: anterior (cystocele, urethrocele), middle (uterine, vault), posterior (rectocele, enterocele). Degrees: first, cervix descends within the vagina; second, cervix comes out through the introitus on straining; third (procidentia), the whole uterus outside; POP-Q stages 0–IV. Management: supervised PFMT for at least 16 weeks for stage I–II (POPPY trial; one-stage improvement in 19% vs 8%, Brækken 2010), the knack, lifestyle (weight, constipation, cough, safe lifting), pessary by the gynaecologist, surgery for large prolapse with pre- and post-operative PT.

LAQ **Explain the physiology of micturition. Describe the principles of incontinence assessment. Write the objective methods for incontinence management. (5 + 5 + 5, W19)**

Answer
Physiology: sympathetic T10–L2 relaxes the detrusor and closes the bladder neck (storage); parasympathetic S2–S4 contracts the detrusor (voiding); pudendal S2–S4 controls the external sphincter; the pontine micturition centre coordinates, the cortex decides when. **Assessment:** history (type, triggers, obstetric, bowel, drugs), bladder diary, questionnaire, pad test, cough stress test, Q-tip test, PERFECT, urine test and residual volume, urodynamics if surgery is planned. **Objective methods:** PFMT with measured dose, biofeedback (perineometer, EMG), electrical stimulation, vaginal cones, bladder training with a diary, and re-measurement with the diary, pad test and PERFECT.

MCQ **The nerve supply of the external urethral sphincter is:**

A. Hypogastric nerve, T10–L2

B. Pelvic splanchnic nerves only

C. Pudendal nerve, S2–S4

D. Obturator nerve

Answer
C.

MCQ **The first-line treatment for stress urinary incontinence is:**

A. Surgery

B. Supervised pelvic floor muscle training for at least 3 months

C. Reducing fluid intake

D. Electrical stimulation alone

Answer
B.

MCQ **In third-degree uterine prolapse:**

A. The cervix stays inside the vagina

B. The cervix reaches the introitus only on straining

C. The whole uterus lies outside the introitus

D. Only the bladder bulges

Answer
C. Also called procidentia.

Last recall check. Draw the pelvic floor sling, write PERFECT from memory, list three types of incontinence with one treatment each, and give the degrees of prolapse.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Community Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. National Institute for Health and Care Excellence. Urinary incontinence and pelvic organ prolapse in women: management (NG123), 2019; and quality standard QS77, statement 4.
3. Dumoulin C, Cacciari LP, Hay-Smith EJC. Pelvic floor muscle training versus no treatment, or inactive control treatments, for urinary incontinence in women. *Cochrane Database Syst Rev* 2018;(10):CD005654.
4. Laycock J, Jerwood D. Pelvic floor muscle assessment: the PERFECT scheme. *Physiotherapy* 2001;87:631–42.
5. Hagen S, et al. Individualised pelvic floor muscle training in women with pelvic organ prolapse (POPPY): a multicentre randomised controlled trial. *Lancet* 2014;383:796–806 (abstract read; the five appointments over 16 weeks checked in the trial registration, ISRCTN35911035, and in Bø et al., 2022).
6. Brækken IH, Majida M, Engh ME, Bø K. Can pelvic floor muscle training reverse pelvic organ prolapse and reduce prolapse symptoms? An assessor-blinded, randomized, controlled trial. *Am J Obstet Gynecol* 2010;203:170.e1–7.
7. Brækken IH, Majida M, Engh ME, Bø K. Morphological changes after pelvic floor muscle training measured by 3-dimensional ultrasonography: a randomized controlled trial. *Obstet Gynecol* 2010;115:317–24.
8. Bø K. Can pelvic floor muscle training prevent and treat pelvic organ prolapse? *Acta Obstet Gynecol Scand* 2006;85:263–8.
9. Bø K, Anglès-Acedo S, Batra A, et al. International Urogynecology Consultation chapter 3 committee 2; conservative treatment of patient with pelvic organ prolapse: pelvic floor muscle training. *Int Urogynecol J* 2022;33:2633–67.
10. Hagen S, Stark D. Conservative prevention and management of pelvic organ prolapse in women. *Cochrane Database Syst Rev* 2011;(12):CD003882.
11. Li C, Gong Y, Wang B. The efficacy of pelvic floor muscle training for pelvic organ prolapse: a systematic review and meta-analysis. *Int Urogynecol J* 2016;27:981–92.

12. Kashyap R, Jain V, Singh A. Comparative effect of 2 packages of pelvic floor muscle training on the clinical course of stage I–III pelvic organ prolapse. *Int J Gynaecol Obstet* 2013;121:69–73.
13. Wiegersma M, Panman CM, Kollen BJ, et al. Effect of pelvic floor muscle training compared with watchful waiting in older women with symptomatic mild pelvic organ prolapse: randomised controlled trial in primary care. *BMJ* 2014;349:g7378.
14. Cong Z, Wang Z, Sun J, Tang Z, Yan W. Pelvic floor muscle training for pelvic organ prolapse: a systematic review and meta-analysis of randomized controlled trials. *BMC Urol* 2026;26:232.
15. Haylen BT, et al. An International Urogynecological Association (IUGA)/International Continence Society (ICS) joint report on the terminology for female pelvic floor dysfunction. *Neurourol Urodyn* 2010;29:4–20.
16. Bump RC, et al. The standardization of terminology of female pelvic organ prolapse and pelvic floor dysfunction. *Am J Obstet Gynecol* 1996;175:10–7.
17. Laycock J, Green RJ. Interferential therapy in the treatment of genuine stress incontinence. *Physiotherapy* 1988 (original not seen; electrode placement checked through later descriptions, including Dumoulin et al., 1995).
18. Dumoulin C, Seaborne DE, Quirion-DeGirardi C, Sullivan SJ. Pelvic-floor rehabilitation, part 1: comparison of two surface electrode placements during stimulation of the pelvic-floor musculature in women who are continent using bipolar interferential currents; part 2: pelvic-floor reeducation with interferential currents and exercise in the treatment of genuine stress incontinence in postpartum women, a cohort study. *Phys Ther* 1995;75:1067–74 and 1075–81 (abstracts read; the exact site of the Dumoulin electrode taken from its description in Sekandari et al., 2026).
19. Demirtürk F, Akbayrak T, Karakaya IC, et al. Interferential current versus biofeedback results in urinary stress incontinence. *Swiss Med Wkly* 2008;138:317–21.
20. Sekandari R, Pourali L, Sheikhi Z, et al. Adding interferential current to pelvic floor exercises reduces weekly frequency of stress urinary incontinence. *Sci Rep* 2026;16:31938.
21. Trivedi S, Dhingra P. Efficacy of interferential therapy in women with stress urinary incontinence. International Continence Society annual meeting, 2011, abstract 838 (Lady Hardinge Medical College, New Delhi).
22. Barmadisatrio, Wiyogo IO, Kloping YP. The role of interferential current electrical stimulation in pediatric urology: a systematic review of randomized controlled trials. *Surg Neurol Int* 2023;14:404 (carrier frequency of about 4000–5000 Hz on most machines; 4000 Hz as the usual setting also checked in a secondary teaching summary).
23. Stewart F, Berghmans B, Bø K, Glazener CMA. Electrical stimulation with non-implanted devices for stress urinary incontinence in women. *Cochrane Database Syst Rev* 2017;(12):CD012390.
24. Royal College of Obstetricians and Gynaecologists. The management of third- and fourth-degree perineal tears (Green-top Guideline No. 29), 2015.
25. Teede HJ, et al. Recommendations from the 2023 international evidence-based guideline for the assessment and management of polycystic ovary syndrome. *J Clin Endocrinol Metab* 2023;108:2447–69.

Part 3

Geriatrics

20 hours theory, 20 field

3(a–c)	Senior citizens in India: policies, NGOs, living arrangements and communication	85
3(d)	Theories of ageing	92
3(e)	Physiology of ageing and the effects of exercise	101
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3(a–c) Senior citizens in India: social issues, national policies and schemes, NGOs, institutional vs community living, and communication barriers

> This part was asked in 6 of 20 papers: national policies for the elderly, NGOs, communication barriers three times, and two long cases asking whether an older person should go to an institution or stay at home. There is no single right answer to that question; the examiner wants the advantages and disadvantages of each, weighed against this person's needs, wishes and supports.

Syllabus	Ch 3(a) Senior citizens in India; 3(b) NGOs and health-related legal rights and benefits for the elderly; 3(c) institutionalised and community-dwelling elders
Priority	**Tier 2.** In 6 of 20 papers (2016–2026): 2 long cases, 4 seven-mark and 2 three-mark questions.
Asked as	LAQ S18 · S20 · 7m S18 · S20 · W21 · W24 · 3m W18 · W19
Usual split	15 (functional diagnosis 5, goals and management 6, institution vs home 4; or diagnosis 5, goals 5, hospital vs home 5); 7 (national policies; communication barriers); 3 (policies; NGOs)
Checked against	Census of India 2011; India Ageing Report 2023 (UNFPA and IIPS); National Policy on Older Persons (1999); Maintenance and Welfare of Parents and Senior Citizens Act (2007); National Programme for Health Care of the Elderly (2010); expansion of AB PM-JAY to all aged 70 and over (2024). Full list under Sources (p. 91). Content checked 20 October 2026.

1 Senior citizens in India

- **Senior citizen:** a person aged 60 or over (Maintenance and Welfare of Parents and Senior Citizens Act, 2007).
- **Numbers:** 104 million people aged 60 and over in the 2011 Census (8.6%); projected to reach about 227 million (14.9%) by 2036, with more elderly than children under 15 by about 2046 (India Ageing Report, 2023).
- **Features of ageing in India:** most elderly live in villages; more widows than widowers (feminisation of ageing); fast growth of the "oldest old" (80 and over); many in the poorest groups; most worked in the informal sector without pensions.

Social issues

- Break-up of joint families and migration of children to cities or abroad; elders living alone or as couples.
- Financial insecurity and dependence on children; few pensions.
- Multiple chronic diseases, disability, poor access to health care and rehabilitation.
- Loneliness, depression, loss of role after retirement or widowhood.
- Elder abuse and neglect: physical, emotional, financial.
- Unsafe housing (stairs, Indian toilets) and few elder-friendly public spaces.

2 National policies, laws and schemes

National policies that facilitate the elderly (S18, 7 marks; W18, 3 marks); also part of the W19 and W20 long answers.

Policy or scheme	Main points
National Policy on Older Persons (1999)	State support for financial and food security, health care and nutrition, shelter, education, welfare, and protection of life and property against abuse and exploitation
Maintenance and Welfare of Parents and Senior Citizens Act (2007)	Children and heirs must maintain parents and senior citizens; claims through maintenance tribunals (allowance up to ₹10,000 a month); old age homes for indigent elders in districts; medical care; protection of life and property; abandoning a senior citizen is punishable

Policy or scheme	Main points
National Programme for Health Care of the Elderly (2010)	Geriatric clinics and wards at district hospitals, rehabilitation units and physiotherapy at CHCs, home-based care through PHCs and sub-centres, regional geriatric centres and national centres of ageing
Ayushman Arogya Mandirs	Elderly and palliative care in the package of primary health care (see health care delivery (p. 6))
AB PM-JAY for 70 and over (2024)	₹5 lakh a year hospital cover for every person aged 70 and over, whatever their income (Ayushman Vay Vandana card)
Indira Gandhi National Old Age Pension Scheme	Monthly pension for elderly persons below the poverty line, topped up by states
Rashtriya Vayoshri Yojana (2017)	Free assistive devices (walking sticks, crutches, walkers, wheelchairs, hearing aids, spectacles, dentures) for poor senior citizens with age-related disability, through camps
Atal Vayo Abhyuday Yojana	Grants to NGOs for old age homes, day care centres, mobile medical units; Elderline 14567 helpline
Other benefits	Higher income-tax exemption, savings schemes with higher interest, priority counters, reserved seats in buses

3 NGOs in elderly care

Role of NGOs for elderly care in India (W19, 3 marks).

- **Examples:** HelpAge India, Dignity Foundation, Agewell Foundation, Nightingales Medical Trust, Silver Innings, local religious and community trusts.
- **Roles:** mobile medical units and health camps (eye, hearing, physiotherapy); old age homes, day care centres and senior citizens' clubs; helplines and protection from abuse; livelihood and pension help; sponsorship of destitute elders; awareness and advocacy for policy; training caregivers; disaster relief for elders; linking elders to government schemes.

4 Institutionalised and community-dwelling elders

Should she be institutionalised or stay at home (S18, 4 marks); hospitalisation versus home management (S20, 5 marks); old age home versus own home with an attendant (W17, 4 marks).

Option	Advantages	Disadvantages
Living at home (community-dwelling)	Familiar surroundings, family and neighbours, independence and dignity, own routine, cheaper, keeps roles and social contact	Unsafe home (stairs, toilet), loneliness if alone, no help in emergencies, carer burden, harder access to therapy
Old age home or care institution	Supervision and help with daily care, safer environment, regular meals and medicines, company of peers, quicker help in emergencies	Loss of independence and privacy, separation from family, routine set by the institution, cost, risk of depression and inactivity, variable quality
Hospital	Needed for acute illness, surgery and intensive therapy	Deconditioning, delirium, infections, cost; not a place to live

How to decide: the person's wishes; medical stability; function (ADL, cognition, falls risk); safety and accessibility of the home; availability of family or a paid attendant; money; and community services (home physiotherapy, day care, helplines). The usual aim is home with support where safe, an institution when needs cannot be met safely at home.

5 Communication barriers in the elderly

Communication barriers in elderly people: 7 marks in S20, W21 and W24.

Barrier	Examples	What to do
Hearing	Presbycusis (high tones lost first), difficulty in noise	Face the person, quiet room, speak slowly and clearly in a lower pitch, do not shout, hearing aid in and working
Vision	Poor near vision, cataract, glare	Good light without glare, glasses on, large-print, high-contrast handouts
Cognition	Slower processing, poor memory, dementia, delirium	Short sentences, one instruction at a time, repeat, demonstrate, written or picture reminders, involve family
Speech and language	Aphasia or dysarthria after stroke, loose dentures	Yes/no questions, gestures, picture boards, patience
Emotional	Depression, anxiety, fear, embarrassment, low self-esteem	Build trust, listen, respect, privacy
Physical	Pain, fatigue, breathlessness	Short sessions, comfortable position
Cultural and educational	Different language, low literacy, beliefs about illness	Local language, simple words, culturally fitting examples
Professional	Jargon, hurry, talking to the relative instead of the patient, "elderspeak" (baby talk), ageist attitudes	Address the person directly, allow time, check understanding (ask them to repeat in their own words)
Environment	Noise, poor lighting, interruptions	Quiet, well-lit room, sit at the same level

6 Worked long case: alone on a fourth floor after hip surgery

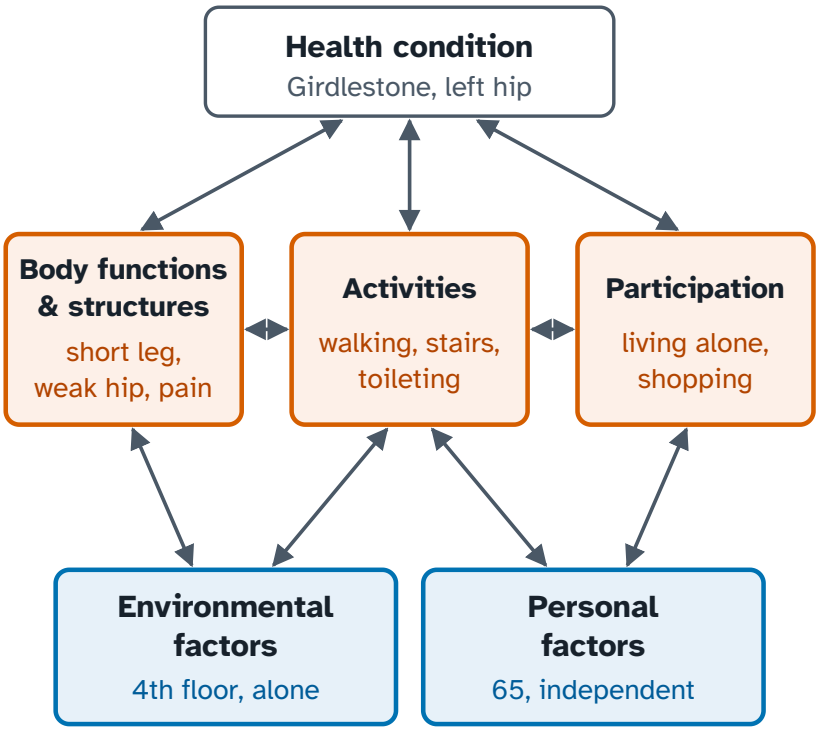
From a real paper (S18, 5 + 6 + 4)

> "A 65 years old, female, living alone on 4th floor in a chawl, underwent girdle stone arthoplasty of left hip joint. i) Explain the functional diagnosis ii) Discuss the long term goals with its relevant management (6) iii) Discuss giving reason if she should be institutionalised or continue to stay at home. (4)"

Spot the clues before you write

Girdlestone arthroplasty: the femoral head and neck are removed without a replacement, so the leg is shorter and the hip unstable; she will need a walking aid, a shoe raise and strong hip muscles. living alone: no help at home. 4th floor of a chawl: many stairs, often a shared toilet: major barriers.

The ICF model filled for this case



Draw the six boxes with two-way arrows, then write two or three points from the case in each box.

One-line functional diagnosis

A 65-year-old woman living alone on a fourth floor after left Girdlestone arthroplasty, with limb shortening, hip weakness and instability, unable to manage stairs and outdoor walking safely without help.

Long-term goals and management (6 marks)

- **Goals (3–6 months):** walk independently indoors and outdoors with a stick or walker and shoe raise; manage stairs with rail and aid; independent in self-care and toileting; no falls.
- **Strength:** hip abductors, extensors and quadriceps, progressive, plus the other leg and arms for aids.
- **Gait:** shoe raise for the shortening; walker progressing to a stick in the right hand; gait training to reduce the lurch.
- **Stairs:** up with the good leg, down with the operated one, rail and stick; practise many times before discharge.
- **Balance and falls prevention;** home advice: raised toilet seat or commode, grab bars, clear floors, phone within reach.
- **Endurance:** walking programme; rest stops on landings.

Institution or home (4 marks)

- **For home:** her familiar place, independence, neighbours in the chawl, lower cost; possible if she can manage stairs and toilet safely, with a helper, a meal service or neighbours checking daily, and home physiotherapy.
- **For an institution:** four floors of stairs without a lift, a shared toilet, and living alone make falls and isolation likely in the early months; an institution or a family member's ground-floor home gives supervision.
- **Suggested answer:** a short stay with family or in a care facility (or a ground-floor arrangement) during rehabilitation, then return home once stairs, toileting and outdoor walking are safe, with support and follow-up. Say that the final choice is hers.

Practice case (S20)

A 60-year-old retired executive with metabolic and heart disease after an Austin Moore prosthesis for a fractured neck of femur, in a joint family whose members cannot give time, with a domestic help and frequent admissions:

functional diagnosis, short- and long-term goals, and hospitalisation versus home management (section 4; home management with a trained helper, home physiotherapy and early detection of deterioration usually wins unless he is medically unstable).

7 Exam corner

Past questions (MUHS, 2016–2026)

Long cases

65-year-old alone on a fourth floor after Girdlestone arthroplasty: institution or home (S18); 60-year-old after Austin Moore prosthesis: hospital or home (S20).

7 marks

National policies that facilitate the elderly (S18); communication barriers in the elderly (S20, W21, W24).

3 marks

Policies that facilitate an elderly person (W18); role of NGOs in elderly care in India (W19).

Keywords to underline

60 years and over | feminisation of ageing | National Policy on Older Persons 1999 | Maintenance and Welfare Act 2007 | NPHCE 2010 | PM-JAY 70+ | Rashtriya Vayoshri Yojana | HelpAge India | institution vs home | presbycusis | elderspeak | teach-back

Answer skeleton: communication barriers (7 marks)

1. **[Why it matters, 1]** Safety, adherence, dignity.
2. **[Barriers, 4]** Hearing, vision, cognition, speech, emotional, physical, cultural, professional, environment.
3. **[Strategies, 2]** One remedy for each.

Draw these

- The ICF model for an elderly case (section 6)
- A table of home versus institution

Viva questions

1. Who is a senior citizen under Indian law?
2. What does the 2007 Act require of children?
3. What is NPHCE?
4. Name two NGOs working for the elderly.
5. Give three ways to communicate with a person with presbycusis.
6. What is a Girdlestone arthroplasty and why does the leg shorten?

Linked practical spots

- Role-play explaining a home exercise programme to an older person with hearing loss; visit a day care centre or old age home for your journal.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Role of NGOs for elderly care in India. (W19)**

Answer
Examples: HelpAge India, Dignity Foundation, Agewell. Roles: mobile medical units and camps, old age homes and day care, helplines and protection from abuse, livelihood and pension help, sponsorship of destitute elders, awareness and advocacy, caregiver training, linking to government schemes.

7 m **National policies that facilitate the elderly. (S18)**

Answer
National Policy on Older Persons 1999 (financial and food security, health, shelter, welfare, protection); Maintenance and Welfare of Parents and Senior Citizens Act 2007 (maintenance tribunals, old age homes, medical care, penalty for abandonment); NPHCE 2010; PM-JAY for all aged 70 and over (2024); IGNOAPS pension; Rashtriya Vayoshri Yojana; NGO grants and Elderline; tax and savings benefits. Add the physiotherapist's role in NPHCE units.

7 m **Communication barriers in the geriatric population. (W21, W24)**

Answer
Barriers: hearing, vision, cognition, speech and language, emotional, physical, cultural and literacy, professional (jargon, haste, elderspeak), environmental. Strategies: face-to-face, quiet and well-lit room, slow clear speech, aids in place, short instructions, demonstration, written large-print handouts, teach-back, family involvement, respect and patience.

MCQ **The Maintenance and Welfare of Parents and Senior Citizens Act was passed in:**

A. 1995
B. 1999
C. 2007
D. 2016

Answer
C.

MCQ **Free assistive devices for poor senior citizens are given under:**

A. Rashtriya Vayoshri Yojana
B. RBSK
C. ICDS
D. ESI

Answer
A.

MCQ **When talking to an older person with presbycusis, you should:**

A. Shout loudly

B. Speak slowly and clearly while facing them

C. Talk only to the relative

D. Use a high-pitched voice

Answer

B.

Last recall check. List six social issues of the elderly in India, eight policies and schemes, NGO roles, a home-versus-institution table, and nine communication barriers with remedies.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Community Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. UNFPA and International Institute for Population Sciences. India Ageing Report 2023: caring for our elders, institutional responses.
3. Ministry of Social Justice and Empowerment. National Policy on Older Persons, 1999.
4. The Maintenance and Welfare of Parents and Senior Citizens Act, 2007.
5. Ministry of Health and Family Welfare. National Programme for Health Care of the Elderly (NPHCE), operational guidelines.
6. Ministry of Health and Family Welfare. Expansion of AB PM-JAY to all senior citizens aged 70 and above (Ayushman Vay Vandana card), October 2024.

3(d) Theories of ageing: definition and types of ageing, biological (programmed and damage) and psychosocial theories

> "Theories of ageing" was asked in 9 of 20 papers, almost always as a 7-mark question that says "enumerate and explain any three". Learn the classification tree first, then three theories in detail: one programmed, one damage, one psychosocial, each with a line on what it means for physiotherapy.

Syllabus	Ch 3(d) Theories of ageing
Priority	**Tier 2.** In 9 of 20 papers (2016–2026): 7 seven-mark and 2 three-mark questions.
Asked as	7m W16 · S17 · S18 · W19 · W20 · S21 · S24 · 3m S22 · W22
Usual split	7 (enumerate the theories 2, explain three 5; or define ageing, types and genetic theories); 3 (enumerate three)
Checked against	Primary papers for each theory: Harman (1956, 1972), Hayflick and Moorhead (1961), Harley et al. (1990), Bodnar et al. (1998), Bjorksten (1942), Orgel (1963), Cumming and Henry (1961), Havighurst (1961), Atchley (1989), Rowe and Kahn (1987, 1997); hallmarks of ageing (López-Otín et al., 2013, 2023); meta-analysis of antioxidant trials (Bjelakovic et al., 2007). Full list under Sources (p. 100). Content checked 21 October 2026.

1 Ageing and its types

Define ageing, mention types of ageing and enumerate genetic theories (W16, 7 marks).

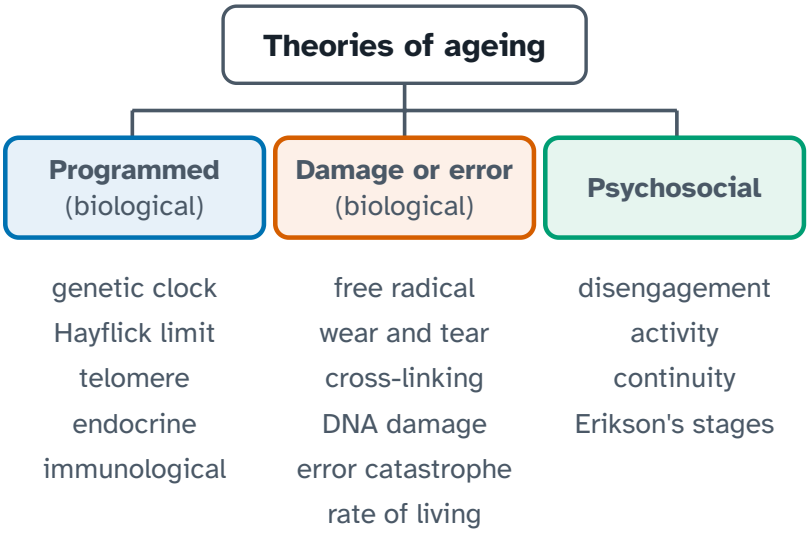
- **Ageing:** the progressive, time-related decline in the structure and function of the body that reduces the ability to adapt to stress and raises the risk of disease and death. It is universal, progressive, intrinsic (from within) and harmful.

Type	Meaning
Chronological	Age in years since birth
Biological (physiological)	Age of the body's systems; two people of 70 may differ greatly
Psychological	Changes in memory, learning, personality and coping
Social	Changing roles: retirement, grandparenthood, widowhood
Functional	What the person can do compared with others of the same age
Primary and secondary	Primary: changes due to age itself. Secondary: changes added by disease, inactivity and lifestyle

- **Usual and successful ageing (Rowe and Kahn):** successful ageing means a low probability of disease and disability, high physical and cognitive function, and active engagement with life. Physiotherapy aims for successful ageing (see the fun-to-failure model (p. 101)).

2 Classification of the theories

Theories of ageing: the classification tree



Programmed theories say ageing follows a built-in plan; damage theories say it is the sum of accumulated harm; psychosocial theories explain how people adapt to growing old.

Classic answer vs current view: one theory or many causes

Classic	Current	Write in the exam
Separate, competing theories: ageing is either programmed or the result of damage.	Ageing is described as a set of interacting hallmarks of ageing: nine in 2013, twelve in the 2023 update. They are genomic instability, telomere attrition, epigenetic alterations, loss of proteostasis, disabled autophagy, deregulated nutrient-sensing, mitochondrial dysfunction, cellular senescence, stem cell exhaustion, altered intercellular communication, chronic inflammation and dysbiosis (López-Otín et al., 2023). The old theories live on inside them: free radical damage within mitochondrial dysfunction, the telomere theory within telomere attrition and senescence, the immunological theory within chronic inflammation.	"No single theory explains ageing. Modern research sees it as interacting hallmarks such as DNA damage, telomere shortening, mitochondrial dysfunction and chronic inflammation."

3 Programmed (genetic) theories

"Enumerate genetic theories of ageing" (W16).

Theory	Idea	Evidence or example
Genetic clock (programmed longevity)	Lifespan is set by genes that switch on and off in sequence	Each species has a typical maximum lifespan; long-lived families; progeria shows early ageing from one gene
Hayflick limit (cellular senescence)	Normal cells divide a limited number of times and then stop (senescence)	Human fibroblasts in culture stop after about 40–60 divisions (Hayflick and Moorhead, 1961); senescent cells build up in old tissues
Telomere theory	Telomeres (protective ends of chromosomes) shorten at each division; when too short, the cell stops dividing	Explains the Hayflick limit; the enzyme telomerase restores telomeres in germ and cancer cells
Endocrine (neuroendocrine; Dilman, 1971)	A biological clock in the hypothalamus and pituitary drives ageing through hormones	Fall in growth hormone, sex hormones and melatonin; menopause; insulin and IGF-1 signalling affects lifespan in animals
Immunological (Walford, 1969)	The immune system declines on a programme (thymus shrinks; immunosenescence), so infections, cancer and autoimmunity rise	Older people get more pneumonia and shingles and respond less to vaccines. The chronic low-grade inflammation of old age was named "inflammaging" (Franceschi et al., 2000)

Telomere theory in full (any one theory, 7 marks in S21)

- **Background:** Hayflick and Moorhead (1961) found that normal human fetal fibroblasts stop dividing after about **50 population doublings**: the Hayflick limit. Cells from adults divide fewer times.
- **Telomeres:** repeats of the DNA sequence TTAGGG, with their proteins, that cap the ends of each chromosome like the plastic tip of a shoelace. They stop the chromosome ends being treated as broken DNA.
- **End-replication problem:** DNA polymerase cannot copy the very end of a chromosome, so a little telomere is lost at every division (predicted by Olovnikov, 1971–73, and Watson, 1972).
- **Shortening:** telomeres of cultured fibroblasts shorten with each division (Harley et al., 1990). In blood cells they are about **9.5 kb** long at birth (Factor-Litvak et al., 2016) and shortened by about **27 base pairs a year** in adult women (Valdes et al., 2005).
- **Senescence:** when telomeres become critically short, the cell stops dividing or dies. Senescent cells build up in old tissues, which lose their power to renew and repair.
- **Telomerase:** the enzyme that rebuilds telomeres. It is active in germ cells and most cancers but low in most body cells. Adding telomerase to normal human cells let them pass their normal lifespan by at least 20 doublings (Bodnar et al., 1998). Blackburn, Greider and Szostak won the 2009 Nobel Prize for showing how telomeres and telomerase protect chromosomes.

Evidence for	Evidence against and limits
Telomeres shorten with cell division and with age	Mice have telomeres up to 150 kb long, far longer than ours, yet live only about 3.5 years
Telomeres of obese women were 240 base pairs shorter, and each pack-year of smoking was equal to about 5 base pairs lost (Valdes et al., 2005)	Telomerase is switched on in about 90% of tumours (Kim et al., 1994), so lengthening telomeres carries a cancer risk
Women under the highest stress had telomeres shorter by the equal of at least 10 years of ageing (Epel et al., 2004)	A meta-analysis of 9 exercise trials found no overall effect on telomere length (Sánchez-González et al., 2024)
In twins, the most active had telomeres 200 nucleotides longer than the least active (Cherkas et al., 2008)	Short telomeres explain the limit on cell division, not every feature of ageing

Meaning for physiotherapy: regular activity, not smoking, weight control and stress management go with longer telomeres. Trials have not yet shown that exercise lengthens them, so the main reason to exercise is still function.

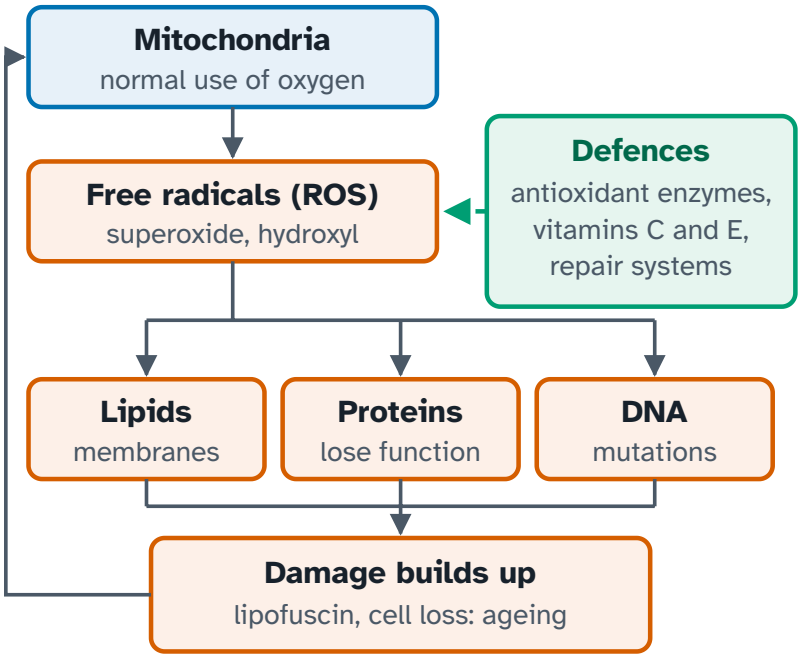
4 Damage or error (stochastic) theories

Theory	Idea	Evidence or example
Free radical (Harman, 1956)	Reactive oxygen species made during normal metabolism damage DNA, proteins and cell membranes; damage to mitochondria makes more radicals (mitochondrial theory)	Oxidative damage rises with age; lipofuscin (age pigment) builds up; antioxidant enzymes protect
Wear and tear (Weismann, 1882)	Cells and tissues wear out with repeated use, like a machine	Cartilage wear in osteoarthritis; but the body also repairs, so this is only part of the story
Cross-linking or glycation (Bjorksten, 1942)	Proteins such as collagen and elastin form cross-links (helped by glucose: advanced glycation end products), so tissues stiffen	Stiff skin, arteries, tendons, lens (cataract), more in diabetes
DNA damage (somatic mutation)	Mutations build up in body cells and are not fully repaired	Cancer risk rises with age; DNA-repair disorders age early
Error catastrophe (Orgel, 1963)	Errors in making proteins accumulate until cells fail	Faulty proteins in old cells; largely not supported as the main cause
Rate of living (Rubner, 1908; Pearl, 1928)	The faster the metabolism, the shorter the life	Small animals with fast metabolism live shorter; birds are an exception

Free radical theory in full (7 marks in S21)

- **Proposed by** Denham Harman in 1956. In 1972 he pointed to the mitochondria as the main site, the mitochondrial free radical theory.
- **Free radical:** an atom or molecule with an unpaired electron, so it reacts readily with whatever it meets. The main reactive oxygen species (ROS) are the superoxide anion, hydrogen peroxide (not itself a radical, but it forms one) and the hydroxyl radical.
- **Source:** normal metabolism. Electrons leak from the respiratory chain in the mitochondria and react with oxygen; enzymes such as NADPH oxidase also make superoxide. The often-quoted 1–2% of oxygen turned into superoxide comes from isolated mitochondria; the true rate in the body is lower and debated (Murphy, 2009).
- **Damage:** peroxidation of membrane lipids; oxidised proteins (carbonyls) that lose function; DNA damage such as 8-oxo-deoxyguanosine, which causes mutations. Damaged mitochondria leak more radicals, a vicious circle.
- **Defences:** the antioxidant enzymes superoxide dismutase, catalase and glutathione peroxidase; dietary antioxidants such as vitamin C, vitamin E and beta-carotene; repair systems.
- **Ageing** follows when damage outruns defence and repair. Lipofuscin, the "age pigment", is oxidised and cross-linked protein and lipid that lysosomes cannot digest; it builds up in old cells.

Free radical theory: the vicious circle



Loop: damaged mitochondria leak more radicals

Orange boxes are the damage, green is the body's defence. Regular exercise strengthens the defences; antioxidant pills do not.

Text version of the flowchart

> Mitochondria (normal use of oxygen) → free radicals (superoxide, hydroxyl), partly neutralised by antioxidant enzymes, vitamins C and E and repair → damage to membrane lipids, proteins and DNA → damage builds up (lipofuscin, cell loss) → ageing. Damaged mitochondria leak more radicals, so the circle repeats.

Evidence for	Evidence against and limits
Oxidative damage to lipids, proteins and DNA rises with age	Antioxidant pills do not prolong life: in 68 trials with 232,606 people, beta-carotene, vitamin A and vitamin E slightly raised mortality (Bjelakovic et al., 2007)
Lipofuscin builds up in old cells	Naked mole-rats live more than 28 years, about 8 times longer than mice, despite more oxidative damage (Andziak et al., 2006)
Cells have a whole system of antioxidant enzymes to limit the damage	In mice, only 1 of 18 genetic changes to antioxidant defences altered lifespan (Pérez et al., 2009)

Conclusion: oxidative damage contributes to ageing but is not its single cause.

Exercise and free radicals: a single hard bout raises radical production for a short time, but regular training lowers the oxidative challenge and raises the body's own antioxidant enzymes, an adaptive (hormesis) response (Radak et al., 2008). High-dose antioxidant pills blunt this: vitamin C 1000 mg with vitamin E 400 IU a day blocked the training gain in insulin sensitivity (Ristow et al., 2009). Advise regular moderate exercise, a diet rich in fruit and vegetables, no smoking, and no antioxidant pills for this purpose.

Wear-and-tear theory in full

- **Proposed by** August Weismann, in a lecture given in 1881 and printed in 1882 ("On the duration of life").
- **Idea:** the body is like a machine. Cells and tissues wear out with repeated use, and the damage of daily living builds up until organs fail. Weismann held that body cells have only a limited power to divide and replace worn parts.

- **Examples:** worn articular cartilage in osteoarthritis of the weight-bearing joints; stiffer, less elastic tendons and discs.
- **Related idea:** the rate-of-living theory (Rubner, 1908; Pearl, 1928): the faster the metabolism, the faster the wear and the shorter the life.
- **Limits:** unlike a machine, the body repairs itself, and tissues get stronger with use: bone, muscle and tendon adapt to load. Disuse causes more loss than use. The theory does not explain why species age at such different rates.
- **Meaning for physiotherapy:** protect joints from excess load (weight control, good technique, aids), but keep them moving and loaded within limits.

5 Psychosocial theories

Theory	Idea	Comment
Disengagement (Cumming and Henry, 1961)	Older people and society withdraw from each other naturally and by mutual agreement	Criticised: withdrawal is not inevitable and often harms health
Activity (Havighurst, 1961)	Staying physically and socially active and replacing lost roles leads to successful ageing and life satisfaction	Basis of exercise groups, day centres and volunteering
Continuity (Atchley, 1989)	People adapt best by keeping the habits, interests and relationships of their earlier life	Plan activities that fit the person's past (gardening, temple visits, walking group)
Erikson's eighth stage	Old age is the stage of ego integrity versus despair: looking back on life with acceptance or regret	Reminiscence, respect and purposeful roles support integrity

6 What the theories mean for physiotherapy

- Ageing has many causes acting together; no single theory explains it, and much of the decline is secondary (inactivity, disease), which physiotherapy can change.
- **Free radical:** regular moderate exercise increases the body's own antioxidant defences; antioxidant pills do not slow ageing and may blunt training gains; avoid smoking and excess sun.
- **Telomere:** regular activity, no smoking, weight control and stress management go with longer telomeres.
- **Cross-linking:** daily stretching and range of motion counter tissue stiffness; good diabetes control slows glycation.
- **Wear and tear:** joint protection, weight control and strengthening reduce joint load; but disuse is worse than use.
- **Immunological and endocrine:** exercise improves immune function and insulin sensitivity; vaccination advice; care with infection in groups.
- **Activity and continuity:** group exercise, walking clubs and meaningful roles; plan programmes around the person's past interests.
- **Disengagement:** recognise withdrawal and low mood as risks, not a normal part of ageing; encourage social participation.

7 Exam corner

Past questions (MUHS, 2016–2026)

7 marks

Define ageing, types of ageing, genetic theories (W16); theories of ageing (S17, S18, W19, S24); enumerate theories and explain any three (W20); enumerate theories and describe one (S21).

3 marks

Enumerate theories of ageing (S22); three theories of ageing (W22).

Keywords to underline

programmed vs damage Hayflick limit telomere shortening free radicals superoxide dismutase, catalase lipofuscin telomerase cross-linking of collagen immunosenescence disengagement activity continuity successful ageing hallmarks of ageing

Answer skeleton: theories of ageing (7 marks)

1. **[Definition, 1]** Ageing and its features.
2. **[Classification, 2]** Draw the tree with all names.
3. **[Three theories, 3]** One programmed (Hayflick or telomere), one damage (free radical), one psychosocial (activity), each with idea and evidence.
4. **[PT meaning, 1]** Exercise, stretching and activity act on the changes the theories describe.

Draw these

- The classification tree (section 2)
- A chromosome with telomere caps shortening over divisions
- The free radical vicious circle (section 4)

Viva questions

1. What is the Hayflick limit?
2. Who proposed the free radical theory, and when?
3. Name three antioxidant enzymes.
4. Why do antioxidant pills not slow ageing?
5. How does cross-linking explain stiff joints in older people?
6. Contrast disengagement and activity theories.
7. What is successful ageing?
8. Which theory explains why exercise groups help older people?

Linked practical spots

- Plan a weekly group exercise and social session for a senior citizens' club and say which theory supports each part.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Enumerate three theories of ageing. (W22)**

Answer
Programmed: Hayflick limit or telomere theory (cells divide a limited number of times as telomeres shorten). Damage: free radical theory (Harman 1956: reactive oxygen species damage cell components). Psychosocial: activity theory (Havighurst: staying active brings successful ageing). Any three with one line each.

7 m **Enumerate theories of ageing and explain any three. (W20)**

Answer
Define ageing. Tree: programmed (genetic clock, Hayflick, telomere, endocrine, immunological), damage (free radical, wear and tear, cross-linking, DNA damage, error catastrophe, rate of living), psychosocial (disengagement, activity, continuity, Erikson). Explain free radical, telomere and activity theories with evidence and PT relevance.

7 m **Enumerate theories of ageing and describe any one. (S21)**

Answer
Define ageing; draw the tree. Describe the free radical theory: Harman 1956, mitochondrial version 1972; free radical defined; superoxide, hydrogen peroxide, hydroxyl; made by normal metabolism in mitochondria; damage to membrane lipids, proteins and DNA; vicious circle; defences (superoxide dismutase, catalase, glutathione peroxidase, vitamins C and E); lipofuscin; evidence for and against (antioxidant trials, naked mole-rat); PT meaning (regular exercise raises own defences). Or describe the telomere theory: Hayflick limit about 50 doublings, TTAGGG caps, end-replication problem, senescence, telomerase, Nobel 2009, evidence and limits.

7 m **Define ageing. Mention types of ageing. Enumerate genetic theories of ageing. (W16)**

Answer
Definition and features of ageing. Types: chronological, biological, psychological, social, functional; primary and secondary. Genetic (programmed) theories: genetic clock, Hayflick limit, telomere, endocrine, immunological, with one line each.

MCQ **The Hayflick limit refers to:**

A. Maximum heart rate in old age
B. The limited number of divisions of normal cells
C. The upper limit of free radicals
D. The age of retirement

Answer
B.

MCQ **Stiffening of collagen with age is best explained by the:**

A. Disengagement theory
B. Cross-linking theory
C. Continuity theory
D. Rate of living theory

Answer
B.

MCQ **The theory that staying active and socially involved leads to successful ageing is the:**

A. Activity theory
B. Wear and tear theory
C. Disengagement theory
D. Error catastrophe theory

Answer
A.

Last recall check. Draw the tree from memory with all fifteen names, then explain one theory from each branch with its PT meaning.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Community Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Harman D. Aging: a theory based on free radical and radiation chemistry. *J Gerontol* 1956;11:298–300.
3. Hayflick L, Moorhead PS. The serial cultivation of human diploid cell strains. *Exp Cell Res* 1961;25:585–621.
4. Bjorksten J. Chemistry of duplication. *Chem Ind* 1942;50:68–72 (cross-linkage theory).
5. Orgel LE. The maintenance of the accuracy of protein synthesis and its relevance to ageing. *Proc Natl Acad Sci USA* 1963;49:517–21.
6. Cumming E, Henry WE. Growing old: the process of disengagement. New York: Basic Books; 1961.
7. Havighurst RJ. Successful aging. *Gerontologist* 1961;1:8–13.
8. Atchley RC. A continuity theory of normal aging. *Gerontologist* 1989;29:183–90.
9. Rowe JW, Kahn RL. Human aging: usual and successful. *Science* 1987;237:143–9; Successful aging. *Gerontologist* 1997;37:433–40.
10. Avers D, Wong RA, eds. Guccione's Geriatric Physical Therapy. 4th ed. St Louis: Elsevier; 2020.
11. Harman D. The biologic clock: the mitochondria? *J Am Geriatr Soc* 1972;20:145–7.
12. Liguori I, et al. Oxidative stress, aging, and diseases. *Clin Interv Aging* 2018;13:757–72 (reactive oxygen species, antioxidant enzymes, damage markers); Murphy MP. How mitochondria produce reactive oxygen species. *Biochem J* 2009;417:1–13.
13. Terman A, Brunk UT. Lipofuscin. *Int J Biochem Cell Biol* 2004;36:1400–4; Jung T, Bader N, Grune T. Lipofuscin: formation, distribution, and metabolic consequences. *Ann N Y Acad Sci* 2007;1119:97–111.
14. Bjelakovic G, et al. Mortality in randomized trials of antioxidant supplements for primary and secondary prevention: systematic review and meta-analysis. *JAMA* 2007;297:842–57.
15. Andziak B, et al. High oxidative damage levels in the longest-living rodent, the naked mole-rat. *Aging Cell* 2006;5:463–71; Pérez VI, et al. Is the oxidative stress theory of aging dead? *Biochim Biophys Acta* 2009;1790:1005–14.
16. Radak Z, et al. Exercise, oxidative stress and hormesis. *Ageing Res Rev* 2008;7:34–42; Ristow M, et al. Antioxidants prevent health-promoting effects of physical exercise in humans. *Proc Natl Acad Sci USA* 2009;106:8665–70.
17. Hayflick L. The limited in vitro lifetime of human diploid cell strains. *Exp Cell Res* 1965;37:614–36 (doubling numbers checked through secondary summaries).
18. Moyzis RK, et al. A highly conserved repetitive DNA sequence, (TTAGGG)n, present at the telomeres of human chromosomes. *Proc Natl Acad Sci USA* 1988;85:6622–6; Olovnikov AM. A theory of marginotomy. *J Theor Biol* 1973;41:181–90 (first proposed in Russian in 1971); Watson JD. Origin of concatemeric T7 DNA. *Nat New Biol* 1972;239:197–201.
19. Harley CB, Futcher AB, Greider CW. Telomeres shorten during ageing of human fibroblasts. *Nature* 1990;345:458–60; Bodnar AG, et al. Extension of life-span by introduction of telomerase into normal human cells. *Science* 1998;279:349–52; Nobel Prize in Physiology or Medicine 2009 press release.
20. Factor-Litvak P, et al. Leukocyte telomere length in newborns: implications for the role of telomeres in human disease. *Pediatrics* 2016;137:e20153927; Valdes AM, et al. Obesity, cigarette smoking, and telomere length in women. *Lancet* 2005;366:662–4.
21. Epel ES, et al. Accelerated telomere shortening in response to life stress. *Proc Natl Acad Sci USA* 2004;101:17312–5; Cherkas LF, et al. The association between physical activity in leisure time and leukocyte telomere length. *Arch Intern Med* 2008;168:154–8; Sánchez-González JL, et al. Systematic review and meta-analysis of randomised trials of exercise and telomere length. *JMIR Public Health Surveill* 2024;10:e46019 (abstract).
22. Kipling D, Cooke HJ. Hypervariable ultra-long telomeres in mice. *Nature* 1990;347:400–2; Kim NW, et al. Specific association of human telomerase activity with immortal cells and cancer. *Science* 1994;266:2011–5.
23. Weismann A. Ueber die Dauer des Lebens. Jena: G. Fischer; 1882 (lecture of 1881); Kirkwood TB, Cremer T. Cytogerontology since 1881. *Hum Genet* 1982;60:101–21; Rubner M. Das Problem der Lebensdauer und seine Beziehungen zu Wachstum und Ernährung. Munich and Berlin; 1908; Pearl R. The Rate of Living. New York: Knopf; 1928.
24. Dilman VM. Age-associated elevation of hypothalamic threshold to feedback control, and its role in development, ageing, and disease. *Lancet* 1971;1:1211–9; Walford RL. The Immunologic Theory of Aging. Copenhagen: Munksgaard; 1969; Franceschi C, et al. Inflamm-aging: an evolutionary perspective on immunosenescence. *Ann N Y Acad Sci* 2000;908:244–54.
25. López-Otín C, et al. The hallmarks of aging. *Cell* 2013;153:1194–217; López-Otín C, et al. Hallmarks of aging: an expanding universe. *Cell* 2023;186:243–78.

3(e) Physiology of ageing: musculoskeletal, neurological, sensory, cardiorespiratory and metabolic changes, and the effects of exercise

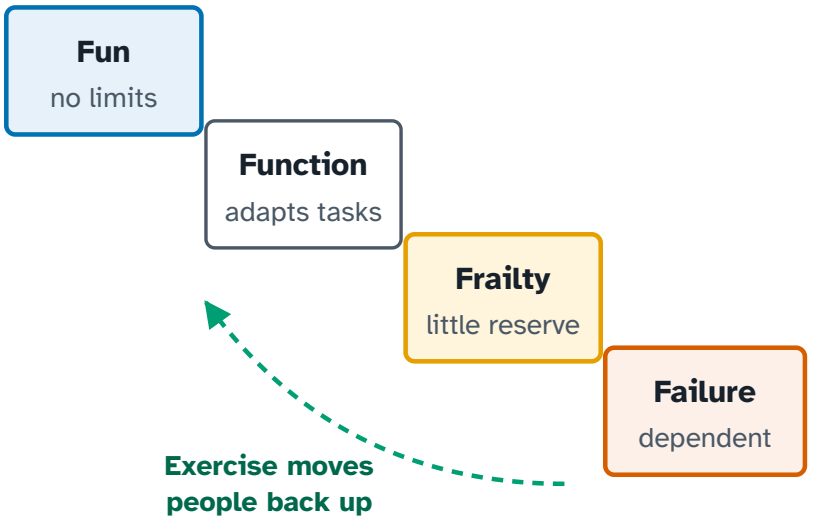
> Ageing changes were asked in 14 of 20 papers, usually one system at a time: "age-related changes in the nervous system", "six cardiopulmonary changes", "musculoskeletal changes". Learn each system as a short list of changes, add what each change does to function, and finish with what exercise can still improve. That last part turns a 7-mark list into a physiotherapy answer.

Syllabus	Ch 3(e) Physiology of ageing: musculoskeletal, neurological, cardio-respiratory and metabolic changes
Priority	**Tier 1.** In 14 of 20 papers (2016–2026): 3 long answers, 5 seven-mark and 7 three-mark questions.
Asked as	LAQ W20 · W21 · W22 · 7m S17 · S22 · S23 · W25 · S26 · 3m S16 · W16 · S19 · S20 · W24 · S25 · S26
Usual split	15 (musculoskeletal and neurological changes 9, effects of exercise 7; or cardiorespiratory and metabolic changes 6, national policies 6, fun to failure 3); 7 (changes in one system, with or without effects of exercise)
Checked against	Fleg et al. (2005) on decline in aerobic capacity; reviews of respiratory ageing (Sharma and Goodwin, 2006; Lowery et al., 2013), the ATS workshop report on asthma in the elderly (2016) and a systematic review of FEV_1 decline (Thomas et al., 2019), with the primary studies cited in section 4; Fiatarone et al. (1990) on strength training in people in their nineties; Cochrane review of exercise to prevent falls (Sherrington et al., 2019); WHO physical activity guidelines (2020); Guccione's Geriatric Physical Therapy on the fun-to-failure model. Full list under Sources (p. 109). Content checked 21 October 2026.

1 Ageing in general

- **Ageing** is the gradual, progressive decline in the body's reserve and its ability to adapt to stress, which raises the risk of disease and death with time. Theories of why it happens are on theories of ageing (p. 92).
- **Primary ageing** is the change that happens to everyone; **secondary ageing** is the change added by disease, inactivity, smoking and poor nutrition. Much of what looks like ageing is secondary, which is why exercise helps.
- **Reserve:** at rest an older person may look normal; the loss shows when the body is stressed (illness, surgery, heat, exercise, a fall).
- People age at different rates, and different systems in the same person age at different rates.

The slippery slope of ageing: fun, function, frailty, failure



Four levels of function described in Guccione's Geriatric Physical Therapy, coloured from blue (fit) to orange (dependent). The physiotherapist's job is to keep people on the higher steps and help them climb back after illness.

Level	What the person can do
Fun	Takes part in work, home and leisure without restriction
Function	Still does most work and home tasks, but changes how they are done; leisure is limited by falling capacity
Frailty	Basic daily activities use up most of the reserve; community activities are limited; needs help with home or outside tasks
Failure	Needs help with basic and instrumental daily activities; may be bedbound

2 Musculoskeletal changes

3 marks in W16 and S25; 9 marks with the nervous system in W22.

Tissue	Change	Effect on function
Muscle	Sarcopenia: loss of muscle mass, especially fast-twitch (type II) fibres; fewer motor units; fat infiltrates muscle; strength and power fall faster than mass	Weak legs, slow sit-to-stand and stairs, poor recovery from a trip, falls
Bone	Loss of bone mass and quality, fast in women after menopause; thinner cortex, fewer trabeculae	Osteoporosis, fractures of the hip, wrist and spine, kyphosis, height loss
Cartilage	Less water and proteoglycan, thinner and stiffer	Osteoarthritis of knees, hips and spine; pain, stiffness
Tendons, ligaments, capsule	Stiffer collagen with more cross-links, less elastic, heal slowly	Reduced range of motion, tendon injuries
Intervertebral discs	Dehydrate and lose height	Stiff spine, spondylosis, stenosis
Posture and gait	Forward head, thoracic kyphosis, flexed hips and knees; slower walking, shorter steps, wider base, more time on both feet	Poor balance, falls, breathing restricted by kyphosis

3 Neurological and sensorimotor changes

Nervous system: 7 marks in S17, W25 and S26; sensorimotor system: 7 marks in S23; visual changes: 3 marks in S16.

Central nervous system

- Brain volume falls (more in the frontal lobes and hippocampus); loss of neurons in some areas and of synapses and dendrites; white matter changes.
- Fall in neurotransmitters, especially dopamine (slower movement) and acetylcholine (memory).
- **Cognition:** slower processing speed and reaction time; weaker short-term and working memory and divided attention; knowledge, vocabulary and judgement are kept.
- **Motor control:** slower, less smooth movements, poorer fine coordination, slower anticipatory postural adjustments.
- **Sleep:** lighter, broken sleep; earlier waking.

Peripheral and autonomic nervous system

- Loss of motor and sensory nerve fibres; slower nerve conduction; ankle jerks may be reduced or absent.
- Autonomic: weaker baroreflex (dizziness on standing, postural hypotension); poor temperature control; bladder changes (urgency, nocturia).

Sensory systems

Sense	Change	What it means
Vision	Presbyopia (cannot focus near); smaller pupil and yellower lens, so less light reaches the retina; poor dark adaptation and glare tolerance; lower contrast sensitivity and depth perception; narrower visual field; cataract, glaucoma and macular degeneration more common	Misjudges steps and kerbs, falls at night, reading difficulty: good lighting, contrast edges on steps, no glare, regular eye checks
Hearing	Presbycusis: loss of high-pitched sounds first; difficulty with speech in noise	Face the person, speak slowly and clearly in a quiet room
Touch, vibration, proprioception	Fewer receptors; reduced sense of position and vibration, most in the feet	Poor balance on uneven ground and in the dark
Vestibular	Loss of hair cells and nerve fibres	Dizziness, unsteadiness on turning
Taste and smell	Reduced	Poor appetite and nutrition

Sensorimotor summary (S23): slower sensing (fewer receptors and fibres), slower processing (central changes) and slower, weaker responses (fewer motor units, sarcopenia) together make balance reactions late and weak, which is why falls rise with age (see falls (p. 119)).

4 Cardiovascular and respiratory changes

Cardiorespiratory changes: 3 marks in S19, S20 and W24, and part of the W20 long answer; respiratory changes: 7 marks in S22, 3 marks in S26.

Heart and vessels

- Arteries stiffen: systolic blood pressure and pulse pressure rise.
- The left ventricle wall thickens and relaxes slowly, so filling depends more on the atrial contraction (atrial fibrillation is poorly tolerated).
- Maximum heart rate falls (roughly 220 minus age); the heart responds less to adrenaline; resting heart rate changes little.
- Maximum cardiac output and VO_2max fall: about 3–6% per decade in the twenties and thirties, more than 20% per decade from the seventies (Fleg et al., 2005).
- Weaker baroreflex: postural hypotension; veins less efficient: varicose veins, swelling.

Respiratory system (7 marks in S22)

An older person usually breathes normally at rest. The losses show as reduced reserve in exertion, pneumonia, surgery or bed rest. Smoking, inactivity and lung disease add secondary loss on top of these primary changes.

Area	Change with age	Effect
Chest wall	Stiffer: costal cartilages calcify, discs narrow and the thoracic kyphosis increases. Rib cage compliance was about **30% lower** at 55–75 than at 24–39 years (Estenne et al., 1985)	Less chest expansion; more work of breathing; the flattened diaphragm works at a mechanical disadvantage
Lung tissue	Loss of elastic recoil; lung compliance is unchanged or slightly raised. Overall respiratory system compliance falls, because the stiffer chest wall outweighs the floppier lung	Small airways lose their support and close earlier in expiration
Respiratory muscles	Maximal diaphragm pressure about **25% lower** in men aged 65–75 than 19–28 (Tolep et al., 1995); maximal inspiratory and expiratory pressures fall	Breathlessness on effort; weaker cough; earlier fatigue when the load rises in illness
Lung volumes	Total lung capacity about the same. Residual volume rises by about **50%** between 20 and 70; functional residual capacity rises slightly; vital capacity falls	More air trapped, less usable volume
Flow rates	FEV_1 (forced expiratory volume in one second) falls by about **20–30 ml a year** from about age 30 in healthy non-smokers (median 22 ml a year in 16 cohorts); the FEV_1/FVC (forced vital capacity) ratio also falls	About 35% of healthy never-smokers over 70 had a ratio below 0.70, so a fixed cut-off over-diagnoses COPD (chronic obstructive pulmonary disease); use the age-adjusted lower limit of normal
Airway closure	Closing volume rises. Airways start to close in normal breathing at about 44 years lying down and about 65 sitting	Basal airways close in lying, so oxygenation is better sitting up
Gas exchange	Alveolar surface area falls from about **75 m^2** at 30 to **60 m^2** at 70; diffusing capacity falls; ventilation–perfusion mismatch rises. Resting arterial oxygen tension, $PaO_2 \approx$ **109 – 0.43 × age** mmHg (lying); arterial carbon dioxide tension ($PaCO_2$) does not change	Lower resting oxygen (about 79 mmHg at 70 by the formula) and a wider alveolar–arterial gradient; less margin in pneumonia
Control of breathing	Ventilatory response to low oxygen down about **50%** and to high carbon dioxide down about **40–50%** (Kronenberg and Drage, 1973; Peterson et al., 1981). Less awareness of airway narrowing; more sleep-disordered breathing	Late warning of hypoxia or an asthma attack; older people may not feel breathless until illness is advanced
Airway defences	Slower mucociliary clearance; less sensitive cough reflex and weaker cough; poorer swallow coordination; weaker immunity	Silent aspiration and pneumonia: aspiration was found in 71% of older patients with community-acquired pneumonia against 10% of matched controls (Kikuchi et al., 1994)
Exercise	Larger dead space and lower maximal voluntary ventilation, so ventilation for a given workload is higher; VO_2max falls (above)	Breathlessness at lower workloads; lower exercise tolerance

Physiotherapy implications

- **Surgery and illness:** age is an independent risk factor for lung complications after surgery: odds ratio about **2.1** at 60–69 and **3.0** at 70–79 (Smetana et al., 2006). Teach deep breathing and coughing before surgery and mobilise early.
- **Position:** sit or stand upright rather than lying flat. In people aged 67–88, PaO_2 was about 5 mmHg higher sitting than lying (Hardie et al., 2002).
- **Inspiratory muscle training:** in healthy adults over 60 it raised maximal inspiratory pressure by about **26 cmH_2O**, against about 4 in controls, but did not clearly improve walking distance (Manifield et al., 2021).
- **Aerobic training:** raised VO_2max by about **16%** in older adults (Huang et al., 2005). It improves exercise tolerance even though it does not stop the decline in resting lung function.
- **Posture and chest mobility:** thoracic extension, rib cage mobility and breathing exercises against kyphosis.

- **Airway clearance and prevention:** active cycle of breathing and huffing in chest infection; avoid long bed rest; stop smoking; advise vaccination; watch for silent aspiration in people with swallowing problems.
- **Monitoring:** do not rely on breathlessness alone; check oxygen saturation and the Borg scale.

Six cardiopulmonary changes for a 3-mark answer

Change	Result
Stiffer arteries	Raised systolic BP
Lower maximum heart rate	Lower maximum cardiac output and VO_2max
Weaker baroreflex	Postural hypotension
Stiffer chest wall, weaker muscles	More work of breathing
Loss of lung recoil	Lower FEV_1 and vital capacity, higher residual volume
Weaker cough and clearance	More chest infections

5 Metabolic and other changes

Metabolic changes were part of the W20 long answer.

- **Body composition:** less muscle and body water, more fat (especially around the abdomen).
- **Basal metabolic rate** falls, mostly because muscle mass falls; weight gain is easy if food intake stays the same.
- **Glucose:** insulin resistance and glucose intolerance; type 2 diabetes more common.
- **Hormones:** fall in growth hormone, sex hormones (menopause, lower testosterone) and vitamin D made in the skin.
- **Temperature control:** less sweating and shivering, weaker thirst: risk of heat stroke, cold injury and dehydration.
- **Kidneys:** lower filtration; drugs are cleared more slowly.
- **Skin:** thin, dry, fragile, slow to heal: pressure sores, tears, care with electrotherapy and heat.
- **Immune system:** weaker, so infections are more common and vaccines work less well.

6 Effects of exercise in older people

Effects of exercise on the neuromuscular system (W21, long answer; W25, 7 marks) and on the musculoskeletal and nervous systems (W22, 7 marks).

System	Effect of training
Muscle	Progressive resistance training increases strength and power at any age: frail nursing-home residents aged about 90 gained 174% in strength and 9% in thigh muscle area, and tandem gait speed improved 48%, after 8 weeks (Fiatarone et al., 1990). Early gains are neural (better recruitment and firing), later gains add muscle size
Nervous system	Faster reaction time, better coordination and balance reactions; aerobic exercise improves blood flow to the brain, memory and mood, and lowers the risk of cognitive decline
Balance and falls	Exercise reduced the rate of falls in older people living in the community by 23%; programmes with balance and functional exercise worked best (Sherrington et al., 2019)
Bone and joints	Weight-bearing and resistance exercise slow bone loss; exercise reduces osteoarthritis pain and stiffness; range of motion improves with stretching
Heart and lungs	Higher VO_2max, lower resting heart rate and blood pressure, better exercise tolerance
Metabolism	Better insulin sensitivity and blood sugar, less abdominal fat, better sleep
Function and mind	Faster walking, easier stairs and sit-to-stand, independence; less depression and anxiety, more social contact

WHO recommendations for adults aged 65 and over (2020)

- 150–300 minutes a week of moderate aerobic activity, or 75–150 minutes of vigorous activity.

- Muscle-strengthening activities on 2 or more days a week.
- Multicomponent activity with balance and strength training on 3 or more days a week, to improve function and prevent falls.
- Do as much as abilities allow; some activity is better than none; reduce sitting.

Precautions

- Screen for heart disease, uncontrolled blood pressure, diabetes, osteoporosis and falls risk; check drugs (beta-blockers blunt heart rate, so use the Borg scale).
- Longer warm-up and cool-down; avoid breath-holding; rise slowly from lying (postural hypotension); keep hydrated and avoid heat.
- Start low, progress slowly; supervise balance work; good footwear and a safe floor.

7 Model long answer: ageing changes and exercise

From a real paper (W22, 9 + 7 marks as printed)

> "Explain the various changes occurring in the Musculoskeletal & Neurological systems due to aging (9 Marks). Also write effects of exercises in the same systems in this population. (7 Marks)"

1. **[Introduction, 1]** Define ageing; primary and secondary ageing; loss of reserve.
2. **[Musculoskeletal, 4]** Muscle, bone, cartilage, tendons, discs, posture and gait, each with its effect on function (section 2).
3. **[Neurological, 4]** Brain, cognition, motor control, peripheral nerves, autonomic, senses (section 3).
4. **[Exercise, 7]** Resistance, balance, aerobic and flexibility training with their effects on muscle, bone, nerves, balance and falls, with two evidence lines (Fiatarone; Sherrington), WHO doses and precautions (section 6).

The W20 long answer pairs cardiorespiratory and metabolic changes (sections 4 and 5) with national policies for senior citizens (see elderly in India (p. 85)) and the fun-to-failure model (section 1). No patient case has been set on this topic alone; use these changes to explain impairments in any elderly long case (see geriatric evaluation (p. 110)).

8 Exam corner

Past questions (MUHS, 2016–2026)

Long answers

Cardiorespiratory and metabolic changes, national policies, fun, function, frailty and failure (W20); physiological effects of ageing and effects of exercise on the neuromuscular system (W21); musculoskeletal and neurological changes and effects of exercise (W22).

7 marks

Age-related changes in the nervous system (S17, S26), with effects of exercise (W25); in the respiratory system (S22); in the sensorimotor system (S23).

3 marks

Visual changes (S16); musculoskeletal changes (W16, S25); cardiorespiratory changes (S19); six cardiopulmonary changes (S20, W24); respiratory changes (S26).

Keywords to underline

loss of reserve | primary and secondary ageing | sarcopenia | type II fibres | osteoporosis | slower reaction time | presbyopia | presbycusis | postural hypotension | lower maximum heart rate | lower FEV_1, higher residual volume | airway closure | blunted response to hypoxia | insulin resistance | exercise reverses

Answer skeleton: changes in one system (7 marks)

1. **[Changes, 4]** Six to eight changes in a table.
2. **[Effects on function, 2]** What each does to daily life (falls, breathlessness, slowness).
3. **[PT implication, 1]** Exercise or adaptation that helps.

Draw these

- The fun-to-failure steps (section 1)
- Young versus old posture: upright versus forward head, kyphosis and flexed knees

Viva questions

1. What is sarcopenia, and which fibres are lost first?
2. Why do older people get dizzy on standing?
3. Which lung volume rises with age, and by how much?
4. Why should an older patient sit up rather than lie flat after surgery?
5. What is presbycusis?
6. Can a 90-year-old gain strength? Give the evidence.
7. What does WHO recommend for balance training in older adults?

Linked practical spots

- Measure postural blood pressure; test vibration sense with a 128 Hz tuning fork at the big toe; perform the Timed Up and Go and 30-second chair stand.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Visual changes due to ageing. (S16)**

Answer
Presbyopia; smaller pupil and yellow lens (less light, colour change); poor dark adaptation and glare tolerance; reduced contrast sensitivity and depth perception; narrower field; cataract, glaucoma, macular degeneration. Effect: falls on steps and at night; advice on lighting, contrast edges and eye checks.

3 m **Enumerate six cardiopulmonary changes in the elderly. (S20, W24)**

Answer
Stiffer arteries with raised systolic BP; lower maximum heart rate and VO_2max; weaker baroreflex (postural hypotension); stiffer chest wall and weaker respiratory muscles; loss of lung recoil with lower FEV_1 and VC and higher RV; weaker cough and mucociliary clearance.

7 m **Age-related changes in the respiratory system. (S22)**

Answer

Normal at rest, less reserve. Chest wall stiffer (rib cage compliance about 30% lower); lung recoil lost; diaphragm strength about 25% lower; TLC same, RV up about 50%, VC down, FEV_1 down 20–30 ml a year; airways close in normal breathing (about 44 lying, 65 sitting); alveolar area 75 to 60 m^2; $PaO_2 \approx 109 - 0.43 \times$ age, $PaCO_2$ unchanged; hypoxic drive down about 50%; weaker cough, slower clearance, silent aspiration. PT: pre-operative breathing and early mobilisation (age raises lung complications), upright positioning, inspiratory muscle training, aerobic training, posture work, airway clearance, monitor saturation.

7 m **Age-related changes in the nervous system. (S17, S26)**

Answer

Central: brain volume loss (frontal, hippocampus), fewer synapses, white matter change, less dopamine and acetylcholine; slower processing and reaction time, weaker working memory, kept knowledge; slower, less coordinated movement; lighter sleep. Peripheral: fewer fibres, slower conduction, reduced ankle jerks, reduced vibration and position sense in the feet. Autonomic: postural hypotension, poor temperature control, bladder urgency. Senses: vision, hearing, vestibular loss. Result: slower balance reactions and falls; PT: balance, strength, dual-task and aerobic training.

LAQ **Write physiological effects of ageing. Describe effects of exercise in detail on the neuromuscular system for the aged population. (W21)**

Answer

Define ageing, primary and secondary. Changes by system: musculoskeletal, neurological, sensory, cardiovascular, respiratory, metabolic (sections 2–5). Effects of exercise on the neuromuscular system: strength and power gains with resistance training (neural then muscle size; Fiatarone 1990), faster reaction time and better coordination, balance training cutting falls by 23% (Sherrington 2019), aerobic exercise for cognition and mood; WHO doses; precautions.

MCQ **With ageing, which lung volume increases?**

A. Vital capacity

B. FEV_1

C. Residual volume

D. Inspiratory reserve volume

Answer

C.

MCQ **Sarcopenia mainly affects:**

A. Type I fibres

B. Type II fibres

C. Smooth muscle

D. Cardiac muscle

Answer

B.

MCQ **In the fun-to-failure model, a person whose basic daily activities use up most of their reserve is at the level of:**

A. Fun

B. Function

C. Frailty

D. Failure

Answer
C.

Last recall check. List six changes for each of muscle and bone, nervous system, senses, heart, lungs and metabolism, then draw the fun-to-failure steps and write what exercise improves.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Community Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Avers D, Wong RA, eds. Guccione's Geriatric Physical Therapy. 4th ed. St Louis: Elsevier; 2020. Chapter 1.
3. Fleg JL, et al. Accelerated longitudinal decline of aerobic capacity in healthy older adults. *Circulation* 2005;112:674–82.
4. Fiatarone MA, et al. High-intensity strength training in nonagenarians: effects on skeletal muscle. *JAMA* 1990;263:3029–34.
5. Sherrington C, et al. Exercise for preventing falls in older people living in the community. *Cochrane Database Syst Rev* 2019;(1):CD012424.
6. World Health Organization. Guidelines on physical activity and sedentary behaviour, 2020.
7. Sharma G, Goodwin J. Effect of aging on respiratory system physiology and immunology. *Clin Interv Aging* 2006;1:253–60; Lowery EM, et al. The aging lung. *Clin Interv Aging* 2013;8:1489–96.
8. Skloot GS, et al. An official American Thoracic Society workshop report: evaluation and management of asthma in the elderly. *Ann Am Thorac Soc* 2016;13:2064–77 (residual volume, FEV_1 decline, diaphragm strength, airway closure by 65).
9. Estenne M, Yernault JC, De Troyer A. Rib cage and diaphragm-abdomen compliance in humans: effects of age and posture. *J Appl Physiol* 1985;59:1842–8 (the 30% figure is calculated from its reported values); Mittman C, et al. Relationship between chest wall and pulmonary compliance and age. *J Appl Physiol* 1965;20:1211–6.
10. Tolep K, et al. Comparison of diaphragm strength between healthy adult elderly and young men. *Am J Respir Crit Care Med* 1995;152:677–82.
11. Thomas ET, et al. Rate of normal lung function decline in ageing adults: a systematic review of prospective cohort studies. *BMJ Open* 2019;9:e028150; Hardie JA, et al. Risk of over-diagnosis of COPD in asymptomatic elderly never-smokers. *Eur Respir J* 2002;20:1117–22.
12. Closing capacity ages (Ruff, 1974) checked only through secondary teaching summaries (Deranged Physiology, LITFL, OpenAnesthesia); alveolar surface area (Thurlbeck, 1967) checked only through Lesauskaite V, Ebejer MJ. Age-related changes in the respiratory system. *Maltese Med J* 1999;11:25–9, and Deranged Physiology.
13. Sorbini CA, et al. Arterial oxygen tension in relation to age in healthy subjects. *Respiration* 1968;25:3–13 (formula checked through two later papers that quote it); Cardús J, et al. Increase in pulmonary ventilation-perfusion inequality with age in healthy individuals. *Am J Respir Crit Care Med* 1997;156:648–53; Gold WM, Koth LL. Pulmonary function testing. In: Murray and Nadel's Textbook of Respiratory Medicine. 6th ed. Elsevier; 2016 ($PaCO_2$ unchanged; dead space and maximal voluntary ventilation).
14. Kronenberg RS, Drage CW. Attenuation of the ventilatory and heart rate responses to hypoxia and hypercapnia with aging in normal men. *J Clin Invest* 1973;52:1812–9; Peterson DD, et al. Effects of aging on ventilatory and occlusion pressure responses to hypoxia and hypercapnia. *Am Rev Respir Dis* 1981;124:387–91.
15. Connolly MJ, et al. Reduced subjective awareness of bronchoconstriction provoked by methacholine in elderly asthmatic and normal subjects. *Thorax* 1992;47:410–3; Senaratna CV, et al. Prevalence of obstructive sleep apnea in the general population: a systematic review. *Sleep Med Rev* 2017;34:70–81.
16. Newnham DM, Hamilton SJ. Sensitivity of the cough reflex in young and elderly subjects. *Age Ageing* 1997;26:185–8; Ho JC, et al. The effect of aging on nasal mucociliary clearance, beat frequency, and ultrastructure of respiratory cilia. *Am J Respir Crit Care Med* 2001;163:983–8; Kikuchi R, et al. High incidence of silent aspiration in elderly patients with community-acquired pneumonia. *Am J Respir Crit Care Med* 1994;150:251–3.
17. McClaran SR, et al. Longitudinal effects of aging on lung function at rest and exercise in healthy active fit elderly adults. *J Appl Physiol* 1995;78:1957–68.
18. Smetana GW, Lawrence VA, Cornell JE. Preoperative pulmonary risk stratification for noncardiothoracic surgery: systematic review for the American College of Physicians. *Ann Intern Med* 2006;144:581–95.
19. Hardie JA, Mørkve O, Ellingsen I. Effect of body position on arterial oxygen tension in the elderly. *Respiration* 2002;69:123–8.
20. Manifield J, et al. Inspiratory muscle training for improving inspiratory muscle strength and functional capacity in older adults: a systematic review and meta-analysis. *Age Ageing* 2021;50:716–24.
21. Huang G, et al. Controlled endurance exercise training and VO_2max changes in older adults: a meta-analysis. *Prev Cardiol* 2005;8:217–25.

3(f) Geriatric evaluation and the role of physiotherapy: assessment scales, functional and fitness assessment, principles of geriatric rehabilitation

> Geriatric evaluation was asked in 9 of 20 papers, with 5 long cases about older people with knee pain, stairs, an Indian toilet, falls or a fracture. In every one, the examiner wants a structured assessment (medical, functional, cognitive, mood, social, home), named scales with cut-offs, and a plan that keeps the person independent where they live.

Syllabus	Ch 3(f) Scheme of evaluation and role of PT in geriatrics
Priority	**Tier 1.** In 9 of 20 papers (2016–2026): 5 long cases, 5 seven-mark and 1 three-mark question.
Asked as	LAQ W16 · S17 · S19 · W19 · S21 · 7m W17 · W18 ×2 · W19 · S25 · 3m S23
Usual split	15 (assessment method, impairments, management; or functional diagnosis 5, PT 10; or evaluation 8, protocol 7; or principles 7, national policies 8); 7 (assessment scales; functional assessment; principles of geriatric rehabilitation; fitness assessment)
Checked against	Senior Fitness Test (Rikli and Jones): procedures, normal ranges (Jones and Rikli, 2002) and criterion-referenced standards (Rikli and Jones, 2013); Fried frailty phenotype (2001); Timed Up and Go cut-off (Shumway-Cook et al., 2000); Berg Balance Scale and Tinetti POMA cut-offs as used in geriatric assessment toolkits; Short Physical Performance Battery (Guralnik et al., 1994). Full list under Sources (p. 118). Content checked 21 October 2026.

1 The scheme of geriatric evaluation

Functional assessment of the elderly (W18, W19, 7 marks); the assessment method in the S17 long case.

- **Comprehensive geriatric assessment:** a multidimensional, team-based assessment of an older person's medical, functional, psychological and social needs, used to plan care that keeps them independent.

Domain	What to assess	Tools
History	Complaints, illnesses, drugs (number, sedatives, BP drugs), falls in the last year, continence, sleep, nutrition and weight change, the person's goals	Interview, family report
Medical and physical	BP lying and standing, heart rate, vision, hearing, feet and footwear, skin, pain, range of motion, strength (grip), sensation	Postural BP, dynamometer, monofilament, Snellen chart
Basic ADL	Feeding, bathing, dressing, toileting, transfers, continence, walking	Katz Index (6 items), Barthel Index (0–100), FIM
Instrumental ADL	Telephone, shopping, cooking, housework, laundry, transport, medicines, money	Lawton IADL scale (8 items)
Mobility and balance	Transfers, gait, stairs, balance, falls risk	Timed Up and Go, Berg, Tinetti, gait speed, SPPB
Cognition	Memory, orientation, attention	MMSE, Mini-Cog, MoCA
Mood	Depression, anxiety, loneliness, fear of falling	Geriatric Depression Scale, FES-I
Nutrition and frailty	Weight loss, muscle mass, exhaustion	MNA, Fried frailty phenotype, Clinical Frailty Scale
Social and home	Who lives with them, carers, money, floor level, stairs, toilet type, lighting, hazards, access to services	Home visit and checklist

2 Geriatric assessment scales

Geriatric assessment scales (W17, 7 marks).

Scale	What it measures	Scoring and cut-off
Katz Index of ADL	Independence in bathing, dressing, toileting, transferring, continence, feeding	0–6; 6 fully independent, 2 or less very dependent
Lawton IADL	Telephone, shopping, food, housekeeping, laundry, transport, medicines, finances	0–8; higher is more independent
Barthel Index	Ten basic ADL and mobility items	0–100; higher is more independent
MMSE	Orientation, registration, attention, recall, language, copying	0–30; 23 or less suggests cognitive impairment (adjust for education)
Geriatric Depression Scale (15 items)	Depressive symptoms, yes/no answers	More than 5 suggests depression
Timed Up and Go	Rise from a chair, walk 3 m, turn, return, sit	13.5 seconds or more: raised falls risk in community-dwelling older people
Berg Balance Scale	14 balance tasks from sitting to standing on one leg	0–56; below about 45: raised falls risk
Tinetti POMA	Balance and gait items	0–28; below 19 high falls risk
SPPB	Standing balance, 4 m gait speed, five chair stands	0–12; lower scores predict disability and admission
Gait speed	Usual walking speed over 4–10 m	Below about 0.8 m/s suggests poor function and frailty
Fried frailty phenotype	Weight loss, exhaustion, weak grip, slow walking, low activity	3 or more frail; 1–2 pre-frail
FES-I	Concern about falling in 16 activities	16–64; higher means more fear

3 Fitness assessment of the elderly

Components of fitness assessment in the geriatric population (S25, 7 marks).

Functional fitness is the physical capacity to do normal daily activities safely and independently without undue fatigue. A fitness assessment measures each component, finds the weak ones, sets goals and shows change after training.

Components of fitness assessment (7 marks in S25)

Component	Why it matters in daily life	Test
Lower-body strength	Rising from a chair or an Indian toilet, climbing stairs, getting on a bus	30-second chair stand
Upper-body strength	Carrying shopping or water, lifting a grandchild, housework	30-second arm curl; grip dynamometer
Aerobic endurance	Walking to the market or temple, stairs, shopping	6-minute walk, or 2-minute step where space is short
Lower-body flexibility	Posture, stride length, putting on footwear	Chair sit-and-reach
Upper-body flexibility	Combing hair, dressing, reaching behind the back	Back scratch
Agility and dynamic balance	Getting up quickly to answer the door, stepping off a bus, avoiding a fall	8-foot up-and-go; Timed Up and Go
Static balance	Standing to dress or wash, standing on one leg on stairs	Single-leg stance
Body composition	Both excess fat and low muscle mass limit function	Height, weight, BMI, waist

Senior Fitness Test (Rikli and Jones)

The Senior Fitness Test was built for community-living adults aged 60–94. Its norms come from **7,183** people at 267 sites in 21 US states (Rikli and Jones, 1999). Test–retest reliability of the items is **0.80–0.98**.

How to do each item

Item (component)	Procedure	Score
30-second chair stand (lower-body strength)	Chair seat **43 cm (17 in)** high, set against a wall. Arms crossed at the wrists on the chest. Stand up fully and sit down fully as many times as possible in 30 s.	Number of full stands; the last counts if more than halfway up at 30 s
30-second arm curl (upper-body strength)	Seated, dominant arm, with a dumbbell of **2.27 kg (5 lb)** for women and **3.63 kg (8 lb)** for men. Full curls for 30 s.	Number of curls
6-minute walk (aerobic endurance)	Walk as fast as is safe around a **45.7 m (50-yard)** course marked every 4.57 m (5 yards) for 6 minutes.	Distance, to the nearest 5 yards
2-minute step (aerobic endurance, alternative)	Step on the spot, raising each knee to a mark midway between the kneecap and the iliac crest.	Number of times the right knee reaches the mark in 2 minutes
Chair sit-and-reach (lower-body flexibility)	Sit at the front of the chair. One leg straight, heel on the floor, ankle at 90°. Reach slowly towards the toes.	Distance from the middle fingertips to the tip of the toe: minus if short, plus if beyond; best of two
Back scratch (upper-body flexibility)	One hand reaches over the shoulder and the other up the middle of the back.	Distance between the middle fingers: minus for a gap, plus for an overlap; best of two
8-foot up-and-go (agility and dynamic balance)	From sitting, rise, walk as quickly as possible round a cone **2.44 m (8 ft)** away and sit down again.	Time in seconds; best of two
Height and weight (body composition)	Measure both.	BMI

Normal ranges

The normal range is the middle 50% of scores (25th to 75th percentile) for age and sex. A score below the range is below average for age. Distances are in yards (1 yard = 0.91 m) and reaches in inches (1 inch = 2.54 cm).

Normal ranges by age group 60–64, 65–69, 70–74, 75–79, 80–84, 85–89, 90–94 (Jones and Rikli, 2002)

Item	Women	Men
Chair stand (stands in 30 s)	12–17, 11–16, 10–15, 10–15, 9–14, 8–13, 4–11	14–19, 12–18, 12–17, 11–17, 10–15, 8–14, 7–12
Arm curl (curls in 30 s)	13–19, 12–18, 12–17, 11–17, 10–16, 10–15, 8–13	16–22, 15–21, 14–21, 13–19, 13–19, 11–17, 10–14
6-minute walk (yards)	545–660, 500–635, 480–615, 430–585, 385–540, 340–510, 275–440	610–735, 560–700, 545–680, 470–640, 445–605, 380–570, 305–500
2-minute step (steps)	75–107, 73–107, 68–101, 68–100, 60–91, 55–85, 44–72	87–115, 86–116, 80–110, 73–109, 71–103, 59–91, 52–86
Chair sit-and-reach (inches)	−0.5 to +5.0, −0.5 to +4.5, −1.0 to +4.0, −1.5 to +3.5, −2.0 to +3.0, −2.5 to +2.5, −4.5 to +1.0	−2.5 to +4.0, −3.0 to +3.0, −3.5 to +2.5, −4.0 to +2.0, −5.5 to +1.5, −5.5 to +0.5, −6.5 to −0.5
Back scratch (inches)	−3.0 to +1.5, −3.5 to +1.5, −4.0 to +1.0, −5.0 to +0.5, −5.5 to 0.0, −7.0 to −1.0, −8.0 to −1.0	−6.5 to 0.0, −7.5 to −1.0, −8.0 to −1.0, −9.0 to −2.0, −9.5 to −2.0, −10.0 to −3.0, −10.5 to −4.0
8-foot up-and-go (seconds; lower is better)	6.0–4.4, 6.4–4.8, 7.1–4.9, 7.4–5.2, 8.7–5.7, 9.6–6.2, 11.5–7.3	5.6–3.8, 5.7–4.3, 6.0–4.2, 7.2–4.6, 7.6–5.2, 8.9–5.3, 10.0–6.2

Criterion-referenced standards for staying independent

Norms compare a person with peers of the same age. The 2013 criterion-referenced standards instead give the score needed at each age to stay physically independent into late life, about age 90 (Rikli and Jones, 2013). A score below the standard marks a risk of losing independence and a target for training.

Standards by age group 60–64, 65–69, 70–74, 75–79, 80–84, 85–89, 90–94 (Rikli and Jones, 2013)

Item	Women	Men
Chair stand (stands)	15, 15, 14, 13, 12, 11, 9	17, 16, 15, 14, 13, 11, 9
Arm curl (curls)	17, 17, 16, 15, 14, 13, 11	19, 18, 17, 16, 15, 13, 11
6-minute walk (yards)	625, 605, 580, 550, 510, 460, 400	680, 650, 620, 580, 530, 470, 400
2-minute step (steps)	97, 93, 89, 84, 78, 70, 60	106, 101, 95, 88, 80, 71, 60
8-foot up-and-go (seconds, at or under)	5.0, 5.3, 5.6, 6.0, 6.5, 7.1, 8.0	4.8, 5.1, 5.5, 5.9, 6.4, 7.1, 8.0

There are no 2013 standards for the two flexibility items. The earlier manual (2001) gave simple risk zones for losing functional mobility: fewer than 8 chair stands, fewer than 11 arm curls, under 350 yards in the 6-minute walk, fewer than 65 steps, or over 9 s on the up-and-go. For both flexibility tests, the risk zone is –4 inches or worse in men and –2 inches or worse in women (Jones and Rikli, 2002).

Add to the battery

- **Grip strength** with a hand dynamometer.
- **Static balance:** single-leg stance.
- **Gait speed:** below about 0.8 m/s suggests poor function and frailty (section 2).
- **Falls risk and performance:** Timed Up and Go (13.5 s or more) and SPPB (0–12).

Safety and use of the results

- **Screen first:** BP, heart disease, recent falls, pain, drugs; get medical clearance when needed.
- **During testing:** warm up, demonstrate and allow a practice trial; set the chair against a wall; stand close for the balance items; good footwear and a safe floor.
- **Stop** if there is chest pain, dizziness or unusual breathlessness.
- **Use:** compare each score with the normal range and the standard; train the weakest components first; retest to show progress and motivate the person.

4 Principles of geriatric rehabilitation and the role of PT

Principles of geriatric rehabilitation (W18, 7 marks; W19, 7 marks of the long answer).

1. **Assess comprehensively,** including cognition, mood, home and carers; treat the person, not the age.
2. **Function is the goal:** independence in the tasks that matter to the person, set with them and their family.
3. **Teamwork:** doctor, nurse, physiotherapist, occupational therapist, social worker, family.
4. **Mobilise early;** avoid unnecessary bed rest (section 5).
5. **Start low, go slow,** but progress: older muscle still adapts to overload; allow longer recovery.
6. **Prevent the next problem:** falls, pressure sores, contractures, deconditioning, incontinence.
7. **Adapt the environment and use aids:** walking aids, raised toilet seat, rails, lighting.
8. **Consider multiple illnesses and drugs:** BP, sugar, heart rate response, pain, sensory loss.
9. **Simple instructions, repetition and written or pictorial home programmes;** involve carers.
10. **Respect dignity and choice;** motivation through meaningful goals.
11. **Continuity:** link to community programmes, senior citizens' groups and follow-up.

Role of the physiotherapist in geriatrics

- Health promotion: exercise groups, walking clubs, fall-prevention classes.
- Assessment and screening (sections 1–3).
- Treatment: osteoarthritis, back and neck pain, stroke, Parkinson's, fractures and joint replacement, COPD and heart disease, incontinence.
- Fall prevention (see falls (p. 119)); osteoporosis care (see menopause and osteoporosis (p. 63)).
- Home visits, aids and home modification; carer training; palliative care.

5 Adverse effects of immobility in the elderly

Adverse effects of immobility in the elderly (S23, 3 marks).

- **Muscle and function:** rapid loss of strength and walking ability (hospital-associated disability), contractures; older people regain strength more slowly than the young.
- **Heart and lungs:** postural hypotension, falls on first standing, deep vein thrombosis, pneumonia.
- **Bone and skin:** bone loss, pressure sores.
- **Bladder and bowel:** incontinence, retention, constipation and faecal impaction.
- **Mind:** delirium, depression, loss of confidence and fear of falling.
- **Metabolic:** insulin resistance, poor appetite and weight loss.
- Full table on bed rest (p. 18). Prevention: sit out and walk early, exercise in bed, turning, toileting schedule.

6 Worked long case: a widow in an old age home

From a real paper (S17)

> "65-years old female, who lost her husband year ago moved into old age home, comes with chief complaints of pain in her left knee and finds difficulty in climbing stairs to reach her room and using Indian toilet. She also complains of loss of balance twice in the past one month. She expresses to lead a quality of life in her remaining days. i) What is your method of assessment in this case? ii) Enumerate the impairments and plan the management."

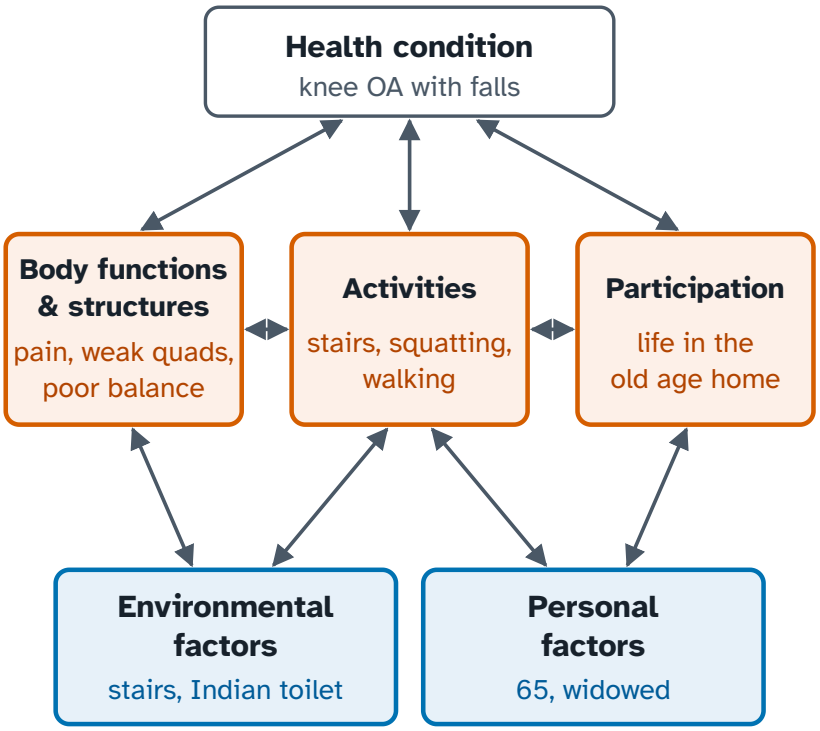
Spot the clues before you write

left knee pain, stairs, Indian toilet: knee osteoarthritis with weak quadriceps. loss of balance twice in a month: high falls risk; needs a falls assessment. lost her husband, moved to an old age home: grief, possible low mood, new environment. wants quality of life: her goal leads the plan.

Method of assessment

- History: knee pain pattern, morning stiffness, the two falls (where, when, how, injuries, could she get up), dizziness, vision, drugs, sleep, mood and grief, continence.
- Examination: BP lying and standing; knee swelling, crepitus, range, quadriceps strength and lag; hip strength; feet and footwear; sensation and vision.
- Function: Timed Up and Go, Berg, 30-second chair stand, gait, stairs, squatting and rising (Indian toilet), Barthel or Katz.
- Cognition and mood: MMSE, Geriatric Depression Scale, FES-I for fear of falling.
- Environment: stairs to her room (rails, lighting), toilet type, bed and chair heights, floor surface.

The ICF model filled for this case



Draw the six boxes with two-way arrows, then write two or three points from the case in each box.

Impairments

- Left knee pain and stiffness; reduced knee range; weak quadriceps and hip abductors.
- Poor static and dynamic balance; slow gait; possible postural hypotension or vision problem.
- Fear of falling; low mood after bereavement; possibly reduced fitness.

Management

Problem	Technique and dose	Why
Knee pain	Heat or ice, TENS for flares; education on OA	Allows exercise
Weak quadriceps and hips	Static quadriceps, straight-leg raise, sit-to-stand from a raised chair, mini-squats, side-lying hip abduction; 10 × 2–3 sets, 3 times a week, progressing with ankle weights	Stairs and rising from the toilet need leg strength
Poor balance	Progressive balance training: narrow stance, tandem stance, single-leg stance at a rail, stepping, weight shifts, walking with turns; more than 3 hours a week in total across the programme	Most effective fall-prevention exercise
Stairs and toilet	Stair training with a rail (up with the good leg, down with the painful one); raised commode seat or western toilet with grab bar; walking stick in the right hand	Reduces knee load and falls
Fear and mood	Group exercise in the home, teach how to get up from the floor, encourage social activities and counselling for grief	Confidence and participation
Fitness	Daily walks building to 30 minutes, chair-based aerobic exercise	Endurance and well-being

Ask the home to add rails on the stairs, a night light, non-slip flooring in the bathroom, and if possible a ground-floor room.

Practice cases from the papers

- **S19:** 69-year-old man of low income, strained family, in a charitable geriatric centre on the second floor, pain on walking and stairs: functional diagnosis and PT (as above; add low-cost aids and group programmes).

- **W16:** 75-year-old with wrist stiffness 6 months after a Colles' fracture from a fall: evaluation (range, grip, swelling, complex regional pain, function, osteoporosis and falls risk) and rehabilitation (mobilisation, strengthening, functional use, fall prevention, bone health).
- **S21:** 62-year-old with neck and upper back pain for 2 years, living with her husband with domestic help: functional diagnosis and PT for chronic neck pain, posture and activity.
- **W19:** principles of geriatric rehabilitation (section 4) and national policies for the elderly (see senior citizens in India (p. 85)).

7 Exam corner

Past questions (MUHS, 2016–2026)

Long cases

Wrist stiffness after Colles' fracture (W16); widow in an old age home with knee pain and falls (S17); 69-year-old in a charitable centre (S19); principles of geriatric rehabilitation and national policies (W19); 62-year-old with neck pain (S21).

7 marks

Geriatric assessment scales (W17); principles of geriatric rehabilitation (W18); functional assessment of the elderly (W18, W19); components of fitness assessment in the elderly (S25).

3 marks

Adverse effects of immobility in the elderly (S23).

Keywords to underline

comprehensive geriatric assessment ADL and IADL Katz, Lawton, Barthel MMSE 23 GDS TUG 13.5 s Berg 45 Senior Fitness Test functional fitness criterion-referenced standards frailty start low, go slow function is the goal

Answer skeleton: functional assessment of the elderly (7 marks)

1. **[Why, 1]** Function predicts independence, care needs and outcome.
2. **[Domains, 4]** ADL, IADL, mobility and balance, cognition, mood, nutrition, social and home, each with a scale.
3. **[Use, 2]** Goals, risk (falls, frailty), referral, review.

Draw these

- The ICF model for an elderly case (section 6)
- The Timed Up and Go course: chair, 3 m line, turn

Viva questions

1. What is the difference between ADL and IADL?
2. What does a TUG of 16 seconds suggest?
3. Name the five Fried frailty criteria.
4. List the Senior Fitness Test items.
5. What MMSE score suggests cognitive impairment?
6. Give four principles of geriatric rehabilitation.

Linked practical spots

- Perform the TUG, 30-second chair stand and Berg items; administer the Katz and Lawton scales; measure postural BP.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Adverse effects of immobility in the elderly. (S23)**

Answer

Rapid loss of strength and walking ability, contractures; postural hypotension, DVT, pneumonia; bone loss and pressure sores; incontinence, constipation; delirium, depression, fear of falling; insulin resistance and weight loss. Prevent by early mobilisation.

7 m **Geriatric assessment scales. (W17)**

Answer

ADL: Katz (6 items), Barthel (0–100). IADL: Lawton (8 items). Cognition: MMSE (30; 23 or less impaired). Mood: GDS-15 (more than 5). Balance and falls: TUG (13.5 s), Berg (0–56; below 45), Tinetti (28; below 19). Physical performance: SPPB, gait speed. Frailty: Fried criteria. Fear of falling: FES-I. Fitness: Senior Fitness Test. One line on use of each.

7 m **Principles of geriatric rehabilitation. (W18)**

Answer

Comprehensive assessment; function-based goals set with the person; teamwork; early mobilisation; start low, go slow, but progress; prevent secondary problems; environment and aids; account for comorbidity and drugs; simple instructions and carer involvement; dignity and choice; continuity with community services.

7 m **Components of fitness assessment in the geriatric population. (S25)**

Answer

Define functional fitness. Components with a test each: lower-body strength (30-s chair stand, 43 cm chair), upper-body strength (30-s arm curl, 5 lb women, 8 lb men), aerobic endurance (6-minute walk on a 50-yard course, or 2-minute step), flexibility (chair sit-and-reach, back scratch), agility and dynamic balance (8-foot up-and-go), body composition (BMI); add grip, single-leg stance, gait speed. Interpret against the normal range and the 2013 standards (for example women 60–64 need 15 chair stands and 625 yards). Screen first; stop for chest pain, dizziness or unusual breathlessness; retest.

LAQ **69-year-old man of low income in a charitable geriatric centre on the second floor with pain on walking and stairs: functional diagnosis and PT plan. (5 + 10, S19)**

Answer

Assessment as in section 6; likely knee or hip OA with deconditioning. ICF: pain, weakness, balance; walking, stairs; participation in the centre; second floor without lift and low income (barriers), centre staff and peers (facilitators). Plan: pain relief, strengthening, balance, stair training with rail, walking aid, group exercise sessions in the centre, low-cost aids, staff training, fall prevention, follow-up.

MCQ **The Lawton scale assesses:**

A. Basic ADL

B. Instrumental ADL

C. Cognition

D. Depression

Answer
B.

MCQ **The 30-second chair stand in the Senior Fitness Test measures:**

A. Upper-body strength

B. Lower-body strength

C. Flexibility

D. Aerobic endurance

Answer
B.

MCQ **A person meeting 3 of the 5 Fried criteria is classed as:**

A. Robust

B. Pre-frail

C. Frail

D. Disabled

Answer
C.

Last recall check. List the domains of geriatric assessment with a scale for each, the Senior Fitness Test, ten principles of geriatric rehabilitation and six effects of immobility.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Community Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Rikli RE, Jones CJ. Senior Fitness Test Manual. Champaign, IL: Human Kinetics; 2001; 2nd ed. 2013 (the 2013 percentile tables were seen only through a reproduction by the StrongPeople Program; they differ from the 2002 normal ranges in a few cells, and this page uses the 2002 version throughout).
3. Rikli RE, Jones CJ. Development and validation of a functional fitness test for community-residing older adults. *J Aging Phys Act* 1999;7:129–61; Functional fitness normative scores for community-residing older adults, ages 60–94. *J Aging Phys Act* 1999;7:162–81 (abstracts).
4. Jones CJ, Rikli RE. Measuring functional fitness of older adults. *The Journal on Active Aging* 2002 (March–April):24–30 (item descriptions, normal ranges and risk zones). Procedure details also checked against the Shirley Ryan AbilityLab measure pages, the CDC STEADI chair stand form and the PhenX Toolkit 6-minute walk protocol.
5. Rikli RE, Jones CJ. Development and validation of criterion-referenced clinically relevant fitness standards for maintaining physical independence in later years. *Gerontologist* 2013;53:255–67 (tables read in an archived reprint of the full text).
6. Fried LP, et al. Frailty in older adults: evidence for a phenotype. *J Gerontol A Biol Sci Med Sci* 2001;56:M146–56.
7. Shumway-Cook A, Brauer S, Woollacott M. Predicting the probability for falls in community-dwelling older adults using the Timed Up and Go test. *Phys Ther* 2000;80:896–903.
8. Guralnik JM, et al. A short physical performance battery assessing lower extremity function. *J Gerontol* 1994;49:M85–94.
9. Geriatric Examination Tool Kit, University of Missouri: norms and cut-offs for Berg, Tinetti and other tests.
10. Avers D, Wong RA, eds. Guccione's Geriatric Physical Therapy. 4th ed. St Louis: Elsevier; 2020.

3(f) Falls in the elderly: risk factors, pathomechanics of balance, fear of falling and fall prevention

> Falls were asked in 8 of 20 papers, with 5 long answers, and they appear inside most elderly cases too. Every answer has the same skeleton: why older people fall (intrinsic and extrinsic factors, failing balance systems), how to find who will fall (history and tests), and what prevents it (balance and strength exercise first, then the home, drugs, eyes, feet and fear).

Syllabus	Ch 3(f) Scheme of evaluation and role of PT in geriatrics (falls and fall prevention)
Priority	**Tier 1.** In 8 of 20 papers (2016–2026): 5 long answers, 1 seven-mark and 2 three-mark questions.
Asked as	LAQ W17 · S20 · S23 · S24 · W24 · 7m W22 · 3m W20 · W25
Usual split	15 (pathomechanics of balance 7, fall-prevention programme 8; or causes 5, rehabilitation 10; or intrinsic and extrinsic factors 7, prevention 8; or impairments 7, home vs institution 4, prevention 4); 7 (causes and environmental modifications)
Checked against	WHO global report on falls prevention in older age (2007); Cochrane review of exercise for preventing falls (Sherrington et al., 2019) and meta-analysis of exercise dose (Sherrington et al., 2017); Otago meta-analysis (Thomas et al., 2010); Cochrane review of home safety (Gillespie et al., 2012); World Falls Guidelines (Montero-Odasso et al., 2022); TUG cut-off (Shumway-Cook et al., 2000). Full list under Sources (p. 125). Content checked 20 October 2026.

1 Falls: definition and burden

- **Fall (WHO):** an event which results in a person coming to rest inadvertently on the ground or floor or other lower level.
- About 28–35% of people aged 65 and over fall each year, rising to 32–42% over 70 (WHO, 2007); a previous fall is the strongest predictor of another.
- **Consequences:** fractures (hip, wrist, spine, pelvis), head injury, long lie on the floor (pressure sores, dehydration, pneumonia), fear of falling, activity restriction, loss of independence, admission to care, death.

2 Risk factors for falls

Risk factors for falls (W20) and extrinsic risk factors (W25), 3 marks; causes and environmental modifications (W22, 7 marks); intrinsic and extrinsic factors with two prevention measures each (S24, long answer); causes of falls (S23, 5 marks).

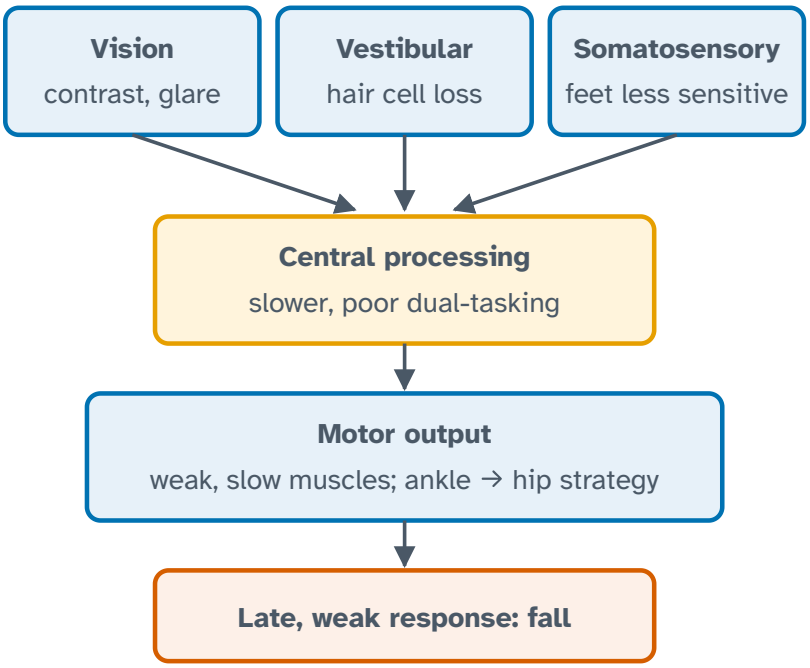
Factor	Examples	Prevention
Intrinsic: musculoskeletal	Leg weakness (quadriceps, hip abductors, ankle dorsiflexors), arthritis, foot problems, reduced ankle range	Progressive strength training; foot care and footwear
Intrinsic: balance and gait	Poor balance reactions, slow gait, shuffling, previous falls	Balance and gait training; walking aid if needed
Intrinsic: sensory	Poor vision (cataract, bifocals), neuropathy in the feet, vestibular loss	Eye check and cataract surgery; single-vision glasses for walking outdoors; vestibular exercises
Intrinsic: medical	Postural hypotension, stroke, Parkinson's, dementia, depression, diabetes, urinary urgency and nocturia, acute illness	Medical review; rise slowly; bladder management; treat illness
Intrinsic: psychological	Fear of falling, cognitive impairment, risk-taking	Confidence building, supervision, education
Extrinsic: environment	Poor lighting, slippery or wet floors, loose rugs, clutter, cords, stairs without rails, low beds and chairs, Indian (squat) toilets, bathrooms without grab bars, uneven roads, animals	Home safety check and modification (section 5)

Factor	Examples	Prevention
Extrinsic: medicines	Four or more drugs; sedatives and sleeping pills, antidepressants, antipsychotics, BP drugs and diuretics	Medication review by the doctor; reduce psychotropic drugs
Extrinsic: footwear and aids	Loose slippers, high heels, worn soles; wrong-height or faulty walking aids	Firm, well-fitting shoes with thin non-slip soles; check and fit aids

3 Pathomechanics of balance problems in the elderly

Pathomechanics of balance issues in the elderly (S20, W24, 7 marks of the long answer).

Balance as a system, and where ageing affects it



Each part of the balance system declines with age (see sensorimotor changes (p. 102)); disease and drugs add to the decline.

Text version of the flowchart

1. Sensory input: vision (poor contrast, glare), vestibular (hair cell loss), somatosensory (less sensitive feet).
2. Central processing: slower, poorer at doing two tasks at once.
3. Motor output: weak, slow muscles; shift from ankle to hip strategy; slow stepping.
4. Result: late, weak balance response, so a slip or trip becomes a fall.

- **Sensory:** less reliable information from eyes, inner ear and feet; older people depend more on vision, so dim light and busy patterns cause falls.
- **Central:** slower reaction time and integration; walking while talking or carrying (dual-tasking) worsens gait.
- **Motor:** loss of fast-twitch fibres and power, so the corrective step is too slow or too short; reduced ankle range and strength shift control from the ankle strategy to the hip strategy, and stepping becomes late.
- **Posture and gait:** flexed posture moves the centre of mass forward; smaller limits of stability; slower, shorter, wider steps; more sway.
- **Anticipatory control:** late postural adjustments before reaching or lifting.
- **Other:** postural hypotension (brain under-perfused on standing), fear of falling (stiff, cautious gait), drugs (sedation, low BP).

4 Assessing falls risk

- **Screening questions:** Have you fallen in the past year? Do you feel unsteady? Do you worry about falling?
- **History of each fall:** place, time, activity, dizziness or blackout, injury, could they get up, how long on the floor.
- **Tests:** Timed Up and Go (13.5 seconds or more means higher risk), Berg Balance Scale, gait speed, 30-second chair stand, single-leg stance, postural BP, vision, feet and sensation, cognition, FES-I for fear (see assessment scales (p. 110)).
- **Medication review** and **home visit**.
- **Risk level** decides the response: low risk gets education and exercise; high risk (a fall with injury, repeated falls, inability to get up, or poor gait and balance) gets a full multifactorial assessment and tailored plan (World Falls Guidelines, 2022).

5 Fall prevention

Fall-prevention programme (S20, W24, 8 marks); environmental modifications (W22); how can falls be prevented (W17, 4 marks).

Exercise: the most effective single measure

- Exercise reduced the rate of falls in community-dwelling older people by 23%; balance and functional exercise reduced it by 24%, and programmes adding resistance training by about a third (Sherrington et al., 2019).
- Larger effects come from programmes that challenge balance and give more than 3 hours of exercise a week in total (Sherrington et al., 2017).
- **Otago Exercise Programme:** home-based progressive leg strengthening (ankle weights) and balance exercises about 3 times a week plus walking, taught by a physiotherapist over several visits; it reduced falls by about a third (Thomas et al., 2010).
- **Balance training content:** standing with a narrower base (feet together, tandem, single leg), reducing hand support, moving the centre of mass (reaching, weight shifts), stepping in all directions, walking with turns, heel and toe walking, sit-to-stand, stair practice; tai chi; progress by reducing support and adding dual tasks.
- **Strength:** sit-to-stand, mini-squats, heel and toe raises, hip abduction and extension, knee extension with weights, 2–3 sets of 8–12, 2–3 times a week.

Environmental modifications

Area	Modification
Lighting	Bright, even light; switches at room entrances and by the bed; night lights to the toilet
Floors	Remove loose rugs, cords and clutter; non-slip mats; wipe spills; mark step edges with contrasting colour
Stairs	Rails on both sides, good light at top and bottom, no objects on steps
Bathroom and toilet	Grab bars; western toilet or a raised commode over the Indian toilet; non-slip floor; bathing stool; hand shower
Bedroom	Bed and chair at knee height; firm chair with armrests; urinal or commode at night if urgency
Kitchen and storage	Common items between waist and shoulder height; no climbing on stools
Outdoors	Even paths, rails at entrances, avoid walking on wet or dark roads

Home safety assessment and modification reduced the rate of falls (rate ratio 0.81), with more benefit in people at high risk and when done by an occupational therapist (Gillespie et al., 2012).

Other measures

- Medication review: stop or reduce sedatives and other risky drugs.
- Vision: cataract surgery, updated glasses; care with bifocals on stairs.
- Feet: podiatry, proper footwear.

- Postural hypotension: rise in stages, fluids, drug review.
- Bone health: calcium, vitamin D if deficient, osteoporosis treatment; hip protectors in care homes.
- Education: what causes falls, what to do after a fall, using a call bell or phone; keep active.

Getting up after a fall (teach it)

1. Stay calm and check for injury; if hurt, call for help and keep warm.
2. Roll onto the side, then onto hands and knees.
3. Crawl to a firm chair or bed.
4. Put hands on the seat, bring the stronger leg forward with the foot flat (half-kneeling).
5. Push up with arms and leg, turn and sit; rest before standing.

6 Worked long case: a man with two falls and fear of falling

From a real paper (S23, 5 + 10)

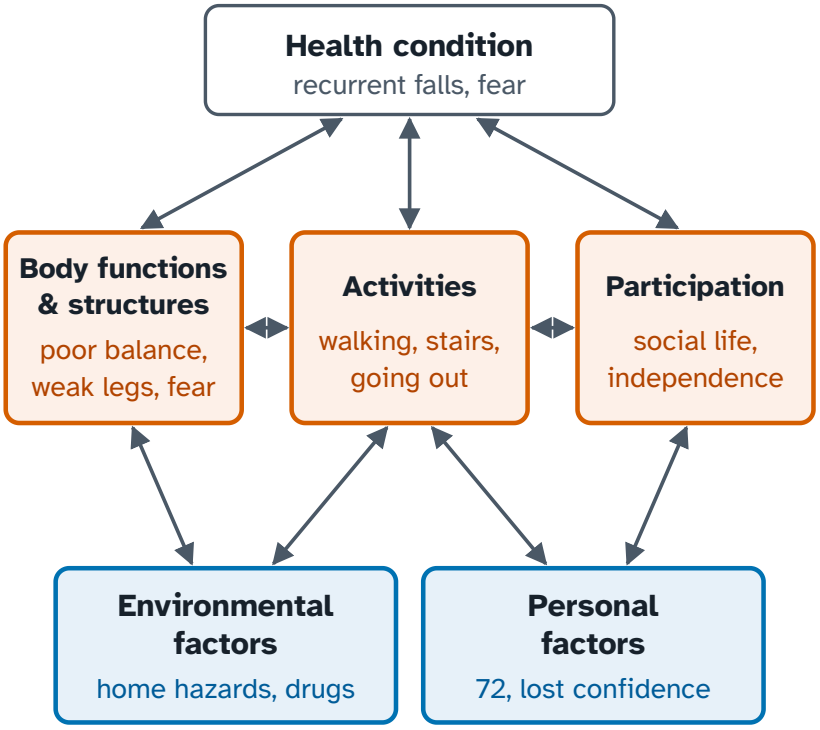
> "Enumerate different causes of falls in elderly people. Discuss rehabilitation of a 72 years old male patient with a history of two falls in last one year and a positive fear of fall."

Spot the clues before you write

two falls in a year: recurrent faller, high risk: needs a full multifactorial assessment. positive fear of falling: he restricts activity, loses strength and confidence, and falls more: break this cycle. 72, male: check BP drugs, prostate symptoms with night toilet trips, vision.

Causes (5 marks): intrinsic and extrinsic factors from section 2, grouped under musculoskeletal, balance and gait, sensory, medical, psychological, environment, medicines, footwear.

The ICF model filled for this case



Draw the six boxes with two-way arrows, then write two or three points from the case in each box.

Rehabilitation (10 marks)

Problem	Technique and dose	Why
Assessment	Fall history, TUG, Berg, chair stand, gait speed, postural BP, vision, feet, cognition, FES-I, drugs, home visit	Finds his own risk factors
Poor balance	Progressive balance training (section 5), 3 times a week supervised plus daily home practice, building to more than 3 hours a week	Strongest evidence for fewer falls
Weak legs	Otago-type strengthening with ankle weights: sit-to-stand, knee extension, hip abduction, heel and toe raises, 2–3 sets of 8–12	Power for recovery steps
Gait	Walking practice with turns, obstacles, uneven surfaces and dual tasks; walking aid if needed; daily walks	Safer walking in real settings
Fear of falling	Explain causes and what is being done; set small goals; graded exposure to feared activities; teach getting up from the floor; group classes; family encouragement	Breaks the cycle of fear, inactivity and weakness
Home and drugs	Home modification (lighting, rails, toilet, rugs); medication review with the doctor; urinal at night if nocturia; footwear	Removes extrinsic causes
Safety	Phone or alarm within reach; tell family his routine	Prevents a long lie

Review every 4–6 weeks with TUG, Berg and FES-I; continue exercise long term because benefits fade when it stops.

Practice case (W17)

An 88-year-old widower living alone on a second floor without a lift, a month after hip replacement for a fall-related fractured neck of femur, with an attendant: impairments (pain, hip precautions, weak hip muscles, poor balance, fear, stairs), home versus old age home (see senior citizens in India (p. 85)), and how to prevent further falls (this page).

7 Exam corner

Past questions (MUHS, 2016–2026)

Long answers

88-year-old after hip replacement: impairments, home vs old age home, fall prevention (W17); pathomechanics of balance and fall-prevention programme (S20, W24); causes of falls and rehabilitation of a 72-year-old with fear of falling (S23); intrinsic and extrinsic factors and two prevention measures each (S24).

7 marks

Causes of falls and environmental modifications (W22).

3 marks

Risk factors for falls (W20); extrinsic risk factors (W25).

Keywords to underline

intrinsic and extrinsic · previous falls · postural hypotension · psychotropic drugs · sensory, central, motor · ankle to hip strategy · dual-tasking · balance training more than 3 hours a week · Otago · home modification · fear of falling · getting up from the floor

Answer skeleton: fall-prevention programme (8 marks)

1. **[Screening and assessment, 2]** Three questions; tests; risk level.
2. **[Exercise, 3]** Balance, strength, gait, dose and evidence.
3. **[Other measures, 2]** Home, drugs, vision, feet, BP, bone health.

④ **[Education, 1]** Fear, getting up, alarms.

Draw these

- The balance system and its ageing (section 3)
- A safe bathroom: grab bars, raised seat, non-slip floor
- Steps of getting up from the floor

Viva questions

1. Define a fall.
2. Give four intrinsic and four extrinsic risk factors.
3. Which drugs increase falls?
4. What is the Otago programme?
5. Why does fear of falling increase falls?
6. How would you teach an older person to get up from the floor?

Linked practical spots

- Perform TUG, Berg and postural BP; teach getting up from the floor; do a home hazard checklist for a classmate's home.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Enumerate the extrinsic risk factors for falls in the elderly. (W25)**

Answer
Poor lighting; slippery or wet floors; loose rugs, cords, clutter; stairs without rails; low beds and chairs; Indian toilets and bathrooms without grab bars; uneven outdoor surfaces; poor footwear; faulty or wrong walking aids; medicines (sedatives, antidepressants, BP drugs, four or more drugs).

7 m **Enumerate the causes of falls in the elderly and explain environmental modifications for prevention. (W22)**

Answer
Causes: intrinsic (weakness, balance and gait, vision and sensation, postural hypotension, neurological and other diseases, cognition and mood, fear) and extrinsic (environment, drugs, footwear, aids). Modifications by area: lighting, floors, stairs, bathroom and toilet, bedroom, kitchen, outdoors; home safety checks reduce falls, more in high-risk people.

LAQ **Describe the pathomechanics of balance issues in the elderly and a physiotherapy programme for fall prevention. (7 + 8, S20, W24)**

Answer
Pathomechanics: sensory decline (vision, vestibular, somatosensory), slower central processing and dual-task interference, motor decline (power, type II fibres, ankle range, strategy shift, slow stepping), flexed posture and smaller limits of stability, anticipatory control, postural hypotension, fear, drugs. Programme: screening and tests, progressive balance and strength (Otago; more than 3 hours a week), gait and dual-task training, home modification, drug review, vision and footwear, education and getting up from the floor, review.

MCQ **The single most effective intervention to prevent falls in community-dwelling older people is:**

A. Bed rest

B. Balance and strength exercise

C. Sedatives for anxiety

D. Wearing slippers

Answer
B.

MCQ **Which is an intrinsic risk factor for falls?**

A. Loose rugs

B. Poor lighting

C. Postural hypotension

D. No grab bars

Answer
C.

MCQ **With ageing, balance control shifts from:**

A. Hip strategy to ankle strategy

B. Ankle strategy to hip strategy

C. Stepping to ankle strategy

D. Vision to somatosensation

Answer
B.

Last recall check. List eight intrinsic and eight extrinsic risk factors, draw the balance system, write a fall-prevention programme with doses, and the five steps of getting up from the floor.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Community Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. World Health Organization. WHO global report on falls prevention in older age. Geneva: WHO; 2007.
3. Sherrington C, et al. Exercise for preventing falls in older people living in the community. *Cochrane Database Syst Rev* 2019; (1):CD012424.
4. Sherrington C, et al. Exercise to prevent falls in older adults: an updated systematic review and meta-analysis. *Br J Sports Med* 2017;51:1750–8.
5. Thomas S, Mackintosh S, Halbert J. Does the 'Otago exercise programme' reduce mortality and falls in older adults? A systematic review and meta-analysis. *Age Ageing* 2010;39:681–7.
6. Gillespie LD, et al. Interventions for preventing falls in older people living in the community. *Cochrane Database Syst Rev* 2012; (9):CD007146.
7. Montero-Odasso M, et al. World guidelines for falls prevention and management for older adults: a global initiative. *Age Ageing* 2022;51:afac205.
8. Shumway-Cook A, Brauer S, Woollacott M. Predicting the probability for falls in community-dwelling older adults using the Timed Up and Go test. *Phys Ther* 2000;80:896–903.

Part 4

Concepts of rehabilitation

11 hours theory, 20 field

4(a)	Disability: definitions, models, types, evaluation and prevention	127
4(b, d)	Rehabilitation: types, IBR vs CBR and the rehabilitation team	133
4(c)	National policies and legal rights for persons with disability	144
4(e, f)	Community-based rehabilitation: principles, strategies and the physiotherapist	150
4	Rehabilitation of an amputee in the community	160

4(a) Disability: impairment, disability and handicap, models, types, evaluation and levels of prevention

> Disability was asked in 9 of 20 papers, mostly as 3-mark definitions and 7-mark "levels of disability prevention". Learn the ICIDH chain (disease, impairment, disability, handicap) with one example, and hang the three levels of prevention on it: primary stops the impairment, secondary stops it becoming a disability, tertiary stops the disability becoming a handicap.

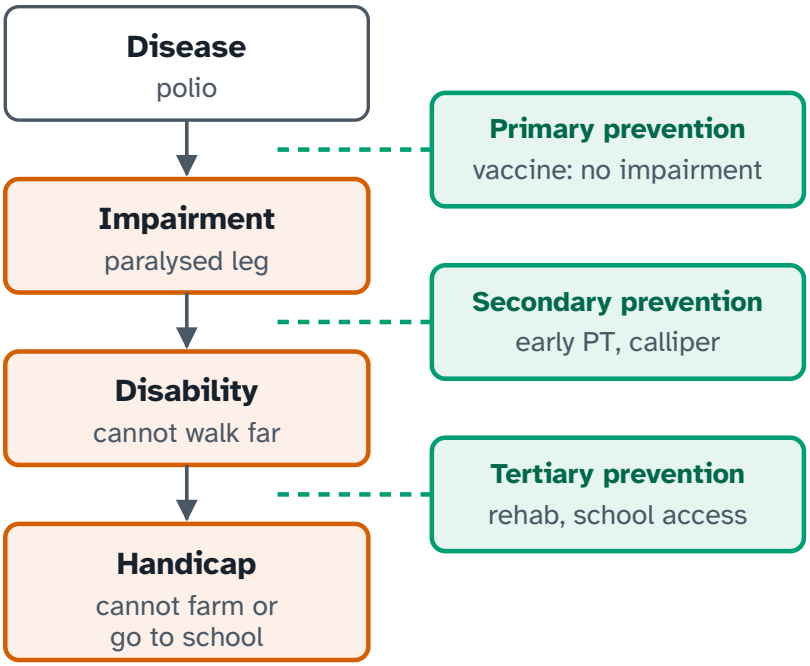
Syllabus	Ch 4(a) Disability: evaluation, types, prevention
Priority	**Tier 1.** In 9 of 20 papers (2016–2026): 3 seven-mark and 10 three-mark questions.
Asked as	7m W20 · S23 · S26 · 3m S16 · W16 ×2 · W17 ×2 · S19 · W20 · W22 · W25 ×2
Usual split	7 (levels of disability prevention; types of disability with evaluation of upper-extremity impairment); 3 (definitions; ICIDH; models; types; primary and secondary disability)
Checked against	WHO International Classification of Impairments, Disabilities and Handicaps (1980) and ICF (2001); Rights of Persons with Disabilities Act, 2016 (schedule of specified disabilities); Ministry of Social Justice and Empowerment guidelines for assessing permanent physical impairment (2001, carried into the 2018 notification); WHO report on disability prevention and rehabilitation (1981). Full list under Sources (p. 132). Content checked 20 October 2026.

1 Impairment, disability and handicap (ICIDH)

Define impairment, disability and handicap (W16, W25); full form of "ICFDH", meaning ICIDH (W17); define disability (W20). All 3 marks.

- **ICIDH:** International Classification of Impairments, Disabilities and Handicaps, published by WHO in 1980; replaced by the ICF in 2001.
- **Impairment:** any loss or abnormality of psychological, physiological or anatomical structure or function (organ level).
- **Disability:** any restriction or lack, resulting from an impairment, of ability to perform an activity in the manner or within the range considered normal for a human being (person level).
- **Handicap:** a disadvantage for a given individual, resulting from an impairment or a disability, that limits or prevents the fulfilment of a role that is normal for that individual, depending on age, sex, and social and cultural factors (society level).

The ICIDH chain and the three levels of prevention



Each level of prevention breaks one link in the chain (green dashed lines). The ICF later replaced this one-way chain with two-way interactions and added environmental and personal factors.

The ICF (2001)

- Body functions and structures (impairments), activities (limitations), participation (restrictions), with environmental and personal factors; all interact both ways.
- **Disability** in the ICF is the umbrella term for impairments, activity limitations and participation restrictions, arising from the interaction between a health condition and contextual factors.
- Full explanation on the ICF page in Neuro.

2 Models of disability

Describe various models of disability (W17, 3 marks).

Model	Sees disability as	Response
Medical (individual)	A problem of the person, caused by disease or injury	Cure or treat the impairment; professionals decide
Social	Created by barriers in society (steps, attitudes, laws), not by the impairment	Remove barriers: ramps, inclusive schools, laws
Biopsychosocial (ICF)	An interaction of health condition, body, activity, participation and context	Treat the person and change the environment together
Rights-based	A human rights issue; persons with disability have equal rights	UN Convention on the Rights of Persons with Disabilities (2006), RPWD Act 2016
Charity	A tragedy needing pity and help	Donations and care; criticised for creating dependence
Nagi model	Pathology → impairment → functional limitation → disability	Used in rehabilitation research

3 Types of disability

Enumerate types of disabilities (S19), define and classify disability (W20), primary and secondary disability (S16), 3 marks; types of disability with evaluation of upper-limb impairment (S23, 7 marks).

21 specified disabilities (RPWD Act, 2016)

Group	Disabilities
Physical: locomotor	Locomotor disability, leprosy-cured persons, cerebral palsy, dwarfism, muscular dystrophy, acid attack victims
Physical: sensory	Blindness, low vision, deaf and hard of hearing, speech and language disability
Intellectual and developmental	Intellectual disability, specific learning disabilities, autism spectrum disorder
Mental behaviour	Mental illness
Chronic neurological conditions	Multiple sclerosis, Parkinson's disease (and other chronic neurological conditions)
Blood disorders	Haemophilia, thalassaemia, sickle cell disease
Multiple	Multiple disabilities including deaf-blindness

- **Benchmark disability:** 40% or more of a specified disability, certified by a medical authority; needed for most reservations and benefits.
- **Other ways to classify:** congenital or acquired; temporary or permanent; visible or invisible; mild, moderate, severe, profound.

Primary and secondary disability (S16)

- **Primary disability:** the loss of function caused directly by the disease or injury (paraplegia after spinal cord injury; hemiplegia after stroke).
- **Secondary disability (secondary conditions):** further problems that develop later because of the primary disability, inactivity or poor care: contractures, pressure sores, obesity, osteoporosis, pain, urinary infections, depression, deconditioning. They are largely preventable, which is a major role of physiotherapy.

4 Evaluation of permanent physical impairment of the upper extremity

S23, 7 marks (with types of disability).

- Done when the condition has reached maximum improvement from treatment; based on measured function, not opinion.
- The upper limb has two parts: the **arm component** and the **hand component**, each worth 90%.

Component	What is measured	Values
Arm: range of motion	Shoulder, elbow and wrist, each joint 30%; mean percentage loss of range in each arc × 0.3, added across joints	Up to 90%
Arm: muscle strength	MRC grades 0–5 converted to percentage loss; mean loss around each joint × 0.3	Up to 90%
Arm: coordinated activities	Ten set activities, 9% each	Up to 90%
Hand: prehension	Opposition 8% (2% for each finger), lateral pinch 5% (holding a key), cylindrical grasp 6%, spherical grasp 6% (large and small objects), hook grasp 5% (lifting a bag)	30%
Hand: sensation	Complete loss: thumb ray 9%, index 6%, middle, ring and little 5% each; partial loss in proportion	30%
Hand: strength	Grip 20%, pinch 10% (dynamometer or clinical)	30%

- **Combining formula:** a + b(90 – a) ÷ 90, where a is the higher and b the lower value. Arm values (motion, strength, coordination) are combined this way; hand values (prehension, sensation, strength) are added; then arm and hand

are combined by the formula. Example: arm 27% and hand 64% give 64 + 27 × 26 ÷ 90 = 71.8%.

- **Extra weightage** of up to 10% for persistent pain, infection, deformity, malalignment, contracture, disfigurement, dominant limb (4%) or shortening (2% for each inch beyond the first); the total never exceeds 100%.

5 Levels of disability prevention

Levels of disability prevention: 3 marks in W16, W22 and W25; 7 marks in W20 and S26.

Level	Aim	Measures
Primary (first level)	Prevent the impairment from occurring	Immunisation (polio, measles, rubella), nutrition (iodine, vitamin A, folic acid, iron), antenatal care and safe delivery, avoiding consanguineous marriage and genetic counselling, road and workplace safety, helmets and seat belts, control of leprosy and diabetes, healthy lifestyle and exercise, health education
Secondary (second level)	Stop an impairment becoming a disability by early detection and treatment	Newborn and developmental screening (RBSK), early treatment of infections and injuries, early physiotherapy, positioning and splinting, early surgery (clubfoot, cataract), hearing and vision screening, screening for diabetic foot
Tertiary (third level)	Stop a disability becoming a handicap	Rehabilitation (medical, vocational, social), aids and appliances, prevention of secondary conditions, inclusive education, accessible buildings and transport, employment, legal rights and schemes, community awareness, CBR

These levels come from the WHO report on disability prevention and rehabilitation (1981) and match the general levels of prevention (see health and disease (p. 5)). The physiotherapist works at all three: health education and safety (primary), early detection and early therapy (secondary), and rehabilitation and inclusion (tertiary).

6 Exam corner

Past questions (MUHS, 2016–2026)

7 marks

Levels of disability prevention (W20, S26); types of disability and evaluation of permanent physical impairment of the upper extremity (S23).

3 marks

Primary and secondary disability (S16); define impairment, disability and handicap (W16, W25); levels of disability prevention (W16); full form of ICIDH and models of disability (W17); types of disabilities (S19); define disability and classify (W20); levels of prevention in community rehabilitation (W22); levels of prevention (W25).

Keywords to underline

ICIDH 1980 organ, person, society levels ICF 2001 umbrella term medical vs social model 21 specified disabilities benchmark 40% secondary conditions arm and hand components, 90% each combining formula primary, secondary, tertiary prevention

Answer skeleton: levels of disability prevention (7 marks)

1. **[Chain, 2]** Define the ICIDH terms; draw the chain with the levels.
2. **[Three levels, 4]** Aim and five measures for each.
3. **[PT role, 1]** One physiotherapy action at each level.

Draw these

- The ICIDH chain with the three levels of prevention (section 1)

- The ICF model with two-way arrows

Viva questions

1. Define handicap.
2. What does ICIDH stand for, and what replaced it?
3. Contrast the medical and social models.
4. How many specified disabilities are in the RPWD Act 2016? Name five.
5. What is a benchmark disability?
6. Give the combining formula for permanent physical impairment.

Linked practical spots

- Calculate an upper-limb impairment from given range, strength and hand values; fill a disability assessment proforma for a classmate's simulated case.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Define impairment, disability and handicap. (W16, W25)**

Answer
ICIDH (WHO, 1980). Impairment: any loss or abnormality of psychological, physiological or anatomical structure or function. Disability: any restriction or lack, resulting from an impairment, of ability to perform an activity in the normal manner or range. Handicap: a disadvantage, resulting from an impairment or disability, that limits or prevents fulfilment of a role normal for that individual. Example: polio, paralysed leg, cannot walk, cannot farm.

3 m **Note on primary and secondary disability. (S16)**

Answer
Primary: loss of function caused directly by the disease or injury (paraplegia after SCI). Secondary: later, preventable problems arising from the primary disability or inactivity (pressure sores, contractures, obesity, osteoporosis, UTI, depression). Physiotherapy prevents secondary disability by positioning, exercise, education and early mobility.

7 m **Write on levels of disability prevention. (W20, S26)**

Answer
Chain and definitions; primary (prevent impairment: immunisation, nutrition, antenatal care, safety, genetic counselling, health education); secondary (prevent impairment becoming disability: screening, early treatment, early PT and splinting, early surgery); tertiary (prevent disability becoming handicap: rehabilitation, aids, education, accessibility, employment, rights, CBR); PT at each level.

MCQ **In the ICIDH, handicap is a problem at the level of:**

A. The organ
B. The person
C. Society (role)
D. The cell

Answer
C.

MCQ **The social model sees disability mainly as caused by:**

A. Disease

B. Barriers in society

C. Genes

D. Old age

Answer
B.

MCQ **Early splinting and physiotherapy after polio paralysis is an example of:**

A. Primary prevention

B. Secondary prevention

C. Tertiary prevention

D. Primordial prevention

Answer
B. It stops the impairment becoming a disability.

Last recall check. Write the three ICIDH definitions, draw the chain with prevention levels, list six models and the 21 specified disabilities by group, and state the combining formula.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Community Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. World Health Organization. International Classification of Impairments, Disabilities, and Handicaps. Geneva: WHO; 1980.
3. World Health Organization. International Classification of Functioning, Disability and Health (ICF). Geneva: WHO; 2001.
4. World Health Organization. Disability prevention and rehabilitation: report of the WHO Expert Committee. Technical Report Series 668. Geneva: WHO; 1981.
5. Rights of Persons with Disabilities Act, 2016, Schedule (specified disabilities) and Section 2(r) (benchmark disability).
6. Ministry of Social Justice and Empowerment. Guidelines for evaluation of permanent physical impairment of the upper limb (2001), as compiled by the Office of the Chief Commissioner for Persons with Disabilities; superseded and carried forward by the notification of 4 January 2018 (S.O. 76(E)) under Section 56 of the RPWD Act.
7. United Nations. Convention on the Rights of Persons with Disabilities, 2006.

4(b, d) Rehabilitation: types, institution-based vs community-based rehabilitation and the rehabilitation team

> IBR versus CBR was asked six times in ten years (five 7-mark questions and a long answer), and the team members (NGO, social worker, occupational therapist, vocational counsellor) almost every year. Learn one comparison table, one team diagram and two or three lines on each member, and this page covers 15 of 20 papers.

Syllabus	Ch 4(b) Rehabilitation: definition, types (institutional, reach-out and community); 4(d) Rehab team work: medical practitioner, PT/OT, audiologist and speech therapist, prosthetist and orthotist, clinical psychologist, vocational counsellors and social workers
Priority	**Tier 1.** In 15 of 20 papers (2016–2026): 1 long answer, 11 seven-mark and 13 three-mark questions.
Asked as	LAQ S23 · 7m W16 · W17 ×3 · S19 · S21 · W21 · S23 · S24 · W24 · S25 · 3m S16 · W17 · S18 · W19 · S20 ×2 · W20 · S21 · W21 · S23 · S25 · S26 ×2
Usual split	15 (define rehabilitation 2, team effort in CBR 3, CBR vs IBR 10); 7 (IBR vs CBR; role of one team member; vocational guidance; ADL training)
Checked against	WHO definition of rehabilitation; ILO, UNESCO and WHO joint position paper on CBR (2004); ILO Convention 159 on vocational rehabilitation (1983); ADAPT (formerly the Spastics Society of India); Indian Red Cross Society Act, 1920; Rights of Persons with Disabilities Act, 2016; NLEP operational guidelines on disability prevention and medical rehabilitation (2007); British Association of Hand Therapists guidance on splinting. Full list under Sources (p. 143). Content checked 21 October 2026.

1 What rehabilitation is

Define rehabilitation: 3 marks in S20 and the first 2 marks of the S23 long answer.

> **Definition (WHO)**
>
> Rehabilitation is "a set of interventions designed to optimize functioning and reduce disability in individuals with health conditions in interaction with their environment".

- **Older WHO wording** (still quoted in Indian textbooks): all measures that reduce the impact of disabling and handicapping conditions and enable disabled people to achieve social integration.
- **Aim:** the highest possible independence in daily life, school, work and community, with the least restriction.
- **Kinds of rehabilitation:** medical (treatment, therapy, aids), vocational (work), social (family and community life), educational, and psychological.

Types by where it is given (syllabus 4b)

Type	What it means
Institution-based (IBR)	Hospitals, rehabilitation centres and special schools with specialist staff and equipment; the person comes to the centre
Outreach (reach-out)	Staff from the institution go out to the community: camps, mobile vans, home visits, satellite clinics
Community-based (CBR)	Rehabilitation within the person's own community, run with the person, the family and local workers, using local resources, with referral to institutions when needed (see CBR (p. 150))

2 Institution-based vs community-based rehabilitation

Asked in W16, W21, S23 (10 marks), S24, W24 and S25. Write the table, then two lines on why they complement each other.

> **CBR definition (ILO, UNESCO and WHO, 2004)**
>
> "CBR is a strategy within general community development for the rehabilitation, equalization of opportunities, poverty reduction and social inclusion of all people with disabilities." It is carried out "through the combined efforts of people with disabilities themselves, their families, organizations and communities, and the relevant governmental and non-governmental health, education, vocational, social and other services".

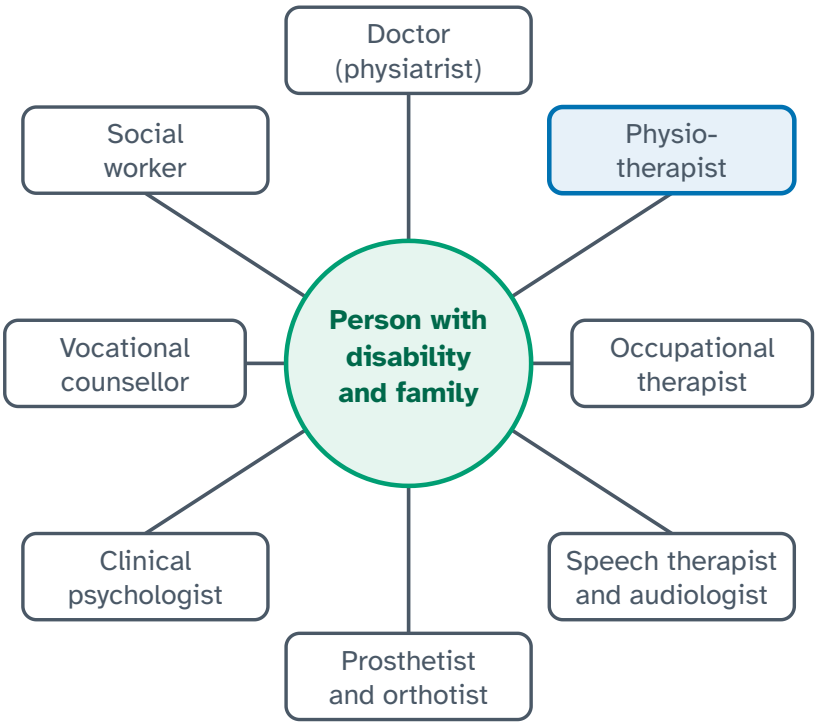
Feature	IBR	CBR
Place	Hospital, rehabilitation centre, special school, mostly in cities	Home, village, school, workplace
Who delivers	Specialists: doctors, therapists, prosthetists, psychologists	CBR workers, family, ASHA and anganwadi workers, teachers, volunteers, supported by professionals
Approach	Medical model: treats the impairment	Social and rights model: health, education, livelihood, social life and empowerment
Decisions	Made by professionals; the person is a patient	Made with the person and family, who are partners
Cost	High (buildings, equipment, staff, travel and stay for the family)	Low; uses local materials and existing services
Coverage	Few people, mainly urban and those who can travel	Many people, including rural and poor families
Technology	Advanced equipment, investigations and surgery	Simple, local, appropriate (aids made from bamboo or wood, home adaptations)
Skills learnt	In an artificial setting; may not carry over to home	In the real home and village, so they carry over
Continuity	Ends at discharge; follow-up is hard	Ongoing, sustained by the community
Social inclusion	May separate the person from family and community	Keeps the person in the family; changes community attitudes
Best for	Acute and complex care: surgery, intensive therapy, fitting of prostheses and orthoses, training of staff	Long-term needs, common disabilities, prevention, school and work inclusion
Limits	Expensive, crowded, far away, dependency	Limited specialist skill; depends on training, supervision and community commitment

Conclusion line for the exam: they are not rivals. CBR handles most needs locally and refers complex cases to institutions; institutions train CBR workers and provide specialist care and back-up. Outreach programmes link the two.

3 The rehabilitation team

Role of the occupational therapist (S26), the social worker (W20, S21, S26) and team effort in CBR (S23).

The rehabilitation team



The person and family (green) are members of the team and set the goals; the physiotherapist is shown in blue. Add the nurse, the special teacher and, in CBR, the CBR worker and community.

Member	Main role
Doctor (physiatrist)	Diagnosis, medical and surgical care, prognosis, disability certification, leads the team in a hospital
Physiotherapist	Movement and function: assessment, exercise, mobility and gait training, pain relief, chest care, aids and their training, home programmes, prevention
Occupational therapist	See below
Speech therapist and audiologist	Speech, language and swallowing therapy; hearing tests and hearing aids; alternative communication
Prosthetist and orthotist	Measures, makes, fits and repairs artificial limbs, callipers, splints and spinal braces
Clinical psychologist	Cognitive and IQ testing, counselling for adjustment, depression and anxiety, behaviour therapy, family counselling
Vocational counsellor	Assesses work ability and interests, guides to training and suitable jobs, helps placement (section 5)
Social worker	See below
Nurse	Day-to-day care, skin, bladder and bowel care, medicines, teaching the family

The occupational therapist (S26)

- Assesses and trains activities of daily living: feeding, dressing, bathing, toileting, cooking, using money and transport.
- Hand function: fine motor training, hand splints, sensory re-education.
- Adaptive devices (built-up spoons, button hooks, reachers, sock aids) and home, school and workplace modification.
- Cognitive and perceptual retraining (memory, attention, neglect) after brain injury and stroke.
- Pre-vocational and work skills, wheelchair seating, energy conservation, play and sensory integration in children.

The medical social worker (W20, S21; in CBR, S26)

- Psychosocial assessment: family, income, housing, education, work, community support.
- Counselling the person and family to accept the disability and take part in rehabilitation.
- Linking to benefits: disability certificate and UDID card, pensions, travel concessions, free aids and appliances (ADIP), scholarships, loans for self-employment (see policies (p. 144)).
- Arranging financial help through hospital funds, government schemes and NGOs.
- Discharge planning, home and community resources, return to school or work, follow-up.
- Advocacy against discrimination and abuse.
- **In CBR:** community survey to identify persons with disability, awareness meetings, forming self-help groups and CBR committees, linking families to schemes, livelihood and education, mobilising local resources.

How a team works

- **Multidisciplinary:** each professional assesses and treats separately with their own goals.
- **Interdisciplinary:** professionals meet, share findings and set common goals with the person.
- **Transdisciplinary:** one key worker (often the CBR worker or family) delivers the programme, trained and supported by the others; the usual model in CBR.

Why team effort matters in CBR (S23)

- Disability affects health, education, work and social life, so no single worker can meet all needs.
- Shared goals avoid duplication and conflicting advice; scarce professionals train and support many local workers.
- Referral in both directions between village and institution; continuity of care.
- The community owns the programme, which makes it cheaper and sustainable; the person and family are empowered.

4 Vocational rehabilitation, guidance and counselling

Vocational rehabilitation (W19, S21, 3 marks); significance of vocational guidance and counselling (W17, S23, 7 marks).

> **Purpose (ILO Convention 159, 1983)**
>
> "To enable a disabled person to secure, retain and advance in suitable employment and thereby to further such person's integration or reintegration into society."

1. **Vocational evaluation:** medical and physical capacity (functional capacity evaluation), education, skills, interests, aptitude, work history, local job market.
2. **Vocational guidance and counselling:** choose a realistic goal that matches ability and interest.
3. **Vocational training:** skill training in industrial training institutes, Vocational Rehabilitation Centres for Handicapped (Ministry of Labour and Employment), NGOs; work conditioning and work hardening (see disability management (p. 186)).
4. **Placement:** return to the old job with changes, open employment (4% of government posts reserved under the RPWD Act, 2016), sheltered workshop, supported employment, or self-employment with concessional loans (National Divyangjan Finance and Development Corporation).
5. **Follow-up** to solve problems at work and keep the job.

Significance of vocational guidance and counselling

- Matches the job to the person's remaining abilities, so training is not wasted and failure is avoided.
- Builds motivation, self-esteem and acceptance of disability; reduces depression.
- Leads to earning and economic independence; reduces dependence on family and state.

- Gives a social role and dignity, the final aim of rehabilitation (participation).
- Guides the family and employer; identifies adaptations, aids and job modifications.
- Informs about schemes, reservation, loans and training centres.

5 ADL training and the disabled hand

Training for activities of daily living with examples, and management of the disabled hand (both W17, 7 marks).

ADL training

- **Basic ADL:** feeding, grooming, bathing, dressing, toileting, transfers, walking or wheelchair mobility. **Instrumental ADL:** cooking, shopping, housework, money, phone, transport.
- **Steps:** assess (Barthel Index, FIM); set goals with the person; analyse the task and break it into steps; teach step by step (forward or backward chaining), in the real setting and at the usual time; use the easiest method first and fade help; add aids and change the environment; train the family; review.
- **Examples:** hemiplegia: dress the affected arm first and undress it last, sit to dress, elastic laces; paraplegia: sliding-board transfer to the toilet, long-handled mirror for skin checks; rheumatoid hands: built-up handles, lever taps; low vision: tactile marks on the stove.
- Principles of energy conservation and joint protection for long-term conditions.

Management of the disabled hand (7 marks in W17)

The hand is the main tool for self-care and work, and it is also a sense organ. A hand that cannot move, grip or feel limits eating, dressing, washing and earning. Common causes in the community are nerve injury, leprosy, burns, crush and machine injuries, stroke, cerebral palsy, rheumatoid arthritis and amputation.

Management follows three steps: restore what can recover, protect what cannot feel or is at risk of deformity, and adapt the task or tool for what cannot be restored.

- **Treat:** pain and oedema control (elevation, retrograde massage), mobilisation and stretching, strengthening, splints, sensory re-education and protection of anaesthetic hands, prehension and dexterity training (pegboards, picking up objects of different sizes).
- **Adapt:** built-up and lever handles, universal cuff, one-handed techniques, tools for work; vocational adjustment.

Assessment of hand function

Area	How to assess
History and look	Cause, hand dominance, job and tools, pain; posture and deformity (claw hand, wrist drop, flat "ape" thumb), wasting, scars, skin cracks, swelling
Movement	Goniometry of the wrist and each finger and thumb joint; oedema with a tape or a water-displacement volumeter
Strength	Grip with a hand dynamometer: sitting, arm by the side, elbow at **90°**, forearm neutral, mean of **3 trials**; pinch gauge for tip, key and three-jaw pinch; muscle testing of little finger abduction (ulnar), thumb abduction (median) and wrist extension (radial)
Sensation	Light touch with Semmes–Weinstein monofilaments; static and moving two-point discrimination; protective sensation of the palm in leprosy and diabetes
Dexterity and function	Nine-Hole Peg Test, Jebsen–Taylor Hand Function Test; each grip tested on real objects; the DASH questionnaire for daily and work activities

Functional grips

Napier (1956) divided grasp into two patterns. In a power grip the object is clamped between the flexed fingers and the palm, with the thumb adding counter-pressure. In a precision grip it is held between the pads of the fingers and the opposed thumb for fine control.

Grip	Everyday example	Lost in
Cylindrical (power)	Glass, tool handle, broom	Weak finger flexors, stiff fingers, burn contracture
Spherical (power)	Ball, fruit, door knob	Weak intrinsics, stiff fingers
Hook	Carrying a bag or bucket	Weak finger flexors
Tip-to-tip pinch (precision)	Picking up a pin or a grain	Median palsy, loss of fingertip sensation
Three-jaw (pad-to-pad) pinch	Holding a pen	Median palsy, stiff fingers
Lateral (key) pinch	Turning a key, holding a card	Ulnar palsy (weak adductor pollicis)

Train each grip with graded objects: large to small, light to heavy, soft to hard. The same grips are scored for the disability certificate (see evaluation of the upper limb (p. 129)).

Splinting

- **Purposes:** relieve pain; position the hand for function (nerve palsy); protect after injury or surgery; prevent or correct deformity; control oedema; keep or regain range; remodel burn scars.
- **Static splints** allow no movement (for example a plaster slab). **Dynamic splints** use elastic or springs and allow movement; they need skilled making and close checking.
- In villages, simple splints can be made from plastic or rubber tubing, wood or bamboo. Check the skin under every splint daily, especially if the hand cannot feel.

Problem	Splint	Why
Radial nerve palsy (wrist drop)	Cock-up (wrist extension) splint	Holds the wrist up so the fingers can grip
Ulnar palsy or leprosy claw hand	Knuckle-bender (anti-claw) splint holding the knuckles bent	With the knuckles blocked in flexion, the long extensors can straighten the fingers; suits a mobile claw
Median nerve palsy	Thumb web spacer (C-bar) or opposition splint	Keeps the first web space open so the thumb can oppose
Burns, fractures, hand infection	Position of safe immobilisation: wrist `20–30°` extension, knuckles `70–90°` flexion, finger joints straight, thumb web open	Keeps the collateral ligaments and capsules long, so the hand does not stiffen
Rheumatoid arthritis	Resting splint at night; wrist support for work	Pain relief, rest of inflamed joints, prevention of deformity
Leprosy reaction with painful nerves	Well-padded resting splint	Rests the nerve and relieves pain; unsplinted limbs develop contractures

Adaptive devices and ADL training for the hand

- **Weak grip:** built-up or padded handles (foam, cloth, bamboo) on spoons, pens and tools; lever taps and door handles; a non-slip mat under the plate.
- **No grip:** a universal cuff holds a spoon, pen or toothbrush in the palm.
- **Fine tasks:** button hook, Velcro fastenings, large-handled keys, a mobile phone with large keys or voice use.
- **One hand only:** one-handed methods (stabilise the object with the body, a clamp or a non-slip mat), a rocker knife.
- **Train the real tasks** she does each day, in her home: eating with the fingers, cooking, washing clothes, toileting, dressing. Use task analysis and chaining as above.

Sensory re-education and the anaesthetic hand

- **Sensory re-education, stage 1 (localisation):** starts once protective sensation returns after nerve repair. With eyes closed, a pen is rubbed on one spot and she names the spot. She opens her eyes to check, feels it again while watching, then repeats with eyes closed.

- **Stage 2 (textures and shapes):** she identifies objects of different size, weight and texture with eyes closed or inside a bag, large ones first. She checks with the other hand and by looking, then tries again. Practise daily as the therapist advises. Earlier methods, such as mirror therapy from the first weeks, are being tried; the evidence for sensory re-education is limited (Oud et al., 2007; Xia et al., 2021).
- **Self-care of the anaesthetic hand** (leprosy programme, NLEP): inspect the hands daily for hot, tender spots and injuries; soak them in water for about **half an hour**; rub off hard skin with a smooth stone; apply oil while the skin is still wet; protect from heat, pressure and friction (padded or cloth-wrapped handles, protective aids).
- **Blister or wound:** do not burst a blister; pad it and rest the part; rest a wound with a splint. Self-care groups help people keep up these habits.
- **Paralysed hand:** oil massage and movement to keep joints mobile; for a claw hand, hold the knuckles bent with the other hand and straighten the fingers repeatedly.
- **Reconstructive surgery** (tendon transfer) is offered for a mobile claw hand at recognised NLEP centres when paralysis has lasted at least **6 months**, drug treatment is complete and there is no reaction. Exercise before and after surgery is essential.

Work and vocational aspects

- Analyse the job: grips, force, repetition, tools and heat (see job analysis (p. 179)).
- Graded work simulation and conditioning before return to work (see work hardening (p. 186)).
- Modify the tool or the task: thicker padded handles, lighter tools, insulated handles for a hand that cannot feel heat.
- If the old job is not possible: vocational guidance, training and placement (section 4), including one-handed work or self-employment.
- Certify the hand disability when maximum improvement is reached, to claim schemes, concessions and reservation (see evaluation (p. 129)).

6 NGOs, the Spastics Society of India and the Red Cross

Role of NGOs: S16, W17, S23 and S25 (3 marks) and S19 (7 marks); Spastics Society of India (S18); Red Cross (S20, W21).

- **NGO:** a voluntary, non-profit organisation run independently of the government, registered as a society or trust. Examples: ADAPT (formerly the Spastics Society of India), Indian Red Cross Society, HelpAge India, Bhagwan Mahaveer Viklang Sahayata Samiti (Jaipur foot), Amar Seva Sangam, Mobility India, CBM.
- **Role in rehabilitation and community welfare:**
  - Surveys and early identification; awareness to reduce stigma.
  - Services where government services do not reach: therapy, camps, aids and appliances, special and inclusive schools.
  - Training CBR workers, parents and volunteers.
  - Vocational training, self-help groups and livelihood.
  - Advocacy for rights, accessibility and policy; linking people to government schemes.
  - Raising funds; research; disaster relief.
  - Flexible and close to the community, so they often pilot new programmes that government later adopts.

Spastics Society of India (now ADAPT)

- Started on 2 October 1972 in Mumbai by Dr Mithu Alur to provide education and treatment for children with cerebral palsy; now called ADAPT (Able Disabled All People Together).
- **Functions:** special and inclusive education; therapy (physiotherapy, occupational and speech therapy); training courses for teachers and therapists; vocational training and placement of young adults; early intervention; advocacy for inclusive education; outreach in the slums and villages; research and publications.

Red Cross Society

- The Red Cross movement began with Henry Dunant after the battle of Solferino; the International Committee was founded in Geneva in 1863. The Indian Red Cross Society was set up by an Act of 1920; the President of India is its president.
- **Seven principles:** humanity, impartiality, neutrality, independence, voluntary service, unity, universality.
- **Functions:** disaster preparedness and relief; blood donation and blood banks; first aid training; ambulance and hospital services; health education and maternal and child welfare; help for war victims and tracing of missing family members; care of the sick and disabled; Junior and Youth Red Cross in schools and colleges.

7 Model long answer: rehabilitation, team effort and CBR vs IBR

From a real paper (S23, 2 + 3 + 10 marks)

> "Define rehabilitation. Importance of team effort in Community Based Rehabilitation. Differentiate between Community Based Rehabilitation and Institutional Based Rehabilitation."

1. **[Definition, 2]** The WHO definition (section 1) and the kinds: medical, vocational, social, educational.
2. **[Team effort, 3]** Team members around the person and family (draw the diagram); the transdisciplinary model; four reasons team effort matters (section 3).
3. **[CBR vs IBR, 10]** Define CBR (2004 wording); the comparison table with at least ten rows; the conclusion that the two work together through referral and outreach.

No patient case has been set on this topic; in a long case about any disabled person, use section 3 to name who else the person needs, and section 4 for return to work.

8 Exam corner

Past questions (MUHS, 2016–2026)

IBR and CBR

Difference between IBR and CBR (W16, S24, S25, 7 marks); compare institution-based with community-based rehabilitation (W21, W24, 7 marks); define rehabilitation, team effort in CBR, CBR vs IBR (S23, long answer).

Team

Role of the medical social worker in rehabilitation (W20, 3 marks; S21, 7 marks); role of the social worker in CBR and of the occupational therapist in the team (S26, 3 marks); role of NGOs in the rehabilitation team (S19, 7 marks; S25, 3 marks).

Work and daily life

Vocational guidance and counselling (W17, S23, 7 marks); vocational rehabilitation (W19, S21, 3 marks); ADL training with examples and management of the disabled hand (W17, 7 marks).

Organisations

Role of NGOs in community welfare (S16) and in CBR (S23); what is an NGO (W17); Spastics Society of India (S18); functions of the Red Cross (S20, W21); define rehabilitation (S20).

Keywords to underline

optimize functioning institution-based outreach community-based local resources coverage and cost social inclusion referral transdisciplinary vocational evaluation placement advocacy

Answer skeleton: IBR vs CBR (7 marks)

1. **[Definitions, 1]** Rehabilitation; CBR.
2. **[Table, 5]** At least eight rows: place, who, approach, decisions, cost, coverage, technology, continuity, inclusion.
3. **[Conclusion, 1]** They complement each other through referral and outreach.

Draw these

- The rehabilitation team around the person and family (section 3)
- IBR, outreach and CBR as three circles from the institution to the village

Viva questions

1. Define rehabilitation.
2. Give three differences between IBR and CBR.
3. What does an occupational therapist do in the team?
4. What is the difference between interdisciplinary and transdisciplinary teams?
5. What are the steps of vocational rehabilitation?
6. Name two NGOs working for persons with disability in India.
7. Name three power grips and three precision grips, with an everyday example of each.

Linked practical spots

- Identify ADL aids (button hook, reacher, sock aid, built-up spoon, universal cuff) and say who uses them; describe the team for a patient in your long case.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Role of the medical social worker in rehabilitation. (W20)**

Answer

Psychosocial assessment of family, income and housing; counselling the person and family; linking to disability certificate, UDID, pensions, concessions and free aids; arranging finance through schemes and NGOs; discharge planning and return to school or work; advocacy; follow-up.

3 m **Functions of the Red Cross Society. (S20, W21)**

Answer

Founded on Henry Dunant's idea (ICRC, Geneva, 1863); Indian Red Cross by an Act of 1920 with the President of India as president. Functions: disaster relief and preparedness, blood banks, first aid training, ambulance and hospital services, health education and maternal and child welfare, help for war victims and tracing, Junior and Youth Red Cross. Seven principles: humanity, impartiality, neutrality, independence, voluntary service, unity, universality.

7 m **Significance of vocational guidance and counselling in rehabilitation. (W17, S23)**

Answer
Define vocational rehabilitation (ILO 159: to secure, retain and advance in suitable employment). Steps: evaluation, guidance and counselling, training, placement, follow-up. Significance: matches the job to remaining ability; prevents failure and wasted training; builds motivation and acceptance; leads to earning and independence; gives social role and dignity; guides family and employer about adaptations; informs about reservation (4% under the RPWD Act), training centres and loans.

LAQ **Define rehabilitation. Importance of team effort in CBR. Differentiate between CBR and IBR. (2 + 3 + 10, S23)**

Answer
Follow section 7: the WHO definition; the team diagram with the transdisciplinary model and reasons for team effort (needs span health, education, work and social life; shared goals; training local workers; referral; community ownership); the CBR definition and a ten-row table (place, who, approach, decisions, cost, coverage, technology, carry-over, continuity, inclusion, best use, limits); conclusion that IBR and CBR complement each other.

MCQ **Which is a feature of community-based rehabilitation?**

A. High cost and advanced technology

B. Decisions made only by specialists

C. Use of local resources with the family as partners

D. Services only in city hospitals

Answer
C.

MCQ **The Spastics Society of India was founded mainly for children with:**

A. Polio

B. Cerebral palsy

C. Blindness

D. Hearing loss

Answer
B. Founded in 1972 by Mithu Alur; now ADAPT.

MCQ **In a transdisciplinary team:**

A. Each professional works alone

B. One key worker delivers the programme with support from the others

C. Only doctors set goals

D. The family is excluded

Answer
B.

Last recall check. Write the WHO definition, draw the team, and fill an IBR vs CBR table with eight rows from memory.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Community Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. World Health Organization. Rehabilitation (health topic page): definition of rehabilitation.
3. ILO, UNESCO and WHO. CBR: a strategy for rehabilitation, equalization of opportunities, poverty reduction and social inclusion of people with disabilities. Joint position paper, 2004.
4. International Labour Organization. Vocational Rehabilitation and Employment (Disabled Persons) Convention, 1983 (No. 159), Article 1.
5. ADAPT (Able Disabled All People Together), formerly the Spastics Society of India: history.
6. Indian Red Cross Society Act, 1920 (Act XV of 1920); Fundamental Principles of the International Red Cross and Red Crescent Movement.
7. Rights of Persons with Disabilities Act, 2016, Section 34 (reservation in government posts).
8. Napier JR. The prehensile movements of the human hand. *J Bone Joint Surg Br* 1956;38-B:902–13 (power and precision grip; checked through later descriptions).
9. Roberts HC, Denison HJ, Martin HJ, et al. A review of the measurement of grip strength in clinical and epidemiological studies: towards a standardised approach. *Age Ageing* 2011;40:423–9; American Society of Hand Therapists test position (seated, elbow at 90°, forearm neutral, mean of three trials) checked through published studies that use it.
10. Jebsen RH, Taylor N, Trieschmann RB, et al. An objective and standardized test of hand function. *Arch Phys Med Rehabil* 1969;50:311–9; Hudak PL, Amadio PC, Bombardier C. Development of an upper extremity outcome measure: the DASH. *Am J Ind Med* 1996;29:602–8.
11. British Association of Hand Therapists. Principles of splinting (training handout), 2019: purposes of splints and the position of safe immobilisation.
12. National Leprosy Eradication Programme, Ministry of Health and Family Welfare. Disability prevention and medical rehabilitation: operational guidelines, primary level care. New Delhi, 2007 (self-care of anaesthetic hands, static and dynamic splints, exercises for claw hand, criteria for reconstructive surgery, voluntary muscle testing).
13. Oud T, Beelen A, Eijffinger E, Nollet F. Sensory re-education after nerve injury of the upper limb: a systematic review. *Clin Rehabil* 2007;21:483–94; Xia W, Bai Z, Dai R, et al. The effects of sensory re-education on hand function recovery after peripheral nerve repair: a systematic review. *NeuroRehabilitation* 2021;48:293–304.
14. Royal Devon and Exeter NHS Foundation Trust. Sensory re-education (patient information leaflet RDE 21 086 001), 2021; Paula MH, et al. Early sensory re-education of the hand after peripheral nerve repair based on mirror therapy: a randomized controlled trial. *Braz J Phys Ther* 2016;20:58–65.

4(c) National policies and legal rights for persons with disability: the Acts, national programmes, schemes, concessions and architectural barriers

> Policies were asked in 7 of 20 papers: the PWD Act 1995 three times, three national programmes three times, barriers, difficulties, and a long case asking for transport and concessions for a paraplegic farmer. Learn the two Acts side by side, five programmes with their years, and a concession list you can apply to any disabled patient in a long case.

Syllabus	Ch 4(c) National policies for rehabilitation of persons with disability
Priority	**Tier 1.** In 7 of 20 papers (2016–2026): 1 long case, 1 seven-mark and 7 three-mark questions.
Asked as	LAQ S18 · 7m S18 · 3m W17 · S18 · S20 · W21 · S23 · S24 · W24
Usual split	15 (functional diagnosis 5, transport 5, concessions 5); 7 (architectural barriers and modifications); 3 (PWD Act; three national programmes; difficulties of a person with disability)
Checked against	Persons with Disabilities Act, 1995; Rights of Persons with Disabilities Act, 2016 (Sections 31, 32, 34); Indian Railways concession rules for persons with disability; Harmonised Guidelines and Standards for Universal Accessibility in India (2021); Income Tax Act Section 80U. Full list under Sources (p. 149). Content checked 20 October 2026.

1 The Acts

Persons with Disability Act (S18, S23, S24, 3 marks).

Feature	PWD Act, 1995	RPWD Act, 2016
Full name	Persons with Disabilities (Equal Opportunities, Protection of Rights and Full Participation) Act	Rights of Persons with Disabilities Act (in line with the UN Convention, 2006); replaced the 1995 Act
Disabilities covered	7: blindness, low vision, leprosy-cured, hearing impairment, locomotor disability, mental retardation, mental illness	21 specified disabilities (see types (p. 128))
Benchmark	40% or more	40% or more of a specified disability
Government jobs	3% reserved	4% reserved
Education	Free education up to 18 years; 3% seats in government-aided institutions	Free education for children with benchmark disability aged 6–18; at least 5% seats in higher education; inclusive education
Other provisions	Prevention and early detection, non-discrimination in transport and buildings, aids, unemployment allowance, Chief and State Commissioners	Rights to equality, non-discrimination, accessibility (buildings, transport, ICT), health care, social security; guardianship; special courts; penalties for offences

Other laws

- **Rehabilitation Council of India Act, 1992:** regulates training and registration of rehabilitation professionals (special educators, audiologists, prosthetists, clinical psychologists).
- **National Trust Act, 1999:** welfare of persons with autism, cerebral palsy, intellectual disability and multiple disabilities; legal guardianship; schemes such as Niramaya health insurance.
- **Mental Healthcare Act, 2017:** rights of persons with mental illness.
- **National Policy for Persons with Disabilities, 2006:** prevention, rehabilitation, education, employment, barrier-free environment.

2 National programmes for prevention of disability

Three national programmes for prevention of disability (S20, W21, W24, 3 marks).

Programme	Prevents
Universal Immunisation Programme (1985) and Pulse Polio (1995)	Polio paralysis, and disability from measles, rubella (congenital rubella), diphtheria, tetanus
National Programme for Control of Blindness and Visual Impairment (1976)	Blindness from cataract, refractive error, glaucoma, childhood blindness
National Programme for Prevention and Control of Deafness (2006)	Hearing loss: early detection, treatment, hearing aids
National Leprosy Eradication Programme (1983)	Deformity from leprosy through early detection, multidrug therapy and disability care
National Iodine Deficiency Disorders Control Programme	Intellectual disability and goitre from iodine deficiency (iodised salt)
Rashtriya Bal Swasthya Karyakram (2013)	Disability in children through screening for defects, diseases, deficiencies and developmental delays, with early intervention centres
National Programme for Health Care of the Elderly (2010)	Disability in older people: geriatric clinics, rehabilitation, home care
NCD programme and National Mental Health Programme (1982)	Stroke and diabetic complications; mental illness

3 Schemes and concessions

Concessions offered by government (S18 long case, 5 marks; the W18 amputee case asks the same).

Area	Benefit
Identity and certificate	Disability certificate from the medical board and the Unique Disability ID (UDID) card, needed for most benefits
Aids and appliances	ADIP scheme: free or subsidised wheelchairs, tricycles, motorised tricycles, callipers, prostheses, hearing aids through camps and centres
Rail travel	Orthopaedically handicapped or paraplegic persons who cannot travel without an escort: 75% off the basic fare in second class, sleeper, first class, 3-tier AC and chair car, 50% in first and second AC, the escort gets the same; reserved berths
Road travel	State bus fare concessions or free passes (vary by state); concession for adapted vehicles
Employment	4% reservation in government posts; vocational rehabilitation centres; age relaxation
Education	Free education to 18, 5% seats in higher education, scholarships, inclusive education support
Money	Disability pension (Indira Gandhi National Disability Pension Scheme for BPL persons with severe disability; state pensions such as Sanjay Gandhi Niradhar Anudan Yojana in Maharashtra); concessional loans (NDFDC); income tax deduction under Section 80U (₹75,000, or ₹1,25,000 for severe disability)
Housing and access	Priority in housing schemes; Accessible India Campaign (Sugamya Bharat Abhiyan, 2015) for accessible buildings, transport and websites
Rehabilitation services	District Disability Rehabilitation Centres, composite regional centres, national institutes; Deendayal Disabled Rehabilitation Scheme grants to NGOs

4 Difficulties and architectural barriers

Difficulties faced by a community-dwelling person with disability (W17, 3 marks); architectural barriers and their modifications (S18, 7 marks).

Difficulties in the community

- **Physical:** steps, narrow doors, inaccessible toilets, uneven roads, no ramps or lifts.
- **Transport:** buses and autos hard to board; cost of private transport.
- **Attitudes:** stigma, pity, discrimination, overprotection; difficulty in marriage.
- **Economic:** unemployment, poverty, cost of aids and treatment.
- **Services:** distant health and rehabilitation centres; lack of information about schemes; paperwork for certificates.
- **Education and communication:** inaccessible schools; communication barriers for deaf and blind persons.
- **Social:** isolation, dependence on family, risk of abuse.

Architectural barriers and modifications

Barrier	Modification
Steps at the entrance	Ramp with a gradient no steeper than 1:12, clear width at least 1,200 mm, handrails on both sides (about 760–900 mm high) extending beyond the ends, landing at top and bottom
Narrow doors	Clear width of about 900 mm; lever handles; no high thresholds
No space to turn a wheelchair	A clear turning circle of 1,500 mm in rooms, toilets and corridors
Inaccessible toilet	Western toilet at wheelchair seat height, grab bars, washbasin without pedestal, outward-opening door, non-slip floor
Upper floors	Lift with wide doors and low buttons; ground-floor rooms
High counters and switches	Lowered counters, switches and handles within reach from a wheelchair
Pavements and roads	Kerb cuts, even paths, tactile paving for blind persons, audible signals
Signs and information	Large clear signs with contrast, Braille, sign-language information

Measurements from the Harmonised Guidelines and Standards for Universal Accessibility in India (2021) and standard wheelchair space needs.

5 Worked long case: a farmer with paraplegia going home

From a real paper (S18, 5 + 5 + 5)

> "A 35 years old male, farmer, diagnosed as a case of T12-L1 injury is discharged from the hospital. i) Mention the functional diagnosis (5) ii) Discuss the modes of transport for him in the community iii) Explain the concessions offered to him (5)"

Spot the clues before you write

T12–L1 injury: paraplegia with conus or cauda equina involvement, so bladder and bowel problems; upper limbs and trunk strong. farmer: rural home, uneven fields, needs a new way to earn. discharged: community phase; transport and benefits now matter.

Functional diagnosis (5 marks)

A 35-year-old farmer with paraplegia after a T12–L1 spinal cord injury (classify with the ASIA scale), independent in upper limbs and trunk, with neurogenic bladder and bowel, at risk of pressure sores; he can be independent in wheelchair mobility and transfers and may walk short distances with orthoses; his rural home, fields and roads limit mobility, and he cannot continue his usual farm work (see spinal cord injury).

Modes of transport (5 marks)

Mode	Use and training
Manual wheelchair	Home and village; rigid or folding, with pneumatic tyres and a pressure-relief cushion for rough roads; wheelie skills for kerbs
Hand-propelled tricycle	Longer distances on village roads and to the fields; cheap and available through ADIP camps
Motorised tricycle or scooter with side wheels	Travel to the market, PHC or town; hand controls; driving licence for an adapted vehicle
Car with hand controls	If affordable; transfer and wheelchair loading training
Public transport	Low-floor buses where available; train travel with concession, station wheelchairs and help; autorickshaw transfers practised
Walking with orthoses	Knee–ankle–foot orthoses and crutches or a walker for short distances and standing, not for daily long distances

Concessions (5 marks)

- Disability certificate and UDID card (benchmark 40% or more).
- Free or subsidised wheelchair, tricycle, motorised tricycle, orthoses and cushions under ADIP.
- Railway concession (75% in second class and sleeper with the same for his escort) and state bus concessions.
- Disability pension if eligible; income tax deduction; concessional loans from NDFDC for self-employment (dairy, poultry, shop).
- Vocational training and 4% job reservation; priority in housing and accessible toilet schemes; panchayat help to build a ramp and toilet.

6 Exam corner

Past questions (MUHS, 2016–2026)

Long case

35-year-old farmer with T12–L1 injury: functional diagnosis, transport in the community, concessions (S18).

7 marks

Architectural barriers faced by persons with disability and modifications (S18).

3 marks

Difficulties faced by a community-dwelling person with disability (W17); Persons with Disability Act (S18, S23, S24); three national programmes for prevention of disability (S20, W21, W24).

Keywords to underline

PWD Act 1995: 7 disabilities, 3% RPWD Act 2016: 21 disabilities, 4% benchmark 40% UDID ADIP NPCBVI, NPPCD, NLEP, UIP, RBSK railway concession 75% ramp 1:12 turning circle 1,500 mm Accessible India Campaign

Answer skeleton: architectural barriers (7 marks)

1. **[Introduction, 1]** Barriers as a cause of handicap (social model); RPWD Act accessibility duty.
2. **[Barriers and modifications, 5]** Entrance, doors, turning space, toilet, floors, lifts, counters, roads, signs, with measurements.
3. **[Role of PT, 1]** Home and building access audits and advice.

Draw these

- A ramp with 1:12 gradient, handrails and landings
- An accessible toilet with grab bars and turning circle

Viva questions

1. How many disabilities did the PWD Act 1995 cover? Name them.
2. What is the job reservation under the RPWD Act 2016?
3. Name three national programmes that prevent disability.
4. What is the ADIP scheme?
5. What ramp gradient is recommended?
6. What railway concession is available to a paraplegic person?

Linked practical spots

- Do an accessibility audit of your college entrance and toilet; measure a ramp gradient; fill a disability certificate application with a classmate.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Persons with Disability Act 1995. (S23, S24)**

Answer

Full name; 7 disabilities (blindness, low vision, leprosy-cured, hearing impairment, locomotor, mental retardation, mental illness); 40% benchmark; 3% job reservation; free education to 18; prevention and early detection; non-discrimination in transport and buildings; aids and unemployment allowance; Chief and State Commissioners. Replaced by the RPWD Act 2016 (21 disabilities, 4%).

3 m **Three national programmes for prevention of disability. (S20, W21, W24)**

Answer

Universal Immunisation Programme (1985) with Pulse Polio (1995): prevents polio and other vaccine-preventable disability. National Programme for Control of Blindness and Visual Impairment (1976): cataract surgery, spectacles. National Programme for Prevention and Control of Deafness (2006): early detection, hearing aids. (Also NLEP, NIDDCP, RBSK.)

7 m **Architectural barriers faced by persons with disability and their modifications. (S18)**

Answer
Barriers: steps, narrow doors, no turning space, inaccessible toilets, upper floors without lifts, high counters, uneven roads without kerb cuts, poor signs. Modifications: ramps (1:12, 1,200 mm wide, handrails both sides), 900 mm doors with lever handles, 1,500 mm turning circle, accessible toilets with grab bars, lifts, lowered counters, kerb cuts and tactile paving, clear and Braille signage. Mention the RPWD Act duty and the Accessible India Campaign.

MCQ **Under the RPWD Act 2016, reservation in government jobs for persons with benchmark disabilities is:**

A. 1%

B. 3%

C. 4%

D. 5%

Answer
C.

MCQ **The maximum recommended gradient for a wheelchair ramp is:**

A. 1:4

B. 1:8

C. 1:12

D. 1:30

Answer
C.

MCQ **The scheme that provides free aids and appliances to persons with disability is:**

A. ADIP

B. RBSK

C. ICDS

D. NPHCE

Answer
A.

Last recall check. Compare the two Acts in six rows, name five programmes with years, list eight concessions, and draw a ramp with its measurements.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Community Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. The Persons with Disabilities (Equal Opportunities, Protection of Rights and Full Participation) Act, 1995.
3. The Rights of Persons with Disabilities Act, 2016: Sections 31 (free education), 32 (higher education), 34 (reservation in posts).
4. Indian Railways. Concessions for persons with disabilities (Railway Board commercial circulars).
5. Ministry of Housing and Urban Affairs and CPWD. Harmonised Guidelines and Standards for Universal Accessibility in India, 2021.
6. Income Tax Act, 1961, Section 80U.
7. Department of Empowerment of Persons with Disabilities. ADIP scheme, UDID project and Accessible India Campaign.

4(e, f) Community-based rehabilitation: principles, the CBR matrix, strategies, the anganwadi worker and the physiotherapist

> CBR was asked in 11 of 20 papers: principles four times, the anganwadi worker three times, strategies, approaches, teamwork, and a long answer on a child with cerebral palsy in a village. CBR is not a cheaper copy of hospital rehabilitation; it is rehabilitation that the person, the family and the village own, with professionals as trainers and back-up.

Syllabus	Ch 4(e) CBR: role of physiotherapy and the physiotherapist; 4(f) CBR strategies in urban areas (UHC, community centres, clubs, mahila mandals, social centres, schools, industries, sports centres) and rural areas (PHC, rural hospital and district hospital infrastructure; locomotor aids from local resources)
Priority	**Tier 1.** In 11 of 20 papers (2016–2026): 1 long answer, 7 seven-mark and 7 three-mark questions.
Asked as	LAQ S22 · 7m S17 · W19 · S20 · W20 · W22 · S26 ×2 · 3m S16 · S17 · W21 · S22 ×2 · W24 · S25
Usual split	15 (local resources 5, CBR plan 10); 7 (principles and team; strategies; teamwork; approaches; anganwadi worker; PT in a special school)
Checked against	ILO, UNESCO and WHO joint position paper on CBR (2004); WHO CBR guidelines and matrix (2010); Ministry of Women and Child Development norms for anganwadi centres; Rashtriya Bal Swasthya Karyakram (RBSK) and District Early Intervention Centres; Rights of Persons with Disabilities Act, 2016 (sections 16, 17 and 31); Samagra Shiksha inclusive education component (Ministry of Education, 2024); WHO 2020 guidelines on physical activity. Full list under Sources (p. 159). Content checked 21 October 2026.

1 What CBR is

Aspects of CBR (S16), three features of CBR (W24), 3 marks; a note on CBR (W20, 7 marks).

- **Definition (ILO, UNESCO and WHO, 2004):** "a strategy within general community development for the rehabilitation, equalization of opportunities, poverty reduction and social inclusion of all people with disabilities", carried out by people with disabilities, their families and communities together with health, education, vocational and social services.
- **History:** WHO began promoting CBR after the Alma-Ata declaration on primary health care (1978), because most disabled people in low-income countries had no access to institutions; the joint position papers (1994, 2004) widened it from medical care to rights and inclusion; the WHO CBR guidelines (2010) set out the CBR matrix.
- Compare with institution-based rehabilitation on IBR vs CBR (p. 133).

Features (aspects) of CBR

- Takes place in the person's own home, village or town, using local resources and simple technology.
- The person with disability and the family are active partners and decide goals.
- Community workers (CBR workers, anganwadi workers, ASHAs, teachers, volunteers) deliver most of the work, trained and supported by professionals.
- Covers health, education, livelihood, social life and empowerment, not only therapy.
- Works with existing services (primary health centres, schools, panchayats, government schemes) and refers to institutions when needed.
- Low cost, wide coverage, sustained by the community.

2 Principles of CBR

Principles of CBR: 3 marks in W21, S22 and S25; with team members, 7 marks in S17.

Principle	Meaning in practice
Community participation and ownership	The village identifies needs, plans and runs the programme (CBR committee, gram panchayat)
Inclusion and equal opportunity	Persons with disability in ordinary schools, jobs, festivals and decisions; no discrimination
Empowerment and self-advocacy	People with disabilities speak for themselves through self-help groups and disabled people's organisations
Use of local resources and appropriate technology	Aids made by the village carpenter or blacksmith; home exercise with household objects
Multisectoral approach	Health, education, social welfare, livelihood and panchayat work together
Integration with existing services	Through primary health care, anganwadis and schools rather than separate systems
Referral system	Two-way links with PHC, district hospital, DDRC and national institutes
Sustainability	Low cost, trained local people, community funds; continues after outside support ends
Gender equality and respect for children	Equal attention to women and girls with disability; children's evolving abilities respected

3 The CBR matrix (WHO, 2010)

Five components of CBR and their elements



In the exam, draw the five components as five columns side by side with their elements underneath; empowerment runs across all of them.

A programme chooses the components and elements that its community needs most; it does not have to do all 25.

4 The CBR team and teamwork

Team members of CBR (S17) and benefits of teamwork in CBR (S20), 7 marks.

Level	People and places
Family and village	Person with disability, family, CBR worker or volunteer, anganwadi worker, ASHA, teacher, self-help group, gram panchayat, local craftsmen
Block	Primary health centre and community health centre staff (doctor, ANM), multipurpose rehabilitation worker or CBR supervisor, block resource centre for inclusive education, RBSK mobile teams
District	District hospital, District Early Intervention Centre (children 0–6), District Disability Rehabilitation Centre, physiotherapist, occupational therapist, prosthetist and orthotist, speech therapist, psychologist, social worker, NGOs
State and national	National institutes and composite regional centres for training, research and complex care; government departments that fund schemes

Benefits of teamwork in CBR

- Disability touches health, school, work and social life, so many sectors must act together.
- Few professionals can train and support many local workers (transdisciplinary model, see the team (p. 134)).
- Shared goals avoid repeated assessments and conflicting advice; families get one consistent plan.
- Early identification by village workers and quick referral; follow-up close to home.
- Local resources and schemes are pooled; costs fall and the programme lasts.
- Community attitudes change when many local people are involved.

5 Strategies and approaches

CBR strategies in rural and urban areas (W19, 7 marks); approaches to CBR (S26, 7 marks).

Steps of a CBR programme

1. **Situation analysis:** meet leaders, map resources, survey households to identify persons with disability.
2. **Plan with the community:** form a CBR committee, set priorities using the matrix.
3. **Train local workers** and families in simple assessment, home programmes and referral.
4. **Implement:** home visits, camps, school inclusion, aids, livelihood links, self-help groups.
5. **Monitor and evaluate:** records, outcomes, community feedback; hand over to local ownership.

Setting	Strategies (syllabus 4f)
Urban	Urban health centres and community centres for clinics and camps; clubs, mahila mandals and social centres for awareness, volunteers and funds; ordinary schools for inclusion and screening; industries for vocational training, jobs and workplace adaptation; sports centres for adapted sport; NGOs and corporate social responsibility funds
Rural	Primary health centres, rural hospitals and district hospital infrastructure for screening, treatment and referral; anganwadi workers and ASHAs for identification and home programmes; locomotor aids made from local materials (bamboo crutches, wooden walkers, parallel bars and standing frames, CP chairs) by village craftsmen; panchayat support; self-help groups for livelihood; camps for certificates and aids

Approaches to CBR (7 marks in S26)

CBR began after Alma-Ata as a way of delivering rehabilitation through primary health care. It has since widened into community development and rights. Programmes differ in two ways: who starts and controls the programme, and where the work is done. Most real programmes mix several approaches.

By who starts and controls it

Approach	How it works	Strengths and drawbacks

Approach	How it works	Strengths and drawbacks
Top-down (government or agency-led)	Planned and run from outside the village by a ministry, an international agency or experts; local workers follow set training packages. Government-led examples: national programmes in Iran and Viet Nam; in India, schemes run through DDRCs and camps	**For:** easy to start, run and evaluate in many places at once; more money, wider reach and better sustainability. **Against:** weak local leadership and commitment; "participation" may mean doing what was decided above
Bottom-up (led by the community, an NGO or disabled people's groups)	Started, planned and run locally, with disabled people and families making the decisions; an outsider may help at the start. Examples: PROJIMO, a village programme run by disabled villagers in Mexico; NGO programmes such as Mobility India in Bengaluru	**For:** flexible, fits local culture, better participation and ownership, works in difficult areas. **Against:** fewer resources and smaller reach; hard to fund without outside help

The WHO guidelines conclude that CBR works best with government support that stays sensitive to local culture, resources and disabled people's organisations (WHO, 2010).

By where the work is done

Approach	How it works	Strengths and drawbacks
Home-based	A trained local worker (the "local supervisor" of the WHO 1989 manual) visits the home and teaches a family member to train the person, using simple training packages	**For:** reaches remote homes; skills learnt where they are used; family as partner; low cost. **Against:** when aids or specialist help are needed the worker may have nowhere to turn, and referrals to the city often fail; heavy load on poor or troubled families
Community centre-based	A small village or ward centre run by local people, an anganwadi or an early intervention centre, where people come for therapy, aids, groups and play	**For:** visible low-cost base and back-up for home work; aids made quickly from local materials; work for disabled villagers; care when home care is too hard. **Against:** people must travel; reaches fewer homes
Outreach from institutions	Camps and mobile teams from hospitals, national institutes, DDRCs and NGOs bring assessment, certificates and aids to the village	**For:** specialist skill and equipment close to home. **Against:** visits are periodic, so day-to-day work still needs local workers and families
Integrated with primary health care	Existing PHC staff, ASHAs, anganwadi workers and RBSK teams identify, refer and follow up	**For:** uses staff and buildings that already exist, so coverage is wide and cheap. **Against:** early programmes of this kind stayed mainly medical (therapy, devices, surgery)
Community development and rights-based	All five matrix components; self-help groups and disabled people's organisations lead; the twin-track approach includes disabled people in mainstream programmes and adds specific support where needed; WHO now presents CBR as a strategy for community-based inclusive development	**For:** tackles poverty, school, work and inclusion, not only impairment. **Against:** complex and multisectoral, so results are hard to measure and evidence is limited

- **Home and centre together:** home-based work goes better with a local centre behind it (Werner, Disabled Village Children).
- **Indian example:** in Amar Seva Sangam's rural Tamil Nadu programme, community rehabilitation workers see children at home or at a local early intervention centre, guided by specialists through a mobile app. Of 1,245 children, about 91% chose home-based care; children seen at the centre had higher motor and independence scores (Bhrasadiya et al., 2025, an observational study).
- **What CBR achieves:** reviews describe CBR-type programmes as effective, with more independence, mobility and communication, and some signs of lower cost (WHO, 2010).

6 The anganwadi worker in CBR

Role of the anganwadi worker in CBR: 3 marks in S17 and S22, 7 marks in S26.

- **Who:** the village-level worker of the Integrated Child Development Services (ICDS, started in 1975), running an anganwadi centre for about 400–800 people in rural and urban areas (300–800 in tribal and hilly areas) with one helper; she serves children under 6, pregnant women and nursing mothers.

Area	Role in CBR
Identification	House-to-house survey; growth monitoring and simple milestone checks; spotting children with delay or disability early
Referral	To the PHC, RBSK mobile team and District Early Intervention Centre; helps with disability certificates and UDID
Prevention	Immunisation (polio, measles), supplementary nutrition, vitamin A, iodised salt, anaemia control, antenatal care, safe delivery, birth spacing
Home programmes	Carries out simple exercises, positioning and stimulation taught by the physiotherapist; checks aids
Pre-school inclusion	Includes children with disability in pre-school play and learning; prepares them for school
Education and awareness	Counsels mothers, reduces stigma, mothers' meetings
Records and links	Keeps registers; links families with schemes, ASHA, ANM and the panchayat

7 The physiotherapist in CBR and in a special school

Role of physiotherapy and the physiotherapist in CBR (syllabus 4e); the physiotherapist in a school for physically handicapped children (W22, 7 marks).

In CBR

- Trainer: teaches CBR workers, anganwadi workers, ASHAs, teachers and families simple assessment, positioning, exercises, transfers and when to refer.
- Assessment and planning at camps and centres; writes simple home programmes with pictures.
- Designs low-cost aids from local materials; checks fit and safety.
- Prevention and health promotion: ergonomics, fall prevention, exercise for chronic disease, antenatal and postnatal advice.
- Referral to institutions for surgery, orthoses and complex care; follow-up.
- Advocacy for accessibility and inclusion; supervision and evaluation of the programme.

In a school for physically handicapped children (7 marks in W22)

Children in such a school may have cerebral palsy, spina bifida, muscular dystrophy, limb deficiency or the after-effects of polio or injury. The physiotherapist's aim is not therapy alone. It is to help each child move, learn and take part in school life, working with the special educator, class teachers, parents, occupational and speech therapists and the doctor.

Role	What the physiotherapist does
Screening and assessment	Assess each child on admission and at regular reviews: GMFCS level, tone, range and contractures, strength, posture and spine, sitting and standing balance, gait and mobility, hand function, self-care. Set goals with teachers and parents within the individual education plan
Individual programmes	Individual and small-group therapy fitted into the school day: stretching, strengthening, balance, gait training, play-based activity; a written home programme for each child
Seating and positioning	Adapted chair and table so the child sits upright with the pelvis level and feet supported; change position through the day (standing frame, side-lying, prone); standing frame **30–60 min a day** for a child who cannot stand
Mobility aids and orthoses	Prescribe, fit, train and check ankle–foot orthoses, walkers, crutches and wheelchairs; teach wheelchair skills and safe moving around the campus

Role	What the physiotherapist does
Accessibility	Audit the school: ramps (no steeper than 1:12) with handrails, accessible toilets, ground-floor classes, wide aisles, safe playground; advise the management on changes (see barriers and modifications (p. 145))
Training teachers, attendants and parents	Safe lifting and transfers, positioning in class, toileting and feeding positions, use and care of aids, the home programme, and when to call the physiotherapist
Sports, play and fitness	Adapted physical education, games and sport for every child; aim for the WHO target, adapted to ability (below); limit long sitting
Prevention of secondary problems	Contractures, hip and spine deformity, obesity, pressure sores, falls; refer early for orthopaedic review
Inclusion and transition	Prepare children who can move to the neighbourhood school: advise that school on seating near the door and blackboard, a peer buddy, extra time and adapted physical education; later, pre-vocational and vocational training
Links to schemes	Help families get the disability certificate and UDID card, free aids (ADIP and Samagra Shiksha camps), RBSK and DEIC review, scholarships and transport support

Physical activity target: WHO (2020) advises children and adolescents aged 5–17, *including those living with disability*, to do an average of **60 min a day** of moderate-to-vigorous, mostly aerobic activity across the week. Vigorous activity and muscle- and bone-strengthening should be included on at least **3 days a week**. Children should do what they can at their level of ability. Wheelchair users can stay active with activities that do not use the legs.

Laws and schemes the school physiotherapist uses

- **RPWD Act 2016, section 16:** schools funded or recognised by government must provide inclusive education. They must admit children with disability without discrimination and give equal sports and recreation. They must make buildings accessible, provide reasonable accommodation and individual support, monitor each child's progress, and provide transport with an attendant for children with high support needs.
- **Section 17:** survey school children every **5 years** to identify disability; train teachers and staff; set up resource centres; give free books, learning materials and assistive devices to students with benchmark disability up to **18 years**; allow extra time and a scribe in examinations.
- **Section 31:** every child with benchmark disability aged **6–18 years** has the right to free education in a neighbourhood school or a special school of choice.
- **Samagra Shiksha** (Ministry of Education, from 2018–19) covers children with special needs from pre-school to class XII. Its inclusive education component funds identification and assessment camps, aids and assistive devices, transport and escort allowance, Braille and large-print books, a stipend for girls with special needs, and therapy at block level. It also funds ramps with handrails and disabled-friendly toilets. Children with severe disability can receive home-based education from special educators.
- **Accessibility Code for Educational Institutions** (notified January 2024, added to the RPWD Rules in June 2024): child-friendly access standards for school buildings.
- **RBSK:** mobile block health teams screen children aged 6–18 in government and government-aided schools for the 4Ds (defects at birth, diseases, deficiencies, developmental delays including disabilities) and refer them.

8 Worked long answer: CBR for a child with cerebral palsy

From a real paper (S22, 5 + 10)

> "6 years old CP child with locomotor disability has been identified during a screening at a village. His educated parents intend to send him school. Enumerate the local resources to be used and plan community based rehabilitation."

Spot the clues before you write

6-year-old with CP and locomotor disability: needs mobility, positioning and aids. village screening: rural CBR, PHC and anganwadi links. educated parents want school: education component is the priority; parents are strong partners.

Local resources (5 marks)

Resource	Use
Family	Daily home programme, school travel, advocacy
Anganwadi worker, ASHA, ANM	Follow-up visits, nutrition, immunisation, referral
PHC, RBSK team, District Early Intervention Centre	Medical review, therapy, hearing and vision checks, referral for orthoses and surgery
Village school, teachers, block resource centre	Enrolment under inclusive education, special educator support, classroom adaptations
Gram panchayat	Ramp and toilet at school, funds, transport
Village carpenter and blacksmith	CP chair, standing frame, parallel bars, walker
Self-help and parents' groups, NGOs	Support, awareness, funds, advocacy
Government schemes and camps	Disability certificate and UDID card, free aids (ADIP), scholarships, travel concessions

CBR plan (10 marks)

1. **Assessment at home and school:** type of CP, GMFCS level, tone, contractures, sitting and walking, hand use, communication, vision and hearing, feeding, toileting; ICF summary; family goals (school).
2. **Health:** home programme taught to parents and the anganwadi worker: stretching of calves, hamstrings and hip adductors daily; sitting balance, sit-to-stand and supported walking practice; positioning in sitting and lying; standing frame 30–60 minutes a day if he cannot stand; ankle–foot orthoses and a walker through the DEIC or an ADIP camp; review every month by the physiotherapist at the PHC or camp.
3. **Education:** enrol in the village school; seat near the door and blackboard; adapted chair and table; ramp, railing and an accessible toilet; teacher training on handling and toileting; peer buddy; extra time and adapted physical education; special educator from the block resource centre.
4. **Social:** play with peers, festivals and outings; counselling to remove stigma; parents' group.
5. **Empowerment:** parents learn rights under the RPWD Act (free education, accessibility, reservation); link to a disabled people's organisation.
6. **Livelihood (family now, child later):** schemes for the family; plan for vocational training as he grows.
7. **Referral and follow-up:** to the district hospital for orthopaedic review, botulinum toxin or surgery if needed; record goals and review every 3 months.

9 Exam corner

Past questions (MUHS, 2016–2026)

Long answer

6-year-old with CP identified at a village screening: local resources and CBR plan (S22).

7 marks

Principles of CBR and team members (S17); CBR strategies in rural and urban areas (W19); benefits of teamwork in CBR (S20); a note on CBR (W20); the physiotherapist in a school for physically handicapped children (W22); approaches to CBR and the anganwadi worker (S26).

3 marks

Aspects of CBR (S16); the anganwadi worker in CBR (S17, S22); principles of CBR (W21, S22, S25); three features of CBR (W24).

Keywords to underline

community development participation and ownership local resources appropriate technology multisectoral referral sustainability CBR matrix health, education, livelihood, social, empowerment anganwadi worker inclusive education

Answer skeleton: principles of CBR (7 marks)

1. **[Definition, 1]** 2004 definition.
2. **[Principles, 4]** Eight principles, each with one practical line.
3. **[Matrix or team, 2]** Draw the matrix or list the team by level.

Draw these

- The CBR matrix as five columns (section 3)
- Levels of referral from village to national institute
- A bamboo or wooden standing frame and parallel bars

Viva questions

1. Define CBR.
2. Name the five components of the CBR matrix.
3. What does an anganwadi worker do for a child with a disability?
4. What aids can be made from local materials?
5. Where does a child aged 0–6 with developmental delay get referred under RBSK?
6. Give two urban CBR strategies from the syllabus.
7. Give one strength and one drawback of home-based CBR.
8. Which section of the RPWD Act gives free education to children with benchmark disability, and up to what age?

Linked practical spots

- Identify a CP chair, a standing frame and a bamboo walker; teach a parent a calf stretch; plan a home programme with pictures.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Principles of CBR. (W21, S22, S25)**

Answer
Community participation and ownership; inclusion and equal opportunity; empowerment and self-advocacy; use of local resources and appropriate technology; multisectoral approach; integration with existing services; two-way referral; sustainability; gender equality.

3 m **Role of the anganwadi worker in CBR. (S17, S22)**

Answer
Survey and early identification (growth and milestones); referral to PHC, RBSK and DEIC; prevention (immunisation, nutrition, antenatal care); carrying out simple home programmes taught by the physiotherapist; pre-school inclusion; counselling mothers and reducing stigma; records and links to schemes.

7 m **CBR strategies in rural and urban areas. (W19)**

Answer
Steps of a programme (survey, committee, training, implementation, evaluation). Urban: UHCs and community centres, clubs and mahila mandals, social centres, schools, industries, sports centres, NGOs and CSR. Rural: PHC, rural and district hospitals, anganwadi workers and ASHAs, local-material aids made by village craftsmen, panchayat, self-help groups, camps. End with referral links and sustainability.

LAQ **A 6-year-old with CP found at a village screening; parents want him in school. Local resources and a CBR plan. (5 + 10, S22)**

Answer
Resources: family, AWW, ASHA, ANM, PHC and RBSK team, DEIC, village school and block resource centre, panchayat, craftsmen, SHGs and NGOs, schemes (UDID, ADIP). Plan by matrix: assessment and ICF; health (home programme, positioning, standing frame, orthoses and walker, monthly review); education (enrolment, seating, access, teacher training, special educator); social; empowerment (RPWD rights, parents' group); referral and 3-monthly review.

MCQ **Which is NOT a component of the WHO CBR matrix?**

A. Health

B. Livelihood

C. Surgery

D. Empowerment

Answer
C. The five are health, education, livelihood, social and empowerment.

MCQ **The anganwadi worker belongs to which programme?**

A. RBSK

B. ICDS

C. ESI

D. NPPCD

Answer
B. Integrated Child Development Services.

MCQ **A key principle of CBR is:**

A. Treatment only in hospitals

B. Use of local resources

C. Professionals decide all goals

D. Separate schools for all disabled children

Answer

B.

Last recall check. Write the CBR definition, eight principles, the matrix, three levels of the CBR team, urban and rural strategies, and the anganwadi worker's role.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Community Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. ILO, UNESCO and WHO. CBR: a strategy for rehabilitation, equalization of opportunities, poverty reduction and social inclusion of people with disabilities. Joint position paper, 2004.
3. World Health Organization, UNESCO, ILO and IDDC. Community-based rehabilitation: CBR guidelines. Geneva: WHO; 2010 (introductory booklet: history of CBR, twin-track approach, government-led and civil-society programmes, Viet Nam and Iran national programmes, evidence on CBR).
4. Helander E, Mendis P, Nelson G, Goerdt A. Training in the community for people with disabilities. Geneva: WHO; 1989.
5. Werner D. Disabled village children: a guide for community health workers, rehabilitation workers, and families. Palo Alto: Hesperian Foundation; chapter 45, Starting village-based rehabilitation activities (top-down and bottom-up programmes; the case for community-run centres).
6. Bhrasadiya N, Coutinho F, Ramachandran S, et al. Effect of multidisciplinary interventions using the Enabling Inclusion program on gross motor function and functional independence in children with disabilities in rural South India. *BMC Pediatr* 2025;25:960.
7. Ministry of Women and Child Development, Government of India. Norms for setting up anganwadi and mini-anganwadi centres.
8. Ministry of Health and Family Welfare, Government of India. Rashtriya Bal Swasthya Karyakram (RBSK) operational guidelines; District Early Intervention Centres (school screening by mobile block health teams checked on state National Health Mission pages).
9. Rights of Persons with Disabilities Act, 2016, sections 16, 17 and 31 (text checked on an online copy of the Act and in a Delhi government order).
10. Press Information Bureau, Ministry of Education. Accessibility and inclusivity in education for Divyangs. 29 July 2024 (Samagra Shiksha inclusive education component; Accessibility Code for Educational Institutions). Start year 2018–19 and home-based education for severe disability checked through Ministry replies in Parliament.
11. Bull FC, Al-Ansari SS, Biddle S, et al. World Health Organization 2020 guidelines on physical activity and sedentary behaviour. *Br J Sports Med* 2020;54:1451–62; Carty C, et al. The first global physical activity and sedentary behavior guidelines for people living with disability. *J Phys Act Health* 2021;18:86–93.

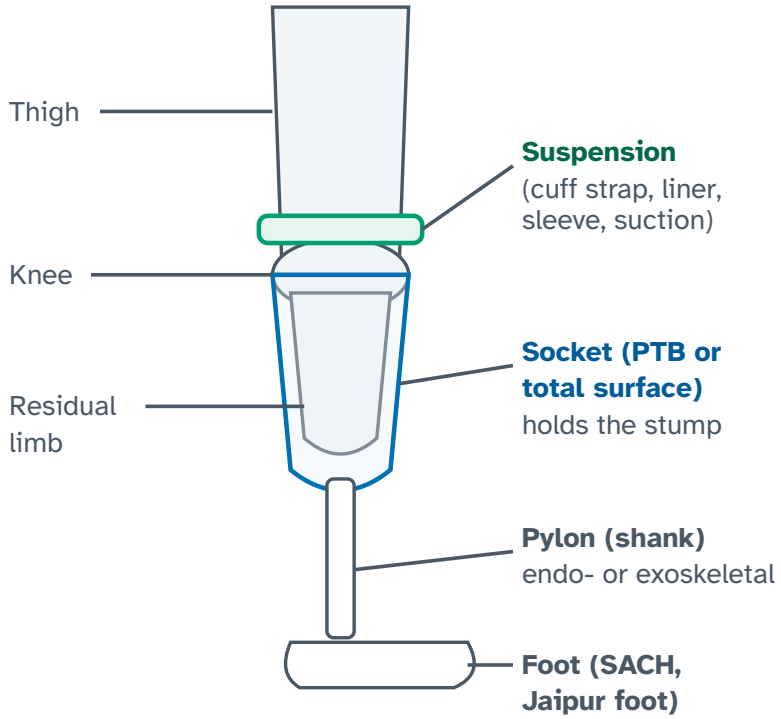
4 Rehabilitation of an amputee in the community: below-knee prosthesis, check-outs, psychosocial issues, disability evaluation and CBR

> The amputee was asked in 5 of 20 papers, with 3 long cases (a teacher found at a village camp, and two truck drivers) and two 7-mark questions (prosthetic check-out and psychosocial issues). Community questions are less about surgery and more about life after it: a prosthesis that fits and is affordable, a person who copes, and a job or role to return to.

Syllabus	Ch 4 Concepts of rehabilitation applied to the amputee: CBR, locomotor aids using local resources, disability evaluation, national policies
Priority	**Tier 2.** In 5 of 20 papers (2016–2026): 3 long cases and 2 seven-mark questions.
Asked as	LAQ S17 · W18 · W19 · 7m W17 · S18
Usual split	15 (functional diagnosis and CBR programme; or difficulties and concessions; or principles of disability evaluation 5 and disability management 10); 7 (prosthetic check-out; psychosocial issues)
Checked against	Ministry of Social Justice and Empowerment guidelines for evaluation of permanent physical impairment in amputees (2001, compiled by the Chief Commissioner for Persons with Disabilities); Bhagwan Mahaveer Viklang Sahayata Samiti (Jaipur foot); standard below-knee prosthetic check-out and gait deviation lists from orthotics and prosthetics texts. Full list under Sources (p. 166). Content checked 20 October 2026.

1 The below-knee prosthesis

Parts of a below-knee prosthesis



The socket (blue) carries weight on pressure-tolerant areas; the suspension (green) keeps it on; the pylon and foot replace the leg and ankle.

- **Socket:** patellar tendon bearing (PTB) or total surface bearing (TSB), usually with a soft liner.

- **Suspension:** supracondylar cuff strap, supracondylar wedge (PTB-SC), sleeve, silicone liner with pin lock, suction; thigh corset for a very short or painful stump.
- **Shank:** exoskeletal (hard outer shell) or endoskeletal (pylon tube with a cosmetic cover; adjustable alignment).
- **Foot:** SACH (solid ankle cushion heel), single-axis, dynamic-response feet; the **Jaipur foot** (developed in Jaipur by Dr P K Sethi and craftsman Ram Chandra Sharma, distributed free by Bhagwan Mahaveer Viklang Sahayata Samiti) is cheap, waterproof, and allows squatting, sitting cross-legged and walking barefoot, which suits Indian life.
- **Weight-bearing areas:** tolerant: patellar tendon, medial tibial flare, shaft of the fibula, pretibial muscles, popliteal area (lightly). Sensitive (relieved): tibial crest and tubercle, fibular head, distal end of tibia and fibula, hamstring tendons, patella.

2 Pre-prosthetic and prosthetic training

- **Stump care:** wound healing, skin inspection and hygiene, desensitisation (tapping, rubbing, textures).
- **Shaping:** figure-of-eight elastic bandaging (tighter distally) or a shrinker sock, worn day and night, to reduce swelling and shape a cone.
- **Positioning:** prevent knee flexion and hip flexion contractures: no pillow under the knee, knee straight in sitting (stump board), lie prone 20–30 minutes a few times a day.
- **Exercise:** knee extensors and flexors, hip extensors and abductors, trunk and arms (for crutches); balance on one leg; endurance.
- **Mobility:** transfers, crutch or walker walking, wheelchair if needed.
- **With the prosthesis:** donning and doffing, wearing time built up gradually, skin check after each session, weight shifting in parallel bars, stepping, gait training, stairs, slopes, uneven ground, getting up from the floor, squatting with a Jaipur foot.

3 Prosthetic check-out for a below-knee amputee

Prosthetic check-outs for a below-knee amputee (S18, 7 marks).

Stage	What to check
Before donning	Prosthesis as prescribed; finish and weight; socket smooth inside; suspension and foot correct; skin of the stump intact
Standing	Comfortable with equal weight on both legs; pelvis level (correct length); knee in slight flexion; foot flat with slight toe-out like the other foot; socket total contact without gaps; patellar tendon bar at the right level; suspension holds; pylon vertical in both views
Sitting	Can sit with the knee bent about 90° comfortably; no pinching behind the knee by the posterior wall; prosthesis stays on; shoe flat on the floor
Walking	Smooth gait without the deviations below; no pistoning; no pain; equal step length; stairs and slopes
After removal	Redness only over pressure-tolerant areas and fading within about 10–20 minutes; no abrasions or blisters; stump sock not wrinkled

Below-knee gait deviations

Deviation	Prosthetic causes
Excessive knee flexion at heel strike	Foot too dorsiflexed or too far back, socket too flexed, heel cushion too hard
Too little knee flexion (knee stays straight)	Foot too plantarflexed or too far forward, heel cushion too soft; also quadriceps weakness or pain
Lateral thrust of the pylon in stance	Foot set too far in (inset)
Medial thrust	Foot set too far out (outset)
Early drop-off in late stance	Keel too short or foot too far back
Pistoning	Poor suspension or a loose socket (stump has shrunk: add socks or new socket)

4 Psychosocial issues of being an amputee

Psychosocial issues of being an amputee (W17, 7 marks).

- **Grief and loss:** shock, denial, anger, sadness, then gradual acceptance; more severe after sudden trauma.
- **Depression and anxiety;** post-traumatic stress after accidents.
- **Body image:** embarrassment, hiding the limb, low self-esteem.
- **Phantom sensation and phantom limb pain;** stump pain.
- **Loss of role and income:** unable to return to work (drivers, farmers, labourers), financial strain on the family.
- **Dependence** on others; strain on family relationships; sexual and marital concerns.
- **Stigma** and social isolation; difficulty in marriage for young people; cultural beliefs (punishment, bad luck).
- **Practical worries:** cost and repair of the prosthesis, travel to centres, inaccessible homes and toilets.

Helping

- Counselling and psychology referral; meeting a successful amputee (peer support); amputee groups.
- Early prosthetic fitting and good training, which restore mobility and confidence.
- Manage phantom pain: explanation, desensitisation, mirror therapy, TENS, medication by the doctor.
- Involve the family; vocational rehabilitation and adapted work; information on concessions and schemes; sport and leisure.

5 Disability evaluation, difficulties and concessions

Principles of disability evaluation (W19, 5 marks); difficulties faced and concessions offered by government (W18, long answer).

Principles of disability evaluation

- Done by a certified medical board after maximum medical improvement; based on measured loss of function, not opinion.
- For amputations, fixed percentages by level (Indian guidelines): below-knee up to 8 cm (short stump) 70%, below-knee up to the lower third of the leg 60%, through the knee 75%, above-knee 80–85%, Syme's 50%.
- Add 5% if the stump is unfit for a prosthesis, and up to 10% for complications (stiff proximal joint, neuroma, infection); for amputation of more than one limb, combine the values with the formula a + b(90 – a) ÷ 90 and add 10%; the total cannot exceed 100%.
- 40% or more is a benchmark disability, qualifying for most benefits (see disability evaluation (p. 129)).

Difficulties faced by a below-knee amputee

- **Physical:** walking long distances, uneven roads, stairs, squatting and the Indian toilet, carrying loads, stump pain and skin problems, more energy used in walking.
- **Work:** driving with a right-leg amputation (adapted controls or automatic vehicle needed), heavy jobs, loss of income.
- **Psychosocial:** section 4.
- **Practical:** cost, repair and replacement of the prosthesis, distance to centres.

Concessions and benefits

- Disability certificate and UDID card; free or subsidised prosthesis under ADIP and from BMVSS (Jaipur foot) camps and DDRCs.
- Railway and state bus concessions; disability pension if eligible; income tax deduction; concessional loans for self-employment.
- 4% job reservation, vocational training, and the RPWD Act's protection against losing a job because of disability (the employer must shift the employee to a suitable post).

- Accident compensation from the Motor Accident Claims Tribunal; employees' compensation if injured at work; ESI benefits for insured workers.
- Details on schemes and concessions (p. 145).

6 Worked long case: a teacher with a below-knee amputation in a village

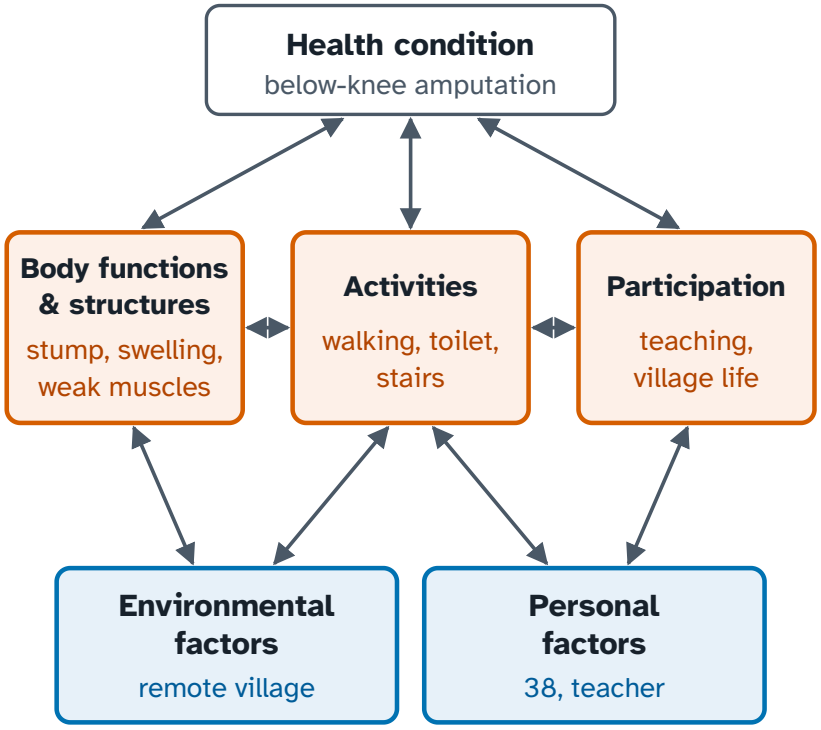
From a real paper (S17)

> "Following recent camp in the remote village, Mr. X 38 years old male, teacher by profession, below knee amputee was identified. Mr. X was discharged following selective amputation three weeks before. i) Give your functional diagnosis. ii) Plan your Community based rehabilitation program."

Spot the clues before you write

3 weeks after amputation: pre-prosthetic stage: wound, shaping, contracture prevention. remote village: CBR with local resources and camps. teacher: can return to work early with a prosthesis and accessible school; standing and walking in class.

The ICF model filled for this case



Draw the six boxes with two-way arrows, then write two or three points from the case in each box.

One-line functional diagnosis

A 38-year-old teacher, 3 weeks after a below-knee amputation, in the pre-prosthetic stage with a healing, swollen stump and weak, deconditioned muscles, walking with crutches and unable to teach or manage the Indian toilet, in a remote village without a prosthetic centre.

CBR programme

CBR component	Plan
Health: pre-prosthetic (weeks 3–8)	Teach the family and CBR worker stump care, figure-of-eight bandaging, prone lying and no pillow under the knee; exercises for knee, hip, trunk and arms; crutch walking; home visits weekly by the CBR worker, monthly by the physiotherapist
Health: prosthesis	Disability certificate at the camp; referral to the DDRC or a Jaipur foot camp (BMVSS) or ADIP camp for a free below-knee prosthesis; gait training, then follow-up for fit and repair
Local resources	Village carpenter for parallel bars, a stump board and a raised commode stool; panchayat for a ramp or handrail at home and school; anganwadi or PHC as a training place
Education and livelihood	Early return to teaching (with a chair in class, ground-floor classroom); school administration informed; job protection under the RPWD Act
Social and empowerment	Counselling and peer support from another amputee; family education; information on concessions; join a self-help or disabled people's group
Follow-up	Stump volume, skin, contractures, prosthetic use; outcome measures (walking distance, TUG, satisfaction)

Practice cases from the papers

- **W18:** 37-year-old truck driver 4 months after below-knee amputation: difficulties (section 5) and concessions.
- **W19:** 32-year-old truck driver, right below-knee amputation after a crush injury in a road accident, sole earner for wife and two children in a village: principles of disability evaluation (section 5) and disability management: prosthesis and gait training, functional capacity evaluation against driving demands, adapted vehicle or alternative work (dispatcher, mechanic), vocational rehabilitation, compensation and schemes (see disability management (p. 186)).

7 Exam corner

Past questions (MUHS, 2016–2026)

Long cases

38-year-old teacher found at a village camp 3 weeks after below-knee amputation: functional diagnosis and CBR programme (S17); 37-year-old truck driver: difficulties and concessions (W18); 32-year-old truck driver: principles of disability evaluation and disability management (W19).

7 marks

Psychosocial issues of being an amputee (W17); prosthetic check-outs for a below-knee amputee (S18).

Keywords to underline

PTB socket · pressure-tolerant areas · SACH and Jaipur foot · figure-of-eight bandaging · prevent knee flexion contracture · static and dynamic check-out · lateral thrust: foot inset · phantom pain · peer support · BK 60–70% · ADIP, BMVSS

Answer skeleton: prosthetic check-out (7 marks)

1. **[Purpose, 1]** Fit, comfort, function, safety before final delivery.
2. **[Static, 3]** Before donning, standing, sitting.
3. **[Dynamic, 2]** Walking, gait deviations with causes.
4. **[After removal, 1]** Skin check.

Draw these

- Below-knee prosthesis parts (section 1)
- Figure-of-eight stump bandaging

Viva questions

1. Name the pressure-tolerant areas in a PTB socket.
2. What are the advantages of the Jaipur foot?
3. What causes a lateral thrust of the pylon?
4. How do you prevent a knee flexion contracture after below-knee amputation?
5. What percentage disability is a below-knee amputation in the lower third of the leg?
6. How would you manage phantom limb pain?

Linked practical spots

- Identify parts of a below-knee prosthesis and a Jaipur foot; demonstrate figure-of-eight bandaging; perform a standing and sitting check-out on a model.

Test yourself

Cover each answer and write yours on paper first, then check.

7 m **Psychosocial issues of being an amputee. (W17)**

Answer

Grief and loss stages; depression, anxiety, PTSD; body image and self-esteem; phantom sensation and pain; loss of role and income; dependence and family strain; sexual and marital concerns; stigma and isolation; practical worries. Management: counselling, peer support, early prosthesis, phantom pain care, family involvement, vocational rehabilitation, schemes, sport.

7 m **Prosthetic check-outs for a below-knee amputee. (S18)**

Answer

Before donning (as prescribed, finish, socket smooth, stump skin); standing (comfort, equal weight, level pelvis, slight knee flexion, foot flat with toe-out, total contact, suspension); sitting (90° knee flexion without pinching); walking (gait deviations: early knee flexion, no knee flexion, lateral or medial thrust, drop-off, pistoning, with causes); after removal (redness only on tolerant areas, fading in 10–20 minutes).

LAQ **A 37-year-old truck driver 4 months after below-knee amputation: difficulties faced and concessions offered by government. (W18)**

Answer

Difficulties: physical (walking, roads, stairs, squatting, toilet, stump problems, energy cost), work (driving with the right leg, heavy work, income), psychosocial, practical (cost and repair of prosthesis). Concessions: certificate and UDID; free prosthesis (ADIP, BMVSS, DDRC); rail and bus concessions; pension, tax, loans; job reservation and job protection; MACT compensation; vocational training.

MCQ **A lateral thrust of the pylon in stance in a below-knee prosthesis is caused by:**

A. Foot outset

B. Foot inset

C. Soft heel

D. Long keel

Answer
B.

MCQ **Which area is pressure-tolerant in a PTB socket?**

A. Fibular head

B. Tibial crest

C. Patellar tendon

D. Distal tibia

Answer
C.

MCQ **The main advantage of the Jaipur foot for Indian users is that it allows:**

A. Running sports only

B. Squatting, cross-legged sitting and barefoot walking

C. Use without a socket

D. Electronic control

Answer
B.

Last recall check. Draw the below-knee prosthesis, list the check-out stages and six gait deviations with causes, eight psychosocial issues, and the disability percentages for amputation levels.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Community Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Ministry of Social Justice and Empowerment. Guidelines for evaluation of permanent physical impairment in amputees (2001), compiled by the Office of the Chief Commissioner for Persons with Disabilities.
3. Bhagwan Mahaveer Viklang Sahayata Samiti, Jaipur: the Jaipur foot.
4. Lusardi MM, Jorge M, Nielsen CC, eds. Orthotics and Prosthetics in Rehabilitation. 4th ed. St Louis: Elsevier; 2020 (transtibial check-out and gait deviations).
5. Rights of Persons with Disabilities Act, 2016, Section 20 (non-discrimination in employment).

5 Disaster management: types of disaster, principles, the disaster management cycle, triage and the role of the physiotherapist

> Disaster management was asked in 3 of 20 papers, but each time as a full 7-mark question or a long answer: types and principles twice, and the cycle, triage and the physiotherapist's role in S26. Learn one diagram (the cycle), one flowchart (START triage) and the physiotherapist's role in each phase.

Syllabus	Ch 5 Introduction to disaster management
Priority	**Tier 2.** In 3 of 20 papers (2016–2026): 1 long answer and 2 seven-mark questions.
Asked as	LAQ S26 · 7m W19 · S24
Usual split	15 (disaster management cycle and triage, role of physiotherapists); 7 (types and principles)
Checked against	Disaster Management Act, 2005 (definition); National Disaster Management Authority structure; Sendai Framework (2015–2030); FEMA principles of emergency management (2007); START triage criteria; WHO minimum technical standards for rehabilitation in emergency medical teams (2016). Full list under Sources (p. 172). Content checked 20 October 2026.

1 What a disaster is, and its types

Types of disaster and principles of disaster management (W19, S24, 7 marks).

- **Definition (Disaster Management Act, 2005):** a catastrophe, mishap, calamity or grave occurrence, from natural or man-made causes, or by accident or negligence, that causes substantial loss of life, human suffering, or damage to property or the environment, and is beyond the coping capacity of the affected community.
- **Hazard** is the threat (an earthquake); **vulnerability** is how easily people are harmed (weak buildings, poverty, disability); **disaster** happens when a hazard meets a vulnerable community; risk = hazard × vulnerability ÷ capacity.

Type	Examples (India)
Natural: geophysical	Earthquakes (Latur 1993, Bhuj 2001), tsunami (2004), landslides
Natural: hydrological and weather	Floods (Kerala 2018), cyclones (Odisha 1999), cloudbursts, droughts, heat and cold waves
Natural: biological	Epidemics and pandemics (COVID-19)
Man-made: technological	Industrial and chemical accidents (Bhopal gas leak 1984), fires, building collapse, nuclear accidents
Man-made: transport and crowds	Rail and road accidents, stampedes
Man-made: conflict	Terrorism, bomb blasts, war, riots

Disasters may also be classified by onset (sudden, like an earthquake, or slow, like drought) and by scale (local, state, national).

2 Principles of disaster management

- **Comprehensive:** all hazards, all phases, all people affected.
- **Progressive:** anticipate future disasters and act before them (preparedness, mitigation).
- **Risk-driven:** priorities set by hazard and vulnerability assessment.

- **Integrated and coordinated:** all levels of government, agencies, NGOs and communities work under one plan and command.
- **Collaborative:** trust and partnership among organisations and the community.
- **Flexible:** adapt to each situation.
- **Professional:** trained people, standards, evidence.
- **Community participation and inclusion:** local people, women, older people and persons with disability involved in planning; "build back better" during recovery (Sendai Framework).

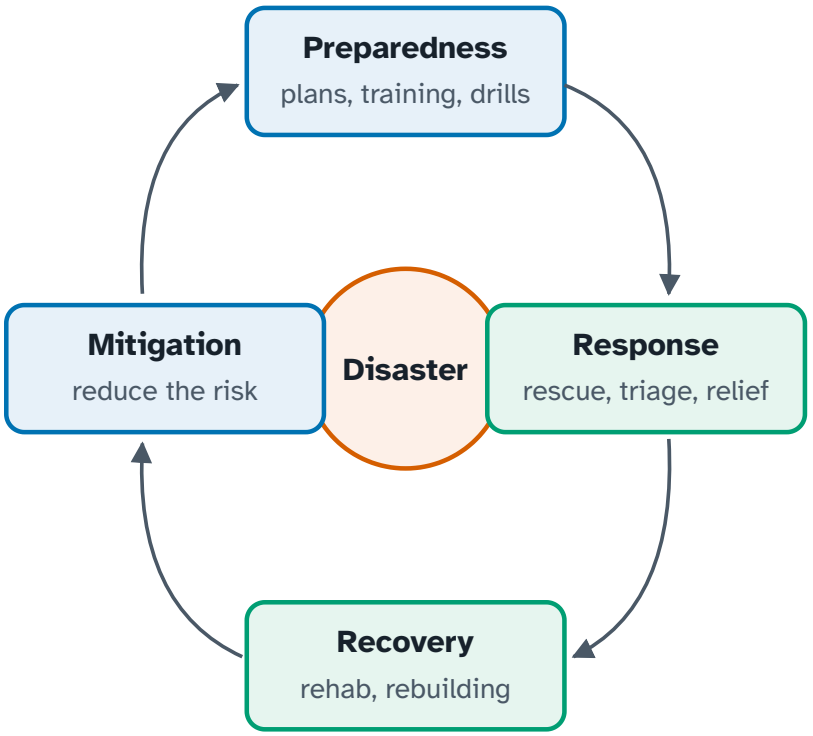
Disaster management in India

- Disaster Management Act, 2005: National Disaster Management Authority (chaired by the Prime Minister), State and District Disaster Management Authorities; National Disaster Response Force (NDRF); National Institute of Disaster Management.
- National Disaster Management Plan (2016) aligned with the Sendai Framework for Disaster Risk Reduction (2015–2030).

3 The disaster management cycle

Different steps of the disaster management cycle (S26, long answer).

The disaster management cycle



Blue phases come before a disaster, green phases after it. Recovery feeds back into mitigation ("build back better").

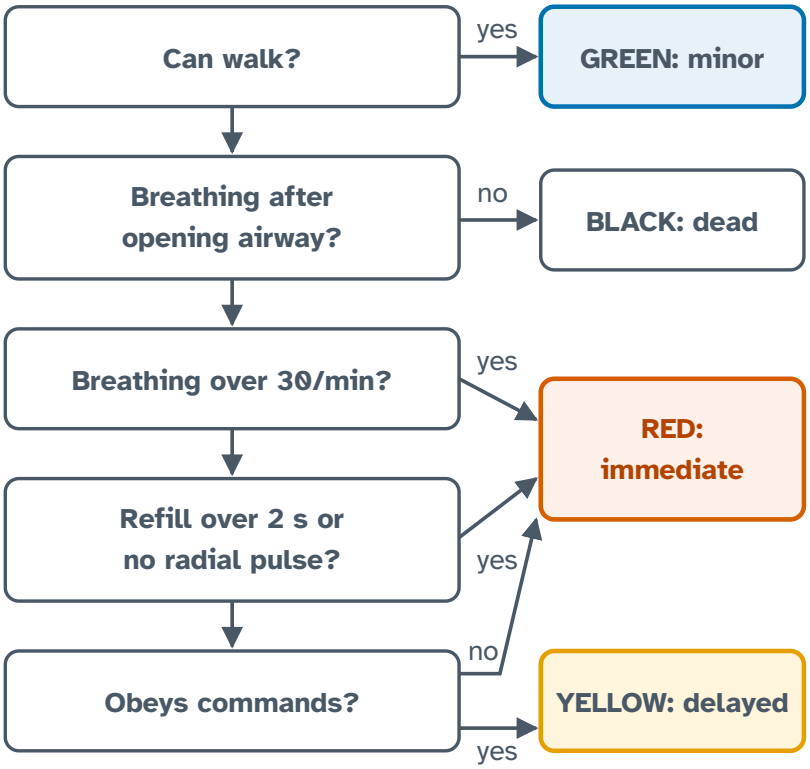
Phase	Activities
Mitigation (prevention)	Earthquake-resistant buildings, flood walls, land-use rules, industrial safety, accessible design, risk mapping
Preparedness	Disaster plans for hospitals and communities, early warning systems, stockpiles, training in first aid and triage, mock drills, lists of vulnerable people (elderly, disabled) and how to evacuate them
Response	Search and rescue, triage, first aid, evacuation, field hospitals, shelter, food, water, sanitation, prevention of epidemics, psychological first aid
Recovery	Rehabilitation of the injured, reconstruction of homes and services, livelihoods, mental health care, long-term disability support

4 Triage

Triage approach (S26, long answer).

- **Triage:** sorting casualties by urgency so the most people survive when resources are limited.
- **START** (Simple Triage and Rapid Treatment) takes under a minute per adult and uses colour tags.

START triage for adults



Memory aid "30-2-can do": breathing 30, refill 2 seconds, can do what you ask. A patient who starts breathing only after the airway is opened is also red.

? Text version of the flowchart

1. Can the person walk? Yes: green (minor, walking wounded).
2. Breathing? Not breathing even after opening the airway: black (dead or expectant). Breathing only after opening the airway: red.
3. Breathing rate over 30 a minute: red.
4. Capillary refill over 2 seconds or no radial pulse: red.
5. Cannot follow simple commands: red. Follows commands: yellow (delayed).

Tag	Meaning	Examples
Red (immediate)	Life-threatening but can be saved with quick treatment	Airway problems, severe bleeding, shock, tension pneumothorax
Yellow (delayed)	Serious but stable; can wait hours	Closed fractures, stable spinal injury, moderate burns
Green (minor)	Walking wounded	Sprains, cuts, bruises
Black (expectant or dead)	Dead, or injuries not survivable with available resources	Not breathing after airway opening

Children are triaged with JumpSTART, which adjusts the breathing criteria.

5 Role of physiotherapists in disaster management

Role of physiotherapists in disaster management (S26, long answer).

- **Common injuries:** fractures, spinal cord injury, traumatic amputation, crush injury, peripheral nerve injury, head injury, burns, soft-tissue injuries; later, chronic pain, disability and psychological trauma. Earthquakes produce many spinal cord injuries and fractures.

Phase	Role of the physiotherapist
Mitigation and preparedness	Be part of hospital and community disaster plans; train in first aid, triage, safe lifting and spinal handling; keep stocks of crutches, walkers, wheelchairs, splints; identify and plan evacuation for persons with disability and older people; teach communities safe evacuation and fall-safe behaviour
Response (acute)	Support triage and casualty handling; safe transfer and positioning of spinal injuries; splinting and positioning of fractures; chest physiotherapy and breathing exercises; early mobilisation; pressure-sore prevention; pre- and post-amputation care; burns positioning; assistive devices at discharge so beds free up; psychological first aid. WHO standards ask field hospitals for at least one rehabilitation professional per 20 beds
Recovery (long term)	Rehabilitation of spinal cord injury, amputation, fractures, nerve and brain injury; prostheses, orthoses and wheelchairs; home and shelter accessibility; community-based rehabilitation; vocational rehabilitation and return to work; follow-up of complications; support for mental health referral
Throughout	Record keeping and data, coordination with the team, care of the rescuers' own health, research and advocacy for inclusive disaster plans

6 Model long answer (S26)

> "Describe disaster management (Different steps of disaster management cycle and triage approach). Discuss the role of physiotherapists in disaster management."

1. **[Definition and types, 2]** DM Act 2005 definition; natural and man-made with Indian examples.
2. **[Cycle, 4]** Draw the cycle; activities in mitigation, preparedness, response and recovery; Indian structure (NDMA, SDMA, DDMA, NDRF).
3. **[Triage, 4]** START flowchart and the four colour tags with examples.
4. **[Role of PT, 5]** By phase: preparedness, acute response, recovery and CBR; WHO standard of one rehabilitation professional per 20 beds.

7 Exam corner

Past questions (MUHS, 2016–2026)

Long answer

Disaster management cycle, triage and the role of physiotherapists (S26).

7 marks

Types and principles of disaster management (W19, S24).

Keywords to underline

beyond the coping capacity, hazard and vulnerability, natural and man-made, mitigation, preparedness, response, recovery, NDMA, Sendai Framework, START, red, yellow, green, black, 30-2-can do, build back better

Draw these

- The disaster management cycle (section 3)

- START triage flowchart (section 4)

Viva questions

1. Define disaster as in the Disaster Management Act 2005.
2. Name the four phases of the disaster management cycle.
3. Which colour tag is given to the walking wounded?
4. What does "30-2-can do" stand for?
5. Who chairs the NDMA?
6. Give three roles of a physiotherapist after an earthquake.

Linked practical spots

- Practise a log roll and spinal board transfer; triage a set of mock casualty cards; demonstrate splinting with improvised materials.

Test yourself

Cover each answer and write yours on paper first, then check.

7 m **Types of disaster and principles of disaster management. (W19, S24)**

Answer
Definition. Types: natural (geophysical, hydrological and weather, biological) and man-made (technological, transport and crowd, conflict) with Indian examples; sudden or slow onset. Principles: comprehensive, progressive, risk-driven, integrated, collaborative, coordinated, flexible, professional; community participation, inclusion and build back better. Mention the DM Act 2005 and NDMA.

LAQ **Describe disaster management (cycle and triage) and the role of physiotherapists. (S26)**

Answer
Follow section 6: definition and types; cycle with activities and Indian structure; START triage with tags; PT role in preparedness, response and recovery with examples of injuries and the WHO staffing standard.

MCQ **In START triage, a casualty breathing 36 times a minute is tagged:**

A. Green
B. Yellow
C. Red
D. Black

Answer
C.

MCQ **Training, drills and early warning systems belong to which phase?**

A. Preparedness
B. Response
C. Recovery
D. Reconstruction

Answer
A.

MCQ **The Bhopal gas tragedy (1984) is an example of a:**

A. Natural disaster

B. Technological (industrial) disaster

C. Biological disaster

D. Hydrological disaster

Answer

B.

Last recall check. Write the definition, six types with examples, eight principles, draw the cycle and the START flowchart, and list the PT role in each phase.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Community Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. The Disaster Management Act, 2005, Section 2(d).
3. National Disaster Management Authority. National Disaster Management Plan, 2016.
4. United Nations. Sendai Framework for Disaster Risk Reduction 2015–2030.
5. FEMA Emergency Management Institute. Principles of emergency management, 2007.
6. START (Simple Triage and Rapid Treatment) adult mass-casualty triage criteria.
7. World Health Organization. Emergency medical teams: minimum technical standards and recommendations for rehabilitation. Geneva: WHO; 2016.

Part 6

Industrial health

20 hours theory, 20 field

6(a)	Industrial health: models, the team and the worker care spectrum	174
6(b i)	Ability management: job analysis, ergonomics and injury prevention	179
6(b ii)	Disability management: functional capacity evaluation, work conditioning and work hardening	186
6(b iii)	Environmental stress at work: physical, chemical and mechanical hazards	193
6(b iv)	Mechanical and psychological stress at work	203

6(a) Industrial health: definition, models of industrial therapy, the industrial rehabilitation team, pre-placement and the role of the physiotherapist

This introduction to Chapter 6 was asked in 3 of 20 papers (the physiotherapist in industry, the industrial rehabilitation team and job analysis, and pre-placement), but it frames the most-asked part of the whole paper: work stress, hazards and the worker care spectrum. The key idea is the shift from the traditional medical model (treat the injury in the clinic) to the industrial model (prevent injury and return the worker to the job, with the employer as a partner).

Syllabus	Ch 6(a) Introduction to industrial health: definition, model of industrial therapy (traditional medical and industrial model)
Priority	**Tier 2.** In 3 of 20 papers (2016–2026): 1 long answer and 2 seven-mark questions.
Asked as	LAQ S22 · 7m W19 · S25
Usual split	15 (define industrial therapy 2, the industrial rehabilitation team 3, job analysis 7, its purpose 3); 7 (role of the PT in industry; pre-placement worker care spectrum)
Checked against	Joint ILO/WHO Committee on Occupational Health definition (1950, revised 1995); Factories Act 1948, ESI Act 1948, Employees' Compensation Act 1923, Occupational Safety, Health and Working Conditions Code 2020; Rights of Persons with Disabilities Act 2016. Full list under Sources (p. 178). Content checked 21 October 2026.

1 Industrial (occupational) health

Definition (Joint ILO/WHO Committee, 1950, revised 1995)

Occupational health should aim at the promotion and maintenance of the highest degree of physical, mental and social well-being of workers in all occupations; the prevention among workers of departures from health caused by their working conditions; the protection of workers from risks adverse to health; and the placing and maintenance of the worker in an environment adapted to his or her physiological and psychological capabilities: the adaptation of work to people and of each person to their job.

- **Industrial therapy:** the application of physiotherapy (with other disciplines) to workers and workplaces: preventing work injury, treating work-related disorders close to the job, and returning injured workers to safe, productive work.
- **Why it matters in India:** most workers are in the informal sector (farms, construction, small units) with little protection; musculoskeletal disorders, injuries and dust diseases are common.

Laws for workers' health in India

- **Factories Act, 1948:** health, safety and welfare in factories; notifiable occupational diseases; working hours.
- **Employees' State Insurance Act, 1948:** medical care and cash benefits for sickness, maternity, employment injury and disablement for insured workers.
- **Employees' Compensation Act, 1923** (formerly Workmen's Compensation): compensation for injury and listed occupational diseases.
- **Mines Act, 1952;** and the **Occupational Safety, Health and Working Conditions Code, 2020**, which brings many of these laws together.

2 Models of industrial therapy

Feature	Traditional medical model	Industrial model
Focus	The injury and its symptoms	The worker's function in the job, and prevention
Place	Hospital or clinic, away from work	At or near the workplace; the therapist visits the job
Who directs	Doctor; the patient is passive	Team with the worker, employer, supervisor and insurer as partners
Rest and return	Time off until "fully healed"	Early return with modified or graded duties
Treatment	Passive modalities, generic exercise	Active, job-specific: work conditioning, work hardening, job simulation, ergonomic change
Outcome measured	Pain, range, strength	Return to work, days lost, productivity, recurrence, cost
Prevention	Little	Central: job analysis, ergonomics, fitness, education

3 The industrial rehabilitation team

The industrial rehabilitation team (S22, 3 marks of the long answer).

Member	Role
Worker (and family)	Active partner; sets goals; reports symptoms early
Employer and supervisor	Provide modified duties, make workplace changes, support return to work
Occupational health physician	Diagnosis, medical care, fitness certification, surveillance
Physiotherapist	Assessment, treatment, FCE, work conditioning and hardening, ergonomics, prevention (section 4)
Occupational therapist	Hand function, ADL, job simulation, adaptive tools
Ergonomist and safety officer	Workplace risk assessment and redesign; safety rules and PPE
Psychologist	Pain coping, stress, fear of re-injury
Vocational counsellor	Alternative jobs and retraining when return to the old job is not possible
Case manager, insurer (ESIC), union representative	Coordinate care, benefits and communication

Job analysis and its purpose (the remaining 10 marks of S22) are on ability management (p. 180).

4 Role of the physiotherapist in industrial health

Role of the physiotherapist in industrial health settings (W19, 7 marks).

Area	What the physiotherapist does
Pre-placement	Screen new workers against job demands; record baseline; advise adaptations (section 5)
Job analysis and ergonomics	Job and task analysis, ergonomic evaluation (REBA, RULA, NIOSH), workstation redesign (see ability management (p. 179))
Prevention and education	Back schools, safe lifting, posture, micro-breaks, workstation exercises, warm-up before heavy shifts
Health promotion	Employee fitness programmes, weight and NCD risk, stress management (see job stress (p. 209))
Early intervention	On-site clinic for early treatment of aches before they become disorders; first aid
Disability management	FCE, work conditioning, work hardening, modified duties, graded return (see disability management (p. 186))
Hazard control	Advise on vibration, manual handling, standing work, PPE (see hazards (p. 193))

Area	What the physiotherapist does
Records and research	Injury data, programme evaluation, cost–benefit reports to management
Inclusion	Adapt work for older workers and persons with disability; reasonable accommodation under the RPWD Act

5 Pre-placement in the worker care spectrum

Describe the pre-placement worker care spectrum (S25, 7 marks).

- **Place in the spectrum:** the first step, before ability management and long before any injury (see the worker care spectrum (p. 179)).
- **Pre-placement (post-offer) screening:** after a job is offered, the worker's health and physical abilities are tested against the essential demands of that job.
- **Aims:** place the worker in a job they can do safely; find needs for training, aids or workplace changes; record a baseline for later comparison if injured; reduce injuries and compensation costs.

Components

1. Job demand analysis of the post (weights, postures, frequency, environment).
2. Health questionnaire and medical examination (history of back or joint problems, heart and lung disease, vision and hearing).
3. Musculoskeletal screen: posture, range, strength, flexibility.
4. Functional tests matched to the job: lifting and carrying to the job's weights and heights, pushing and pulling, positional tolerance (stooping, kneeling, climbing, overhead work), grip, aerobic fitness.
5. Result: fit; fit with adjustments (training, tools, rotation); temporarily unfit (with a plan to improve); report to the employer in functional terms, not diagnoses.

Rules

- Tests must be job-related, standardised and safe; the same tests for all candidates for a post.
- Must not be used to exclude persons with disability unfairly: the employer must consider reasonable accommodation (RPWD Act, 2016).
- Results are confidential.

6 Exam corner

Past questions (MUHS, 2016–2026)

Long answer

Define industrial therapy; the industrial rehabilitation team; job analysis and its purpose (S22, 2 + 3 + 7 + 3).

7 marks

Role of the physiotherapist in industrial health settings (W19); the pre-placement worker care spectrum (S25).

Keywords to underline

adaptation of work to people, industrial therapy, traditional medical vs industrial model, early return to work, employer as partner, industrial rehabilitation team, pre-placement screening, job-related tests, reasonable accommodation, Factories Act, ESI Act

Answer skeleton: role of the PT in industrial health (7 marks)

1. **[Definition, 1]** Occupational health; industrial model.

2. **[Roles, 5]** Pre-placement, job analysis and ergonomics, prevention and education, fitness, early care, disability management, hazard advice, records.

3. **[Outcome, 1]** Fewer injuries and days lost, better productivity.

Draw these

- The worker care spectrum (on ability management (p. 179))
- A two-column comparison of the medical and industrial models

Viva questions

1. Give the ILO/WHO aim of occupational health in one line.
2. Contrast the traditional medical model with the industrial model.
3. Who are the members of the industrial rehabilitation team?
4. What is pre-placement screening, and when is it done?
5. Name two Indian laws protecting workers' health.
6. Why must pre-placement tests be job-related?

Linked practical spots

- Design a pre-placement test battery for a warehouse loader or a nurse; present a short back-school talk to a group of workers.

Test yourself

Cover each answer and write yours on paper first, then check.

7 m **Role of the physiotherapist in industrial health settings. (W19)**

Answer
Define occupational health and the industrial model. Roles: pre-placement screening; job analysis and ergonomic evaluation; injury prevention education; employee fitness and stress management; early on-site care; FCE, work conditioning, work hardening and graded return; hazard advice; records and programme evaluation; inclusion of older and disabled workers. Outcome: fewer injuries, lost days and costs.

7 m **Describe the pre-placement worker care spectrum. (S25)**

Answer
Worker care spectrum (pre-placement, ability management, disability management, return to work). Pre-placement: after job offer; aims (safe placement, needs, baseline, fewer injuries); components (job demand analysis, health questionnaire and examination, musculoskeletal screen, job-matched functional tests, result and report); rules (job-related, standardised, non-discriminatory with reasonable accommodation, confidential).

LAQ **Define industrial therapy. The industrial rehabilitation team. Explain job analysis and its purpose. (2 + 3 + 7 + 3, S22)**

Answer
Definition (section 1). Team (section 3). Job analysis: job description, job demand analysis with frequency categories and physical demand levels, task analysis with risk factors (ability management page). Purpose: prevention and ergonomic redesign, pre-placement, matching with FCE, work hardening design, light duty, training, compensation.

MCQ **Early return to work with modified duties is a feature of the:**

A. Traditional medical model

B. Industrial model

C. Charity model

D. Disengagement model

Answer
B.

MCQ **Pre-placement screening is done:**

A. After an injury

B. After a job offer, before starting work

C. At retirement

D. Only for managers

Answer
B.

MCQ **The Act that provides medical and cash benefits to insured workers in India is the:**

A. ESI Act, 1948

B. PCPNDT Act

C. RCI Act

D. MTP Act

Answer
A.

Last recall check. Write the ILO/WHO definition, compare the two models in seven rows, list the team, nine roles of the PT, and the components and rules of pre-placement screening.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Community Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Joint ILO/WHO Committee on Occupational Health. Definition of occupational health (1950, revised 1995).
3. The Factories Act, 1948; the Employees' State Insurance Act, 1948; the Employees' Compensation Act, 1923; the Mines Act, 1952; the Occupational Safety, Health and Working Conditions Code, 2020.
4. Rights of Persons with Disabilities Act, 2016 (reasonable accommodation, non-discrimination in employment).
5. Isernhagen SJ, ed. The comprehensive guide to work injury management. Gaithersburg: Aspen; 1995.

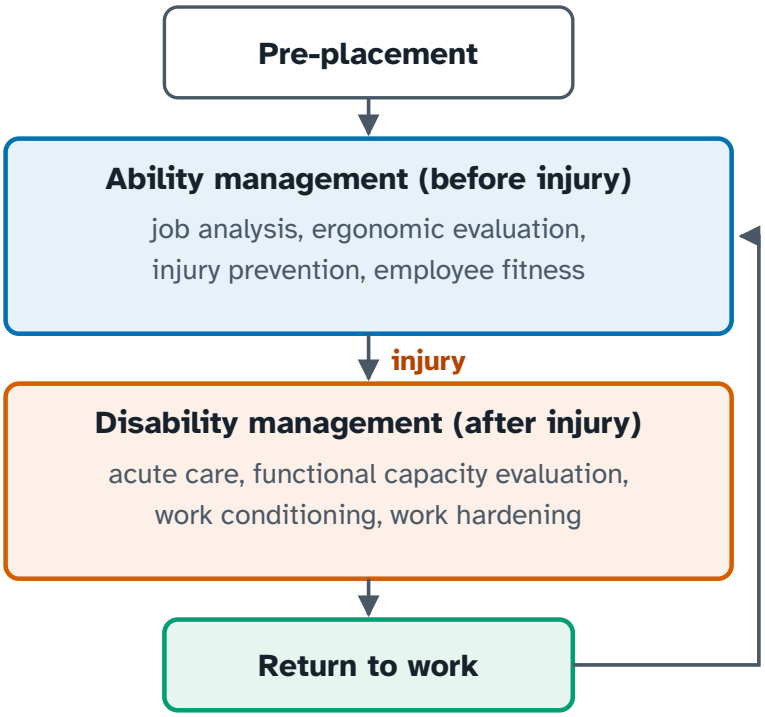
6(b i) Ability management: the worker care spectrum, job analysis, ergonomic evaluation, REBA, injury prevention and employee fitness

> Ability management was asked in 10 of 20 papers, always as short answers: task analysis, job demand analysis, ergonomic evaluation, REBA, cognitive ergonomics, workstation exercises, employee fitness. It is the "before injury" half of the worker care spectrum: understand the job, fit it to the worker, and keep the worker fit for it.

Syllabus	Ch 6(b i) Worker care spectrum: ability management: job analysis (job description, job demand analysis, task analysis), ergonomic evaluation, injury prevention, employee fitness programme
Priority	**Tier 1.** In 10 of 20 papers (2016–2026): 7 seven-mark and 6 three-mark questions.
Asked as	7m S16 · S19 · W19 · S21 · S22 · S23 · S25 · 3m S18 · S19 · W19 · S20 · S21 · S26
Usual split	7 (importance of task analysis; ergonomic evaluation; REBA; cognitive ergonomics; purpose and principles of ergonomics; workstation exercises); 3 (job analysis; job demand analysis; employee fitness; workplace health promotion)
Checked against	International Ergonomics Association definition and domains of ergonomics; REBA (Hignett and McAtamney, 2000); US Dictionary of Occupational Titles strength levels (via Social Security Administration rules); Luxembourg Declaration on workplace health promotion (1997); revised NIOSH lifting equation. Full list under Sources (p. 185). Content checked 20 October 2026.

1 The worker care spectrum

Worker care spectrum: from hiring to return to work



This page covers the blue half; the orange half is on disability management (p. 186). After return to work, the cycle starts again with ergonomics and prevention.

- **Pre-placement screening:** after a job offer, the worker's health, fitness and functional abilities are matched with the job demands; it records a baseline, identifies needs for training or adaptations, and must not be used to exclude persons with disability unfairly (RPWD Act, 2016).
- **Ability management:** keeping healthy workers healthy and productive by fitting the job to the worker and the worker to the job.

2 Job analysis

Job analysis (S26) and job demand analysis (S21), 3 marks; importance of task analysis (S19, 7 marks).

Part	What it records
Job description	Job title and purpose, main duties, tools and machines, materials, hours and shifts, breaks, workplace, supervision, pay system (piece rate or salary)
Job demand analysis	The physical, mental and environmental demands of the job and how often each occurs (below)
Task analysis	Each task broken into steps, with posture, force, repetition, duration, tools and environment for each step, and the risk factors found

Job demand analysis

- **Physical:** lifting, carrying, pushing and pulling (weights and distances); sitting, standing and walking time; bending, kneeling, squatting, crawling, climbing, reaching above shoulder; gripping, fine finger work, repetition; vibration.
- **Frequency (Dictionary of Occupational Titles):** occasionally up to a third of the day, frequently a third to two-thirds, constantly more than two-thirds.
- **Mental and sensory:** attention, memory, decisions, reading, vision, hearing, speed and deadlines.
- **Environment:** heat, cold, noise, light, dust, fumes, heights, wet floors, shift work.

Physical demand levels

Level	Lifting
Sedentary	Up to 10 lb (about 4.5 kg) occasionally; mostly sitting
Light	Up to 20 lb (9 kg) occasionally, 10 lb frequently
Medium	Up to 50 lb (23 kg) occasionally, 25 lb (11 kg) frequently
Heavy	Up to 100 lb (45 kg) occasionally, 50 lb frequently
Very heavy	Over 100 lb occasionally, over 50 lb frequently

Importance of task analysis (S19)

- Finds the exact steps and postures that cause pain or injury, so the fix is specific.
- Basis for ergonomic redesign of tools, workstation and methods.
- Matches the job to the worker in pre-placement and return to work (compare with the functional capacity evaluation).
- Designs job-specific work conditioning and hardening (job simulation).
- Shows what light duties are possible during recovery; guides job rotation.
- Basis for training safe methods and setting fair production rates; supports compensation decisions.

3 Ergonomics

Purpose and principles of ergonomics (S22), cognitive ergonomics (S21), define ergonomics (S23), 7 marks each.

- **Definition (International Ergonomics Association, 2000):** "the scientific discipline concerned with the understanding of interactions among humans and other elements of a system", and the profession that applies

theory and methods to design in order to optimise human well-being and overall system performance. Simply: fitting the job to the person.

Domain	Concerned with
Physical	Anatomy, anthropometry, physiology and biomechanics: postures, manual handling, repetition, musculoskeletal disorders, workplace layout
Cognitive	Mental processes (perception, memory, reasoning, motor response): mental workload, decision-making, skilled performance, human–computer interaction, human error, work stress, training
Organisational	Work systems: shift design, teamwork, communication, work pace, participation

Purpose

- Prevent work-related musculoskeletal disorders and accidents.
- Reduce fatigue and errors; improve comfort, productivity and quality.
- Include older workers and persons with disability by adapting work.

Principles

1. Work in neutral postures (spine in its curves, elbows by the side, wrists straight, neck upright).
2. Reduce excessive force (lighter loads, mechanical aids, good grips).
3. Keep everything within easy reach; work at the right height (around elbow height; lower for heavy work, higher for fine work).
4. Reduce excessive repetition and static loading; vary tasks.
5. Minimise fatigue with rest breaks and job rotation.
6. Minimise contact stress and vibration.
7. Provide clearance and space to move; allow change of posture (sit–stand).
8. Keep a comfortable environment (light, temperature, noise).
9. Make displays and controls easy to understand (cognitive); involve workers in changes (participatory ergonomics).

Cognitive ergonomics (S21)

- **Mental workload:** too much (overload: errors, stress) or too little (boredom, lapses); balance tasks and breaks.
- **Attention and alertness:** monotonous monitoring, night shifts and fatigue reduce vigilance.
- **Display and control design:** clear labels, logical layout, warnings that stand out, controls that move the expected way.
- **Human error:** design so mistakes are hard to make and easy to correct (checklists, forcing functions).
- **Computer work:** simple screens and software, readable fonts, fewer interruptions.
- **Training and decision support** for complex tasks; stress management.
- **Examples:** control rooms, drivers and pilots, surgeons, call centres, health-care workers.

4 Ergonomic evaluation and REBA

Importance of ergonomic evaluation at the workplace (S16, 7 marks); ergonomic physiotherapy evaluation (S19, 3 marks); REBA (W19, 7 marks); one ergonomic scale (S23, 7 marks).

Ergonomic physiotherapy evaluation: steps

1. Meet management and workers; review injury and absence records; symptom survey (Nordic Musculoskeletal Questionnaire).
2. Walk-through observation and video of the job; job and task analysis.
3. Measure the workstation (heights, reaches, seat, screen), loads, forces, frequency and environment (light, noise, temperature).

4. Score risk with a tool: RULA, REBA, NIOSH lifting equation, Strain Index (see risk assessment tools (p. 196)).
5. Recommend changes from the hierarchy of controls: redesign, tools and aids, layout, breaks and rotation, training, exercise.
6. Re-evaluate after changes.

Importance of ergonomic evaluation

- Identifies risk factors before injury occurs; prevents musculoskeletal disorders, accidents and absence.
- Improves comfort, productivity and quality; reduces compensation costs.
- Gives objective scores to prioritise and justify changes, and to show improvement.
- Helps return injured workers and include workers with disability.
- Meets legal duties for a safe workplace.

Rapid Entire Body Assessment (REBA)

- Developed by Hignett and McAtamney (2000) to score whole-body postures in tasks that are unpredictable or varied (health care, service, industry).
- **Group A:** trunk, neck and legs, plus a load or force score → score A.
- **Group B:** upper arm, lower arm and wrist, plus a coupling (grip) score → score B.
- Scores A and B give score C from a table; an activity score is added (static posture over a minute, repeated small actions, or rapid large posture changes) → REBA score 1–15.

REBA score	Risk	Action
1	Negligible	None needed
2–3	Low	May be needed
4–7	Medium	Needed
8–10	High	Needed soon
11–15	Very high	Needed now

RULA is the upper-limb version for seated and standing work with arms (computer, assembly), scored 1–7.

5 Injury prevention, employee fitness and workplace health promotion

Employee fitness programme (W19), workplace health promotion (S18), workstation exercises (S20), 3 marks each; workstation exercises for a sedentary worker (S25, 7 marks).

Injury prevention

- Ergonomic redesign first; then safe work methods, training in handling and tool use, PPE.
- Early reporting of symptoms and early physiotherapy (stage 1 of a cumulative trauma disorder is reversible).
- Warm-up before heavy shifts, micro-breaks, job rotation.

Employee fitness programme

1. Needs assessment and management support; a committee with workers.
2. Health risk appraisal and fitness tests (BP, BMI, waist, aerobic fitness, strength, flexibility).
3. Programme: on-site exercise classes or gym, walking groups, stretching breaks, sports; education on back care, nutrition, weight, smoking, stress and sleep.
4. Incentives and convenient timing; family involvement.
5. Evaluate: participation, fitness changes, injuries, absence, productivity.

Benefits: fewer musculoskeletal injuries and sick days, better fitness, morale and productivity, lower health costs.

Workplace health promotion (S18)

- **Definition (Luxembourg Declaration, 1997):** "the combined efforts of employers, employees and society to improve the health and well-being of people at work".
- Works through better work organisation and environment, active participation of workers, and personal development.
- Examples: ergonomic workplaces, fitness and health checks, healthy canteens, tobacco-free workplace, stress management, flexible hours.

Workstation exercises for a sedentary worker

Region	Exercises (5–10 repetitions, every 30–60 minutes)
Neck	Chin tucks; side bending and turning, holding 10 seconds
Shoulders and upper back	Shoulder rolls and shrugs; squeeze the shoulder blades back; arm reaches overhead
Mid back	Seated back extension over the chair back; seated twists; side bends
Wrists and hands	Wrist flexor and extensor stretches; fist and spread fingers; tendon gliding
Low back and hips	Standing back bends with hands on hips; pelvic tilts in sitting; stand up and walk
Legs	Ankle pumps and circles; seated knee straightening; heel raises in standing; marching
Eyes	Look into the distance regularly; blink; palming

Pair the exercises with the workstation set-up in desk work (p. 203).

6 Exam corner

Past questions (MUHS, 2016–2026)

7 marks

Importance of ergonomic evaluation at the workplace (S16); importance of task analysis (S19); REBA (W19); cognitive ergonomics (S21); purpose and principles of ergonomics (S22); define ergonomics and describe one scale (S23); workstation exercises for a sedentary worker (S25).

3 marks

Workplace health promotion (S18); ergonomic physiotherapy evaluation (S19); employee fitness programme (W19); workstation exercises (S20); job demand analysis (S21); job analysis (S26).

Keywords to underline

worker care spectrum, pre-placement, job description, job demand analysis, task analysis, occasionally, frequently, constantly, physical, cognitive, organisational, neutral posture, REBA 1–15, employee fitness, workplace health promotion

Answer skeleton: REBA or any ergonomic scale (7 marks)

1. **[Ergonomics, 1]** IEA definition; domains.
2. **[The tool, 4]** Who made it, what it scores (group A, group B, load, coupling, activity), the action table.
3. **[Use, 2]** When to use; example job; how scores guide change.

Draw these

- The worker care spectrum (section 1)
- Work heights: precision work above elbow, light work at elbow, heavy work below elbow

Viva questions

1. What are the three parts of job analysis?
2. Define "frequently" in a job demand analysis.
3. Name the three domains of ergonomics.
4. Which body parts are in REBA group A?
5. What REBA score means action is needed now?
6. Give four workstation exercises.

Linked practical spots

- Score a classmate's lifting posture with REBA; do a job demand analysis of a hospital attendant; demonstrate workstation exercises.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Job demand analysis. (S21)**

Answer
A systematic record of the physical (lifting, carrying, postures, hand use), mental and sensory, and environmental demands of a job, with weights, distances and frequency (occasionally up to a third of the day, frequently a third to two-thirds, constantly more than two-thirds). Used for pre-placement, ergonomic change, FCE matching and return to work.

3 m **Employee fitness programme. (W19)**

Answer
Workplace programme to improve workers' health and fitness: needs assessment, health risk appraisal and fitness testing, exercise classes, walking groups and stretch breaks, education (back care, nutrition, smoking, stress), incentives, evaluation of participation, injuries and absence. Benefits: fewer injuries and sick days, better productivity and morale.

7 m **Cognitive ergonomics. (S21)**

Answer
Define ergonomics and its three domains. Cognitive ergonomics deals with perception, memory, reasoning and response: mental workload, attention and vigilance, display and control design, human error, human–computer interaction, decision support, training, work stress. Give examples (control rooms, drivers, surgeons, call centres) and design solutions (clear displays, checklists, breaks, task variety).

7 m **Importance of task analysis. (S19)**

Answer
Define task analysis within job analysis. Importance: identifies harmful steps and postures; basis for ergonomic redesign; matches worker to job in pre-placement and return to work; designs job simulation in work hardening; identifies light duties; guides rotation, training and production rates; supports compensation. Give an example (lifting bags into a truck: bending, twisting, 50 kg, 3 ft height).

MCQ **A REBA score of 11–15 means:**

A. Negligible risk

B. Medium risk, action needed

C. Very high risk, action needed now

D. Low risk, action may be needed

Answer
C.

MCQ **Mental workload belongs to which domain of ergonomics?**

A. Physical

B. Cognitive

C. Organisational

D. Environmental

Answer
B.

MCQ **A job requiring lifting up to 50 lb occasionally and 25 lb frequently is classed as:**

A. Light

B. Medium

C. Heavy

D. Very heavy

Answer
B.

Last recall check. Draw the worker care spectrum, list the three parts of job analysis, the domains and principles of ergonomics, the REBA action table and six workstation exercises.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Community Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. International Ergonomics Association. What is ergonomics? Definition and domains (2000).
3. Hignett S, McAtamney L. Rapid entire body assessment (REBA). *Appl Ergon* 2000;31:201–5.
4. US Department of Labor. Dictionary of Occupational Titles, appendix C: strength levels and frequency; as used in 20 CFR 404.1567 and SSA POMS DI 25001.001.
5. European Network for Workplace Health Promotion. Luxembourg Declaration on workplace health promotion in the European Union, 1997.
6. Rights of Persons with Disabilities Act, 2016.

6(b ii) Disability management: acute care, functional capacity evaluation, work conditioning, work hardening and job simulation

> Disability management was asked in 10 of 20 papers: functional capacity evaluation five times, work conditioning and work hardening six times, and two long cases asking for both. Remember the order: treat the injury, measure what the worker can do (FCE), compare it with what the job needs (job analysis), and close the gap with work conditioning and then work hardening.

Syllabus	Ch 6(b ii) Worker care spectrum: disability management: acute care, concept of functional capacity assessment, work conditioning, work hardening
Priority	**Tier 1.** In 10 of 20 papers (2016–2026): 2 long cases, 4 seven-mark and 7 three-mark questions.
Asked as	LAQ W18 · W20 · 7m W16 · S17 · W18 · S19 · 3m S16 · S17 · W21 · S22 · W22 ×2 · W25
Usual split	15 (job task analysis, then work hardening and conditioning programme; or functional capacity assessment, then work conditioning and hardening); 7 (FCE; work hardening with examples)
Checked against	APTA and CARF descriptions of work conditioning and work hardening as quoted in state workers' compensation standards (Wisconsin DWD 81.14; Washington State work hardening standards); Isernhagen Work System FCE literature; Dictionary of Occupational Titles physical demand levels. Full list under Sources (p. 192). Content checked 20 October 2026.

1 Disability management and acute care

- **Disability management:** the "after injury" half of the worker care spectrum (see the diagram on ability management (p. 179)): a planned process by the employer, worker and health team to limit the effect of an injury or illness on work and bring the worker back safely and early.
- **Steps:** early reporting → acute care → case management and communication with the employer → functional capacity evaluation → work conditioning → work hardening → modified or transitional duties → full return to work → prevention of recurrence.

Acute care

- Diagnosis and medical treatment; pain and swelling control (rest from aggravating tasks, ice, elevation, medication, electrotherapy).
- Early, graded movement and exercise; avoid long bed rest and deconditioning.
- Education and reassurance; set an expected return-to-work date early.
- Contact with the employer about light or modified duties; keep the worker connected to the workplace.
- Screen for "yellow flags" (fear of movement, low mood, job dissatisfaction) that delay recovery.

2 Functional capacity evaluation

Functional capacity evaluation or assessment: 7 marks in W16, S17 and S19; define it for 3 marks in W22 and W25; the W20 long case.

- **Definition:** a standardised set of tests that measures a person's safe maximum ability to perform work-related physical tasks, and compares it with the demands of a job or a physical demand level.
- **Uses:** return-to-work decisions and job matching; planning work conditioning and hardening; deciding on modified duties; disability rating and compensation; pre-placement screening; measuring progress.

Components

Part	What is tested
Intake	History, job description, symptoms, self-rated abilities, medical clearance, consent
Musculoskeletal screen	Posture, range of motion, strength, sensation, heart rate and BP
Material handling	Lifting floor to waist, waist to shoulder and overhead; carrying; pushing and pulling, with weights increased until a safe maximum
Positional tolerances	Sitting, standing, walking, stooping, kneeling, crouching, reaching, climbing stairs and ladders, balance
Hand function	Grip (dynamometer), pinch, dexterity tests
Aerobic capacity	Step or treadmill test
Effort and consistency	Repeated trials, heart rate response, observed body mechanics, consistency between tests
Report	Safe abilities, physical demand level, comparison with the job, projection to an 8-hour day, recommendations

- **Approaches:** the kinesiophysical approach (Isernhagen Work System, 28 work-related tests) watches body mechanics to decide the safe maximum; psychophysical approaches use the worker's own judgement of an acceptable load.
- **Duration:** usually several hours, often over 1–2 days.
- **Limits:** affected by pain, fear and motivation; predicts return to work only modestly; needs a trained evaluator and standard protocol.

3 Work conditioning and work hardening

Work conditioning (S17, S22), definitions of both (W21, W22), 3 marks; work hardening with two examples (W18, 7 marks).

Feature	Work conditioning	Work hardening
Definition	An intensive, work-related, goal-oriented conditioning programme to restore systemic, neuromusculoskeletal and cardiopulmonary function (strength, endurance, mobility, motor control, aerobic capacity) so the person can return to work	A highly structured, goal-oriented, individualised, interdisciplinary programme using real or simulated work tasks to restore physical, behavioural and vocational functions for return to a specific job
Team	Usually one discipline (physiotherapist)	Interdisciplinary: physiotherapist, occupational therapist, psychologist, vocational counsellor, case manager
Focus	Physical capacity: general strength, flexibility, endurance, aerobic fitness	Job-specific tasks, productivity, safety, work tolerance, work behaviours (punctuality, pacing, attitude), pain coping
Hours	Part of a day, several days a week	Builds up towards a full working day, several days a week
Entry	After acute care, when there is a physical deficit	When physical capacity is nearly enough but job tolerance, confidence or behaviour limit return

Work conditioning: content

- Aerobic training (treadmill, cycle, walking) 20–40 minutes.
- Progressive strengthening of the injured area and the whole body (trunk, legs, grip).
- Flexibility and mobility; motor control and core stability.
- Basic body mechanics and lifting practice; education.

Work hardening: content

- Job simulation: real or simulated tasks from the job analysis, graded in weight, height, distance, repetitions and time.
- Physical conditioning continued alongside.
- Body mechanics and pacing during the tasks; ergonomic changes for the real job.
- Pain management, stress and fear-of-movement counselling; work habits (timekeeping, following instructions).
- Regular re-measurement against the job demands; plan for graded return.

Two examples of work hardening (W18)

- **Warehouse loader after low back strain:** lifting crates from floor to waist to shoulder with gradually heavier loads (for example 10 kg to 25 kg), carrying 20 m, pushing a loaded trolley, loading a mock truck bed, building up to the full shift's number of lifts with correct technique.
- **Data-entry clerk after carpal tunnel release or wrist fracture:** graded typing and mouse sessions from 15 to 60 minutes, filing and lifting files, dexterity tasks, workstation set-up and micro-breaks, building up to a full working day.

Job simulation (S16, 3 marks)

- Reproducing the real tasks of a job (same tools, weights, heights, postures and pace) in the clinic or at a mock workstation.
- Used to train (work hardening) and to test whether the worker can meet the job demands (FCE).
- Examples: work simulators with interchangeable tool handles, work sample batteries (such as Valpar work samples), mock construction, assembly or office stations.

4 Return to work

- **Options:** same job full duty; same job with changes (tools, layout, hours); modified or light duty for a period (transitional work); a different job with the same employer; retraining (see vocational rehabilitation (p. 136)).
- **Graded return:** start with part days and lighter tasks, increase weekly as tolerance improves.
- **Prevention of recurrence:** ergonomic changes, home exercise, fitness, early reporting (see ability management (p. 179)).

5 Worked long case: a sugarcane worker with neck and arm pain

From a real paper (W18)

> "A 32 years old, male, 65 kg, 5'8 ft, working in sugar cane industry, complains of neck arm pain since 6 months. Write the job task analysis. Describe the work hardening and work conditioning program for him."

> **Spot the clues before you write**
>
> sugarcane industry: cutting with a sickle, carrying bundles on the head or shoulder, loading, or feeding cane and lifting sugar bags in the mill. neck and arm pain for 6 months: cervical overload with possible nerve root irritation; persistent, so deconditioning and fear are likely. 32, 65 kg: young, light build carrying heavy loads.

Job task analysis

Task	Demands	Risk to neck and arm
Cutting cane	Stooping or squatting for hours, repetitive forceful sickle strokes, heat	Sustained neck flexion, repetitive shoulder and elbow loading

Task	Demands	Risk to neck and arm
Bundling and carrying	Heavy bundles on the head or shoulder, carried over uneven fields	Axial compression of the cervical spine, one-sided shoulder load
Loading trucks or carts	Lifting bundles above shoulder height, twisting, climbing	Overhead lifting with neck extension
Mill work (if in the factory)	Feeding cane into crushers, lifting sugar bags, vibration, noise, heat, shifts	Repetitive lifting, awkward reach
Work organisation	Piece rate, long days in the harvesting season, few breaks	Fatigue, no recovery

Functional diagnosis in one line: a 32-year-old sugarcane worker with 6 months of mechanical neck pain with arm referral (rule out cervical radiculopathy) from head and shoulder loading, stooped cutting and overhead loading, with reduced neck mobility, weak deep neck and scapular muscles and low work tolerance, unable to manage full harvesting or loading work.

Assessment before the programme

- Neck range, neurological screen (myotomes, reflexes, sensation, Spurling's and upper limb tension tests), strength of deep neck flexors, scapular and grip muscles; pain scale and Neck Disability Index.
- Functional capacity evaluation: lifting floor to waist and waist to shoulder, carrying, overhead work tolerance, repetitive arm work, stooping tolerance; compare with the job demands above (very heavy work).

Work conditioning (weeks 1–3)

Impairment	Exercise and dose	Why
Neck stiffness and pain	Neck and thoracic mobility, nerve gliding if arm symptoms, 10 repetitions twice a day; heat or TENS before exercise if needed	Restores movement, calms nerve symptoms
Weak neck and scapular muscles	Deep neck flexor training, isometrics in all directions, scapular retraction and lower trapezius with bands, 3 × 10, progressing to resisted shoulder work	Supports the neck under load
Low general capacity	Aerobic exercise 20–30 minutes; trunk, hip and leg strengthening; grip strengthening	Builds endurance for long harvest days
Poor lifting mechanics	Hip-hinge and squat lifting practice with light loads	Moves load from neck and back to the legs

Work hardening (weeks 3–8)

- **Job simulation:** stooped cutting motions with a weighted handle; lifting and carrying bundles of graded weight (10 kg rising towards his job load) with correct technique; loading onto a raised platform; repeated for longer periods towards a full shift.
- **Load carriage advice:** avoid head-loading; carry on both shoulders with a padded yoke or use a cart or wheelbarrow; share loads; keep loads close.
- **Pacing:** micro-breaks every 30–60 minutes with neck and shoulder stretches; drink water in the heat.
- **Behaviour and confidence:** pain education, graded exposure to feared tasks, punctuality and full-day attendance.
- **Return to work:** lighter tasks first (cutting without carrying), then full work; discuss with the contractor or employer about load size and carts.

Practice case (W20)

A 54-year-old production supervisor in a garment factory, a month after surgery for multiple fractures from a road accident: plan the functional capacity assessment (section 2: intake, musculoskeletal screen, lifting and carrying within fracture healing limits, walking and stairs, standing and sitting tolerance, hand function, aerobic test) and then work conditioning (strength, mobility, endurance for the operated limbs and whole body) and work hardening

(supervising rounds on the factory floor, standing and walking for hours, climbing stairs, carrying files and samples, computer work), building to full days.

6 Exam corner

Past questions (MUHS, 2016–2026)

Long cases

32-year-old sugarcane worker with neck and arm pain: job task analysis, work hardening and conditioning (W18); 54-year-old supervisor after fracture fixation: functional capacity assessment, work conditioning and hardening (W20).

7 marks

Functional capacity evaluation (W16, S17, S19); work hardening with two examples (W18).

3 marks

Job simulation (S16); work conditioning (S17, S22); define work hardening and work conditioning (W21, W22); define functional capacity assessment (W22, W25).

Keywords to underline

worker care spectrum acute care early return to work functional capacity evaluation material handling positional tolerance physical demand level work conditioning: physical, one discipline work hardening: job-specific, interdisciplinary job simulation graded return

Answer skeleton: functional capacity evaluation (7 marks)

1. **[Definition and uses, 2]** What it measures; why it is done.
2. **[Components, 4]** Intake, screen, material handling, positional tolerance, hand function, aerobic, effort, report.
3. **[Approach and limits, 1]** Kinesiophysical vs psychophysical; reliability and motivation.

Draw these

- The worker care spectrum with this page's steps in the orange box (on ability management (p. 179))
- A comparison table of work conditioning and work hardening

Viva questions

1. Define functional capacity evaluation.
2. Name four material-handling tests in an FCE.
3. Who delivers work conditioning, and who delivers work hardening?
4. What is job simulation?
5. Give an example of work hardening for a loader.
6. What are "yellow flags"?

Linked practical spots

- Demonstrate a floor-to-waist lift test with safe stopping criteria; measure grip with a dynamometer; set up a simulated loading task.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Define work hardening and work conditioning. (W21, W22)**

Answer
Work conditioning: an intensive, work-related, goal-oriented programme, usually by one discipline, to restore strength, endurance, mobility, motor control and aerobic capacity for return to work. Work hardening: a highly structured, individualised, interdisciplinary programme using real or simulated work tasks to restore physical, behavioural and vocational functions for a specific job.

3 m **Job simulation. (S16)**

Answer
Reproducing the actual tasks of a job (tools, weights, heights, postures, pace) in the clinic or a mock workstation, to train job-specific tolerance in work hardening and to test job readiness in an FCE. Examples: work simulators, work sample batteries, mock loading or assembly stations.

7 m **Functional capacity evaluation. (W16, S17, S19)**

Answer
Definition; uses (return to work, job matching, programme planning, compensation, pre-placement); components (intake, musculoskeletal screen, lifting floor-waist-shoulder and carrying, pushing and pulling, positional tolerances, hand function, aerobic test, effort and consistency, report with physical demand level and 8-hour projection); approaches (kinesiophysical, Isernhagen; psychophysical); limits.

LAQ **54-year-old production supervisor, a month after fracture fixation from a road accident: functional capacity assessment, work conditioning and work hardening. (W20)**

Answer
Job analysis of a supervisor (standing and walking rounds, stairs, carrying files, computer work). FCA: intake and medical clearance, fracture healing limits, ROM and strength, walking and stair tolerance, standing and sitting tolerance, carrying and lifting within limits, hand function, aerobic test, report. Work conditioning: progressive strengthening, mobility, gait and balance, aerobic training. Work hardening: simulated floor rounds, stair climbing with files, desk work periods, building to full days; graded return with modified duties first.

MCQ **Work hardening differs from work conditioning mainly because it:**

A. Uses only aerobic exercise

B. Is interdisciplinary and uses real or simulated job tasks

C. Is done in the acute stage

D. Does not involve the employer

Answer
B.

MCQ **The Isernhagen Work System is a method of:**

A. Ergonomic posture scoring

B. Functional capacity evaluation

C. Noise measurement

D. Vocational counselling

Answer
B.

MCQ **The best first step in managing an acute work injury is:**

A. Two weeks of bed rest

B. Early care with graded movement and contact with the employer

C. Immediate work hardening

D. Waiting for full pain relief before any activity

Answer

B.

Last recall check. List the steps of disability management, the FCE components, a five-row comparison of work conditioning and work hardening, and one work hardening example.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Community Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. American Physical Therapy Association. Occupational health physical therapy guidelines: work conditioning and work hardening (as quoted in Wisconsin Administrative Code DWD 81.14 and the Rhode Island Workers' Compensation Court work hardening protocols).
3. Washington State Department of Labor and Industries. Work hardening program standards.
4. Isernhagen SJ. Functional capacity evaluation: rationale, procedure, utility of the kinesiophysical approach. *J Occup Rehabil* 1992;2:157–68.
5. US Department of Labor. Dictionary of Occupational Titles: physical demand levels.

6(b iii) Environmental stress at work: physical, chemical and mechanical hazards, pneumoconiosis, vibration and cumulative trauma disorders

> Occupational hazards were asked in 12 of 20 papers, with 4 long cases: a vibration-tool worker, a sand-industry worker, a coal miner and a clerk with carpal tunnel syndrome. For every hazard, write three things: what it does to the body, how the physiotherapist treats the effects, and how the workplace prevents it, starting at the top of the hierarchy of controls.

Syllabus	Ch 6(b iii) Environmental stress in the industrial area: accidents due to physical agents (heat, cold, light, noise, vibration, UV and ionising radiation), chemical agents (inhalation, local action, ingestion) and mechanical hazards (overuse and fatigue injuries; ergonomic evaluation of the workplace)
Priority	**Tier 1.** In 12 of 20 papers (2016–2026): 4 long cases, 5 seven-mark and 6 three-mark questions.
Asked as	LAQ S16 ×2 · W17 · S21 · 7m S18 · W20 · S22 · W22 · S24 · 3m W16 · S18 · W18 · S19 · W20 · W25
Usual split	15 (functional diagnosis 5, goals and management 7, protective gear 3; or functional diagnosis 5, PT 10; or task analysis 5, functional diagnosis 5, prevention 5; or prevention 5, ergonomics 5, orthosis 5); 7 (physical agents; CTD; risk assessment tools; industrial injury)
Checked against	NIOSH hierarchy of controls; EU vibration directive 2002/44/EC; Stockholm Workshop scale for hand-arm vibration syndrome; Schilling grading of byssinosis; RULA and REBA; ICRP dose limits; Tamil Nadu Factories Rules noise limits. Full list under Sources (p. 202). Content checked 20 October 2026.

1 Types of occupational hazards

Types of occupational hazards (W25, 3 marks).

Type	Examples
Physical	Heat, cold, poor lighting and glare, noise, vibration, ultraviolet and ionising radiation, high or low air pressure
Chemical	Dusts (silica, coal, asbestos, cotton), gases and fumes, solvents, metals (lead, mercury), pesticides; acting by inhalation, skin contact or ingestion
Biological	Tuberculosis and hepatitis B in health workers, anthrax and brucellosis in animal handlers, tetanus, leptospirosis
Mechanical and ergonomic	Machinery accidents, falls, lifting, repetitive and forceful work, awkward postures (see mechanical stress (p. 203))
Psychosocial	Job stress, shift work, monotony, harassment (see job stress (p. 209))

2 Physical agents

Influence of physical agents on industrial workers' health (W20, 7 marks).

Agent	Effects on health	Prevention
Heat (furnaces, glass, steel, kitchens, outdoor summer work)	Heat rash, cramps, syncope, heat exhaustion, heat stroke (very high body temperature with confusion: an emergency); dehydration; more accidents	Shielding and ventilation, acclimatisation, water and salt, rest in shade, work in cooler hours, light clothing
Cold (cold stores, high altitude)	Chilblains, frostbite, trench foot, hypothermia; stiffness and more injuries	Insulated clothing and gloves, warm breaks, limit exposure
Light (poor or glaring)	Eye strain, headache, accidents, miner's nystagmus (poor light underground)	Adequate, even lighting without glare; task lights

Agent	Effects on health	Prevention
Noise (textile looms, presses, drilling)	Noise-induced hearing loss (first around 4 kHz), tinnitus, poor communication, stress, accidents	Quieter machines and enclosures, limit exposure (8 hours at 90 dBA under Indian factory rules), ear plugs and muffs, hearing tests
Vibration	Hand-arm: vibration white finger, numbness, carpal tunnel syndrome, grip weakness. Whole-body: low back pain	Section 5; EU daily limits (hand-arm action 2.5, limit 5 m/s^2; whole-body action 0.5, limit 1.15 m/s^2)
Ultraviolet (welding, sun)	Welder's flash (photokeratitis), sunburn, cataract, skin cancer	Welding shields and goggles, screens, covering clothing
Ionising radiation (X-ray, nuclear, radiography)	Skin burns, radiation sickness, cataract, cancer and leukaemia, infertility, birth defects	Time, distance, shielding (lead aprons), dosimeter badges; dose limit 20 mSv a year averaged over 5 years (ICRP)

3 Chemical agents and the pneumoconioses

Three hazards of chemical agents (S19), byssinosis (S18), asbestosis (W18), sequelae of pneumoconiosis (W20), 3 marks each; sand-industry worker (S16) and coal miner (S21) long cases.

- **Chemical hazards by route:** local action on skin and eyes (dermatitis, burns, ulcers, skin cancer); inhalation of dusts, gases, fumes and vapours (pneumoconiosis, asthma, poisoning by carbon monoxide, lead, solvents); ingestion from contaminated hands, food and smoking at work (lead and mercury poisoning).
- **Pneumoconiosis:** lung disease from inhaling mineral or organic dust, causing fibrosis.

Disease	Dust and jobs	Key points
Silicosis	Free crystalline silica: stone quarrying and crushing, mining, sandblasting, sand industry, glass, pottery, slate-pencil and agate work	Upper-lobe nodular fibrosis, progressive massive fibrosis; high risk of tuberculosis (silico-tuberculosis); no cure
Coal workers' pneumoconiosis	Coal dust: underground mining	Simple nodules to progressive massive fibrosis; chronic bronchitis; with rheumatoid arthritis, Caplan's syndrome
Asbestosis	Asbestos fibres: asbestos mining, cement sheets, insulation, brake linings, ship-breaking	Lower-lobe fibrosis, pleural plaques and thickening, clubbing; lung cancer (more with smoking) and mesothelioma, decades later
Byssinosis	Cotton, flax and hemp dust: textile mills (carding and blowing rooms)	Chest tightness and breathlessness on the first day back at work after a break ("Monday chest tightness"), later chronic airflow obstruction
Organic dust lung diseases	Bagassosis (sugarcane bagasse), farmer's lung (mouldy hay)	Hypersensitivity pneumonitis: fever, cough and breathlessness hours after exposure; fibrosis if exposure continues

Schilling grades of byssinosis

Grade	Symptoms
½	Occasional chest tightness or breathlessness on the first day of the working week
1	Chest tightness or breathlessness on every first day of the working week
2	On the first and other days of the working week
3	Grade 2 with permanent incapacity from reduced effort tolerance

Sequelae of pneumoconiosis (W20)

- Progressive fibrosis and restrictive lung disease; progressive massive fibrosis.

- Tuberculosis (especially with silica); chronic bronchitis and airflow obstruction.
- Pulmonary hypertension and cor pulmonale; respiratory failure.
- Lung cancer (asbestos, silica) and mesothelioma (asbestos); pneumothorax.
- Disability, loss of work and income, early death.

Physiotherapy and prevention

- **PT:** breathing control and pursed-lip breathing, airway clearance (active cycle of breathing) when sputum is present, thoracic mobility and posture, graded exercise training as pulmonary rehabilitation, energy conservation, oxygen if desaturating as prescribed, smoking cessation, vaccination advice (see interstitial lung disease and pulmonary rehabilitation).
- **Prevention:** substitute safer materials (asbestos bans); wet methods, enclosure, exhaust ventilation; respirators (N95 or better) correctly fitted; pre-placement and periodic examination with chest X-ray (ILO classification) and spirometry; screening for tuberculosis in silica workers; remove affected workers from exposure; compensation under the Employees' Compensation Act (these are notifiable diseases under the Factories Act).

4 Cumulative trauma disorders and carpal tunnel syndrome

Causative factors for CTD (W16, 3 marks); CTD with one common example (W22, S24, 7 marks); clerk with carpal tunnel syndrome (W17, long case).

- **Cumulative trauma disorders** (work-related musculoskeletal disorders): injuries of muscles, tendons, nerves and joints from repeated small strains without enough recovery.
- **Work factors:** repetition, force, awkward or static posture, vibration, contact pressure from tool edges, cold, long hours without breaks, piece-rate pressure.
- **Personal factors:** age, female sex, pregnancy and menopause, diabetes, hypothyroidism, rheumatoid arthritis, obesity, previous injury, low fitness, smoking; stress.
- **Examples:** carpal tunnel syndrome, de Quervain's tenosynovitis, trigger finger, lateral epicondylitis, rotator cuff tendinopathy, tension neck syndrome, hand-arm vibration syndrome.

Stages of a CTD as commonly described

Stage	Features
1	Ache and tiredness during work that settles overnight; work not affected; reversible
2	Pain starts early in the shift and lasts overnight; sleep may be disturbed; work capacity reduced
3	Pain at rest and with light non-repetitive tasks; poor sleep; unable to do even light duties; may last months to years

Carpal tunnel syndrome (the usual "one common CTD")

- **What:** compression of the median nerve under the flexor retinaculum at the wrist.
- **Features:** tingling and numbness in the thumb, index, middle and half the ring finger, worse at night and with gripping; shaking the hand helps; later weakness and thenar wasting (abductor pollicis brevis).
- **Tests:** Phalen's (wrist flexion 60 seconds), Tinel's over the carpal tunnel, carpal compression test; nerve conduction studies confirm and grade.
- **Treatment:** night splint holding the wrist in neutral; nerve and tendon gliding exercises; ergonomic changes and breaks; steroid injection; surgery (carpal tunnel release) when there is thenar wasting, constant numbness or failed conservative care.

5 Hand-arm vibration syndrome

The S16 long case (see section 8): an automobile worker using vibrating tools without protection.

- **Cause:** regular use of vibrating hand tools (grinders, drills, chipping hammers, riveters, chain saws, polishers).
- **Vascular:** attacks of finger blanching (white finger) set off by cold, then blue and red on rewarming.

- **Sensorineural:** tingling, numbness, poor touch and dexterity; often carpal tunnel syndrome as well.
- **Musculoskeletal:** grip weakness, pain in the hands, wrists and elbows.

Stockholm Workshop scale (summary)

Component	Stages
Vascular	0 no attacks; 1 occasional attacks of the fingertips; 2 occasional attacks of the distal and middle phalanges; 3 frequent attacks of all phalanges of most fingers; 4 as 3 with skin changes at the fingertips
Sensorineural	0SN exposed, no symptoms; 1SN intermittent numbness with or without tingling; 2SN numbness with reduced sensation; 3SN numbness with reduced touch discrimination or dexterity

Protective gear and instruments

- **Tools:** low-vibration tools, anti-vibration handles and mounts, tool balancers and suspension so the worker does not grip hard, sharp and well-maintained tools.
- **Gloves:** anti-vibration gloves reduce little of the vibration at the frequencies that matter, but warm, dry gloves help prevent blanching attacks.
- **Measurement:** vibration meters to estimate daily exposure against the action and limit values; limit time on the tool and rotate jobs.
- Warm clothing and a warm workplace; no smoking (it narrows vessels).

6 Industrial injury, risk assessment and control

Workplace hazards and risk assessment tools (S18, 7 marks); contributing factors for industrial injury (S22, 7 marks).

Factors in industrial injury

Factor	Examples
Worker	Fatigue, lack of training or experience, haste, alcohol, illness, poor vision or hearing, stress, not using protection
Machine and tools	Unguarded moving parts, faulty or poorly maintained equipment, unsuitable tools
Environment	Poor lighting, noise, heat, slippery or cluttered floors, poor ventilation, work at height
Organisation	Long hours, night shifts, production pressure, poor supervision, no safety rules or PPE, poor maintenance
Ergonomic	Heavy lifting, awkward postures, repetition, badly designed workstations

Risk assessment

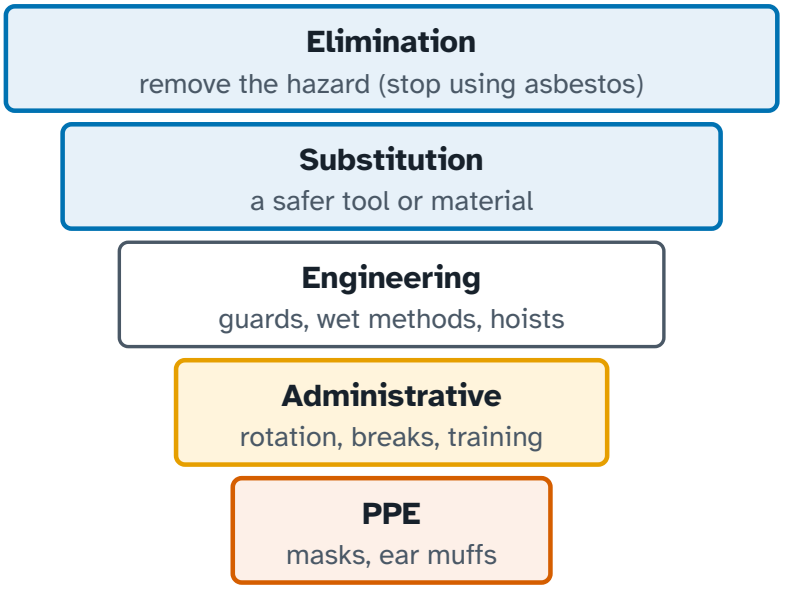
1. Identify hazards (walk-through, job analysis, injury records, worker interviews).
2. Decide who might be harmed and how.
3. Evaluate the risk (likelihood × severity on a risk matrix) and decide controls.
4. Record findings and act.
5. Review regularly and after any change or injury.

Ergonomic risk assessment tools

Tool	What it assesses
RULA (Rapid Upper Limb Assessment)	Posture of neck, trunk and upper limbs with force and repetition; score 1–7 (1–2 acceptable, 7 change immediately)
REBA (Rapid Entire Body Assessment)	Whole-body postures, load and coupling; score 1–15 (11 or more very high risk)
NIOSH lifting equation	Recommended weight limit for a lifting task (23 kg in ideal conditions) and lifting index

Tool	What it assesses
Strain Index, OCRA	Risk of distal upper-limb disorders in repetitive hand work
OWAS, QEC, checklists	Working postures; quick exposure check by observer and worker
Nordic Musculoskeletal Questionnaire	Symptoms in body regions in the past 12 months and 7 days

Hierarchy of controls



Most effective at the top (blue), least effective at the bottom (orange). Answer every prevention question from the top down; PPE is the last line, not the first.

7 Practice cases from the papers

- **S16, 30-year-old in the sand industry** with breathlessness on effort for 6 months, mucoid sputum and patchy consolidation (functional diagnosis 5, PT 10): think silicosis and rule out tuberculosis; ICF with reduced exercise tolerance and job loss risk; PT and prevention from section 3.
- **S21, 40-year-old coal miner for 15 years** with breathlessness and cough (task analysis 5, functional diagnosis 5, prevention 5). **Task analysis:** underground work in cramped space with stooping, kneeling and crawling; drilling (vibration, dust), shovelling and lifting; heat, humidity, poor light, noise; shift work; risk of roof falls, gas and fire. **Diagnosis:** coal workers' pneumoconiosis with chronic bronchitis. **Prevention:** water sprays and wet drilling, ventilation and dust extraction, respirators, periodic X-ray and spirometry, job rotation, smoking cessation, compliance with the Mines Act.
- **W17, 35-year-old clerk** at a computer for 8 hours with thumb and index tingling and thenar wasting (prevention 5, ergonomics 5, orthosis 5): carpal tunnel syndrome, already with wasting, so refer for nerve conduction and a surgical opinion; education on breaks and wrist posture, nerve gliding; workstation (section 2 of desk work (p. 203): keyboard at elbow height, neutral wrists, light touch, mouse close); orthosis: a resting or cock-up splint with the wrist in neutral, worn at night and during flares.

8 Worked long case: an automobile worker with numb hands

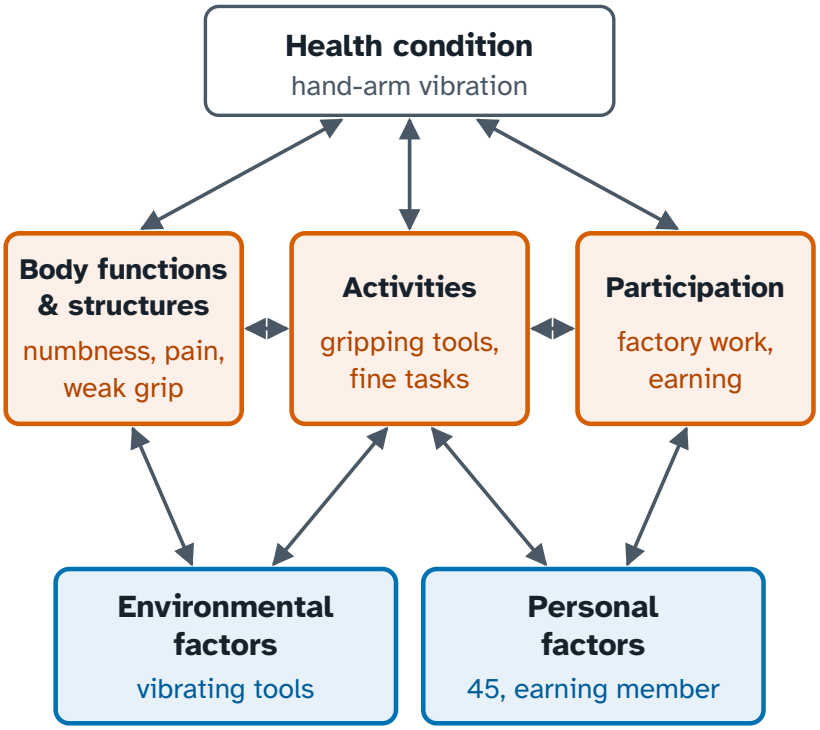
From a real paper (S16, 5 + 7 + 3)

> "A 45 year old male working in an automobile company (vibrations) without any precaution comes to the PT department with complains of tingling numbness & pain in bilateral hands. i) Discuss functional diagnosis with clinical reasoning. (5) ii) Give short term goals and management. (7) iii) Preventive gears and instruments. (3)"

Spot the clues before you write

vibrations without any precaution: hand-arm vibration syndrome. tingling, numbness and pain in both hands: sensorineural component, possible carpal tunnel syndrome; ask about white fingers in the cold. 45, automobile company: years of grinding or drilling; his job is at stake.

The ICF model filled for this case



Draw the six boxes with two-way arrows, then write two or three points from the case in each box.

ICF component	From this case	Clinical reasoning
Health condition	Hand-arm vibration syndrome (sensorineural, possibly vascular) with possible bilateral carpal tunnel syndrome	Years of unprotected exposure to vibrating tools
Body functions and structures	Tingling and numbness of the fingers, hand pain, reduced touch and two-point discrimination, weak grip and pinch, possible finger blanching in cold, positive Phalen's and Tinel's	Vibration damages digital nerves and vessels; tight grip adds median nerve load
Activity limitations	Holding tools steadily, fine tasks (buttons, screws), carrying, working in cold	Loss of sensation and grip
Participation restrictions	Full work at the factory; risk of losing his job and income	The job itself causes the problem
Environmental factors	**Barriers:** vibrating tools, no protective equipment, no exposure limits, cold workshop. **Facilitators:** employer safety officer, ESI and compensation schemes	Prevention must change the workplace

ICF component	From this case	Clinical reasoning
Personal factors	45 years, earning member of the family, possibly a smoker	Needs to keep working safely

One-line functional diagnosis

A 45-year-old automobile worker with hand-arm vibration syndrome from unprotected tool use, with numbness, pain and weak grip in both hands limiting tool handling and threatening his job.

Short-term goals (4–6 weeks)

- Reduce pain and tingling; no blanching attacks at work.
- Improve grip and pinch strength and dexterity.
- Daily vibration exposure reduced below the action value; protective measures in use.
- Understands the condition, warm-hands routine and warning signs.

Management

Impairment	Technique and dose	Why
Ongoing exposure	Reduce or stop vibrating-tool work; job rotation; refer to the occupational physician for nerve conduction and vascular tests	Removing exposure is the main treatment
Tingling and numbness	Median and ulnar nerve gliding, 5–10 repetitions, 3 times a day; night splint in neutral if carpal tunnel signs	Reduces nerve tension and night symptoms
Pain	Warm water soaks (not hot, sensation is poor), TENS as needed	Relief while exposure is reduced
Weak grip and dexterity	Putty and hand-gripper exercises, finger-thumb opposition, pegboard and coin tasks, 10–15 minutes twice a day	Restores hand function
Blanching in cold	Keep the whole body and hands warm, gloves, avoid smoking; hand and arm exercises before work to warm up	Prevents vasospasm
Sensory loss	Protect hands from burns and cuts, check skin daily, sensory re-education	Prevents injury to numb fingers

Preventive gear and instruments (3 marks): low-vibration tools and anti-vibration handles; tool balancers and suspension; regular maintenance; warm, dry gloves; vibration measurement and exposure time limits; health surveillance (section 5).

9 Exam corner

Past questions (MUHS, 2016–2026)

Long cases

Automobile worker with vibration and numb hands (S16); sand-industry worker with breathlessness (S16); clerk with carpal tunnel syndrome (W17); coal miner with breathlessness and cough (S21).

7 marks

Workplace hazards and risk assessment tools (S18); influence of physical agents (W20); contributing factors for industrial injury (S22); cumulative trauma disorder and one common CTD (W22, S24).

3 marks

Causative factors for CTD (W16); byssinosis (S18); asbestosis (W18); three hazards of chemical agents (S19); sequelae of pneumoconiosis (W20); types of occupational hazards (W25).

Keywords to underline

physical, chemical, biological, mechanical, psychosocial heat stroke noise-induced hearing loss vibration white finger pneumoconiosis silico-tuberculosis Monday chest tightness mesothelioma repetition, force, posture Phalen's and Tinel's RULA, REBA hierarchy of controls

Answer skeleton: cumulative trauma disorder (7 marks)

1. **[Definition and factors, 2]** Repeated microtrauma; work and personal factors.
2. **[Stages and examples, 1]** Three stages; list of common CTDs.
3. **[One CTD in detail, 3]** Carpal tunnel syndrome: anatomy, features, tests, treatment.
4. **[Prevention, 1]** Ergonomic redesign, breaks, rotation, early reporting.

Draw these

- The hierarchy of controls (section 6)
- The carpal tunnel in cross-section: flexor retinaculum, median nerve, tendons

Viva questions

1. Name the five types of occupational hazards.
2. What is "Monday chest tightness"?
3. Which pneumoconiosis increases the risk of tuberculosis?
4. What is the vascular component of hand-arm vibration syndrome?
5. How do you perform Phalen's test?
6. Why is PPE the last step in the hierarchy of controls?

Linked practical spots

- Perform Phalen's and Tinel's tests, a grip dynamometer test and two-point discrimination; score a posture with RULA; identify a cock-up splint.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Byssinosis. (S18)**

Answer

Lung disease from cotton, flax or hemp dust in textile mills (carding and blowing rooms). Chest tightness and breathlessness on the first day back at work, improving through the week; later chronic airflow obstruction. Schilling grades ½, 1, 2, 3. Prevention: dust control, ventilation, respirators, periodic spirometry, job transfer; PT: breathing exercises, airway clearance, exercise training.

3 m **Enumerate causative factors for cumulative trauma disorders. (W16)**

Answer

Work: repetition, force, awkward or static posture, vibration, contact pressure, cold, insufficient rest, piece-rate pressure. Personal: age, female sex, pregnancy, diabetes, hypothyroidism, rheumatoid arthritis, obesity, previous injury, low fitness, smoking, stress.

7 m **Write a note on the influence of physical agents on industrial workers' health. (W20)**

Answer
Heat (cramps, exhaustion, heat stroke), cold (frostbite, hypothermia), light (eye strain, accidents, miner's nystagmus), noise (hearing loss at 4 kHz, stress; 90 dBA for 8 hours), vibration (HAVS, back pain), ultraviolet (welder's flash, cataract, skin cancer), ionising radiation (cancer, sickness; 20 mSv a year limit). For each, one or two prevention measures following the hierarchy of controls.

LAQ **40-year-old coal miner for 15 years with breathlessness and occasional cough for a year: task analysis, functional diagnosis, preventive measures. (5 + 5 + 5, S21)**

Answer
Task analysis: underground environment, postures (stooping, kneeling), tools (drills: dust and vibration), lifting and shovelling, heat, noise, poor light, shifts, accident risks. Functional diagnosis (ICF): coal workers' pneumoconiosis with chronic bronchitis; reduced exercise tolerance, cough; limited heavy work and climbing; risk to employment; dusty environment, smoking. Prevention: elimination and engineering (wet drilling, sprays, ventilation, extraction), administrative (rotation, exposure limits, surveillance with X-ray and spirometry), PPE (respirators); smoking cessation; PT for breathing and fitness.

MCQ **Mesothelioma is linked to exposure to:**

A. Coal dust
B. Cotton dust
C. Asbestos
D. Sugarcane bagasse

Answer
C.

MCQ **The most effective level in the hierarchy of controls is:**

A. PPE
B. Administrative controls
C. Engineering controls
D. Elimination

Answer
D.

MCQ **Noise-induced hearing loss first appears at about:**

A. 250 Hz
B. 1 kHz
C. 4 kHz
D. 16 kHz

Answer
C.

Last recall check. List the five hazard types, one effect and one control for each physical agent, the four pneumoconioses with their dusts, the CTD stages, and the hierarchy of controls.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Community Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. National Institute for Occupational Safety and Health. Hierarchy of controls.
3. Directive 2002/44/EC of the European Parliament and of the Council on the minimum health and safety requirements regarding exposure of workers to vibration, Article 3.
4. Gemne G, et al. The Stockholm Workshop scale for the classification of cold-induced Raynaud's phenomenon in the hand-arm vibration syndrome. *Scand J Work Environ Health* 1987;13:275–8; Brammer AJ, Taylor W, Lundborg G. Sensorineural stages of the hand-arm vibration syndrome. *Scand J Work Environ Health* 1987;13:279–83.
5. Schilling grading of byssinosis (grades ½ to 3), as used in cotton-mill studies in *Occupational and Environmental Medicine*.
6. McAtamney L, Corlett EN. RULA: a survey method for the investigation of work-related upper limb disorders. *Appl Ergon* 1993;24:91–9; Hignett S, McAtamney L. Rapid entire body assessment (REBA). *Appl Ergon* 2000;31:201–5.
7. International Commission on Radiological Protection. Publication 103: the 2007 recommendations (occupational dose limits).
8. Tamil Nadu Factories Rules, 1950: permissible exposure to noise (8 hours at 90 dBA), under the Factories Act, 1948.

6(b iv) Mechanical and psychological stress at work: desk workers, drivers, standing jobs, labourers and job stress

> This is the most asked part of the Community paper: 17 of 20 papers, with 16 long cases about an office worker, driver, teacher, goldsmith, watchman, labourer or tailor in pain. Every answer follows the same path: analyse the job, find the posture, load or stress that causes the problem, then treat the person and change the work.

Syllabus	Ch 6(b iii c, iv) Mechanical hazards and stresses: sedentary table work (executives, clerks), inappropriate seating (vehicle drivers), constant standing (watchmen, defence forces, surgeons), over-exertion in labourers, psychological hazards and stress management
Priority	**Tier 1.** In 17 of 20 papers (2016–2026): 16 long cases, 12 seven-mark and 3 three-mark questions.
Asked as	LAQ W16 · S17 · W17 · S19 ×2 · S20 · S21 · W21 · W22 · S23 · S24 ×2 · W24 · S25 · W25 · S26 · 7m S16 · W17 · S18 · W18 · S19 · S20 · S21 · W21 · W22 · W24 · W25 · S26 · 3m W18 · W21 · S24
Usual split	15 (functional diagnosis or task analysis 5, goals 5, management or prevention 5; or 5 + 10; or 7 + 8); 7 (causes, prevention and management of one occupation or of job stress)
Checked against	OSHA computer workstation guidance; NIOSH review of prolonged standing (Waters and Dick, 2015); revised NIOSH lifting equation and its use for patient handling (Waters, 2007); EU vibration directive 2002/44/EC; American College of Surgeons surgical ergonomics recommendations (2025); meta-analysis of work-related disorders in surgeons (Epstein et al., 2018); review of work-related disorders in physiotherapists (Milhem et al., 2016); NICE guideline NG59 on low back pain (2016, updated 2026); review of exercise for office workers with neck pain (Chen et al., 2018); IARC Monograph 124 on night shift work (2019); Cochrane review of manual handling advice (Verbeek et al., 2011). Full list under Sources (p. 215). Content checked 21 October 2026.

1 How work causes pain

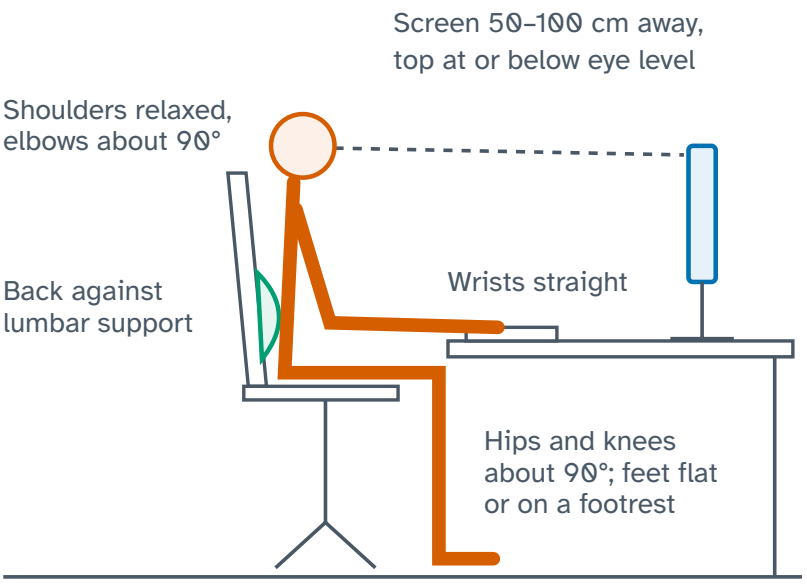
- **Work-related musculoskeletal disorders** build up from many small strains (cumulative microtrauma) rather than one injury; see cumulative trauma disorders (p. 193).
- **Physical risk factors:** awkward or static posture held for long; repetition; force (lifting, gripping); vibration; contact pressure; cold; too little rest.
- **Personal factors:** low fitness, obesity, previous injury, smoking, age.
- **Psychosocial factors:** high demands with little control, tight deadlines, monotony, poor support, job insecurity; they raise muscle tension and the chance that pain persists.
- **The physiotherapist's method:** job and task analysis (see ability management (p. 179)), find the risk factors, treat the impairments, change the workstation or the way of working, and teach self-management and fitness.

2 Sedentary desk work and computer users

Executives, clerks, bank staff, computer operators and engineers: long cases in W16, S21, W21, W25 and S26.

- **Problems:** neck and shoulder pain with a forward head and rounded shoulders; thoracic stiffness; low back pain from slumped sitting; wrist and hand symptoms (carpal tunnel syndrome) from keyboard and mouse use; eye strain and headache; weight gain and low fitness.

Setting up a computer workstation



Draw this for any desk-worker question. Distance and screen height follow the OSHA workstation guidance; the person is drawn in orange and the lumbar support in green.

Ergonomic advice

- **Screen:** directly in front, about an arm's length away (50–100 cm), top line at or slightly below eye level; laptops raised on a stand with a separate keyboard and mouse.
- **Chair:** height so the feet are flat and thighs level; backrest with lumbar support; armrests that let the shoulders relax.
- **Keyboard and mouse:** close, at elbow height, wrists straight; mouse beside the keyboard.
- **Document holder** beside the screen; headset for long calls instead of holding the phone with the neck.
- **Breaks:** change position often; a short movement break every 30–60 minutes; rest the eyes regularly by looking into the distance.

Physiotherapy

- Neck and shoulder strengthening (deep neck flexors, scapular retractors and lower trapezius) and general fitness: in office workers with neck pain, strengthening and general fitness training reduced pain, with moderate-quality evidence, and ergonomic changes alone had weaker support (Chen et al., 2018).
- Thoracic mobility and pectoral stretching; posture awareness; workstation exercises during the day (see workstation exercises (p. 179)).
- Pain relief in the acute phase (heat, TENS, manual therapy) only as a bridge to exercise.
- Regular physical activity, weight control and a plan for flare-ups.

3 Drivers and vehicle seating

Bus driver with cervical spondylosis (S17), tourist bus drivers (W17, 8 + 7), mechanical stresses of truck drivers (S24, 3 marks).

Stress	Effect	Advice
Prolonged sitting in a fixed posture	Low back and neck pain, stiffness, poor circulation in the legs	Break every 2 hours: walk, stretch back, hips and calves

Stress	Effect	Advice
Whole-body vibration	Low back pain and disc degeneration with long exposure	Well-maintained, suspended seat; good tyres and roads; limit hours (EU action value 0.5 m/s², limit 1.15 m/s² over 8 hours)
Poor seat and reach	Twisting, reaching, neck strain on mirrors and turning	Seat close enough to reach the pedals with knees slightly bent; backrest slightly reclined with lumbar support; headrest at head height; mirrors adjusted to avoid neck twisting
Loading, unloading, climbing in and out	Back and knee strain, falls	Use handholds and steps; do not jump down; lift with help
Long hours, night driving, irregular meals	Fatigue, sleep loss, obesity, stress, accidents	Planned sleep and rest, healthy food on the road, exercise

Psychological impairments in drivers (W17): stress from traffic and deadlines, irritability, anxiety, poor sleep and loneliness on long routes; manage with planned rest, relaxation and breathing techniques, fitness and family time.

Bus driver with cervical spondylosis (S17): neck isometrics and deep neck flexor training, thoracic mobility, posture correction, a headrest and lumbar roll, breaks with neck and shoulder stretches, pain relief in flare-ups, and a home programme.

4 Constant standing

Musculoskeletal hazards of prolonged standing (S21, 7 marks); watchman (S23) and traffic police (S24) long cases; surgeons (S26, 7 marks).

- **Effects** reported consistently: low back pain, leg pain and swelling, fatigue and muscle pain, and pooling of blood in the legs (varicose veins); foot pain and plantar heel pain (Waters and Dick, 2015).
- **Why:** static loading of the back and legs, little calf-muscle pumping, hard floors, poor footwear.
- **Prevention:** change position often and walk when possible (movement was the most effective measure); sit–stand options or a leaning stool; anti-fatigue mats; supportive cushioned footwear and insoles; a footrest to alternate feet; compression stockings for those with swelling; calf-raise and ankle-pump exercises during the shift; rest breaks.

Surgeons: ergonomic evaluation and modifications (7 marks in S26)

Surgeons hold one posture for hours, with the neck bent over the field and the arms held still. A meta-analysis of 21 studies (5,828 surgeons and interventionalists) found a pooled prevalence of pain of **35–60%**, depending on the questionnaire. Of those with a work-related disorder, **12%** needed leave, a change in practice or early retirement (Epstein et al., 2018).

Ergonomic evaluation uses the same steps as any workplace evaluation (see ability management (p. 179)):

1. **Symptom survey:** the Nordic Musculoskeletal Questionnaire asks about pain in each body region over the past 12 months and 7 days.
2. **Observe and video** whole operations: the posture of the neck, trunk and arms, how long each posture is held, and when breaks happen.
3. **Measure the set-up:** table height against the surgeon's elbow height, monitor position and height, loupes or microscope, foot pedals, floor and footwear.
4. **Score the postures:** RULA suits static neck and arm work; REBA scores the whole body (see risk assessment tools (p. 193)).
5. **Recommend** changes, equipment and layout first, then **re-evaluate**: a lower score and fewer symptoms show they worked.

Problems and modifications by type of surgery

Type	Main problems	Key modifications
Open surgery (disorders reported in 66–94%)	Neck bent forward over the field for long periods, worse with loupes and headlamps; static shoulders; prolonged standing; twisting towards the assistant	Table so the hands are near elbow height with elbows about 90°; light, well-fitted loupes angled downwards so the head stays upright; lights placed between surgeon and assistant; a step for shorter team members
Laparoscopic surgery (disorders reported in 73–100%)	Monitor to the side or too high, so the neck twists or extends; long-shafted instruments with poorly designed handles that bend the wrist; arms held out from the body; uneven standing to press foot pedals	Table lower than for open surgery, at about **0.7–0.8 × elbow height**, so the handles are near elbow level and elbows at 90–110°; monitor straight ahead in line with the forearms, top edge at eye level and centre slightly below; handle type chosen for a straight wrist; pedals in front at foot level
Microsurgery (operating microscope)	Sustained neck flexion at the eyepieces; very long static postures; heavy high-power loupes; unsupported arms, which raise neck and back strain	Sit with the feet and forearms supported (a chair with arm supports); adjust the eyepieces so the neck is near neutral; keep the wrists straight

Prevalence ranges and problems by type are from Catanzarite et al. (2018); table and monitor settings from the American College of Surgeons (2025), van Veelen et al. (2002) and van Det et al. (2009); microsurgery from Lakhiani et al. (2018) and Statham et al. (2010).

Modifications for every surgeon

- **Micro-breaks:** guided stretching breaks of **1.5–2 min** every **20–40 min** reduced pain in the neck, back, shoulders, wrists, knees and ankles. They did not lengthen the operation, and **87%** of surgeons planned to continue them (Park et al., 2017). The American College of Surgeons advises range-of-motion micro-breaks every **30–60 min**.
- **Standing:** a theatre-grade anti-fatigue mat; weight on both legs, not one; supportive shoes with arch support and a firm heel grip; compression stockings for leg swelling.
- **Instruments:** size and handle shape chosen for the surgeon's hand; ratchets or trigger locks so the hand does not squeeze for long; a power grip rather than a fingertip grip where possible; powered tools to cut hand force; no sharp edges pressing into the hand.
- **Planning:** set the table height, patient position and monitors before the case; an "ergonomics buddy" reminds the team; seated or robotic surgery may suit long cases.
- **Exercise:** deep neck flexor nods (hold **10 s, 10 times**) and scapular retraction during the day; upper trapezius, levator scapulae and pectoral stretches between cases (hold **30 s, twice each side**); back and neck strengthening and general fitness outside theatre.

5 Manual handling and labourers

Low back pain in industry (W22 and W25, 7 marks); labourer loading 50 kg bags (W22 long case); automobile worker who stoops and squats (S19).

- **Factors for low back pain in industry:** heavy, frequent or awkward lifting (bending and twisting), lifting far from the body or from the floor, whole-body vibration, prolonged sitting or standing, slips and falls; low fitness, smoking and obesity; job dissatisfaction and stress.
- **Revised NIOSH lifting equation:** under ideal conditions a load of 23 kg is the recommended limit, and it is reduced for distance from the body, height, travel, twisting, frequency and poor grip. A 50 kg bag lifted to 3 feet is well beyond it, so the task itself must change.
- **Prevention:** redesign first (lighter bags, trolleys, hoists, conveyors, loads at waist height, team lifts), then training and fitness. Training in lifting technique alone did not prevent back pain (Verbeek et al., 2011; see also safe lifting in Neuro).
- **Squatting and stooping work** (automobile workers, farm labour): knee and ankle pain, back strain; low stools or kneeling pads, rotate tasks, strengthen quadriceps and hips, stretch calves and hamstrings.

6 Other occupations asked in the papers

Worker	Main stress	Key advice
Housemaid (S16)	Stooping to sweep and mop, floor-level washing, lifting buckets	Long-handled brooms and mops, sit on a low stool instead of squatting long, carry half-filled buckets close to the body, back and core strengthening
Goldsmith (S20, W24)	Sustained neck flexion over fine work for long hours, repetitive pinch grip, poor light	Raise the work surface and use magnifiers and good lighting, sit with back support, micro-breaks, neck and upper-back exercises, nerve and tendon gliding for arm tingling
Tailor (S25)	Long sitting bent forward, foot pedal work, low activity	Chair height and back support, machine at elbow height, breaks to walk; examine for the cause of walking-limited leg pain (section 8)
Teacher (S19, S24)	Writing on a blackboard with arms raised, standing, voice use	Write at chest to eye height, use the lower part of the board or a projector, alternate sitting and standing, neck and shoulder strengthening
Air hostess (W18)	Prolonged standing, pushing trolleys, overhead lifting of bags, shift work	Push trolleys with both hands close to the body, avoid lifting bags overhead alone, supportive shoes, calf exercises, sleep planning
Physiotherapists (S19)	Manual therapy, patient handling, bending over low plinths	Height-adjustable plinths, hoists and slide sheets, body mechanics, rotate tasks, own fitness and strengthening

The housemaid, the physiotherapist and the air hostess have been asked on their own, so each has a fuller sub-section below.

Housemaid with low back ache (7 marks in S16)

Paid domestic work in India is done mostly at floor level, with simple tools, in several houses a day. In a Bengaluru survey of 408 women domestic workers, the average working day was about **6 h**, and most worked all 7 days of the week (Jebaraj et al., 2022).

- **Pain in the past 12 months** in the same survey: knees **38.2%**, upper back **31.9%**, low back **27.5%**.
- **Risk factors for her low back pain:** more than 10 years in the job and psychological distress each about doubled the odds (Jebaraj et al., 2022). Low fitness, obesity and earlier injury add to them.
- **Knees:** prolonged kneeling or squatting at work nearly doubled the odds of knee osteoarthritis (odds ratio 1.9), and obesity raised the risk much further (Coggon et al., 2000).
- **Washing work:** in women who wash utensils and clothes, symptoms were linked with the working posture, the height of the washing platform and the height of the seat (Tondre et al., 2025).

Task analysis of a housemaid's work

Task	Posture and load	Ergonomic change
Sweeping with a short broom (jhadu)	Trunk bent forward for long periods; repeated reaching	Long-handled broom; sweep close to the body and step with the broom instead of reaching
Mopping the floor with a cloth by hand	Squatting or on hands and knees, trunk bent and twisted; pressure on the knees	Long-handled mop and a bucket with a wringer; if done by hand, kneel on a pad, change sides and switch the leading hand often
Washing utensils and clothes	Deep squat on a wet floor for a long time; neck and back bent; scrubbing and wringing with force	Sit on a low wooden stool (paat) of suitable height, or raise the washing platform or use a sink; keep the work close; wring small loads
Carrying water and buckets	Lifting from the floor; load held on one side, away from the body; twisting	Half-fill buckets; carry two smaller loads, one in each hand, close to the body; a bucket on wheels; turn with the feet
Several houses a day	Long hours, few breaks, hurried work	Alternate stooped and upright tasks; short breaks between houses

Long handles, knee pads, wheeled buckets and task rotation follow the OSHA housekeeping guidance; keeping the load close to the waist and not twisting follow the UK Health and Safety Executive's handling advice.

Physiotherapy for the housemaid

- **Assess:** pain, back and hip mobility, core endurance, hamstring length and the knees; ask her to show each chore, and score the worst posture with REBA.
- **Educate:** back pain is common and rarely serious; stay active and keep working with the changes above.
- **Strengthen** (**8–12 reps, 2–3 days a week** , as in ACSM guidance (p. 11)): abdominal bracing, bridging, four-point arm and leg lifts, and sit-to-stand or wall squats for the quadriceps and hips.
- **Stretch** the hamstrings, hip flexors and calves, each held **10–30 s** ; practise bending from the hips with the back straight.
- **Walk** regularly and control weight, which also protects the knees; relaxation for stress.

Classic answer vs current practice: low back pain

Classic	Current	Write in the exam
Rest, a lumbar belt, and electrotherapy such as TENS, ultrasound or interferential therapy before exercise.	NICE (NG59) advises information to stay active and an exercise programme, with manual therapy only as part of a package that includes exercise. It says not to offer belts or corsets, TENS, ultrasound or interferential therapy.	"Education to stay active, exercise and change of the tasks; belts and electrotherapy are not recommended for low back pain (NICE)."

Ergonomic care for physiotherapists (7 marks in S19)

Physiotherapists get the same work-related disorders they treat. A review found a lifetime prevalence of **55–91%** . The low back is the commonest site, followed by the neck, upper back and shoulders (Milhem et al., 2016).

- **USA:** **45%** had low back symptoms in the past year, followed by the wrist and hand, upper back and neck. Lifting or transferring dependent patients was the factor rated most likely to cause them (Bork et al., 1996).
- **Australia:** lifetime prevalence was **91%** , and about 1 in 6 changed jobs within physiotherapy or left it. Mobilisation and manipulation were linked with thumb symptoms (Cromie et al., 2000).
- **India:** about 9 in 10 physiotherapists in a Delhi survey had pain that began after they started work. Specialty, gender, clinic furniture and the hours of patient contact were related to it (Iqbal and Alghadir, 2015).
- **Risk factors:** lifting and transferring patients, repetitive movements, awkward and static postures, treating many patients in a day, and working while injured. Low back pain is more common in women, younger therapists and rehabilitation settings (Milhem et al., 2016).

High-risk tasks and how to make them safer

Task	Why it strains	Prevention
Transfers and repositioning of dependent patients	Heavy, unpredictable load held away from the body, with bending and twisting. Even for a cooperative patient, the NIOSH equation gives a limit of about **16 kg (35 lb)** for manual handling (Waters, 2007)	Above this weight use hoists, slide sheets, transfer boards or sit-to-stand aids instead of lifting; let the patient help as much as possible; plan the move
Manual therapy	Repeated force through the thumbs and wrists; trunk bent over the patient	Plinth at the right height; use body weight with straight wrists; vary the technique and the contact point (heel of the hand instead of the thumb)
Treating on low plinths, mats or beds	Sustained stooping and kneeling; neck bent	Height-adjustable plinth: about **5–10 cm below elbow** for light work and **20–40 cm below** for heavy pressure; raise the bed before treating; kneel on a pad for floor work

Task	Why it strains	Prevention
Long, busy days	Many patients, few breaks, fatigue	A fair number of patients per day; scheduled breaks; rotate heavy and light tasks

Work heights for light and heavy work are from the Canadian Centre for Occupational Health and Safety guidance on standing work.

- **Equipment first:** knowing correct body mechanics does not by itself prevent injury, so mechanical aids for transfers are needed (Milhem et al., 2016).
- **Body mechanics** as a support: keep the patient close to your waist, take a stable wide base, bend the hips and knees a little, do not twist (turn with the feet), and keep the head up.
- **Work organisation:** a fair caseload, scheduled rest breaks, regular ergonomics training and yearly health checks (Mahmood et al., 2025); rotate tasks; report pain early and do not work through an injury.
- **Own fitness:** daily physical activity; back, core, neck, shoulder and grip strengthening; stretching between patients.
- **Self-check:** a Nordic questionnaire survey of the team, and REBA scores of transfer and treatment postures.

Air hostess (3 marks in W18)

Cabin crew lift, push, pull and reach in a narrow cabin that moves. NIOSH lists lifting, pushing and pulling, reaching, prolonged standing, cramped or twisted postures and repetition. Turbulence adds the risk of falls. In one airline study most injuries were to the back, neck or shoulders. Back injuries were linked most often with turbulence, overhead bins and service trolleys (Logie et al., 1997).

- **Bags:** avoid lifting heavy bags overhead; passengers lift their own bags or check them in; heavy bags go under the seat.
- **Trolleys:** push with both hands, close to the body, using body weight; keep trolleys maintained and report broken wheels.
- **Turbulence:** stop service, secure the trolleys and sit with the belt fastened; use handholds when moving.
- **Galley:** store often-used items where they need no overhead lifting or deep bending.
- **Standing and long duty days:** supportive shoes without high heels, calf raises, compression stockings if the legs swell; a sleep plan for irregular schedules and time-zone changes.
- **Fitness:** back, core and upper-body strengthening.

7 Psychological hazards, job stress and shift work

Asked for 7 marks in W17, S18 and W18 (stress and psychological hazards) and in S20, W21 and W24 (the physiotherapist's role in stress management), and 3 marks in W21 (shift work).

- **Causes:** high demands with little control over the work (the demand–control model); great effort with little reward (effort–reward imbalance); monotony and job dissatisfaction; anxiety about finishing work on time and to quality; long hours and shift work; poor relationships, harassment, insecurity.
- **Effects:** muscle tension and pain, headache, fatigue, poor sleep, raised blood pressure, anxiety, depression, more errors and accidents, absence.
- **Prevention at work:** reasonable workload and deadlines, more control and variety, breaks, support from supervisors, clear roles, safe workplaces.

The physiotherapist's role in stress management

- **Relaxation:** progressive muscle relaxation (Jacobson: tense then release each muscle group), physiological relaxation (Laura Mitchell: move into the opposite of the stress posture), diaphragmatic breathing.
- **Exercise:** regular aerobic exercise and stretching programmes at work; yoga.
- **Ergonomic and task changes** that remove physical strain, and education on posture, sleep and pacing.

- Referral to a counsellor or doctor for anxiety or depression.

Effects of shift work (W21)

- Disturbed body clock: poor sleep, fatigue, more accidents.
- Digestive problems, weight gain and metabolic and heart risk; night shift work is classified as "probably carcinogenic" (IARC, 2019).
- Strain on family and social life.
- **Advice:** forward-rotating shifts, protected sleep in a dark quiet room, regular meals, exercise, avoiding long runs of night shifts.

8 Worked long case: a computer operator with neck pain

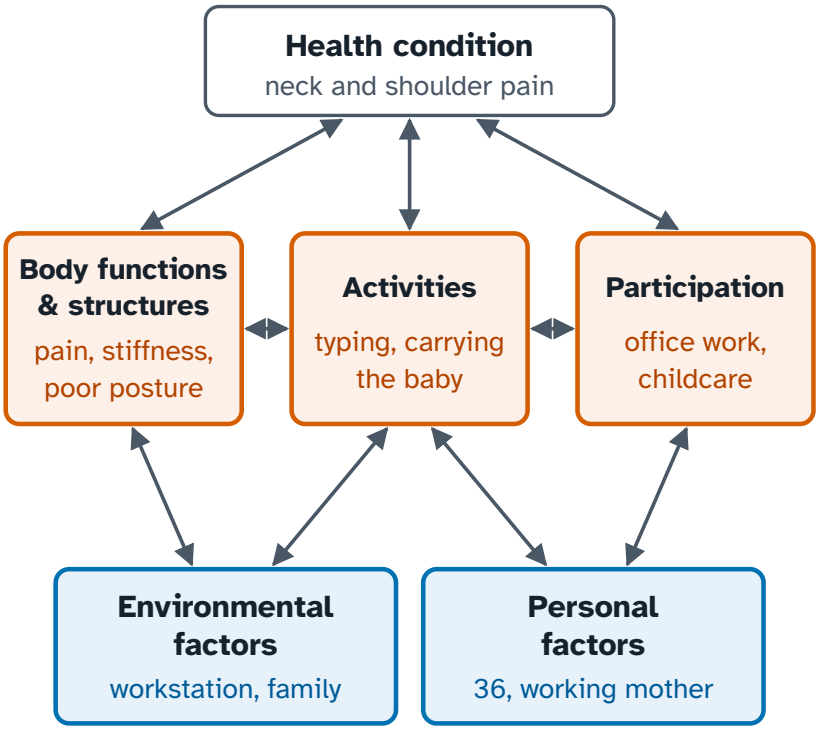
From a real paper (S21, 5 + 5 + 5 marks)

> "A 36 yrs old married female with 1 yr old son, computer operator by occupation since last 3 yrs, complains of neck and shoulder pain since last six months. Discuss functional diagnosis, short term and long term goals of management."

Spot the clues before you write

computer operator for 3 years: static sitting, forward head, keyboard work. 1-year-old son: lifting and carrying, feeding postures, broken sleep, little time for exercise. neck and shoulder pain for six months: persistent, so posture, strength and stress all matter. married, 36: working mother with two jobs, home and office.

The ICF model filled for this case



Draw the six boxes with two-way arrows, then write two or three points from the case in each box.

ICF component	From this case	Clinical reasoning
Health condition	Work-related neck and shoulder pain for six months (mechanical neck pain; rule out radiculopathy)	Static posture and repeated keyboard work over 3 years

ICF component	From this case	Clinical reasoning
Body functions and structures	Neck and upper trapezius pain and tenderness, reduced neck range, forward head and rounded shoulders, weak deep neck flexors and scapular muscles, tight pectorals, fatigue and poor sleep	Sustained neck flexion overloads posterior neck muscles and joints; the baby adds carrying and night waking
Activity limitations	Sitting and typing for long, looking down at the screen or phone, lifting and carrying the child, feeding	Static loading and lifting provoke pain
Participation restrictions	Full working day, childcare, rest and leisure	Pain and fatigue across both roles
Environmental factors	**Barriers:** unadjusted workstation, long hours, no breaks. **Facilitators:** family support, employer willing to adjust the desk	Workstation change removes the cause
Personal factors	36-year-old working mother with little time for herself	Short home exercises that fit her day work best

> **One-line functional diagnosis**
>
> A 36-year-old computer operator and mother of a 1-year-old with work-related mechanical neck and shoulder pain for six months, with poor posture and weak neck and scapular muscles, limiting long typing and childcare.

Goals

Problem	Short-term (2–4 weeks)	Long-term (2–3 months)
Pain	Pain reduced from 6 to 3 out of 10 at work	Pain-free full working day
Posture and strength	Corrected sitting posture; deep neck flexor hold improving	Good posture held through the day; strong neck and scapular muscles
Workstation	Workstation adjusted; breaks every 30–60 minutes	Self-checks her own set-up
Childcare and fitness	Safe lifting and feeding positions	Regular exercise 3 times a week; no recurrence

Management

- **Pain relief** (first 2 weeks): heat, TENS, gentle mobilisation, as a bridge to exercise.
- **Exercise:** deep neck flexor training (chin nods), scapular retraction and lower trapezius strengthening with bands, neck and thoracic mobility, pectoral stretching; progressing to resisted shoulder and neck strengthening and general fitness.
- **Ergonomics:** the workstation set-up in section 2, a headset, micro-breaks with 2–3 workstation exercises.
- **Childcare:** lift the child close to the body with knees bent, feed with the back supported and the baby on a pillow, use a carrier that spreads the load.
- **Health promotion:** sleep, stress management (breathing, relaxation), walking or yoga, weight and posture awareness.

Practice cases from the papers: W21 computer operator with neck and low back pain whose husband is away; S26 computer engineer (with ergonomic modifications); W16 accounts manager (level of CTD and job analysis); S19 overweight teacher at a blackboard; S20 and W24 goldsmiths; S24 teacher with cervical spondylosis; W22 labourer with sciatica; S23 watchman and S24 traffic policeman; S25 tailor who cannot walk more than 15 minutes (examine for spinal stenosis or vascular claudication: lumbar extension tests, foot pulses and ankle–brachial index, neurological examination).

9 Exam corner

Past questions (MUHS, 2016–2026)

Desk workers, long cases

Accounts manager with neck and back pain: CTD level, job analysis, PT (W16); computer operator with neck and shoulder pain (S21, W25); computer operator with neck and low back pain, with health promotion and ergonomics (W21); computer engineer with neck and back pain and ergonomic changes (S26).

Drivers

Bus driver with cervical spondylosis: PT and ergonomic advice (S17, LAQ); tourist bus drivers: physical and psychological impairments and management (W17, LAQ); mechanical stresses of truck drivers (S24, 3 marks).

Standing jobs

Hazards of prolonged standing (S21, 7 marks); work hardening, environmental stresses and the watchman (S23, LAQ) or traffic police (S24, LAQ); ergonomics for a surgeon (S26, 7 marks).

Lifting and other jobs

Housemaid with back ache (S16, 7 marks); low back pain in industry (W22, W25, 7 marks); labourer loading bags (W22, LAQ); automobile worker who squats (S19, LAQ); teacher at a blackboard (S19, LAQ) or with cervical spondylosis (S24, LAQ); goldsmith (S20, W24, LAQ); tailor (S25, LAQ); air hostess (W18, 3 marks); physiotherapists (S19, 7 marks).

Stress

Psychological hazards (W17); stress at work (S18); job stress (W18); PT in industrial stress (S20, W21, W24); shift work (W21, 3 marks).

Keywords to underline

job and task analysis, static posture, repetition and force, forward head posture, workstation set-up, micro-breaks, whole-body vibration, prolonged standing, NIOSH lifting limit, hoists and slide sheets, table near elbow height, redesign before training, demand and control, relaxation techniques, health promotion

Answer skeleton: an occupational long case (15 marks)

1. **[Task analysis or functional diagnosis, 5]** Job demands (posture, load, hours, tools, environment), ICF and one-line diagnosis.
2. **[Goals, 5]** Short- and long-term, measurable, linked to work and home.
3. **[Management and prevention, 5]** Pain relief, exercise with doses, ergonomic changes at the workstation, breaks and pacing, health promotion, return-to-work plan.

Draw these

- The computer workstation (section 2)
- A driver's seat with lumbar support, headrest and knee angles
- Correct and incorrect lifting with the load close to the body
- A surgeon at the table with the hands near elbow height and the monitor straight ahead

Viva questions

1. How far away and how high should a monitor be?
2. What are the hazards of prolonged standing, and how are they prevented?
3. What is the recommended weight limit in the revised NIOSH lifting equation?

4. Name two models that explain job stress.
5. Describe Jacobson's progressive muscle relaxation.
6. What ergonomic advice would you give a truck driver?
7. Above what weight should a patient be moved with a hoist or slide sheet rather than lifted?
8. How should the table and monitor be set for laparoscopic surgery?

Linked practical spots

- Assess and correct a classmate's sitting posture at a desk; teach chin nods and scapular setting; demonstrate safe lifting of a box.

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **Mechanical stresses faced by truck drivers. (S24)**

Answer
Prolonged sitting in a fixed posture; whole-body vibration transmitted through the seat; poor seat and reach causing twisting and neck strain; loading and unloading heavy goods and climbing in and out of the cab; long hours and night driving causing fatigue.

3 m **Ergonomic advice to an air hostess. (W18)**

Answer
Stresses: lifting, pushing and pulling trolleys, reaching overhead, prolonged standing, cramped postures, turbulence and long irregular duty days. Advice: avoid lifting heavy bags overhead (passengers lift their own, heavy bags under the seat); push trolleys with both hands close to the body and report broken wheels; stop service and sit belted in turbulence; store galley items without overhead reach; supportive shoes, calf raises, stockings if legs swell; sleep plan; back, core and upper-body strengthening.

3 m **Effects of shift work on health. (W21)**

Answer
Disturbed circadian rhythm with poor sleep and fatigue; more errors and accidents; digestive problems, weight gain and metabolic and heart risk; night shift work is probably carcinogenic (IARC 2019); strain on family and social life.

7 m **Discuss the role of physiotherapy in stress management in an industrial set-up. (S20, W21, W24)**

Answer
Define work stress and list its causes (demand–control, effort–reward, monotony, deadlines, shift work) and effects (muscle tension, pain, fatigue, sleep problems, anxiety). PT role: screening and education; relaxation (Jacobson's progressive relaxation, Mitchell's physiological relaxation, diaphragmatic breathing); workplace exercise and stretching programmes and aerobic fitness; yoga; ergonomic redesign to remove physical strain; pacing, breaks and sleep advice; referral for psychological help; evaluation of the programme (pain, absenteeism, stress questionnaires).

LAQ **A 27-year-old labourer loads 50 kg bags into a truck at about 3 feet, has back pain radiating to the right leg for 5 months, no neurological deficit, pain 3/10 on bending and loading, and smokes bidis. Functional diagnosis with ICF; goals and plan. (7 + 8, W22)**

Answer

ICF (7): health condition: mechanical low back pain with referred leg pain; body functions: pain on flexion and loading, possible reduced hamstring flexibility and core endurance; activities: bending, lifting, loading; participation: daily wage work; environment: 50 kg bags with no lifting aids (barrier), co-workers (possible team lifting); personal: young smoker. **Goals and plan (8):** short term: pain-free bending, correct lifting habits; long term: full work without recurrence. Plan: education and reassurance, graded activity, core and hip strengthening, hamstring flexibility, aerobic fitness, smoking cessation advice, and above all task change: 25 kg bags or two-person lifts, trolleys or a ramp, loading from waist height, rotating tasks.

MCQ **The top of a computer screen should be:**

A. Well above eye level

B. At or slightly below eye level

C. At chest level

D. At desk level

Answer

B.

MCQ **For prolonged standing, the most effective preventive measure was found to be:**

A. Standing still on a hard floor

B. Regular movement and changes of position

C. Tighter shoes

D. Longer shifts with fewer breaks

Answer

B. Mats, sit–stand options, footwear and stockings also help.

MCQ **For laparoscopic surgery, the operating table is best set at about:**

A. 1.2 × elbow height

B. Shoulder height

C. 0.7–0.8 × elbow height

D. Knee height

Answer

C. Lower than for open surgery, so the instrument handles are near elbow level.

MCQ **The load constant of the revised NIOSH lifting equation is:**

A. 10 kg

B. 23 kg

C. 40 kg

D. 50 kg

Answer

B.

Last recall check. Draw the workstation, list the stresses and advice for drivers, standing workers and labourers, and write five ways a physiotherapist manages job stress.

Sources

1. MUHS BPTh syllabus (2012) and MUHS IV BPTh Community Physiotherapy question papers 2016–2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page).
2. Occupational Safety and Health Administration. Computer workstations eTool: monitors.
3. Waters TR, Dick RB. Evidence of health risks associated with prolonged standing at work and intervention effectiveness. *Rehabil Nurs* 2015;40:148–65.
4. Waters TR, et al. Revised NIOSH equation for the design and evaluation of manual lifting tasks. *Ergonomics* 1993;36:749–76.
5. Directive 2002/44/EC of the European Parliament and of the Council on the minimum health and safety requirements regarding exposure of workers to vibration, Article 3.
6. American College of Surgeons, Surgical Ergonomics Committee. Surgical ergonomics recommendations. Chicago: ACS; 2025.
7. Epstein S, Sparer EH, Tran BN, et al. Prevalence of work-related musculoskeletal disorders among surgeons and interventionalists: a systematic review and meta-analysis. *JAMA Surg* 2018;153(2):e174947.
8. Catanzarite T, Tan-Kim J, Whitcomb EL, Menefee S. Ergonomics in surgery: a review. *Female Pelvic Med Reconstr Surg* 2018;24(1):1–12 (checked from the abstract).
9. van Veelen MA, Kazemier G, Koopman J, et al. Assessment of the ergonomically optimal operating surface height for laparoscopic surgery. *J Laparoendosc Adv Surg Tech A* 2002;12(1):47–52.
10. van Det MJ, Meijerink WJ, Hoff C, et al. Optimal ergonomics for laparoscopic surgery in minimally invasive surgery suites: a review and guidelines. *Surg Endosc* 2009;23(6):1279–85.
11. Park AE, Zahiri HR, Hallbeck MS, et al. Intraoperative "micro breaks" with targeted stretching enhance surgeon physical function and mental focus: a multicenter cohort study. *Ann Surg* 2017;265(2):340–6.
12. Lakhiani C, Fisher SM, Janhofer DE, Song DH. Ergonomics in microsurgery. *J Surg Oncol* 2018;118(5):840–4 (checked from the abstract).
13. Statham MM, Sukits AL, Redfern MS, et al. Ergonomic analysis of microlaryngoscopy. *Laryngoscope* 2010;120(2):297–305.
14. Kuorinka I, et al. Standardised Nordic questionnaires for the analysis of musculoskeletal symptoms. *Appl Ergon* 1987;18:233–7; McAtamney L, Corlett EN. RULA. *Appl Ergon* 1993;24:91–9; Hignett S, McAtamney L. REBA. *Appl Ergon* 2000;31:201–5.
15. Milhem M, Kalichman L, Ezra D, Alperovitch-Najenson D. Work-related musculoskeletal disorders among physical therapists: a comprehensive narrative review. *Int J Occup Med Environ Health* 2016;29(5):735–47.
16. Bork BE, Cook TM, Rosecrance JC, et al. Work-related musculoskeletal disorders among physical therapists. *Phys Ther* 1996;76(8):827–35.
17. Cromie JE, Robertson VJ, Best MO. Work-related musculoskeletal disorders in physical therapists: prevalence, severity, risks, and responses. *Phys Ther* 2000;80(4):336–51.
18. Iqbal Z, Alghadir A. Prevalence of work-related musculoskeletal disorders among physical therapists. *Med Pr* 2015;66(4):459–69.
19. Mahmood T, Afzal W, Mahmood W, et al. Frequency of work-related musculoskeletal disorders among physical therapists: a systematic review. *Rev Bras Med Trab* 2025;23(3):e20251481.
20. Waters TR. When is it safe to manually lift a patient? *Am J Nurs* 2007;107(8):53–8.
21. Canadian Centre for Occupational Health and Safety. Working in a standing position: basic information (OSH Answers).
22. Jebaraj P, Vellaiputhiyavan K, Mohan V, Alex RG. Prevalence and risk factors for work-related musculoskeletal disorder among female domestic workers employed in private homes in South India: a cross-sectional study. *Indian J Med Sci* 2022;74(2):72–8.
23. Tondre S, Pokale RS, Gawande KR, Raja NN. General guidelines for utensil/clothes washing workstation to reduce risk of musculoskeletal disorders among Indian domestic workers. *EPJ Web Conf* 2025;328:01067 (checked from the abstract).
24. Coggon D, Croft P, Kellingray S, et al. Occupational physical activities and osteoarthritis of the knee. *Arthritis Rheum* 2000;43(7):1443–9.
25. Occupational Safety and Health Administration. Hospitals eTool: housekeeping, work-related musculoskeletal disorders.
26. Health and Safety Executive (UK). Manual handling at work: good handling technique.
27. National Institute for Health and Care Excellence. Low back pain and sciatica in over 16s: assessment and management (NG59). 2016, last updated July 2026.
28. Garber CE, et al. American College of Sports Medicine position stand: quantity and quality of exercise for developing and maintaining cardiorespiratory, musculoskeletal, and neuromotor fitness in apparently healthy adults. *Med Sci Sports Exerc* 2011;43:1334–59.
29. National Institute for Occupational Safety and Health. Aircrew and musculoskeletal disorders (aviation safety web page).

30. Logie A, VanDerDoe L, Ryan A. Musculoskeletal injury prevention project: report on the flight attendant group. Workers' Compensation Board of British Columbia, 1997 (checked only through the summary in Flight Safety Foundation, *Cabin Crew Safety* 2002;37(4)).
31. Chen X, et al. Effectiveness of exercise in office workers with neck pain: a systematic review and meta-analysis. *Phys Ther* 2018 (PMC6093121).
32. IARC Monographs Volume 124: Night shift work (2019).
33. Verbeek JH, et al. Manual material handling advice and assistive devices for preventing and treating back pain in workers. *Cochrane Database Syst Rev* 2011;(6):CD005958.

7 The project synopsis: choosing a topic, writing each section, and presenting and defending it in the practical exam

> The synopsis carries 25 of the 80 practical marks (30 in the preliminary exam). It is the plan of the study you will do during internship, on any physiotherapy topic, under a teacher's guidance, and you present and defend it before the examiners. A clear question, a design that fits it, and honest answers about its limits score more than a complicated plan you cannot explain.

Syllabus	Ch 7 Project synopsis: select a study under a teacher's guidance, decide its methodology (to be carried out in internship), and present and defend the synopsis in the university practical examination; introduction, aims and objectives, methods and methodology, and review of literature expected
Priority	**Essential.** 25 practical marks (30 in the preliminary exam).
Asked as	Practical exam: synopsis presentation and defence
Usual split	Introduction, aims and objectives, methods and methodology, review of literature; presentation and viva
Checked against	MUHS BPTh syllabus (2012) practical scheme; ICMR National Ethical Guidelines for Biomedical and Health Research Involving Human Participants (2017); Clinical Trials Registry–India requirements; FINER criteria (Hulley et al., Designing Clinical Research). Full list under Sources (p. 221). Content checked 21 October 2026.

1 What the synopsis is and how it is marked

- **Synopsis:** a short, complete plan of a research study: what you will ask, why, and exactly how you will answer it. You will carry out the study in internship.
- **Topic:** any area of physiotherapy (musculoskeletal, neuro, cardiorespiratory or community).
- **Marks:** 25 in the university practical (with the long case 50 and journal 5); 30 in the preliminary exam. The syllabus expects introduction, aims and objectives, methods and methodology, and review of literature.
- **What examiners look for:** a clear and answerable question, a design that fits it, correct definitions of variables, feasible sample and setting, valid outcome measures, a sensible analysis plan, ethics, and your ability to explain every choice.

2 Choosing the research question

FINER criteria for a good question

Letter	Meaning
F: Feasible	Enough participants, time (internship months), skills, equipment and money
I: Interesting	To you, your guide and clinicians
N: Novel	Adds something: a new population, setting (Indian, rural), outcome or comparison
E: Ethical	Safe, with consent; approvable by the ethics committee
R: Relevant	Useful for practice, policy or future research

- **PICO** shapes an intervention question: Population, Intervention, Comparison, Outcome. Example: in desk workers with neck pain (P), does deep neck flexor training (I) compared with stretching (C) reduce Neck Disability Index scores at 4 weeks (O)?
- **Good internship-sized topics:** prevalence surveys (for example low back pain in bus conductors, incontinence in postnatal women, falls risk in a senior citizens' club), correlation studies (for example BMI and knee pain), and small comparative studies of two physiotherapy techniques.

- **Avoid:** very rare conditions, long follow-up, expensive equipment, vague titles ("a study on back pain").

3 The sections of a synopsis

Section	What to write
Title	Specific: design, intervention or exposure, population, outcome. "Effect of X versus Y on Z in [population]: a randomised controlled trial"
Introduction	The problem and how common it is (with numbers), its impact, current treatments, the gap in knowledge
Need for the study	Why this study, here and now; who will benefit
Review of literature	10–20 recent, relevant studies summarised (author, year, design, sample, result), grouped by theme, ending with the gap your study fills
Research question and hypotheses	Null hypothesis (no difference or no association) and alternative hypothesis
Aim and objectives	One broad aim; specific, measurable objectives ("to find the effect of X on pain using VAS")
Methodology	Section 4
Ethics	Institutional ethics committee approval, written informed consent in the local language, confidentiality, right to withdraw, risks and how they are handled; registration with the Clinical Trials Registry–India for a trial
Timeline and budget	Gantt chart of months (approval, data collection, analysis, writing); costs
References	One style throughout (usually Vancouver, numbered)
Annexures	Consent form, information sheet, assessment proforma, questionnaires, exercise protocol

4 Methodology

Study designs

Design	Use	Example
Cross-sectional survey	Prevalence and associations at one time	Prevalence of neck pain in IT workers
Correlational	Relationship between two variables	BMI and knee pain in homemakers
Case–control	Compare past exposures of people with and without a condition	Occupational factors in women with and without prolapse
Cohort	Follow exposed and unexposed people forward	Falls over 6 months in seniors with and without fear of falling
Quasi-experimental (pre–post)	Effect of an intervention without randomisation	Pelvic floor training in a group of postnatal women
Randomised controlled trial	Compare interventions with random allocation; strongest for cause and effect	Two exercise programmes for desk workers

Population and sample

- **Setting and population:** where and who (for example women 3–6 months postpartum attending a PHC).
- **Inclusion and exclusion criteria:** clear, measurable, with reasons.
- **Sampling:** probability (simple random, systematic, stratified, cluster) or non-probability (convenience, purposive, snowball); say which and why; random allocation method (sealed envelopes, computer list) for trials.
- **Sample size:** calculated from a previous study's result (prevalence, or mean difference and SD) with significance 5% and power 80%, using a formula or software, plus 10–20% for drop-outs; state the source.

Variables and outcome measures

- **Independent variable** (intervention or exposure) and **dependent variables** (outcomes); control likely confounders.
- **Outcome measures** that are valid and reliable, with references: VAS or NPRS (pain), goniometer (range), Neck Disability Index, Oswestry, Berg, TUG, 6MWT, ICIQ, PERFECT, questionnaires validated in the local language.
- **Procedure:** step by step, so another person could repeat it: assessment, intervention dose (frequency, intensity, time, type, duration of the programme), when outcomes are measured, who measures (blinding).

Statistical analysis plan

Question	Test (normal data)	Test (non-normal or ordinal)
Before vs after in one group	Paired t-test	Wilcoxon signed-rank
Two independent groups	Unpaired t-test	Mann–Whitney U
Three or more groups	ANOVA	Kruskal–Wallis
Two categorical variables	Chi-square test	Fisher's exact test (small numbers)
Relationship between two measures	Pearson correlation	Spearman correlation

Describe data with mean and SD (or median and range) and percentages; significance at $p < 0.05$; name the software.

5 Presenting and defending the synopsis

The presentation (about 8–10 minutes)

1. Title, your name, guide.
2. The problem in two or three lines with one number.
3. Gap from the literature (three or four key studies).
4. Aim, objectives, hypotheses.
5. Design, setting, sample and sample size, criteria.
6. Outcome measures and procedure (a flowchart of the study).
7. Analysis, ethics, timeline.
8. Expected outcome and how it will help practice.

Questions examiners often ask

- Why this topic? What is new about it?
- Why this design and not another? What are its limitations?
- How did you calculate the sample size?
- Define your variables. Which is dependent and which independent?
- Why this outcome measure? Is it valid and reliable? How exactly will you measure it?
- How will you reduce bias (randomisation, blinding, standard instructions)?
- What is your null hypothesis? Which test will you use, and why?
- How will you take consent? What if a participant is harmed or drops out?
- Can you finish it in internship? Where will you get the participants?

Tips

- Know every reference you cite (author, year, finding).
- Admit limits honestly and say how you will handle them.
- Carry the consent form, proforma and questionnaire.
- Keep slides simple: one idea per slide, large text, a flowchart of the method.

- Your own work: write it yourself with your guide; copying another student's synopsis is easy for examiners to detect and fails the viva.

6 Exam corner

Keywords to underline

FINER PICO null and alternative hypothesis aim and objectives inclusion and exclusion criteria sample size with power 80% probability and non-probability sampling valid and reliable outcome measures paired and unpaired t-test informed consent ethics committee CTRI

Draw these

- A study flowchart: assessed for eligibility, excluded, randomised, two groups, outcomes, analysis
- A Gantt chart of your internship months

Viva questions

1. What does FINER stand for?
2. Difference between an aim and an objective?
3. Difference between reliability and validity?
4. When would you use a Mann–Whitney U test?
5. What is a confounding variable? Give one from your study.
6. What is informed consent, and what must the information sheet contain?

Test yourself

Cover each answer and write yours on paper first, then check.

3 m **What makes a good research question?**

Answer
FINER: feasible, interesting, novel, ethical, relevant; framed with PICO (population, intervention, comparison, outcome); specific and measurable within the time available.

7 m **List the sections of a research synopsis with one line on each.**

Answer
Title; introduction; need for the study; review of literature; research question and hypotheses; aim and objectives; methodology (design, setting, population, criteria, sampling, sample size, variables, outcome measures, procedure, statistics); ethics; timeline and budget; references; annexures (consent, proforma, questionnaires).

MCQ **To compare pain scores before and after treatment in the same group (normal data), you would use:**

A. Unpaired t-test
B. Paired t-test
C. Chi-square test
D. ANOVA

Answer
B.

MCQ **The design that best shows cause and effect between a treatment and an outcome is:**

A. Cross-sectional survey

B. Case report

C. Randomised controlled trial

D. Correlational study

Answer
C.

MCQ **Choosing every 5th patient from a register is:**

A. Convenience sampling

B. Systematic sampling

C. Snowball sampling

D. Purposive sampling

Answer
B.

Last recall check. Write your own synopsis on one page: title, question in PICO, design, sample size method, two outcome measures, the statistical test and the ethics steps, then answer the nine examiner questions aloud.

Sources

1. MUHS BPTh syllabus (2012): Community Physiotherapy project synopsis and practical examination scheme.
2. Hulley SB, Cummings SR, Browner WS, Grady DG, Newman TB. Designing Clinical Research. 4th ed. Philadelphia: Lippincott Williams & Wilkins; 2013 (FINER criteria).
3. Indian Council of Medical Research. National Ethical Guidelines for Biomedical and Health Research Involving Human Participants. New Delhi: ICMR; 2017.
4. Clinical Trials Registry–India (CTRI): registration of clinical trials.

Exam Answer writing for the Community Physiotherapy paper

> All 20 papers from 2016 to 2026 had the same shape: 80 marks in 3 hours, five of six 3-mark questions, five of six 7-mark questions, and two long answers of 15 marks, usually a patient in a community setting. The Community paper rewards answers that move from the person to their work, home and society, so every template here ends with prevention, ergonomics or health promotion.

Syllabus	Exam skills for the whole Community Physiotherapy paper (51414)
Priority	**Essential.** Applies to every question in the paper.
Asked as	Every paper, 2016–2026
Usual split	Q1 5 × 3, Q2 5 × 7, Q3 and Q4 1 × 15 each (80 marks)
Checked against	MUHS IV BPTh Community Physiotherapy papers 2016–2026 (pattern and question counts from the session lists and question bank); MUHS syllabus (2012) printed scheme. Full list under Sources (p. 224). Content checked 21 October 2026.

1 The paper and a time plan

The syllabus prints a scheme with 20 MCQs and 5-mark questions, but it was never used in 2016–2026. The real paper:

Question	Choice and marks	Time to spend
Read and choose	Tick your choices; plan the two long answers	5 min
Q1 short answers	Any 5 of 6 × 3 = 15	About 6 min each (30 min)
Q2 short answers	Any 5 of 6 × 7 = 35	About 14 min each (70 min)
Q3 and Q4 long answers	Any 1 of 2 each × 15 = 30	About 30 min each (60 min)
Revise	Underline keywords; add doses, measurements and prevention	15 min

- About **2 minutes a mark**.
- The long answer gives its break-up (for example 5 + 5 + 5 or 3 + 5 + 7): use the same parts as your headings and give each part space in proportion to its marks.
- Choose the long case whose setting you know best (worker, woman, elder), not the shortest one.

2 What is asked most (2016–2026)

Topic	Papers (of 20)	Page
Mechanical and psychological stress at work	17, with 16 long cases	Work stress (p. 203)
Incontinence, prolapse, pelvic floor	16	Pelvic floor (p. 70)
Rehabilitation, IBR vs CBR, the team	15	Rehabilitation (p. 133)
Obesity	14, with 10 long cases	Obesity (p. 26)
Labour and postnatal care	14, with 9 long answers	Labour and postnatal (p. 52)
Physiology of ageing	14	Ageing (p. 101)

Topic	Papers (of 20)	Page
Pregnancy	13	Pregnancy (p. 43)
Fitness and exercise prescription	12	Fitness (p. 11)
Occupational hazards	12	Hazards (p. 193)
Community-based rehabilitation	11	CBR (p. 150)
Ability and disability management	10 each	Ability (p. 179), disability management (p. 186)
Health care delivery, theories of ageing, geriatric evaluation, disability	9 each	Delivery (p. 5), theories (p. 92), evaluation (p. 110), disability (p. 127)
Menopause and osteoporosis, falls	8 each	Menopause (p. 63), falls (p. 119)
Policies, elderly in India, women in India, amputee, disaster, industrial health	3–7 each	Policies (p. 144), elderly (p. 85), women (p. 37), amputee (p. 160), disaster (p. 167), industrial (p. 174)

Repeated questions are common: theories of ageing (9 papers), work conditioning and work hardening (8), IBR vs CBR (6), functional capacity evaluation (6), levels of disability prevention (5), communication barriers in the elderly (3). Prepare these as polished model answers.

3 Short answers

3 marks (about half a page)

- A definition, then four to six points as a list or a small table; one example or number; a diagram only if it is quick (the ICIDH chain, the T-score scale).
- "Enumerate" means a list; "define" needs the standard definition word for word where one exists (WHO health, ICIDH, rehabilitation, ergonomics, CBR).

7 marks (one and a half to two pages)

1. Definition or introduction (one or two lines).
2. The body: a table or numbered points, with headings that echo the question.
3. A diagram if the topic has one.
4. The physiotherapy role or prevention.
5. One line of conclusion or evidence (for example "exercise reduces falls by 23%").

4 Long answers: five templates

Type	Headings to use	Worked example
Worker in pain (desk worker, driver, labourer, goldsmith, teacher)	Job and task analysis; ICF and functional diagnosis; goals; treatment with doses; ergonomic changes; health promotion and prevention; return to work	Computer operator (p. 210), vibration worker (p. 197), sugarcane worker (p. 188)
Woman (pregnancy, postnatal, pelvic floor, menopause)	Stage and reasoning (body changes); ICF; goals; exercise and pelvic floor programme; posture and ergonomics of daily tasks; warning signs; education	Stress incontinence (p. 77), postnatal back pain (p. 57), pregnant nurse (p. 47)
Older person	Comprehensive assessment with scales; ICF; goals; strength, balance and gait; falls prevention; home and environment; institution or home; social support and schemes	Widow in an old age home (p. 114), recurrent faller (p. 122)
Weight and fitness	Measurements with working (BMI, waist, body fat, fitness tests) and interpretation; ICF; goals; FITT prescription with heart-rate target; diet and lifestyle; precautions; review	Bride-to-be (p. 30)

Type	Headings to use	Worked example
Disability and CBR	Functional diagnosis (ICF); local resources; CBR plan by matrix (health, education, livelihood, social, empowerment); aids, transport, concessions and schemes; referral and follow-up	Child with CP (p. 155), paraplegic farmer (p. 146), teacher amputee (p. 163)

- **Always include** the ICF (draw the six boxes), a one-line functional diagnosis, short- and long-term goals with numbers, treatment with doses, and a community element (ergonomics, home, schemes, prevention).
- **Theory long answers** (for example "IBR vs CBR", "disaster management cycle", "pathomechanics of balance"): use the question's break-up as headings, a diagram, a table, and the PT role.

5 Diagrams that earn marks

- The ICF model filled for the case (every long case).
- Computer workstation set-up; hierarchy of controls; worker care spectrum.
- Pelvic floor sling; diastasis recti test; stages of labour.
- Fun-to-failure steps; balance system; theories-of-ageing tree.
- Rehabilitation team; CBR matrix; ICIDH chain with levels of prevention.
- Three tiers of health care; disaster management cycle; START triage.
- Apple and pear shapes; energy systems; circuit training.

Draw in pencil, label clearly, give a title, and refer to it in the text ("as shown in the diagram").

6 Common mistakes

- Writing a clinical answer with no community part: no job analysis, ergonomics, home, schemes or prevention.
- "Strengthening exercises" without naming the exercise, dose and progression.
- Ignoring the break-up and writing one long paragraph.
- Missing measurements: BMI calculation, waist cut-offs, TUG, PERFECT, REBA scores.
- Old or wrong facts: PWD Act 1995 is replaced by the RPWD Act 2016 (21 disabilities, 4% jobs); use Asian Indian BMI cut-offs; mention current schemes.
- Forgetting red flags and precautions (pre-eclampsia, osteoporotic fracture, postural hypotension, hypoglycaemia).
- Spending too long on 3-mark questions and rushing the long answers.

7 The last week

1. **Days 1–2:** the four Tier 1 occupational pages (work stress, hazards, ability and disability management); practise one worker long case in 30 minutes.
2. **Days 3–4:** women's health (pregnancy, labour and postnatal, pelvic floor, menopause); practise one woman long case.
3. **Day 5:** geriatrics (ageing, theories, evaluation, falls, elderly in India); practise one elderly long case.
4. **Day 6:** health promotion (delivery system, fitness, obesity) and rehabilitation (disability, rehabilitation, CBR, policies, amputee, disaster).
5. **Day 7:** redraw every diagram from memory; write the repeated 7-mark answers (IBR vs CBR, theories of ageing, levels of prevention, work conditioning vs hardening, FCE, communication barriers) in outline.

Sources

1. MUHS IV BPTh Community Physiotherapy (51414) question papers, Summer 2016 to Summer 2026 (muhsonline.com, unofficial archive; Dhole Patil College of Physiotherapy past-paper page); counts from the question bank compiled for these notes.
2. MUHS BPTh syllabus (2012): Community Physiotherapy scheme of university examination.

Exam The Community practical exam: long case, project synopsis and journal

> The Community practical is 80 marks: one long case (50), the project synopsis (25) and the journal (5). The long case can come from any of five areas, so prepare a proforma that adds the community questions (job, home, family, schemes) to the usual clinical assessment, and practise presenting it in ICF order.

Syllabus	Practical scheme: long case (rehabilitation, women's health, geriatric, industrial health or health promotion) 50; project synopsis 25; journal 5 (one case each of rehabilitation, health promotion, industrial health, women's health and geriatrics, and documentation of at least one visit to an industry, geriatric home or community assessment)
Priority	**Essential.** 80 practical marks plus 20 internal assessment.
Asked as	Long case 50 · Synopsis 25 · Journal 5
Usual split	Long case 50; synopsis 25; journal 5
Checked against	MUHS BPTh syllabus (2012), Community Physiotherapy practical examination scheme, internal assessment and field work list. Full list under Sources (p. 228). Content checked 21 October 2026.

1 The marking scheme

Part	What is examined	Marks
Long case	One case from rehabilitation, women's health, geriatrics, industrial health or health promotion: assessment, ICF diagnosis, goals, plan, demonstration and viva	50
Project synopsis	Presentation and defence of the study planned for internship: introduction, aims and objectives, methods and methodology, review of literature	25
Journal	Five cases (one each of rehabilitation, health promotion, industrial health, women's health and geriatrics) and documentation of at least one visit to an industry, geriatric home or community assessment	5

Internal assessment adds 20 marks. The terminal exam has two long cases of 40 marks each; the preliminary exam has one long case of 50 and the synopsis for 30.

2 Long case: a community assessment proforma

Subjective

- Demographics: age, sex, education, marital status, family (joint or nuclear, who helps), income and socioeconomic status.
- Chief complaints in the person's words; history; past, medical, drug, obstetric and menstrual history (women); falls history (elderly).
- **Occupation:** job title, hours, shifts, postures, loads, tools, breaks, environment; job and task analysis.
- **Home and community:** floor level and stairs, toilet type (Indian or western), floor surface, lighting, distance to services, transport.
- **Lifestyle:** activity, diet, sleep, tobacco and alcohol, stress.
- The person's goals; knowledge of schemes and benefits.

Objective

- Vital signs, height, weight, BMI, waist.
- Posture, range, strength, flexibility, special tests for the region.
- Function: ADL and IADL, gait, balance (TUG, Berg), endurance (6-minute walk), fitness tests.

- Area-specific: pelvic floor (PERFECT) and diastasis recti in women; cognition (MMSE), mood (GDS), frailty in the elderly; ergonomic score (REBA, RULA) and FCE items in workers; disability percentage and aids in rehabilitation cases.
- Outcome measures to repeat later (NPRS, NDI or Oswestry, ICIQ, FES-I, Barthel).

Then

- ICF table and one-line functional diagnosis; problem list; short- and long-term goals; plan; home programme; community and workplace changes; schemes and referrals.

3 Area-specific points for the long case

Area	Add to the case	Pages
Industrial health	Job and task analysis, ergonomic score, workstation drawing, hazards, work conditioning and hardening plan, return to work	Work stress (p. 203), ability (p. 179), disability management (p. 186)
Women's health	Obstetric history, stage of pregnancy or postpartum, pelvic floor and diastasis assessment, feeding and lifting posture, safe exercise and warning signs	Pregnancy (p. 43), postnatal (p. 52), pelvic floor (p. 70), menopause (p. 63)
Geriatrics	Comprehensive assessment with scales, falls risk, home hazards, carer and living arrangement, institution or home	Evaluation (p. 110), falls (p. 119)
Health promotion	Fitness testing, BMI and body composition, risk factors, FITT prescription with heart-rate target, lifestyle plan	Fitness (p. 11), obesity (p. 26)
Rehabilitation	Disability type and percentage, aids, transport, accessibility, CBR plan, schemes and concessions, vocational plans	CBR (p. 150), policies (p. 144), amputee (p. 160)

4 Presenting the long case (5 minutes)

1. One sentence: "Mrs A, 38, homemaker with two children, leaks urine on coughing and lifting for 2 years..."
2. Key findings in ICF order: body functions and structures, activities, participation, environmental and personal factors (draw the ICF if asked).
3. The functional diagnosis in one line.
4. Two or three priority problems with short- and long-term goals.
5. The plan: treatment with doses, home programme, ergonomic or home changes, health promotion, schemes and referrals.
6. Demonstrate one test and one treatment on the patient (for example the PERFECT grading explained and pelvic floor teaching; REBA scoring and a workstation exercise; TUG and a balance exercise).

5 The project synopsis (25 marks)

- Bring two copies, the consent form and the proforma; prepare a short presentation if allowed.
- Be ready to explain the question (FINER, PICO), design, sample size, criteria, outcome measures, statistics and ethics, and to admit limitations.
- Full guidance on the project synopsis (p. 217).

6 The journal (5 marks)

- **Five cases:** one each of rehabilitation, health promotion, industrial health, women's health and geriatrics, written with assessment, ICF diagnosis, goals, intervention and outcome.
- **At least one visit documented:** industry, geriatric home or community assessment (the syllabus field work also lists UHC and PHC visits, adoption of a nearby area and surveys on antenatal care, disability, exercise and health promotion, and prevention of smoking, alcohol and drugs in youth).

- For a visit, record: date, place, purpose, people met, observations (work tasks and hazards, ergonomic problems, facilities for elders, accessibility), physiotherapy recommendations, and what you learnt.
- Neat, complete, signed and up to date; examiners may ask about any entry.

7 What examiners watch

- Introduction, consent, explanation, privacy and draping (especially in women's health cases).
- Safety: BP and warning signs, guarding the elderly, no heavy testing in pregnancy or osteoporosis.
- Practical, affordable advice that fits the person's home, job and income.
- Correct, current facts on schemes and laws; honest answers ("I would refer to the doctor") rather than guesses.

Last two weeks

- Practise the proforma on classmates and family members until it takes under 20 minutes.
- Rehearse one case from each of the five areas aloud in ICF order.
- Rehearse the synopsis presentation and the nine common examiner questions.
- Complete and check the journal; revise the visit report.

Sources

1. MUHS BPTh syllabus (2012): Community Physiotherapy practical examination scheme, internal assessment and clinical field work.

Abbreviations

1RM: One-repetition maximum

6MWT: Six-minute walk test

AAM: Ayushman Arogya Mandir

AB: Ayushman Bharat

ABM: Academy of Breastfeeding Medicine

AC: Air-conditioned

ACE: Angiotensin-converting enzyme

ACSM: American College of Sports Medicine

ADA: American Diabetes Association

ADIP: Assistance to Disabled Persons for purchase of aids and appliances

ADL: Activities of daily living

AIIMS: All India Institute of Medical Sciences

ANM: Auxiliary nurse midwife

ANOVA: Analysis of variance

APTA: American Physical Therapy Association

ASHA: Accredited Social Health Activist

ASIA: American Spinal Injury Association

ATP: Adenosine triphosphate

ATS: American Thoracic Society

AWW: Anganwadi worker

BK: Below-knee

BMI: Body mass index

BP: Blood pressure

BPL: Below poverty line

CARF: Commission on Accreditation of Rehabilitation Facilities

CBM: CBM Global (an international disability charity)

CBR: Community-based rehabilitation

CHC: Community health centre

COPD: Chronic obstructive pulmonary disease

CP: Cerebral palsy

CRPS: Complex regional pain syndrome

CSEP: Canadian Society for Exercise Physiology

CSR: Corporate social responsibility

CT: Computed tomography

CTD: Cumulative trauma disorder

CTRI: Clinical Trials Registry–India

CVR: Cardio-vascular and respiratory physiotherapy (the CVR paper)

DDMA: District Disaster Management Authority

DDRC: District Disability Rehabilitation Centre

DEIC: District Early Intervention Centre

DEXA: Dual-energy X-ray absorptiometry

DM: Diabetes mellitus

DNA: Deoxyribonucleic acid

DVT: Deep vein thrombosis

DWD: Department of Workforce Development (Wisconsin, USA)

ECT: Every contraction timed

EMG: Electromyography

ENT: Ear, nose and throat

EPDS: Edinburgh Postnatal Depression Scale

ESI: Employees' State Insurance

EU: European Union

FCA: Functional capacity assessment

FCE: Functional capacity evaluation

FEMA: Federal Emergency Management Agency (USA)

FES-I: Falls Efficacy Scale–International

FIM: Functional Independence Measure

FINER: Feasible, interesting, novel, ethical, relevant (a good research question)

FITT: Frequency, intensity, time and type

FRAX: Fracture Risk Assessment Tool

FRC: Functional residual capacity

FSH: Follicle-stimulating hormone

FVC: Forced vital capacity

GDS: Geriatric Depression Scale

HAVS: Hand–arm vibration syndrome

HDL: High-density lipoprotein (good cholesterol)

HR: Heart rate

HRR: Heart-rate reserve

IADL: Instrumental activities of daily living

IAP: Indian Academy of Pediatrics

IBR: Institution-based rehabilitation

ICDS: Integrated Child Development Services

ICF: International Classification of Functioning, Disability and Health

ICIDH: International Classification of Impairments, Disabilities and Handicaps (WHO, 1980)

ICIQ: International Consultation on Incontinence Questionnaire

ICIQ-UI: ICIQ urinary incontinence short form

ICMR: Indian Council of Medical Research

ICS: International Continence Society

ICT: Information and communication technology

ICU: Intensive care unit

IEA: International Ergonomics Association

IGF-1: Insulin-like growth factor 1

IGNOAPS: Indira Gandhi National Old Age Pension Scheme

IIPS: International Institute for Population Sciences

ILO: International Labour Organization

IQ: Intelligence quotient

IT: Information technology

JSSK: Janani Shishu Suraksha Karyakram

LAQ: Long-answer question

LH: Luteinising hormone

LPG: Liquefied petroleum gas (cooking gas)

MACT: Motor Accident Claims Tribunal

MET: Metabolic equivalent

METs: Metabolic equivalents (1 MET = resting energy use)

MMSE: Mini-Mental State Examination

MNA: Mini Nutritional Assessment

MRC: Medical Research Council

MRI: Magnetic resonance imaging

MTP: Medical Termination of Pregnancy (Act)

NCD: Non-communicable disease

NDI: Neck Disability Index

NDMA: National Disaster Management Authority

NFHS-4: National Family Health Survey, fourth round (2015–16)

NGO: Non-governmental organisation

NHS: National Health Service (UK)

NICE: National Institute for Health and Care Excellence (UK)

NIDDCP: National Iodine Deficiency Disorders Control Programme

NIOSH: National Institute for Occupational Safety and Health (USA)

NLEP: National Leprosy Eradication Programme

NPCBVI: National Programme for Control of Blindness and Visual Impairment

NPHCE: National Programme for Health Care of the Elderly

NPPCD: National Programme for Prevention and Control of Deafness

OA: Osteoarthritis

OCRA: Occupational Repetitive Actions index

OSHA: Occupational Safety and Health Administration (USA)

OT: Occupational therapy

OWAS: Ovako Working Posture Analysis System

$PaCO_2$: Arterial carbon dioxide pressure

PaO_2: Arterial oxygen pressure

PC: Phosphocreatine (creatine phosphate)

PCOD: Polycystic ovarian disease

PCOS: Polycystic ovary syndrome

PCPNDT: Pre-Conception and Pre-Natal Diagnostic Techniques (Act)

PERFECT: Power, endurance, repetitions, fast contractions, every contraction timed (pelvic floor grading)

PFM: Pelvic floor muscles

PFMT: Pelvic floor muscle training

PHC: Primary health centre

PICO: Population, intervention, comparison, outcome

PM-JAY: Pradhan Mantri Jan Arogya Yojana

PMMVY: Pradhan Mantri Matru Vandana Yojana

PMSMA: Pradhan Mantri Surakshit Matritva Abhiyan

POMA: Performance-Oriented Mobility Assessment (Tinetti)

POP-Q: Pelvic Organ Prolapse Quantification

POSH: Prevention of Sexual Harassment of women at the workplace (Act, 2013)

PPE: Personal protective equipment

PT: Physiotherapy

PTB: Patellar tendon bearing

PTSD: Post-traumatic stress disorder

PWD: Persons with disabilities

PWDVA: Protection of Women from Domestic Violence Act

QEC: Quick Exposure Check

RA: Rheumatoid arthritis

RBSK: Rashtriya Bal Swasthya Karyakram

RCI: Rehabilitation Council of India

REBA: Rapid Entire Body Assessment

ROM: Range of motion

RPE: Rating of perceived exertion (Borg scale)

RPWD: Rights of Persons with Disabilities

RULA: Rapid Upper Limb Assessment

RV: Residual volume

SACH: Solid ankle cushion heel

SCI: Spinal cord injury

SDMA: State Disaster Management Authority

SHG: Self-help group

SIJ: Sacroiliac joint

SPPB: Short Physical Performance Battery

START: Simple Triage and Rapid Treatment

SUI: Stress urinary incontinence

TB: Tuberculosis

TENS: Transcutaneous electrical nerve stimulation

TLC: Total lung capacity

TUG: Timed Up and Go test

UDID: Unique Disability ID

UHC: Urban health centre

UIP: Universal Immunisation Programme

UK: United Kingdom

UN: United Nations

UNESCO: United Nations Educational, Scientific and Cultural Organization

UNFPA: United Nations Population Fund

UTI: Urinary tract infection

UUI: Urge urinary incontinence

UV: Ultraviolet

VAS: Visual analogue scale

VC: Vital capacity

VHSND: Village Health, Sanitation and Nutrition Day

VO_2max: Maximal oxygen uptake

WHO: World Health Organization

YMCA: Young Men's Christian Association